Birla Institute of Technology & Science (BITS), Pilani
Practice School Division
Practice School – I Chronicles (Chemical, Mechanical, Infrastructure, Steel, Others Domain)

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Name: YASH PANDEY .(2021A8PS3194H) 226
PS-I station: Indian Red Cross Society - IT , Mumbai 227
Student 227
Name: HARSHAL CHANDRAKANT GUTHE(2021AAPS2926G) 227
PS-I station: Indian Red Cross Society Gujarat State Branch , Ahmedabad 228
Student 228
Name: NILESH GUPTA .(2021A3PS1699H) 228

PS-I station: Indian Red Cross Society Gujarat State Branch , Ahmedabad 228
Student 228
Name: GAURAV KHURDIA .(2021A4PS2552H) 228
PS-I station: Institute of Seismological Research (Onsite) , Gandhinagar 229
Student 229
Name: PRAKHAR KUMAR .(2020A2PS1739P) 229
PS-I station: Institute of Seismological Research (Onsite) , Gandhinagar 230
Student 230
Name: GUNNAM JAYARAM CHOWDARY .(2021A2PS3056H) 230
PS-I station: Institute of Seismological Research (Onsite) , Gandhinagar 231
Student 231
Name: MAHIN PATEL .(2021B3A41086P) 231
PS-I station: Institute of Seismological Research (Onsite) , Gandhinagar 232
Student 232
Name: MANAN GUPTA .(2021B4A41689P) 232
PS-I station: Institute of Seismological Research (Onsite) , Gandhinagar 233
Student 233
Name: HEMANK JAIN .(2021B5A12505P) 233
PS-I station: Institute of Seismological Research (Onsite) , Gandhinagar 234
Student 234
Name: SHREYAS ASHISH KESKAR(2021B5A13034G) 234
PS-I station: Institute of Seismological Research (Onsite) , Gandhinagar 235
Student 235
Name: TAVISH NATH SHARMA(2021B5A43005G) 235
PS-I station: Institute of Seismological Research (Onsite) , Gandhinagar 236
Student 236
Name: OJAL AGRAWAL(2021B5TS2054P) 236
PS-I station: IOCL - Information Services , Mumbai 236
Student 237
Name: AAYUSH RANJAN JHA .(2021B5A70895P) 237
PS-I station: IOCL - Operation LPG Operation , Delhi 238
Student 238

Name: ANUBHAV SINHA(2021A4PS2893G) 238
PS-I station: IOCL - Operation LPG Operation , Delhi 239
Student 239
Name: VAIBHAV JAIN .(2021B1A42475P) 239
PS-I station: IOCL - Operation/Maintenance , Delhi 240
Student 240
Name: KUSHAGRA SINGH .(2021A3PS2393P) 240
PS-I station: IOCL - Operation/Maintenance , Delhi 241
Student 241
Name: MRITYUNJAY GOSWAMI .(2021A4PS2291H) 241
PS-I station: IOCL - Operation/Maintenance , Delhi 242
Student 242
Name: BHASKAR .(2021B2A42761P) 242
PS-I station: IOCL - Operation/Maintenance , Panipat 243
Student 243
Name: DEVANSH MAHESHWARI .(2021AAPS0984P) 243
PS-I station: IOCL - Operation/Maintenance , Panipat 244
Student 244
Name: ROHIT BIND .(2021ABPS2719P) 244
PS-I station: IOCL - Operation/Maintenance , Vadodara 245
Student 245
Name: RUDRA MATHUR .(2021A1PS2112P) 245
PS-I station: IOCL - Operation/Maintenance , Vadodara 246
Student 246
Name: ABHAY KUMAR CHOUDHARY .(2021A1PS2595P) 246
PS-I station: IOCL - Operation/Maintenance , Vadodara 246
Student 247
Name: ABHIGYA KHANDELWAL(2021A1PS2969G) 247
PS-I station: IOCL - Operation/Maintenance , Vadodara 247
Student 247
Name: SHREYANSH SINGH .(2021A2PS2519P) 247
PS-I station: IOCL - Operation/Maintenance , Vadodara 248

Student 248
Name: SHIVAM HEMANT PACHCHIGAR(2021A3PS2228G) 249
PS-I station: IOCL - Operation/Maintenance , Vadodara 249
Student 250
Name: VRAJ MAHESHBHAI SANARIYA(2021A3PS2658G) 250
PS-I station: IOCL - Operation/Maintenance , Vadodara 251
Student 251
Name: PARTH POL(2021A4PS1556P) 251
PS-I station: IOCL - Operation/Maintenance , Vadodara 251
Student 252
Name: VARUN MANOHAR CHITLANGE(2021B1A12733P) 252
PS-I station: IOCL - Operation/Maintenance , Vadodara 252
Student 252
Name: DAMERLA TARUN SAI(2021B1A12741P) 253
PS-I station: IOCL - Operation/Maintenance , Vadodara 253
Student 253
Name: SIDHANT BHATTACHARJEE .(2021B2A12745P) 253
PS-I station: IOCL - Operations , Mumbai 254
Student 254
Name: VINAYAK BHATTACHARYA .(2021B3A42570P) 254
PS-I station: IOCL - Operations/Maintenance , Mathura 255
Student 255
Name: VINAYAK GUPTA .(2021A1PS2365P) 255
PS-I station: IOCL - Operations/Maintenance , Mathura 256
Student 256
Name: HEMANTI BISWAS .(2021A1PS2406H) 256
PS-I station: IOCL - Operations/Maintenance , Mathura 257
Student 257
Name: SAKSHAM CHABA .(2021A1PS2592P) 257
PS-I station: IOCL - Operations/Maintenance , Mathura 258
Student 259
Name: JYOTIRADITYA SIKDER .(2021B2A12762P) 259

PS-I station: IOCL - Operations/Maintenance , Mathura 260
Student 260
Name: PULKIT TRIPATHI .(2021B4A42779P) 260
PS-I station: IOCL - Retail Automation , Mumbai 261
Student 261
Name: TANISH NEVATIA .(2021AAPS0313P) 261
PS-I station: IRCS - Non Tech , Delhi 262
Student 262
Name: THAREJA DHRITI(2021A8PS1659G) 262
PS-I station: IRCS - Non Tech , Delhi 263
Student 263
Name: SAMAY GOYAL .(2021ABPS2486P) 263
PS-I station: IRCS - Non Tech , Delhi 264
Student 264
Name: YASH GOYAL .(2021B2A12764P) 264
PS-I station: IRCS - Non Tech , Delhi 265
Student 265
Name: VANSH .(2021B4A10841P) 265
PS-I station: IRCS - Non Tech , Delhi 266
Student 266
Name: ANMOL MURTI .(2021B4A82539H) 266
PS-I station: IRCS - Nontech , Bhopal 267
Student 267
Name: AARYAN SINHA .(2021A3PS0098P) 267
PS-I station: IRCS - Nontech , Bhopal 268
Student 268
Name: BHAVESH MAURYA .(2021A4PS2899H) 268
PS-I station: IRCS - Nontech , Bhopal 269
Student 269
Name: KARTIKEYA GUPTA(2021A4PS3043G) 269
PS-I station: IRCS - Nontech , Bhopal 270
Student 270

Name: SHIVANI SAINI .(2021B1A10825P) 270
PS-I station: IRCS - Nontech , Bhopal 271
Student 271
Name: ARINJAY SAXENA .(2021B1A12274P) 271
PS-I station: IRCS - Nontech , Bhopal 272
Student 272
Name: RACHIT CHOUDHARY(2021B3A42147G) 272
PS-I station: IRCS - Nontech , Bhopal 273
Student 273
Name: VAIBHAV SHARMA .(2021B4A12328P) 273
PS-I station: IRCS - Nontech , Chandigarh 274
Student 274
Name: AMAN PHOGAT .(2021A7PS0582P) 274
PS-I station: IRCS - Nontech , Chandigarh 274
Student 274
Name: PRIYANSHU VERMA(2021B1TS2081P) 274
PS-I station: IRCS - Nontech , Chandigarh 276
Student 276
Name: YAKSHAYAM VATS .(2021B2A12746P) 276
PS-I station: IRCS - Nontech , Chandigarh 276
Student 276
Name: ARPITA(2021B3TS2051P) 276
PS-I station: IRCS - Nontech , Chandigarh 277
Student 277
Name: PARTH KHANNA .(2021B5A12794P) 277
PS-I station: IRCS - Nontech , Chandigarh 278
Student 278
Name: NISHA SATYAPAL(2021D2TS2091P) 278
PS-I station: IRCS - Nontech , Lucknow 279
Student 279
Name: VIKHYAT GIRI .(2021A4PS2670P) 279
PS-I station: IRCS - Nontech , Lucknow 280

Student 280
Name: SHUBHASHISH SINHA .(2021B2A12760P) 280
PS-I station: IRCS - Tech , Bhopal 280
Student 280
Name: BHAVYA GARHWAL .(2021A3PS2111H) 280
PS-I station: IRCS - Tech , Bhopal 281
Student 281
Name: SATYAM KUMAR(2021B1A42813G) 281
PS-I station: IRCS - Tech , Chandigarh 283
Student 283
Name: KAVISH .(2021A7PS0559P) 283
PS-I station: IRCS - Tech , Chandigarh 283
Student 283
Name: LAKSHAY GARG(2021A7PS2056G) 283
PS-I station: IRCS - Tech , Chandigarh 285
Student 285
Name: VILAKSHAN GUPTA(2021A7PS2075G) 285
PS-I station: IRCS - Tech , Delhi 285
Student 285
Name: JOY SINHA .(2021A8PS1606H) 285
PS-I station: IRCS - Tech , Delhi 286
Student 286
Name: MAYANK KUMAR SINHA(2021B4TS2078P) 286
PS-I station: IRCS - Tech , Delhi 288
Student 288
Name: VISHNU GOYAL(2021B5TS2068P) 288
PS-I station: IRCS - Tech , Delhi 289
Student 289
Name: TUSHAR SAINI(2021D2TS2086P) 289
PS-I station: IRCS - Tech , Lucknow 290
Student 290
Name: DHRUV VERMA .(2021A8PS2711P) 290

PS-I station: IRCS , Dehradun 291
Student 291
Name: SHIRASKAR SOHAM VIJAY.(2020A4PS2108G) 291
PS-I station: IRCS , Dehradun 292
Student 292
Name: HARSH MALIK .(2021A1PS1661P) 292
PS-I station: IRCS , Dehradun 292
Student 292
Name: ISHIKA BHANDARI(2021A1PS2279G) 293
PS-I station: IRCS , Dehradun 293
Student 293
Name: JUGRAAJ SINGH .(2021A1PS2589P) 293
PS-I station: IRCS , Dehradun 294
Student 294
Name: PURUJIT PADHY .(2021A2PS2629P) 294
PS-I station: IRCS , Dehradun 295
Student 295
Name: YASHVARDHAN SHARMA .(2021A3PS2645P) 295
PS-I station: IRCS , Dehradun 296
Student 296
Name: GURJOT SINGH BINDRA .(2021A4PS2373H) 296
PS-I station: IRCS , Dehradun 296
Student 296
Name: SUGAM CHAUHAN(2021A4PS2405G) 296
PS-I station: IRCS , Dehradun 297
Student 297
Name: ADITYA PANT(2021A4PS2901G) 297
PS-I station: IRCS , Dehradun 298
Student 298
Name: AKHOURI SHAKALYA .(2021ABPS2721P) 298
PS-I station: IRCS , Dehradun 299
Student 299

Name: SANYA AGARWAL .(2021B1A32314P) 299
PS-I station: IRCS , Dehradun 300
Student 300
Name: ABHISHEK BISHT .(2021B2A12485P) 300
PS-I station: IRCS , Dehradun 300
Student 301
Name: MUDDAM SANJITH .(2021B4PS1112H) 301
PS-I station: IRCS , Dehradun 301
Student 301
Name: VISHAL THANGEDA(2021B5A42954G) 301
PS-I station: Johnson Controls (I) Pvt. Ltd. , Pune 302
Student 302
Name: RAMANUJ PRALHAD SOMANI(2021A4PS0311G) 302
PS-I station: JSW Steel , Vijaynagar 303
Student 303
Name: PRAMESH MJ .(2021A1PS3050H) 303
PS-I station: JSW Steel , Vijaynagar 304
Student 304
Name: AKSHITA RAJ PANDITA .(2021A4PS0151P) 304
PS-I station: JSW Steel , Vijaynagar 306
Student 306
Name: SAMARTH JAISWAL .(2021A4PS1352P) 306
PS-I station: JSW Steel , Vijaynagar 307
Student 307
Name: ADITYA GATTANI .(2021A4PS1510P) 307
PS-I station: JSW Steel , Vijaynagar 308
Student 308
Name: ARJUN RAGHURAMAN .(2021A4PS1648H) 308
PS-I station: JSW Steel , Vijaynagar 308
Student 308
Name: SHANTANU SUSANTA PANDA(2021A4PS1727G) 309
PS-I station: JSW Steel , Vijaynagar 309

Student 310
Name: PRADHYUMN SINGH RATHORE(2021A4PS2828G) 310
PS-I station: JSW Steel , Vijaynagar 310
Student 310
Name: M C VRINDA(2021A4PS3077H) 310
PS-I station: JSW Steel , Vijaynagar 311
Student 311
Name: MINAL SHARMA .(2021A4PS3088H) 311
PS-I station: JSW Steel , Vijaynagar 312
Student 312
Name: KUNAL AGGARWAL .(2021AAPS1476P) 312
PS-I station: JSW Steel , Vijaynagar 313
Student 313
Name: SHIVAM GAUTAM .(2021AAPS3018H) 313
PS-I station: JSW Steel , Vijaynagar 313
Student 313
Name: HARSHVARDHAN KEJRIWAL .(2021B2A41713P) 314
PS-I station: Kalpataru Power Transmission Ltd. , Noida 314
Student 314
Name: AARUSH TEWARI(2021A2PS2163P) 314
PS-I station: KEC International Limited , Jabalpur 315
Student 315
Name: PRANAV KUMAR MALAIYA .(2021A4PS1059P) 315
PS-I station: KEC International Limited , Jabalpur 316
Student 316
Name: SRIJAN KOSHTI .(2021ABPS1678P) 316
PS-I station: KEC International Limited , Jabalpur 317
Student 317
Name: ANIKET PANDEY .(2021B2AB2768P) 317
PS-I station: KEC International Limited , Jabalpur 319
Student 319
Name: PRANAV PANDEY .(2021B4A40794P) 319

PS-I station: KEC International Ltd , Jaipur 320
Student 320
Name: ABHISHEK KOCHAR .(2021A4PS2495H) 320
PS-I station: KEC International Ltd , Jaipur 321
Student 321
Name: GAURAV AGARWAL .(2021B4A41715P) 321
PS-I station: KEC International Ltd , Jaipur 322
Student 322
Name: SHREYANSHI GARG .(2021B5A41891H) 322
PS-I station: Kerala Infrastructure and Technogy for Education , Thiruvananthapuram 323
Student 323
Name: GARVIT JAIN(2021A7PS2840G) 323
PS-I station: Kerala Infrastructure and Technogy for Education , Thiruvananthapuram 324
Student 324
Name: LAVANYA SABHARWAL .(2021B4AA2263H) 324
PS-I station: Kotak Education Foundation , Mumbai 325
Student 325
Name: JAIN DIYAN VISHAL .(2021A1PS2121P) 325
PS-I station: Kotak Education Foundation , Mumbai 326
Student 326
Name: VEDANT BHATT .(2021A4PS3080H) 326
PS-I station: Kotak Education Foundation , Mumbai 327
Student 327
Name: RISHI KETAN SHAH(2021A7PS2496G) 327
PS-I station: Kotak Education Foundation , Mumbai 329
Student 329
Name: KHANDELWAL AVICHAL PANKAJ .(2021B3A42317P) 329
PS-I station: Kotak Education Foundation , Mumbai 329
Student 329
Name: VIVEK RAMCHANDRA YEDGE(2021B3A82149G) 330
PS-I station: L&T Defence , Hazira 330
Student 330

Name: ARIN MANGAL .(2021A2PS2153P) 330
PS-I station: L&T Defence , Hazira 331
Student 331
Name: VIJAY KUMAR UDHANI .(2021B5AB1697P) 331
PS-I station: L&T Defence , Powai 332
Student 332
Name: VIKRAMADITYA BHATTACHARJEE(2021A4PS0736P) 332
PS-I station: L&T Defence , Powai 333
Student 333
Name: YASH MANOHAR AGARWAL .(2021A8PS3120H) 333
PS-I station: L&T Defence , Powai 334
Student 334
Name: ANUJ MANOJ SHIKARKHANE(2021B5A71145G) 334
PS-I station: L&T Defence , Talegaon 335
Student 335
Name: ADITYA LALIT SHRIPAD(2021A4PS2711G) 335
PS-I station: L&T Defence , Talegaon 336
Student 336
Name: ARYAN SONTHALIA .(2021A4PS3103H) 336
PS-I station: L&T Defence , Talegaon 337
Student 337
Name: SOHAM ANAND KULKARNI(2021A8PS2756G) 337
PS-I station: L&T Defence , Talegaon 338
Student 338
Name: VISHNU VINAY(2021A8PS2989G) 339
PS-I station: L&T Defence , Talegaon 339
Student 339
Name: LOKHANDE SALONI .(2021ABPS2561P) 339
PS-I station: Lakshyam , New delhi 341
Student 342
Name: AKASH JAIN .(2021AAPS2049H) 342
PS-I station: Lakshyam , New delhi 342

Student 343
Name: RAIYAN KATARIA(2021AAPS2556G) 343
PS-I station: Log9 Materials Scientific Pvt. Ltd. , Bengaluru 344
Student 344
Name: PARAMESWARAN SARAVANAN .(2021A1PS2371H) 344
PS-I station: Log9 Materials Scientific Pvt. Ltd. , Bengaluru 345
Student 345
Name: GARG ACHINTYA JITENDRA .(2021A1PS2373P) 345
PS-I station: Log9 Materials Scientific Pvt. Ltd. , Bengaluru 346
Student 346
Name: ATHARVA SHRIRAM GHAISAS(2021A1PS2802G) 346
PS-I station: Log9 Materials Scientific Pvt. Ltd. , Bengaluru 347
Student 347
Name: VISHNU RAJAGOPAL(2021A3PS1537G) 347
PS-I station: Log9 Materials Scientific Pvt. Ltd. , Bengaluru 348
Student 348
Name: SRIVENKAT VARMA NANDIMANDALAM .(2021A3PS2932H) 348
PS-I station: Log9 Materials Scientific Pvt. Ltd. , Bengaluru 349
Student 349
Name: ATHREYA UPADHYA(2021A3PS3045G) 349
PS-I station: M/s Mosdorfer India Pvt. Ltd. , Nashik 350
Student 350
Name: PRIYANKA SATHEESAN NAMBOODRI .(2020B1AB1901P) 350
PS-I station: M/s Mosdorfer India Pvt. Ltd. , Nashik 351
Student 351
Name: ISHWARI SANTOSH BIDWAI(2021A4PS2903G) 351
PS-I station: M/s Mosdorfer India Pvt. Ltd. , Nashik 352
Student 352
Name: VISHAL DALAL .(2021A4PS3099H) 352
PS-I station: M/s Mosdorfer India Pvt. Ltd. , Nashik 353
Student 353
Name: ARYAN SANTOSH SOLANKE .(2021ABPS2722P) 353

PS-I station: Mahindra Research Valley , Chennai 354
Student 354
Name: SHREERANG SUNIL GUPTE(2021A4PS0254G) 354
PS-I station: Mahindra Research Valley , Chennai 355
Student 355
Name: NAMAN SETH(2021A4PS0722P) 355
PS-I station: Mahindra Research Valley , Chennai 356
Student 356
Name: ADITYA BALASUBRAMANIAN .(2021A8PS1475H) 356
PS-I station: Maker & Asylum , Goa 356
Student 357
Name: ANIMESH KRISHNA TIWARI(2021B2A12300P) 357
PS-I station: Manodayam Pvt. Ltd. , Lucknow 357
Student 358
Name: AKSHAT MUDGAL(2021A8PS2912G) 358
PS-I station: Manodayam Pvt. Ltd. , Lucknow 358
Student 358
Name: PARTH GUPTA .(2021AAPS2053H) 359
PS-I station: Manodayam Pvt. Ltd. , Lucknow 359
Student 359
Name: UJJWAL PANDEY .(2021B1A12744P) 360
PS-I station: Manodayam Pvt. Ltd. , Lucknow 360
Student 361
Name: AYUSH SRIVASTAVA .(2021B5AA0772P) 361
PS-I station: Mormugao Port Trust , Goa 361
Student 362
Name: Y E MAYURESH .(2021A2PS3055H) 362
PS-I station: Mormugao Port Trust , Goa 362
Student 362
Name: TANMAY AGRAWAL .(2021A4PS2409P) 362
PS-I station: Mormugao Port Trust , Goa 363
Student 363

Name: KARTIK SHARMA .(2021A4PS3085H) 363
PS-I station: National Aluminium Company -A central PSU , Angul 364
Student 364
Name: SATWIK RATH .(2021A4PS2668P) 364
PS-I station: National Aluminium Company -A central PSU , Angul 365
Student 365
Name: AUM JENA .(2021A4PS2759H) 365
PS-I station: National Aluminium Company -A central PSU , Angul 366
Student 366
Name: SASWAT SUMAN DWIBEDI .(2021B1A42293P) 366
PS-I station: National Metallurgical Laboratory Madras Centre , Chennai 366
Student 366
Name: SHRINANDHAN B .(2021A1PS1680P) 367
PS-I station: National Metallurgical Laboratory Madras Centre , Chennai 368
Student 368
Name: ADITI CHANDRAMOULEE .(2021A1PS2584P) 368
PS-I station: National Metallurgical Laboratory Madras Centre , Chennai 371
Student 371
Name: ABEL OOMMEN LUKE(2021A1PS2963G) 371
PS-I station: NCC(Nagarjuna Construction Company) , Hyderabad 372
Student 372
Name: MUDUNURI SUBRAMANYA VARAHA ABHIRAM VARMA(2021A2PS2361P) 372
PS-I station: NCC(Nagarjuna Construction Company) , Hyderabad 373
Student 373
Name: NARKULA SRIKAR REDDY .(2021A2PS2624P) 373
PS-I station: NHPC Ltd , Faridabad 374
Student 374
Name: SHRUTI CHATURVEDI .(2021A2PS2637P) 374
PS-I station: NHPC Ltd , Faridabad 375
Student 375
Name: ANUSHA SINGH(2021A4PS2312G) 375
PS-I station: NHPC Ltd , Faridabad 376

Student 376
Name: KESHAV PRANSHU(2021A4PS2410G) 376
PS-I station: NHPC Ltd , Faridabad 377
Student 377
Name: ADITYA SRIVASTAVA .(2021A4PS2436H) 377
PS-I station: NHPC Ltd , Faridabad 378
Student 378
Name: ADITYA BANDYOPADHYAY(2021A4PS2708G) 378
PS-I station: NHPC Ltd , Faridabad 379
Student 379
Name: AMAN ALI .(2021B1A22735P) 379
PS-I station: NHPC Ltd , Faridabad 381
Student 381
Name: AMRIT KUMAR JHA .(2021B4A41696P) 381
PS-I station: Nirmaan Organization , Madhapur 381
Student 381
Name: MANTRI SRI KRUTHI .(2021A7PS2006P) 382
PS-I station: Pilani AtmaNirbhar Resource Center (PARC) , Pilani 382
Student 382
Name: OM MISHRA .(2021A8PS3123H) 382
PS-I station: Preto Tooling Systems , Hyderabad 383
Student 384
Name: LAKSHMI NARAYANA KOCHERLAKOTA .(2021A2PS2150P) 384
PS-I station: Preto Tooling Systems , Hyderabad 385
Student 385
Name: VISHNU AKUMALLA .(2021A4PS1894H) 385
PS-I station: Preto Tooling Systems , Hyderabad 386
Student 386
Name: BURUGU VENKATA SAKETHA REDDY .(2021A4PS1986H) 386
PS-I station: Preto Tooling Systems , Hyderabad 387
Student 387
Name: ANKIT SHARMA .(2021A4PS2343H) 387

PS-I station: Preto Tooling Systems , Hyderabad 388
Student 388
Name: PAVAN VISWANADHA .(2021A4PS2774H) 388
PS-I station: Preto Tooling Systems , Hyderabad 389
Student 389
Name: MARRI THANVITA REDDY .(2021B2A43134H) 389
PS-I station: Public Work Department , Lucknow 390
Student 390
Name: SHRIYANSH AGARWAL .(2021A2PS2381P) 390
PS-I station: Public Work Department , Lucknow 391
Student 391
Name: YASHWARDHAN RAI .(2021A2PS2517P) 391
PS-I station: Public Work Department , Lucknow 392
Student 392
Name: SHIVENDRA PAL SINGH .(2021A2PS2618P) 392
PS-I station: Public Work Department , Lucknow 394
Student 394
Name: VAIBHAV KRISHNA .(2021A2PS3214H) 394
PS-I station: Putzmeister , Goa 395
Student 395
Name: KASAT HORAJI SUSHIL .(2021ABPS2560P) 395
PS-I station: Rajasthan Housing Board , Jaipur 396
Student 396
Name: LOHIT GAGGAR .(2021A2PS2617P) 396
PS-I station: Rajasthan Housing Board , Jaipur 397
Student 397
Name: NAMAN SHARMA .(2021A2PS2632P) 397
PS-I station: Rajasthan Housing Board , Jaipur 399
Student 399
Name: ANIRUDH JAIMINI .(2021B3A21192P) 399
PS-I station: Rajasthan State Industrial Development and Investment Corporation (RIICO) , Jaipur 400
Student 400

Name: ISHA SINGHAL .(2021A1PS2379P) 400
PS-I station: Rajasthan State Industrial Development and Investment Corporation (RIICO) , Jaipur..... 401
Student..... 401
Name: DIVYE SRIVASTAVA .(2021A2PS2135P) 401
PS-I station: Rajasthan State Industrial Development and Investment Corporation (RIICO) , Jaipur..... 402
Student..... 402
Name: PARTH RANA .(2021B1A12473P)..... 402
PS-I station: Rajasthan State Industrial Development and Investment Corporation (RIICO) , Jaipur..... 403
Student..... 403
Name: TANISHQ RULANIA .(2021B4A21566P)..... 403
PS-I station: Realty Advance Structure Enterprises Pvt Ltd , Patna 404
Student..... 404
Name: HARSH SINHA .(2021A2PS2609P)..... 404
PS-I station: Realty Advance Structure Enterprises Pvt Ltd , Patna 405
Student..... 405
Name: ISHA APURVA .(2021A2PS2635P) 405
PS-I station: Roads and Buildings Department - Govt of Telangana , Hyderabad 406
Student..... 406
Name: R NISHANT .(2021A2PS2557H) 406
PS-I station: Rourkela Steel Plant , Rourkela..... 407
Student..... 407
Name: VIDIT AGARWAL .(2021A4PS2408P)..... 407
PS-I station: Rourkela Steel Plant , Rourkela..... 408
Student..... 408
Name: ARYAN THAKUR .(2021B2A22164P) 408
PS-I station: SARDAR PATEL RENEWABLE ENERGY RESEARCH INSTITUTE (SPRERI) , Anand..... 408
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Name: PILLAI AYYAPPAN RAJESH(2021A1PS2370P) 408
PS-I station: SARDAR PATEL RENEWABLE ENERGY RESEARCH INSTITUTE (SPRERI) , Anand..... 409
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PS-I station: Starflex Sealing India Pvt. Ltd. , Goa 410

Student 410
Name: ADDYA KASHYAP .(2021A4PS2805H) 410
PS-I station: Starflex Sealing India Pvt. Ltd. , Goa 411
Student 411
Name: CHAITANYA SANJAY NAYAK .(2021A4PS3075H) 411
PS-I station: Starflex Sealing India Pvt. Ltd. , Goa 412
Student 412
Name: SIDDHANTH IYENGAR .(2021AAPS1181P) 412
PS-I station: Starflex Sealing India Pvt. Ltd. , Goa 413
Student 413
Name: SIDDHARTH GUPTA .(2021ABPS1677P) 413
PS-I station: State Council of Education Research and Training , Thiruvananthapuram 415
Student 415
Name: RITHU PRAKASH .(2021ABPS1663P) 415
PS-I station: State Council of Education Research and Training , Thiruvananthapuram 416
Student 416
Name: ATHREYAKRISHNA J N(2021B5A42705G) 416
PS-I station: STEM4ALL , Usa 417
Student 417
Name: ANSHUL SINGH CHAUHAN .(2021A3PS2639P) 417
PS-I station: STEM4ALL , Usa 418
Student 418
Name: AMAAN SAJID NALAKATH(2021A7PS0517G) 418
PS-I station: STEM4ALL , Usa 419
Student 419
Name: SAUBHAGYA ANAND .(2021AAPS0539H) 419
PS-I station: STEM4ALL , Usa 420
Student 420
Name: RAJDEEP SINGH(2021B4A72945G) 420
PS-I station: STEM4ALL , Usa 421
Student 421
Name: SAYANTAN BERA(2021B4AA1828G) 421

PS-I station: STEM4ALL , Usa 422
Student 422
Name: PRADYUMAN SINGH .(2021B5A71204P) 422
PS-I station: STEM4ALL , Usa 422
Student 422
Name: MANAV JAMUAR(2021B5A72638G) 422
PS-I station: Sud-Chemie India Pvt. Ltd - Engineering , Vadodara 423
Student 423
Name: ADITYA CHAKRANI .(2021A1PS1344P) 423
PS-I station: Sud-Chemie India Pvt. Ltd - Engineering , Vadodara 424
Student 424
Name: SHIVANG RAI .(2021B2A42292P) 424
PS-I station: Survey of India , Dehradun 425
Student 425
Name: SNEHA VIVEK .(2021A2PS2627P) 425
PS-I station: Survey of India , Dehradun 426
Student 426
Name: SHREYA NAG .(2021A2PS2636P) 426
PS-I station: Survey of India , Dehradun 427
Student 427
Name: NAKUL SINGH ADHIKARI(2021A8PS2988G) 427
PS-I station: Survey of India , Dehradun 427
Student 427
Name: PALAASH PATEL(2021B1A82938G) 427
PS-I station: Survey of India , Dehradun 428
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PS-I station: Suzlon Foundation , Pune 429
Student 429
Name: CHAITANYA JAIN(2021A3PS1155G) 429
PS-I station: Suzlon Foundation , Pune 430
Student 430

Name: DESHMUKH SANMESH MANISH .(2021B5A12576P) 430
PS-I station: Tamil Nadu eGovernance Agency , Chennai 431
Student 431
Name: YASH AGRAWAL .(2021A1PS2119P) 431
PS-I station: Tamil Nadu eGovernance Agency , Chennai 431
Student 432
Name: HIREMATH NIRANJAN VEERANNA .(2021A1PS2583P) 432
PS-I station: Tamil Nadu eGovernance Agency , Chennai 432
Student 432
Name: SWAHA PATI .(2021A3PS0027P) 432
PS-I station: Tamil Nadu eGovernance Agency , Chennai 433
Student 433
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Student 438
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PS-I station: Tamil Nadu eGovernance Agency , Chennai 439
Student 439
Name: AYUSH SHARMA .(2021B2A20842P) 439
PS-I station: Tamil Nadu eGovernance Agency , Chennai 440
Student 440
Name: SAI ANKIT SAHOO .(2021B3A82571P) 440
PS-I station: Tamil Nadu eGovernance Agency , Chennai 442

Student 442
Name: SARVAKSH SHUKLA .(2021B5A31378P) 442
PS-I station: Tamil Nadu eGovernance Agency , Chennai 443
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Name: AYUSH MAHAJAN .(2021A8PS1411P) 446
PS-I station: Tata Motors Ltd. , Pune 447
Student 447
Name: NAVIN VENKAT .(2021B3AA0983P) 447
PS-I station: Tata Motors Ltd. , Pune 448
Student 448
Name: SHIVANGI PACHAURI .(2021B4A31708P) 448
PS-I station: Tata Motors Ltd. , Pune 449
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Name: ROHAN GUPTA(2021A4PS2661G) 450
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Name: AMITESH SAHA(2021AAPS2340G) 452
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Name: SHUBHRANSH SHARMA(2021B2A33016G) 453

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Name: VISHWAM BAGRODIA .(2021B3A70940P) 456
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Name: RAHUL DAS(2021A8PS2060H) 463
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PS-I station: T-Works Foundation - Mechanical/Mechatronics , Hyderabad 468
Student....468
Name: MANISHA JAKKULA .(2021A4PS3064H)....468
PS-I station: UltraTech Cement Limited, Maihar Cement Works , Maihar 470
Student....470
Name: YASH HRISHIKESH PHADKE(2021A4PS1942G)....470
PS-I station: UltraTech Cement Limited, Maihar Cement Works , Maihar 471
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Name: PURSHOTTAM SRIVASTAVA .(2021A4PS3076H) 471
PS-I station: UltraTech Cement Limited, Maihar Cement Works , Maihar 471
Student....471
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PS-I station: Vikram Cement Works , Neemuch....472
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Name: VEDAANT RATH(2020A8PS1755G) 472
PS-I station: Vikram Cement Works , Neemuch....473
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Name: ABHAY GARG(2021A4PS3050G) 473
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Name: BHUMAN PANDITA .(2021A2PS2605P) 476
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Name: ADITYA UPADHYAY .(2021A4PS2397H) 478
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Name: SAMARTH NASNODKAR .(2021A7PS3204H) 479
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Name: ANUSHKA AGGARWAL .(2021B1A32269P) 480
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Student 481
Name: PRIYANSHU SHEKHAR SINGH .(2021B5AB2797P) 481
PS-I station: Western Irrigaation System Pvt. Ltd. - Core , Rajkot 482
Student 482
Name: SANKHARVA SHYAM RAMESHBHAI(2021A1PS2563G) 482
PS-I station: Zuari Industries Ltd (Paradeep Phosphates Ltd) , Goa 483
Student 483
Name: MISHIKA NARULA .(2021A1PS1327P) 483
PS-I station: Zuari Industries Ltd (Paradeep Phosphates Ltd) , Goa 484
Student 484
Name: SACHIKA LALA .(2021A1PS2375P) 484
PS-I station: Zuari Industries Ltd (Paradeep Phosphates Ltd) , Goa 486
Student 486
Name: Vaibhav Biresh Prasad(2021B5AB2271P) 486

PS-I station: A.V. Engineers , Gurugram

Student

Name: SOHAM MUKHERJEE .(2020B5AB2004P)

Student Write-up:

PS-I Project Title: Roller-track conveyor

Short Summary of work done: First we were shown around the factory divided into 3 groups of 6 each. We were told to build something in the domain of "Robotics in Warehousing". We then chose to build a roller-track conveyor belt. We were told to design it on Solidworks as best as we could, and the manufacturing part would be done later by the company after our PS-1 gets over (since it would take time to order the parts for the roller track conveyor and for the orders to be dispatched and delivered). So me and my team of 5 people started designing the roller track. First we designed the roller, then tapered frustum, then vertebrate, then sheet metal, then railings, but we realised sheet metal and railings are not required.. Then we designed the full thing on blender, but it was of no use since the design couldn't be imported to Solidworks successfully. Finally we designed the full thing on Solidworks.

Objectives of the project: Designing a Roller-track conveyor on Solidworks

Tool used: Solidworks, Google meet

Details of Papers/patents: None

Brief description of the working environment: The working environment was hard to judge because most of my PS-1 happened in hybrid mode, and it was mostly online. Our major work was designing the roller-track conveyor belt on CAD softwares, which could be done online without visiting the office/factory regularly. We only went once a week or so to see the actual parts and tools available in the factory. In fact, this PS-station can be made an Online PS station. Moving on to expectations from company, the major expectation from the company boss was that we should do something during our PS-1 that counts as a productive achivement (so that the company also benefits and so that we can also say we did something during PS-1). Other than that the people at the company were very relaxed, in fact they themselves said we need not come to office every day. Finally, the major learning I had during PS-1 was how to use Solidworks, but I also learnt invaluable lessons in teamwork, as working in a team of 6 people is not easy and it was difficult to coordinate the times when we would work together. What worked best is dividing up the work among people and letting everyone do their job at their own pace.

Academic courses relevant to the project: Manufacturing Second year first semester course - Machine Drawing

Learning Outcome: CAD on Solidworks, Manufacturing processes

PS-I station: A.V. Engineers , Gurugram

Student

Name: ADITYA PANDEY(2021A4PS2343G)

Student Write-up:

PS-I Project Title: Roller track conveyor system

Short Summary of work done: The 1st day was a general factory visit ,all the tools and machine were showed to us.mostly our work was done from home as we had to design a rollertrack conveyor.

Objectives of the project: To design a working roller track conveyor in solidworks 2020

Tool used: Solidworks

Details of Papers/patents:N/a

Brief description of the working environment: The 1st day was a general factory visit ,all the tools and machine were showed to us.mostly our work was done from home as we had to design a rollertrack conveyor.

Academic courses relevant to the project: Material science ,autocad

Learning Outcome: Learned to use solidworks software and its application

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PS-I station: A.V. Engineers , Gurugram

Student

Name: SAIDEEP VERMA .(2021A4PS2493H)

Student Write-up:

PS-I Project Title: Automating Trolleys using Sensory Paths

Short Summary of work done: The aim of this project is to develop an autonomous line following robot that is attached to a trolley. The robot will be able to navigate along a designated path while efficiently transporting goods. The ultimate goal of this project is to provide a practical and reliable solution that enhances automation, efficiency, and safety in material transport tasks, thereby improving operations and cost-effectiveness in various industries.

Objectives of the project: Use advanced sensors for automating transport in industrial areas.

Tool used: Fusion360, ROS, Sensor Integration, Research

Details of Papers/patents: None

Brief description of the working environment: Working environment was not well supported but comfortable. Faculty was helpful. Very informative and experience gaining station. Very supportive faculty from AV Engineers.

Academic courses relevant to the project: Robotics, Mechanisms and Machines, Material Science

Learning Outcome: Robotics, CAD, Coding

PS-I station: A.V. Engineers , Gurugram

Student

Name: PRATHAM GB .(2021A4PS2666P)

Student Write-up:

PS-I Project Title: Automation of stackable trolleys

Short Summary of work done: Designed a prototype for the robot and scaled it to a larger model. Researched on AGVs and AMRs for the project.

Objectives of the project: Design a robot to automate trolleys that can transport materials inside warehouses without human intervention

Tool used: Autodesk FUSION 360

Details of Papers/patents: None

Brief description of the working environment: The environment is accommodating and friendly. The faculty from the company were very inviting and helpful.

Academic courses relevant to the project: Robotics, engineering design.

Learning Outcome: Robotics

PS-I station: A.V. Engineers , Gurugram

Student

Name: ASHMIKA JAIN(2021A4PS2829G)

Student Write-up:

PS-I Project Title: Automation of Trolleys

Short Summary of work done: As an Intern in A.V Engineers, I am designing a line-following robot that can take care of the transportation of goods within the factory to reduce human interference and hence, errors and accidents. We are working on automating a stackable trolley with this concept of a Line-following robot to enable it to carry different payloads and sizes efficiently. The ultimate goal of this project is to provide

a practical and reliable solution that enhances automation, efficiency, and safety in material transport tasks, thereby improving operations and cost- effectiveness in various industries.

Objectives of the project: To automate the stackable trolleys in the factory in order to increase efficiency in in-factory transportation thereby increasing the overall productivity of the factory

Tool used: Fusion 360, TinkerCAD

Details of Papers/patents:-

Brief description of the working environment: Got to work with really nice people from all the 3 campuses.

Academic courses relevant to the project: Mechanisms and Machines, CAD

Learning Outcome: Arduino, TinkerCAD, CAD

PS-I station: A.V. Engineers , Gurugram

Student

Name: ATIBHAV SAXENA .(2021A4PS3105H)

Student Write-up:

PS-I Project Title: Roller Track Conveyor Belt

Short Summary of work done: Project = Roller track conveyor Belt system. First we were given few subptopics and areas from which to think of a project to pick. Then after research and discussions we chose to work on rollertrack conveyor belt systems. The aim was to design the rollertrack conveyor belt system using certain template used in industry. The concept design was divided into components among the team of 5 in our project. After designing the component and holding regular update discussions we assembled the different components in solidworks 2020 for the initial design. Then after presenting this design to our mentor at the company, we were given some feedback and suggested some changes. We chose dimensions according to load and torque calculations and industry load standards. We then worked on creating a prototype design

in Blender to minimize error. We used the same dimensions and this time switch our focus to work on a single roller component which would be duplicated to form the roller conveyor chain. After successfully creating the design in blender we showcased it and were suggested some minor tweaks. We were provided a template design from our mentor for the interlocking system. We then attempted to import the Blender design into solidworks using third party software. Then due to limited freedom of operation on the imported file, we made the final design by dividing it into three components and designing from scratch in Solidworks 2020 incorporating all the changes and appropriate dimensions.

Objectives of the project: Develop a more efficient and cost effective model of roller-track conveyor belt system meeting the load and other requirements of a system used in the current industry

Tool used: Solidworks 2020, Blender

Details of Papers/patents:-

Brief description of the working environment: Working was hybrid, for software designing purposes we were given liberty of WFH. On site facilities were more than adequate to suffice our needs with a large warehouse consisting of a fully equipped workshop and industry standard systems for designing (CAD).The company also recently acquired and installed a conveyor loop for storing and sifting packages involving AI sensors and camera based detection. They also house a small scale robot for the same. Expectations from the company before the start of PS was to get insight on the working of a warehousing company and get to know the production phase of a product and the machinery required to carry out the production. The company definitely met these expectations and gave us a lot of freedom to utilize various teams and facilities available on site. All the people working on site were inviting and friendly and were happy to entertain our questions and eager to help whenever asked. My takeaways and learnings from this PS were satisfactory to me. I learnt how to create a timeline for a project/product which would at the industry level, I learnt the requirements, dedication and expertise needed to acheive success while undertaking a project of such scale. I also learnt how to apply concepts learnt in academics in a practical setting. Throughout the course I also learnt how to use and create projects using industry standard softwares like Solidworks.

Academic courses relevant to the project: Engineering Graphics, Workshop Practice, Manufacturing Processes and Mechanics of Solids(to a degree)

Learning Outcome: My takeaways and learnings from this PS were satisfactory to me. I learnt how to create a timeline for a project/product which would at the industry level, I learnt the requirements, dedication and expertise needed to achieve success while undertaking a project of such scale. I also learnt how to apply concepts learnt in academics in a practical setting. Throughout the course I also learnt how to use and create projects using industry standard software like Solidworks 2020.

PS-I station: A.V. Engineers , Gurugram

Student

Name: JAI AVASTHI .(2021ABPS2716P)

Student Write-up:

PS-I Project Title: Roller track conveyor

Short Summary of work done: Got to learn the tools used in industry

Objectives of the project: To build a roller track conveyor to be used in warehouses

Tool used: Solid works

Details of Papers/patents: None

Brief description of the working environment: Got to learn the tools used in industry

Academic courses relevant to the project: Machine Drawing

Learning Outcome: Got to learn the tools used in industry

PS-I station: A.V. Engineers , Gurugram

Student

Name: AKSHAT JAIN .(2021ABPS2725P)

Student Write-up:

PS-I Project Title: EV battery cabinet

Short Summary of work done: Designed and prototyped EV battery cabinets in collaboration with a cross-functional team of engineers, effectively integrating usability and safety features. Utilized CAD software to create 3D models of battery cabinets, streamlining the product development process and ensuring precision in design. Assisted in the development of new product specifications and requirements based on market analysis and customer feedback. Conducted rigorous testing scenarios to verify product quality and reliability under various environmental conditions. Actively participated in design reviews, providing constructive feedback and innovative solutions to improve overall product performance.

Objectives of the project: To develop efficient EV battery cabinet

Tool used: Fusion 360, AutoCAD

Details of Papers/patents: Submitted final project report which was analysed by the company's design team and scheduled for prototyping.

Brief description of the working environment: During my tenure as an intern at AV Engineers, a prominent manufacturing company, I was immersed in a dynamic and fast-paced working environment where innovation, diligence, and precision were highly valued.
The company's workplace was an embodiment of efficient industrial design where state-of-the-art machinery and innovative technology coalesced with the skilled human effort. I was stationed in different departments throughout my internship, from design and production to quality control, which provided me with a holistic view of the manufacturing process. It was fascinating to witness raw materials transforming into high-quality products through well-coordinated efforts and advanced technology.
The management team at AV Engineers nurtured an inclusive and collaborative work culture. Teamwork was not only encouraged but celebrated. I had the chance to interact with professionals from diverse backgrounds, which enriched my perspective and understanding of the sector. The open-door policy meant that everyone, regardless of their position, had a voice and was encouraged to share their ideas and concerns.

Academic courses relevant to the project: Machine drawing, Engineering drawing

Learning Outcome: Team work, Manufacturing, Market Research

PS-I station: A.V. Engineers , Gurugram

Student

Name: SHREYASI BHARATIYA .(2021B1A40793P)

Student Write-up:

PS-I Project Title: Affordable Robotics In Everyday Life - AUTOMATING TROLLEYS THROUGH SENSORY PATHS/ROBOTICS

Short Summary of work done: My major scope of work at AV Engineers is research and development (R&D) for the project “Affordable Robotics in everyday life”. My work included gaining a comprehensive understanding of automation of trolleys to work on my project of automating trolleys through sensory paths/robotics. Robotics involves manufacturing intelligent machines and includes mechanical construction powered by electrical circuitry and controlled by computer programming. I did thorough research from research papers of various colleges and researchers that have worked in this field with specific attention to integrating a small scale idea like sensory paths to a larger scale. Then I worked on the development of a small scale prototype and then scaling it up to achieve the objective of automating trolleys solely through the utilization of sensory paths. Finally, expand our project by exploring AGVs (Automated Guided Vehicles) and AMRs (Automated Mobile Robots) and their future implications and usefulness in factory settings and production processes and submitted a project report highlighting our progress and outcomes in the report for the project. Prepared the challenges and solutions faced by our project and did deep analysis and research on mounting motors and sensors on an automated trolley using robotics.

Objectives of the project: The primary objective of this project is to develop a robot-trolley system that can autonomously navigate along a designated path while efficiently transporting goods. Robotics involves manufacturing intelligent machines and includes mechanical constructi

Tool used: Fusion 360, SolidWorks, MS Excel, MS Word.

Details of Papers/patents: Final Project Report on the PSMS Portal.

Brief description of the working environment: A.V Engineers was established in 1984 as a TOOL ROOM FOR IMPORT SUBSTITUTION of
machinery parts for various imported plants and machinery. Its vision is to provide satisfaction and gain trust of the customers and provide adaptability and flexibility.

Academic courses relevant to the project: Manufacturing Processes, Mechanisms of Machines, Computer Programming, Artificial Intelligence

Learning Outcome: The project involves delving into the fundamental workings of a basic line follower robot and identifying the necessary components for building a robot capable of line following. The project requires understanding of: 1. Arduino UNO 2. L293D motor driver 3. Infra-Red sensor 4. Electrical knowledge of BO motors/ Servo motors along with their power ratings 5. Fusion 360/ SolidWorks software for 3D modeling. The project also requires doing research and development (R&D) and submit a project report highlighting our progress and outcomes. Finally, the project is expanded by exploring AGVs (Automated Guided Vehicles) and AMRs (Automated Mobile Robots) and their future implications and usefulness in factory settings and production processes.

PS-I station: A.V. Engineers , Gurugram

Student

Name: YUVRAJ SINGH(2021B1A42937G)

Student Write-up:

PS-I Project Title: EV battery cabinet

Short Summary of work done: Collaborating with a diverse team of engineers, I successfully designed and prototyped electric vehicle (EV) battery cabinets, seamlessly incorporating user-friendly and safety elements. Leveraging CAD software, I created precise 3D models of the cabinets, expediting the product development process. Additionally, I contributed to crafting new product specifications and requirements through market analysis and customer input. My active involvement in design reviews included offering valuable feedback and creative solutions to enhance overall product performance.

Objectives of the project: To develop EV Battery Cabinet and improve its efficiency and safety in a cost effective way.

Tool used: Autocad, Sketchup

Details of Papers/patents: Submitted final project report.

Brief description of the working environment: During my internship at AV Engineers, a prominent manufacturing company, I thrived in a dynamic and fast-paced environment that highly valued innovation, diligence, and precision. The workplace exemplified

efficient industrial design, where cutting-edge machinery and technology harmoniously merged with skilled human effort. Throughout the internship, I gained valuable insights by working in various departments, including design, production, and quality control, providing me with a comprehensive understanding of the manufacturing process. Observing raw materials being transformed into high-quality products through well-coordinated efforts and advanced technology was truly captivating.
At AV Engineers, the management fostered an inclusive and collaborative work culture, emphasizing the importance of teamwork. I had the opportunity to interact with professionals from diverse backgrounds, which broadened my perspective and deepened my understanding of the industry. The open-door policy promoted a sense of equality, enabling everyone to contribute ideas and address concerns regardless of their position within the company.

Academic courses relevant to the project: Engineering Graphics

Learning Outcome: Market Analysis, Working with various teams, Manufacturing processes.

PS-I station: A.V. Engineers , Gurugram

Student

Name: RAGHAV SHARMA .(2021B1AB2727P)

Student Write-up:

PS-I Project Title: Automation for stackable trolleys

Short Summary of work done: During our practical summer internship at AV Engineers, our team undertook the development of automated stackable line following trolleys, demonstrating resilience and adaptability amidst unexpected challenges. Through extensive online research, we explored robotics engineering principles, control systems, and sensor integration to design the trolleys. Precise simulations and feasibility studies refined the virtual 3D model. Our comprehensive online report encompassed research findings, design concepts, and algorithm simulations. Though prevented from physical prototyping due to flooding, virtual discussions with experts validated our design. During virtual evaluations, we confidently presented the project to the PS instructor, showcasing dedication and resourcefulness in navigating adversities. The successful project completion honed technical expertise in robotics, problem-solving, and project

management. The internship fostered professional growth, reinforcing teamwork, adaptability, and innovation.

Objectives of the project: The aim of this project is to develop an autonomous line following robot that is attached to a trolley. The robot will be able to navigate along a designated path while efficiently transporting goods.

Tool used: Hardware: Microcontrollers (Arduino, Raspberry Pi), Motors, Actuators, Sensors (Proximity, Infrared, Ultrasonic), 3D Printer (for prototype). Software: Programming Languages (Python, C++, Arduino IDE), Simulation Software (MATLAB, ROS, V-REP), 3D

Details of Papers/patents: No papers published

Brief description of the working environment: During the PS-I at AV Engineers, we thrived in a collaborative and innovative working environment. The company encouraged active participation, providing access to cutting-edge tools and resources for hands-on learning. Our technical proficiency improved through robotics engineering principles, programming, and sensor integration in designing automated trolleys. AV Engineers set high expectations, urging us to take ownership, exhibit problem-solving skills, and meet deadlines. We honed communication and teamwork abilities while facing unforeseen challenges, fostering adaptability and resilience. Virtual evaluations enhanced our presentation and communication skills. Overall, the internship enriched our technical knowledge, project management skills, and understanding of real-world applications, leaving a profound impact on our professional growth and engineering perspectives.

Academic courses relevant to the project: Machine drawing, electrical sciences

Learning Outcome: The major learning outcomes of the project are as follows:

Technical Proficiency: Gain expertise in robotics engineering, control systems, sensor integration, and 3D modeling for designing automated trolleys.

Problem-Solving Skills: Develop critical thinking and troubleshooting abilities to address challenges during design, testing, and implementation phases.

Collaboration: Enhance teamwork and communication skills by working collaboratively with team members, stakeholders, and experts.

Project Management: Acquire project management skills in planning, organizing, and executing tasks to achieve project goals efficiently.

Innovation: Foster creativity and innovative thinking in developing cutting-edge solutions for real-world applications.

Adaptability: Learn to adapt to unforeseen circumstances and overcome challenges to ensure project success.

Industry Relevance: Understand the practical applications of robotics in industries and the potential impact of automated systems on processes.

PS-I station: A.V. Engineers , Gurugram

Student

Name: DHRUV SINGH SHEKHAWAT(2021B2A43065G)

Student Write-up:

PS-I Project Title: EV Battery Cabinet

Short Summary of work done: We prepared a solution and conducted market research about what EV Battery Cabinets

Objectives of the project: Finding a better solution for EV battery cabinets

Tool used: AutoCAD, Solidworks

Details of Papers/patents: None

Brief description of the working environment: They provided us with a holistic and positive learning environment

Academic courses relevant to the project: Engineering Graphics

Learning Outcome: I learned a lot about the Electrical Vehicle Industry while researching for the project. It was an insightful experience finding out about the process of how battery cabinets are manufactured and what the customer needs are.

PS-I station: A.V. Engineers , Gurugram

Student

Name: AZZAAM USMANI .(2021B4A42572P)

Student Write-up:

PS-I Project Title: Roller Track conveyors for warehousing

Short Summary of work done: A team of 6 was created and we were assigned a project to work on during the PS. The project objective was to create a digital design of a cost effective version of flexible roller track coveyors, to be used in warehousing and short distance transport.

Objectives of the project: -To design a flexible roller track conveyor ; -Make the design as cost effective and budget friendly as possible for the Indian Market

Tool used: S/w: SolidWorks 2020, Blender, Rhino 7

Details of Papers/patents:-

Brief description of the working environment: Our PS was hybrid, so during the times we did visit the PS station, we were given a proper area to work and discuss, the mentor from the company was very helpful. The company expected us to give regular updates and was willing to let us participate in their ongoing projects to get useful insights from the students.

Academic courses relevant to the project: -

Learning Outcome: Working and demands of Indian market; effective delivery of product within the time provided; method of communication in office and organisation.

PS-I station: A.V. Engineers , Gurugram

Student

Name: SAMBHAV KUMAR .(2021B5A43181H)

Student Write-up:

PS-I Project Title: ROLLER TRACK CONVEYOR BELT

Short Summary of work done: First of all ,my work was to research major companies in this domain and the materials used by them in making this . After that I used solidworks to design parts of roller track . And then we assembled the individual made parts using

solidworks. Major attention of given to size , circumference and material used to make this as efficient as possible and at the same time not compromising the strength for industrial uses.

Objectives of the project: Designing of roller track conveyor belt for industrial uses.

Tool used: Solidworks , Blender

Details of Papers/patents:N.A

Brief description of the working environment: Working environment was great. If we had any difficulties we could easily contact Faisal sir who was on the designing team in the company.

Academic courses relevant to the project: N.A as I only studied msc phy . Though I can say that workshop helped a little bit.

Learning Outcome: Learned how to use solidworks.
Gained knowledge about industrial designing of roller tracks, materials used in making them.

PS-I station: Act Now India (an NGO) - Nontech , Delhi

Student

Name: UTSAV SETHI .(2021A3PS2908H)

Student Write-up:

PS-I Project Title: Tech Enablement in Skill Development

Short Summary of work done: My project was based on Tech Enablement in Skill Development, where I had to make the resources accessible to the people of rural areas for their skill development. For this, the medium was a landing page. So, I created a landing page which could offer various educational and vocational courses by partnering with different institutions.

Objectives of the project: To create a landing page which can provide various educational courses to needy people by partnering with different educational institutions.

Tool used: VS Code, HTML, CSS, JS

Details of Papers/patents: None

Brief description of the working environment: Working for the company gave a good idea about the corporate world as we worked together as a team, worked according to the deadline and were also being monitored regularly. It was more of a self learning experience as the station did not offer any technicalities to learn. We had to learn on our own from scratch and make things work. Also, the organisation did not co-operate much with the FIC and our university. Also, the mentor sometimes gave unrealistic deadlines. Overall, it was a mixed experience working for the organisation.

Academic courses relevant to the project: None

Learning Outcome: Learnt Web Dev(HTML, CSS, JS) and also came to know about how an NGO works and

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PS-I station: Act Now India (an NGO) - Nontech , Delhi

Student

Name: NITISH BRAT DAS .(2021A8PS3185H)

Student Write-up:

PS-I Project Title: Web Development and Tech-enabled Connect Interface

Short Summary of work done: Work was great and got a lot of things to learn but industry mentor assigned was not technical guy so had lot of miscomunication

Objectives of the project: Build a website for act now

Tool used: vs code , youtube , udemy

Details of Papers/patents:na

Brief description of the working environment: working environment was not conducive with unrealistic deadlines, it would have been better if someone who had some experience in tech was assigned as a mentor for the web development team.

Academic courses relevant to the project: NA

Learning Outcome: web development

PS-I station: Act Now India (an NGO) - Nontech , Delhi

Student

Name: LALITHA KOTIKALAPUDI .(2021AAPS0648H)

Student Write-up:

PS-I Project Title: Tech Enablement in the field of education

Short Summary of work done: I made a landing page for the teachers training workshop which was then integrated woth the main websote of the ngo. I also made few detailed instruction documents on what cpulf be taught in the workshop. After completing the lwnding page, i was assigned to ensure macimum participation in the seminar which was supposed to be conducted by the ngo. I personally sent out mails to 15 colleges with the invite to the seminar, the reigstratikns links and other relevant posters.

Objectives of the project: To create a landing page for the teachers training workshop, ensure participation in a seminar conducted by the ngo

Tool used: Html css bootstrap framework

Details of Papers/patents:-

Brief description of the working environment: The work environment was fine. We had daily meets with our industry mentor Mr Devraj. We also had to submit detailed workplans, update our Gantt chart and update our todo list everyday.

Academic courses relevant to the project: -

Learning Outcome: Learnt HTML, CSS and Bootstrap framework

PS-I station: Act Now India (an NGO) - Nontech , Delhi

Student

Name: ARYAN GUPTA .(2021ABPS1672P)

Student Write-up:

PS-I Project Title: Social Media Engagement and Campaign

Short Summary of work done: My work was to graphic design banners and make content for social media handles on a daily basis.(Around 2-3 posts a day). Also had to publish a blog every week and send formal connection invites on a regular basis. Second part of my project was related to organize a campaign which would end with a seminar. For that daily post, brochures and heavy outreach work was done

Objectives of the project: To increase the social media engagement and the presence of the NGO on various social media platforms

Tool used: Adobe Photoshop, Canva, GIMP, MS excel

Details of Papers/patents:NA

Brief description of the working environment: Working environment was pretty intense with multiple meetings in a day with the mentor and we had tight deadlines and results to get.
Company expectation was we work 9hrs a day including the lunch break and do all PS evaluative work after working hours.

Academic courses relevant to the project: None in my knowledge

Learning Outcome: Graphic designing, photo editing, outreach and communication skills

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PS-I station: Act Now India (an NGO) - Nontech , Delhi

Student

Name: PRIYANSHU KUMAR(2021B3A72865G)

Student Write-up:

PS-I Project Title: Act now fitness

Short Summary of work done: Overall, this comprehensive study demonstrates the vital role of NGOs in India's environmental conservation efforts and showcases the transformative potential of technology-driven solutions like the Act Now Fitness app. The successful execution of Project-2 and Project-3 further underscores the effectiveness of tech tools in streamlining processes and achieving outreach objectives. This work contributes to the understanding of sustainability efforts and technological innovations that can shape a greener and more conscientious future.

Objectives of the project: To create a fitness app that will help save the environment

Tool used: Flutter, Firebase, Octoparse, Dart, Python, Selenium

Details of Papers/patents:N/A

Brief description of the working environment: During my practice school, I gained valuable insights and practical experience in several key areas. Firstly, I deepened my understanding of the significant role NGOs play in environmental conservation and sustainability efforts in India. I learned about their historical development, current challenges, and the potential for impactful contributions.
Additionally, I developed hands-on skills in leveraging technology for positive change. Through the creation and implementation of the Act Now Fitness app, I learned how to use Flutter and Firebase to build a user-friendly front-end interface and handle essential backend functionalities. This experience enhanced my knowledge of mobile app development and its potential for promoting both personal well-being and environmental consciousness.
Furthermore, I acquired proficiency in web scraping using Octoparse and Selenium, enabling me to efficiently collect and manage data from various sources. I also learned how to utilize Google Apps Script for automating tasks, such as mail merges, enabling effective communication with colleges and companies during the distribution of the ACT NOW Seminar itinerary.
Beyond technical skills, my practice school taught me the importance of collaboration and community engagement. Through interactions with NGOs, educational institutions, and

corporations, I realized the significance of building partnerships and involving stakeholders to drive positive change.
Overall, my practice school experience enriched my knowledge, honed my technical abilities, and instilled a sense of responsibility towards environmental conservation. It provided me with valuable real-world exposure and equipped me with versatile skills that will undoubtedly contribute to my future endeavors and aspirations in making a meaningful impact on society and the environment.

Academic courses relevant to the project: Object oriented programming

Learning Outcome: App development, data mining

PS-I station: Act Now India (an NGO) - Nontech , Delhi

Student

Name: MAEDHIKA MALHOTRA .(2021B5A82006H)

Student Write-up:

PS-I Project Title: Web Development and Tech Enabled Connect Interface

Short Summary of work done: We created the official website for the organization, however we were unable to host it unfortunately.

Objectives of the project: Creating the official website for the organization to improve user-experience and engagement

Tool used: HTML, CSS, JavaScript, Canva, Powerpoint

Details of Papers/patents:none

Brief description of the working environment: The working environment was pretty awful. Since it was online, in order to "keep a check on us", our industry mentor would randomly declare meets and would give us half-day leaves if we didn't join within 3-5minutes. This made work really frustrating and robbed us off the joy from our projects since he also had a tendency to be very rude and almost condescending in these meets. Furthermore, we literally didn't interact with anyone else from the organization except for Devraj sir, from HR and that really limited our experience.

Additionally, the nature of our projects was technical (we had to make a website) and we naturally expected some technical lectures on web development or some external resources or insight provided by the company; however they literally shared no such things, there were no lectures, no tutorials and we had to learn everything on our own using YouTube or Udemy. Even then he could not help us when we encountered issues and that made work really difficult. Lack of knowledge in these matters made him set unreasonable deadlines, which he was very adamant and unflinching about.

Academic courses relevant to the project: Web Development related

Learning Outcome: Web Development (UI+UX) includes learning HTML, CSS and JavaScript

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PS-I station: Act Now India (an NGO) - Tech , Delhi

Student

Name: KUSH JAYESH DESAI .(2021B4A73158H)

Student Write-up:

PS-I Project Title: Web development and tech enabled connect interface

Short Summary of work done: We created the prototype website for the ngo.

Objectives of the project: Making a website for the ngo

Tool used: Html, css, javascript

Details of Papers/patents: None

Brief description of the working environment: The working environment in the ps station was overbearing as we were threatened with our marks to meet the industry mentor's deadlines. Moreover, he was insistent on micromanaging us which hindered our work greatly. There was also a lot of lack of clarity and a lot of miscommunication from their side about what needed to be done in the project. He was adamant on making us work in 2-3 projects irrespective of the completion of the original project. They had expected us to be as good as a working professional. I learnt basic HTML, css, js during my ps time.

Academic courses relevant to the project: Computer programming

Learning Outcome: Learnt basic web development (html, css, js)

PS-I station: AERB - Chemical , Mumbai

Student

Name: SAMRUDH D B(2021A1PS0044G)

Student Write-up:

PS-I Project Title: Fire Hazard Analysis of Chemical Plants Storing Flammable Substance

Short Summary of work done: In the first half of the internship program, a Literature review was conducted where, I got an understanding regarding the various types of fire explosions with their models. I also went through the basics of Fire Hazard Analysis- various steps involved and the methodology. With the knowledge obtained through the literature review, in the second half, I was given two simulation problems with respect to fire hazards, where there was leakage of flammable substances, and was asked to carry out the risk assessment failure of the same. While one problem was related to a simulation where the efficient position of the cable tray was to be found, the second problems was related with simulating a pool fire type incident. Through the softwares such as FDS and Pyrosim, I was able to simulate the given problems and carry out the risk analysis successfully.

Objectives of the project: The objective of my project was to understand the basics of fire hazards, and eventually be able to carry out a Fire Hazard Analysis for any given situation (especially in chemical plants).

Tool used: FDS software and Pyrosim software

Details of Papers/patents:NA

Brief description of the working environment: From an organization such as AERB, I expected to enhance my research skills along with my technical knowledge with respect to the chemical industry. The overall working environment was fine, with meetings

between the mentee and mentors being scheduled once in a while. Eventually, through this project, I did learn a few things, such as Fire Hazard analysis, usage of softwares such as FDS and Pyrosim, to name a few.

Academic courses relevant to the project: Heat Transfer

Learning Outcome: Through this project, I learned how to carry out a Fire Hazard Analysis for any given situation. Along with this, an understanding of the various types of fire explosions, and their models was obtained. The use of softwares such as Pyrosim was enhanced through this project, where I could simulate my own compartment or geometry and carry out the risk assessment failure. Through this project, an understanding of how to carry out a Literature Review was obtained.

PS-I station: AERB - Chemical , Mumbai

Student

Name: SARTHAK MAHESHWARI .(2021A1PS1353P)

Student Write-up:

PS-I Project Title: HAZOP Studies of Chemical Plant Containing Toxic Flammable Materials

Short Summary of work done: The project was related to Hazop studies which is a risk assessment methodology used extensively in chemical process plants. The project was mainly divided into 2 parts. The first part involved a rigorous literature review of Hazop studies, it was mostly theoretical covering all aspects of the need, importance and application of Hazop studies. The second part covered the practical implementation of it through a hands-on case study. The project was an interesting one despite being theoretically involved.

Objectives of the project: To learn about the implementation of Hazop studies, emerging trends and technologies in its application.

Tool used: excel, Matlab fuzzy logic designer app

Details of Papers/patents:NA

Brief description of the working environment: The working environment was flexible. Mentors interacted with us through mails, calls and Google Meets imparting valuable practical knowledge and details about the project. A lot of soft skills development took place through regular interaction with such industry experts.

Academic courses relevant to the project: chemical second year CDC's

Learning Outcome: Learned a lot about Hazop studies and gained an insight into the workings of chemical process industries. GD's and seminar presentations conducted improved my soft skills.

PS-I station: AERB - Chemical , Mumbai

Student

Name: AGARWAL RACHIT VIKAS(2021A1PS1551G)

Student Write-up:

PS-I Project Title: Reliability Analysis of typical Engineered Safety Features

Short Summary of work done: We started with learning the theory of the tools and models used to analysis the reliability of a system including quantitative probability analysis of system failure, later I conducted an analysis of myself on one of the chemical process from the industry.

Objectives of the project: To analysis reliability of a system to check its safety

Tool used: NA

Details of Papers/patents:NA

Brief description of the working environment: The working environment was good and easy to fit into. The mentor was helpful and very knowledgeable. The learning outcome was a industry insight and the process of planning of setting up a chemical plant.

Academic courses relevant to the project: --

Learning Outcome: Fault tree Modelling

PS-I station: AERB - Chemical , Mumbai

Student

Name: JOYEETA NATH .(2021A1PS2166P)

Student Write-up:

PS-I Project Title: Risk Assessment of Chemical Plant Using H2S, Ammonia and Chlorine

Short Summary of work done: The project itself was divided into 2 broad parts - a literature review section, and a problem solving section. In between, a seminar took place in order to simulate an interview-like experience and provide us the opportunity to present our findings in a formal setting as well as answer questions posed by experts as well as peers. The second part of the project focussed on solving a given problem by hand as well as suggested software. In the research half of the internship, various risk assessment models were read about and an example of each model being applied to a real world problem was understood. The drawbacks, assumptions and advantages of all the models were then concluded based on the literature review. In the second half, a problem statement was given and had to be solved using the Gaussian Dispersion Model. The problem was first solved by hand to obtain a worst case scenario result and then by the ALOHA software to model the incident that took place. Thus the PS exposed us to both research and practical application of problem solving skills.

Objectives of the project: To understand why risk assessment is a necessity for all chemical plants, especially ones that work with hazardous elements and compounds. To gain an understanding of the various risk assessment models already mentioned in literature. To understand the dr

Tool used: ALOHA

Details of Papers/patents:12 Journal papers and 2 textbooks were used as references over the course of the project

Brief description of the working environment: The working environment although welcoming and foccused was quite erratic in nature. For the first 2 weeks, there were no meetings which was then followed by a great introductory meeting in which the sheer

knowledge and teaching abilities of the mentors were observed. Following this, clear communication took place between the students and the mentors with nearly weekly meetings until the day of seminar 1. However after this, there were again very slow response rates and one problem solution took up nearly an entire week. After the was done, both mentors went on official assignments and resulted in nearly 4-5 days of nearly no communication. The slow response time and drop in regular work assignment in the last weeks of the internship was disappointing as the opportunity to learn from such knowledgable mentors was cut short. Having said that, in whatever meetings did take place, the professors were patient and clear about the topic and it was a great learning experience. Furthermore, Prof. Ankita Choudhary served as an active and approachable POC throughout the course of the internship, providing clarity whenever neccessary.

Academic courses relevant to the project: Separation Processes, Heat Transfer, Probability and Statistics, M3 (Differential Equations)

Learning Outcome: 1. Research
2. Literature Review
3. Presentation skills
4. Preparation of a Project Report
5. Using ALOHA software to simulate real world problems

PS-I station: AERB - Civil , Mumbai

Student

Name: VANSH DESAI .(2021A2PS1361P)

Student Write-up:

PS-I Project Title: Performance Assessment of Nuclear Reactor Containment Structure under Missile Impact Loads

Short Summary of work done: I worked for the Atomic Energy Regulatory Board, which is responsible for regulating all activities involving atomic and nuclear energy. I discovered how radiation is used in a variety of fields, including healthcare and energy generation. I read a total of 11 papers throughout my internship, some of which were challenging to comprehend, especially for an undergraduate. I made an effort to understand a variety of topics, including continuum-damaged plasticity models and finite element analysis. My

understanding of the ABAQUS model, which served as our final submission to the PS Station, was aided by all of these concepts. As my project title asks me to simulate the behavior of RC slab under impact of a steel hammer, this was simulated in ABAQUS with different cases of velocity of impact of the hammer on the RC slab. The real experiment results from one of the well-known works in this field, the Zinneddin and Krauthammer study, were then compared to my results. Throughout the journey, my PS station mentor from AERB, Moloy Chakroborty, also gave us some knowledge on how nuclear energy power plants operate and differ from other power plants and also stressed the significance of our study.

Objectives of the project: Model a simulation of a hammer impacting on RC slab to study the behaviour of the RC slab under impact loading

Tool used: ABAQUS

Details of Papers/patents: None

Brief description of the working environment: PS1 was a wonderful experience for me as I was finally applying what I had studied to my work. My PS mentor was very helpful in making us learn the ABAQUS software and helped us how to approach problems and how can we optimised our project. I also was exposed to other smart colleagues from BITS who also helped me in my PS journey and we learned how to work as a team.

Academic courses relevant to the project: Continuum Damaged Plasticity, Finite Element Analysis, Failure Theories

Learning Outcome: I became familiar with ABAQUS, learned the fundamentals of finite element analysis, and gained in-depth knowledge of the characteristics of various materials' properties, such as their stress-strain curves, and how high strain rates affect steel. Additionally, I learned about impact loads and how software simulates them. Apart from these I also learnt about various failure phenomenon in civil engineering.

PS-I station: AERB - Civil , Mumbai

Student

Name: SANSKAR SHARMA .(2021A2PS1592P)

Student Write-up:

PS-I Project Title: Evaluation of structural capacity of RCC elements subjected to aircraft impact.

Short Summary of work done: We started off by learning the working of Nuclear power plant and gradually advanced to our problem statement. After carefully reading,analysing and presenting a few research papers on this field we were able to get a idea on how we have to go about our project. We started modelling on Abaqus and after it was done, extracted the necessary outputs and inputs to calculate the leakage rate.

Objectives of the project: To prepare a simulation model of a containment structure on Abaqus and calculate the leakage rate on MATLAB.

Tool used: ABAQUS, MATLAB, MS Excel, MS Word, MS PPT, Grapher

Details of Papers/patents:N/A

Brief description of the working environment: The station was an online one, so the description is based on my communication with one person only.
Working with Moloy Sir, one certainly gets to know that he has a lot of experience in this field.

Academic courses relevant to the project: Mechanics of Solids, Structural Analysis

Learning Outcome: We learnt about the wroking of a typical nuclear power plant. We discussed on topics like the integrity of the containment structure and leakage of radiation through cracks. We also learnt how to employ softwares like abaqus to perform finite element analysis on structural elements and determine the variation of various parameters across the element.

PS-I station: AERB - Electronics , Mumbai

Student

Name: SMRITI .(2021A3PS1424P)

Student Write-up:

PS-I Project Title: Dynamic response of in-core neutron detectors

Short Summary of work done: Our topic was - "Dynamic Response of In-core Neutron Detectors'. We worked on developing models to improve the response of Self-powered neutron detectors. We also looked at the response to step change in neutron flux. My study was basically on prompt type SPND, so needed to see whether Cobalt is a worthy emitter material and conclude on the same.

Objectives of the project: To develop mathematical models to improve the response of in-core neutron detectors.

Tool used: MATLAB

Details of Papers/patents:-

Brief description of the working environment: The people at AERB emphasize on understanding the concepts to the core. We had weekly meetings to discuss about our progress. Since it was research oriented, it would have been better if it was offline, but nevertheless, learning here was very fun.

Academic courses relevant to the project: Microelectronics, Signals and systems, control systems

Learning Outcome: Need to work with MATLAB to be able to replicate some of the models, and learn about methods for noise suppression.

PS-I station: AERB - Electronics , Mumbai

Student

Name: GUNNIT AHUJA(2021A8PS2330G)

Student Write-up:

PS-I Project Title: Dynamic Response Assessment of In-Core Neutron Detectors

Short Summary of work done: Studied the Delayed Response Neutron Detectors in detail. Conducted a study to bring out characteristics of Vanadium Self-Powered Neutron Detectors and its dynamic response. Analyzed the response of Vanadium Self-Powered Neutron Detectors to show the change in response and output current of the same, with

changing neutron flux and change in time. Using mathematical models, compared the efficacy of the different Dynamic Compensation Methods available for response improvement of the Vanadium Self-Powered Neutron Detectors.

Objectives of the project: Study of Delayed Response Self-Powered Neutron Detectors

Tool used: MATLAB

Details of Papers/patents: None

Brief description of the working environment: Industry Mentors were very knowledgeable and offered valuable guidance and help at various steps, throughout the course of the internship. This internship helped in improving my research skills, communication and presentation skills.

Academic courses relevant to the project: Mathematics-I,III and Control Systems

Learning Outcome: Studied the Delayed Response Neutron Detectors in detail to bring out characteristics of Vanadium SPNDs and its dynamic response. Enhanced Research Skills, while studying tens of research papers and articles. Also enhanced communication and presentation skills, throughout the process of presenting my work.

PS-I station: AERB - General , Mumbai

Student

Name: ANAGH ADITYA(2021A3PS1567H)

Student Write-up:

PS-I Project Title: Development of a Computational Model to calculate Environmental Radiation dose

Short Summary of work done: First, I read up on Methodologies of Environmental radiation dose assessment and Atmospheric dispersion modelling to understand the different pathways taken by radionuclides when expelled by a nuclear reactor (through the stack mainly) and eventually reaching man. The Gaussian-Plume atmospheric dispersion model helped visualize how the radionuclides spread from the stack. Next, I

compiled all the different radiation doses reaching man through air or aquatic routes. Finally I implemented this as a program on Python which could compute these doses.

Objectives of the project: Ability to convert physical model to a computational model, Verification and validation process involved in developing a computational tool, Data visualization and interpretation

Tool used: S/w - Python

Details of Papers/patents:AERB Safety Guide-1 (based on Atmospheric dispersion modelling), AERB Safety Guide-5 (based on Environmental radiation dose assessment)

Brief description of the working environment: The AERB folks were very friendly and helpful and placed no hard deadlines. My mentor left the work to me and was always available to clarify any doubt. So the working environment was really amiable and cooperative.
My expectations from the company were to improve my technical and soft skills, learn about organizational attitudes and how to work in a professional environment while coordinating with others. All of these were satisfied beyond expectations for me.

Academic courses relevant to the project: Mathematics-1 (MATH F111), Probability and Statistics (MATH F113)

Learning Outcome: Learned how to implement a physical model on software, Improved Python technical skills, Gained a good idea of data visualization and analysis, Improved soft skills and social skills

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PS-I station: AERB - General , Mumbai

Student

Name: RIKHIL GUPTA .(2021A7PS0533P)

Student Write-up:

PS-I Project Title: Development of GUI for probability distribution functions

Short Summary of work done: I developed a python GUI using tkinter that generates samples of distribution functions of Uniform distribution, Normal distribution, exponential

distribution, log normal distribution, Weibull distributions, GEV distributions. Once samples are generated, plot them using matplotlib.

Objectives of the project: Development of a Python GUI to generate samples and plot functions.

Tool used: Tkinter, matplotlib, numpy, python

Details of Papers/patents:NA

Brief description of the working environment: Good and supportive working environment, the company expects you to learn and do the work.

Academic courses relevant to the project: Probability and Statistics

Learning Outcome: GUI development - Tkinter, Probability and Statistics

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PS-I station: AERB - General , Mumbai

Student

Name: ARYAN P SONWATIKAR .(2021A7PS2220P)

Student Write-up:

PS-I Project Title: Efficient Solvers for Systems of Linear Algebraic Equations

Short Summary of work done: I was the only student under my project. The first half of the PS work was understanding the functioning of various solvers, and then coding them in MATLAB (of which I had no previous experience, so a week was devoted to familiarising myself with MATLAB as well). Post-midsem came the actual comparison work, where the codes were tested on a real-life large matrix. The last couple of weeks were also devoted to understanding the eigenvalue problem and the generalised eigenvalue problem, and their connection to linear solvers.

Objectives of the project: To write MATLAB code for and compare various direct and iterative methods to solve linear systems of the type AX=B.

Tool used: MATLAB

Details of Papers/patents:-

Brief description of the working environment: Since it was an online station, work coordination with my AERB mentor happened during regular Gmeets, where sir would review the work I had done and then assign me further work until the next meet.

Academic courses relevant to the project: Mathematics 2, Computer Programming

Learning Outcome: I understood the practical importance of optimising runtimes, in addition to deepening my knowledge of Linear Algebra.

PS-I station: AERB - General , Mumbai

Student

Name: TRISHA HITESH(2021B4A72157G)

Student Write-up:

PS-I Project Title: Methods of extreme value analysis and applications in hazard analysis

Short Summary of work done: I was assigned the project methods of extreme value analysis and instructed to conduct a literature review of the same. After I was exposed to the many methods and their application I then began literature review won the mathematical modeling of the same. Then proceeded to model the methods in MATLAB without using builtin models and by verifying those results with builtin models.

Objectives of the project: To understand eva methods and their modeling and further their application in hazard analysis

Tool used: MATLAB

Details of Papers/patents:-

Brief description of the working environment: Was very supportive and positive interaction with both mentor and faculty

Academic courses relevant to the project: Prob Stats, Mathematical Modelling

Learning Outcome: EVA methods and MATLAB use for modeling

PS-I station: AERB - General , Mumbai

Student

Name: NIMISH SHARMA .(2021B5A71144P)

Student Write-up:

PS-I Project Title: Study of various decision analysis methods for Integrated Risk Informed Decision Making

Short Summary of work done: I gained knowledge about functioning and regulation in nuclear power plants and how the process of IRIDM (Integrated Risk Informed Decision Making) combined with various techniques of decision analysis can result in a efficient process for decision making in nuclear power plants. I finally made a Graphical User Interface which is essentially a Multi criteria decision analysis (MCDA) tool and can be used for general decision making.

Objectives of the project: To understand various Multi Criteria Decision Analysis(MCDA) methods and develop a Graphical User Interface(GUI) for performing MCDA

Tool used: Pycharm (integrated development environment used for programming in Python)

Details of Papers/patents: None

Brief description of the working environment: This PS station was online. My interactions with people involved in PS-1 at AERB were of very friendly nature. My expectation from AERB was to learn about nuclear energy and power plants in general.

Academic courses relevant to the project: Computer Programming, Technical Report Writing

Learning Outcome: Python coding skills(Graphical User Interface development), Knowledge of various Decision Analysis methods like AHP, TOPSIS etc.

PS-I station: AERB - Mechanical , Mumbai

Student

Name: DEBANGSHU CHOWDHURY(2021A4PS0264G)

Student Write-up:

PS-I Project Title: Thermal and Hydraulic Modelling of a Small Modular Reactor

Short Summary of work done: The first part of the internship involved researching the general functioning of pressurized water reactors, and the modified applications of the same principle in smaller scale reactors, such as the NuScale Power Module. The components were simplified and then modelled using the software RELAP5, and once the parameters matched the operational values given in the official documentation, a rupture was introduced to inspect the SMRs ability to deal with accidents.

Objectives of the project: Modelling and simulating the NuScale Power Modulde SMR

Tool used: RELAP5

Details of Papers/patents:none

Brief description of the working environment: It was an online internship, so the mentor regularly gave me work and I was responsible to get back to him once I'm done with said work, within the stipulated amount of time

Academic courses relevant to the project: Fluid Mechanics, Heat Transfer

Learning Outcome: The ability to use the proprietary software RELAP5 and research on the innovations with respect to the development of small modular reactors

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PS-I station: AERB - Mechanical , Mumbai

Student

Name: SINGH HARSHVARDHAN BHARAT SHARAN(2021A4PS1336P)

Student Write-up:

PS-I Project Title: Cfd simulation for temperature distribution in multi compartment geometry.

Short Summary of work done: Work was based on learning cfd simulations on ansys fluent. My project also involved a greater amount of programming for writing user defined functions (UDF) in ansys fluent. With UDF you can provide different boundary conditions which are not available in fluent . Then learnt the basics of simulating problems using cfd.

Objectives of the project: To study temperature distribution.

Tool used: Ansys fluent, C programming

Details of Papers/patents: No other papers involved.

Brief description of the working environment: Work environment was good. Could learn many things , although a new and good experience.

Academic courses relevant to the project: Fluid Mechanics, Heat Transfer

Learning Outcome: Learnt using ansys and user defined functions in ansys fluent using C programming language.

PS-I station: AERB - Mechanical , Mumbai

Student

Name: MANAS AGARWAL .(2021A4PS1349P)

Student Write-up:

PS-I Project Title: Estimation of fuel pin, PT and CT temperature after fuel pins disengagement from end plates during the accident scenario LOCA + ECCS using FEM

Short Summary of work done: Initially I was told to get familiar with COMSOL and learn the various possible operations on it.Then I designed a Formed 37 fuel pin Geometry, Applied Material Properties,Physics,Mesh and Computed the steady state result.Then I designed a deformed geometry where the fuel pins had disengaged from the plate and piled up.After adding the Material Properties,Physics and Mesh,I computed the time dependent temperature variation in the geometry after LOCA and ECCS had occurred.

Objectives of the project: To simulate temperature variation in a 37 fuel pin assembly in Formed and Deformed Case.

Tool used: COMSOL Multiphysics Version 5.6

Details of Papers/patents:-

Brief description of the working environment: Overall it was a great experience to work in such an esteemed organisation.The Mentors were really supportive and guided me throughout the process. The BITS Professor was really kind to schedule regular meetings and helped in every way possible.I got to know a lot of new things related to Nuclear Energy and Safety protocols.

Academic courses relevant to the project: Heat Transfer

Learning Outcome: Got to know a lot about Pressurized Heavy Water Reactor(PHWRs),Loss of Coolant Accident(LOCA) and Emergency Core Cooling System(ECCS).Understood the usage of COMSOL Multiphysics Version 5.6 which is a finite element analyzer, solver, and simulation software package for various physics and engineering applications.

PS-I station: AERB - Mechanical , Mumbai

Student

Name: TEJAS MATHUR SRIGANESH .(2021A4PS1415P)

Student Write-up:

PS-I Project Title: MODELLING AND ANALYSIS OF HEAVY WATER PROPERTIES IN NUCLEAR REACTORS

Short Summary of work done: During PS-I, my main objective was to develop a robust and efficient D2O property estimation code using the IAPWS standard. I aimed to gain a deep understanding of D2O's thermodynamic properties, especially in nuclear reactors like PHWRs. To validate the code's accuracy, I meticulously compared its results with the existing IAPWS data. Throughout the project, I delved into the complexities of D2O's behavior under varying conditions. The identification and resolution of isolated inaccuracies in the IAPWS dataset, particularly concerning random spikes in density values, sharpened my analytical and critical thinking skills. The project underscored the significant role of D2O as a coolant and moderator in nuclear reactors, highlighting its importance in optimizing reactor behavior and design. My commitment to accuracy and reliability led me to continuously monitor and update the code, ensuring it provides reliable property estimates. Overall, PS-I has been a valuable learning experience, empowering me with specialized knowledge in D2O properties and their applications in the nuclear power industry and other scientific fields.

Objectives of the project: The main objectives of this project are to develop a robust D2O property estimation code based on the IAPWS standard. By accurately simulating D2O properties, the code aims to support scientific and industrial applications, especially in nuclear reactors

Tool used: laptop,vscode,iapws package

Details of Papers/patents:none

Brief description of the working environment: During PS-I, I worked in a remote environment from the comfort of my home, utilizing my laptop for all tasks. The working environment was flexible, allowing me to manage my time effectively and strike a balance between work and personal commitments. Despite the distance, I remained connected to my team through virtual meetings and communication tools, ensuring smooth collaboration and knowledge sharing.

Expectations from the company during PS-I were to deliver a robust D2O property estimation code based on the IAPWS standard. This entailed thorough research, meticulous coding, and validation of results through comparisons with existing data. The company expected a high level of accuracy and reliability in the code, as it would be crucial for scientific and industrial applications, particularly in nuclear reactors.

During PS-I, I learned to dive deep into D2O's thermodynamic properties, gaining specialized knowledge in this field. I honed my coding skills and problem-solving abilities while identifying and addressing isolated inaccuracies in the IAPWS dataset. Working remotely taught me self-discipline and time management, enabling me to stay focused

and productive. Additionally, I developed effective communication skills in virtual team environments, collaborating seamlessly with colleagues from different locations.
Overall, the experience of PS-I allowed me to learn valuable technical and soft skills, contributing to my growth as a professional in the field of D2O properties and scientific software development.

Academic courses relevant to the project: Thermodynamics-first year,fluid mechanics-second year

Learning Outcome: The project's major learning outcomes include gaining an in-depth understanding of the thermodynamic properties of deuterium oxide (D2O). By developing the D2O property estimation code based on the IAPWS standard, we have delved into the complexities of D2O's behavior under varying conditions. This exploration has provided valuable insights into its unique physical and chemical properties, particularly in nuclear reactors like Pressurized Heavy Water Reactors (PHWRs). Moreover, the meticulous comparison with the IAPWS data has honed analytical skills and critical thinking, enabling the identification and resolution of inaccuracies. Overall, the project has empowered researchers with specialized knowledge to optimize D2O's application across scientific and industrial domains.

PS-I station: AERB - Mechanical , Mumbai

Student

Name: S. VIBAV .(2021A4PS1899H)

Student Write-up:

PS-I Project Title: ASSESSMENT OF CONTAINMENT MODEL OF PRABHAVINI

Short Summary of work done: Learnt the working the behind PRABHAVINI software of AERB and how it is mathematically modelled to simulate accident conditions. Also learned how to use the software the different modules inside the software and steps to simulate an nuclear containment during accident condtions

Objectives of the project: To learn and use computational software of AERB in modelling accident conditions in nuclear reactors.

Tool used: PRABHAVINI Software, Ubuntu , Excel

Details of Papers/patents:none

Brief description of the working environment: It was a very professional work environment and the mentors were patient and helpful. The company expects the students to be sincere in their work and eager to learn , other than that the company does not expect students to know any prerequisites and mentors provide enough support to learn the prerequisites. Communication with the mentors could have been better as the internship was online.

Academic courses relevant to the project: Fluid Mechanics, Heat Transfer

Learning Outcome: Learn how computational software PRABHAVINI of AERB is used to model accident conditions in nuclear reactors and simulate an experiment with the software.

PS-I station: AERB - Mechanical , Mumbai

Student

Name: Divya Riken Patel(2021A4PS2037G)

Student Write-up:

PS-I Project Title: Integrity assessment of a typical nuclear reactor containment consequent to aircraft crash scenarios.

Short Summary of work done: Undersood working of nuclear containment structures and various loads which can act on it. Learnt how crack formation in nuclear containment structure can lead to leakage of nuclear radiation and formulated a methodology on how to calculate air leakage rate out of the containment based on the loading conditions using p

Objectives of the project: To estimate leakage rate and leakage area of nuclear containment structure consequent to external impact scenario

Tool used: ABAQUS, MATLAB

Details of Papers/patents:no

Brief description of the working environment: Tri weekly google meets with entire team of colleagues, open environment . Expected more detailed work and guidance which was not received.

Academic courses relevant to the project: Mechanics of Solids, Advanced Mechanics of Solids, Fluid Mechanics

Learning Outcome: understood Nuclear Power plant and Nuclear containment structure working, learnt how to use MATLAB and ABAQUS, learnt in detail about material properties of concrete as well. Understood the importance of nuclear safety regulations.

PS-I station: AERB - Mechanical , Mumbai

Student

Name: AMAL PRAKASAN MANATHANATH(2021A4PS2040G)

Student Write-up:

PS-I Project Title: Thermal Hydraulic Analysis of Relocated Channels

Short Summary of work done: I modelled a section of a coolant channel to perform thermal hydraulic analysis of the section to analyse the conditions where enhanced hydrogen generation may occur in the channel.

Objectives of the project: 1) To perform thermal hydraulic analysis of channels near the core region. 2)To study the effects of Hydrogen generation

Tool used: SCDAP/RELAP5/MOD3.3

Details of Papers/patents:-

Brief description of the working environment: The work environment was excellent, my mentor took me through every step of the process and was always open to answering doubts and providing the necessary guidance. However, I am of the opinion that the internship would be much more impactful if it was offline rather than online as the software I used could only be run on AERB computers and this proved to a hindrance.

Academic courses relevant to the project: Fluid mechanics, Thermodynamics, Material Science

Learning Outcome: During my PS, I learned how to use keywords to find relevant research and also to analyze and find gaps in existing literature. Going into the technical details of my work, I learned how to work with input decks and used it in SCDAP/RELAP5/MOD3.3.
Besides technical and soft skills, I was introduced to the internal working of a nuclear reactor. I was able to apply the learnings from my academic course. I studied how flow characteristics in the reactor core affect multiple parameters like core temperature, fission rate, hydrogen generation, etc.

PS-I station: AERB - Mechanical , Mumbai

Student

Name: VISHNU GOPALAKRISHNAN .(2021A4PS2450H)

Student Write-up:

PS-I Project Title: Analysis of overhead crane under dynamic excitation

Short Summary of work done: My project primarily comprised of designing an overhead crane and performing simulations on the integrated model in order to find it's natural frequency via modal analysis. I worked on ABAQUS software to both model and simulate an overhead crane. Over the course of the project I gained an in-depth understanding of how a finite element software works and the various kinds of simulations that are done in the industry in order to create safe and structurally sound components. I was able to successfully model and validate my simulation data with experimental values.

Objectives of the project: Creating a model of an overhead crane composed of multiple part and conducting modal analysis on it using finite element analysis software

Tool used: Dassault System's ABAQUS simulia software

Details of Papers/patents: Nil

Brief description of the working environment: My experience overall working at AERB was great. I was able to get involved in a constructive manner and picked up many new

skills over the course of two months. I was able to successfully model and simulate an overhead crane and validate my data with experimental values. The mentors and faculty advisors were helpful at every step of the way and managed to clarify all our doubts and queries. The learning experience was a highly positive one. I managed to gain a detailed understanding of how FEA solvers work and the core concept behind it's functioning.

Academic courses relevant to the project: Mechanics of solids, Advanced mechanics of solids, Material science

Learning Outcome: Finite element analysis and various types of simulations.

PS-I station: AERB - Mechanical , Mumbai

Student

Name: NINAAD SACHIN KOTASTHANE(2021A4PS2545G)

Student Write-up:

PS-I Project Title: Estimation of Leakage of Nuclear Reactor Containment under Accident Damage conditions

Short Summary of work done: We conducted extensive literature reviews and outlined a comprehensive methodology for the estimation of radiation leakage rates for different types of containments; recieved FEA data as outputs from ABAQUS and implemented the mathematical model in MATLAB.

Objectives of the project: Estimating the leakage rate through cracks in unlined rc Containment

Tool used: MatLab

Details of Papers/patents:none

Brief description of the working environment: The mode was online, so we had regularly scheduled meetings where the status of weekly goal were reviewed. It was a collaborative experience with interns from IITB. Presentations of our research and findings were done.

Academic courses relevant to the project: none

Learning Outcome: Nuclear containment structural analysis, literature review, multiple probability simulation in matlab

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PS-I station: AERB - Mechanical , Mumbai

Student

Name: ISHA BANDYOPADHYAY(2021A4PS2888G)

Student Write-up:

PS-I Project Title: Leakage area and leakage rate prediction for probabilistic risk assesment of concrete containments under severe core conditions

Short Summary of work done: Read a lot of papers regarding the project I was assigned and created a literature review. Made common steps to approach the problem statement from all the papers and made a flow on it. After the methodology was made, I started working on the Matlab code. Once the code was done we started with the simulation process and obtained the values of flow rates with different scenarios. With the results of the simulation, we made graphs and made a general analysis of how to reduce the leakage rate through nuclear power plant containment zone.

Objectives of the project: To make a matlab code which can determine the leakage rate though a nuclear power plant containment zone due to impact from a missile or aircraft crash

Tool used: Matlab

Details of Papers/patents: We didn't publish any paper

Brief description of the working environment: They expect sincerity in our work and if any work is assigned to us, we are expected to be ready with it and present it on time even if it has errors. We are expected to follow basic ethics like joining the meet 2 min prior and we cannot join for any reason we should inform them well before .

Academic courses relevant to the project: Fluid Mechanics, Material science

Learning Outcome: Learned to use matlab, developed a good understanding of fluid mechanics in nuclear engineering , learnt about the construction of nuclear power plant containment zone

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PS-I station: AERB - Mechanical , Mumbai

Student

Name: PRETYUT SUNDARRAJAN .(2021B2A40898P)

Student Write-up:

PS-I Project Title: Ageing assessment studies in chemical industries to tackle corrosion

Short Summary of work done: Initiially i was asked to conduct a literature review on the latest research available on corrosion tackling and how corrosion takes place in other industries. After about 3 weeks I was asked to instead focus on a particular plant which had few specific conditions and estimate the rate of corrosion.

Objectives of the project: To create a model to estimate corrosion in chemical industries and assess how to prevent it

Tool used: Adobe Acrobat reader , Excel

Details of Papers/patents:NA

Brief description of the working environment: Because our PS was online I don't know about the working environment.
But the expectations from the company are very flexible , no fixed working hours etc

Academic courses relevant to the project: NA

Learning Outcome: Learnt how corrosion takes place in different chemical industries and how to tackle it.
Learnt about various materials used to prevent corrosion caused by various corrosive agents.
Learnt about different processes used in industries to tackle corrosion

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PS-I station: AERB - Mechanical , Mumbai

Student

Name: ADARSH MISHRA .(2021B4A42503P)

Student Write-up:

PS-I Project Title: Vibrational Analysis and Isolation of a Turbine

Short Summary of work done: I was given the task to write a brief account of analysis of Vibrations, both regular and irregular, of steam turbines. Then I identified probable causes, like resonance, rotor imbalance, misalignment, mechanical wear-tear etc. I also went through the basic standard procedure

Objectives of the project: Vibrational Analysis of a Steam Turbine used in Nuclear Plant and recommendations of Vibrational isolation methods.

Tool used: None

Details of Papers/patents:1. Allianz et. al. (1974), Vibration and Noise Analysis in Nuclear Power Plants
2. Siniscalchi and Prates (2009), Supervisory System of the Turbine Generator set of Angra 1 NPP
3. EPRI Journal (2016), Real-Time Eye on Turbines
4. Shin, Kim and Kim (1998),

Brief description of the working environment: As it was online, there wasn't much of a working environment. Expectations were moderate and not too demanding. I learned quite a lot about the various divisions that are required to run and maintain a nuclear power plant, and to regulate it.

Academic courses relevant to the project: Vibrational Analysis and Damping

Learning Outcome: How Vibrational Analysis is done of a Turbine, and the types of Vibrational Isolation tools that are used in turbines.

PS-I station: Agromach Engineering Pvt. Ltd. , Faridabad

Student

Name: PRIYANSH AGGARWAL .(2021A4PS0744P)

Student Write-up:

PS-I Project Title: STREAMLINING THE PRODUCT MANUFACTURING DEPARTMENT

Short Summary of work done: I had to work with sales department to make sure that quotations are made using the erp software. This is a major bottleneck since all the processes can function only if the name of customer as well as the item name is saved in the software. I further worked with marketing department in order to reach out to new potential customers. I further worked with store department to improve tracking process.

Objectives of the project: To understand working of Store Department, Production Department & Marketing Department and further working for ERP implementation.

Tool used: AutoSquare, AiSensy

Details of Papers/patents:-

Brief description of the working environment: Working environment was positive with everyone keen to share learning. I learnt a lot of processes involved in manufacturing companies.

Academic courses relevant to the project: Engineering drawing

Learning Outcome: Understanding the working of manufacturing companies.

PS-I station: Agromach Engineering Pvt. Ltd. , Faridabad

Student

Name: GANTAVYA SHARMA .(2021A4PS3100H)

Student Write-up:

PS-I Project Title: Streamline working of industry

Short Summary of work done: The work done by me was understanding the work done by different departments of the organization such as Design & Development, sales, HR, Accounts, purchase, store and production. I learned the functioning of 5 different machines, the work done by the sales executive and coordinator, how an industry manages its store to ensure no shortage of material occurs and different processes used by the company to manufacture its machines.

Objectives of the project: Understand the work of all departments in the company and how they interact with each other at different stages of a project

Tool used: NA

Details of Papers/patents:NA

Brief description of the working environment: The company has a positive employee culture, and all employees perform their tasks diligently. The industrial exposure during the PS-I helped me improve my soft skills and taught me many new things such as lean manufacturing tools and how teams that handle different tasks work together to ensure smooth workflow in an organisation. Apart from this We were able to witness some live interviews at the organisation that will help us in our future endeavors.

Academic courses relevant to the project: Manufacturing Processes, Workshop

Learning Outcome: Responsibilities and roles of different departments and some Lean Manufacturing tools

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PS-I station: Agromach Engineering Pvt. Ltd. , Faridabad

Student

Name: SUYASH KHARE(2021B2A43021G)

Student Write-up:

PS-I Project Title: Streamline the working of Industry

Short Summary of work done: I WAS ABLE TO GRAB DEEP KNOWLEDGE OF FINANCE DEPARTMENT AND HOW THE FINANCES ARE MANAGED IN BIG INDUSTRIES SECONDLY I LEARNT ABOUT THE CODES INVOLVED IN THE WORKING OF MACHINES THERE AND HELPED IN DEVOLOPING ERP PLATFORM OF THE COMPANY PRODUCTION WORK OF VARIOUS MACHINES LIKE CONVEYOR AND LASER COATING MACHINES

Objectives of the project: FINANCE OF THE INDUSTRY PRODUCTION, SALES AND MARKETTING OF A INDUSTRY ALSO THE DESIGN DEVOLOPEMENT OF MACHINERY

Tool used: ERP TALLY PRIME SOLID WORKS AND OTHER MACHINERY USED IN AGRICULTURE AND MILLING FIELD

Details of Papers/patents:NO

Brief description of the working environment: Working environment was fine though In the finance department, I learned about how a company's finances are managed and how to make e-invoices and e-way
billing, as well as why these bills are important. I also learned
how to use the software Tally Prime. I learnt about the company's numerous risks, such as
currency risk, interest rate risk, and credit risk and many
other risks. I learnt the codes involved and helped in creating an erp platform for the company. I learnt various codes used in mechanical machinery also design of the machines and development of the machines.

Academic courses relevant to the project: mechanical workshop, computer programming, principle of economics

Learning Outcome: FINANCE OF AN INDUSTRY HOW TO CHANGED ITS FINANCES HELPED IN DEVOLOPING ERP OF THIS INDUSTRY
LEARNT THE COMPUTER CODE PATTERN OF THE MACHINES
PRODUCTION OF A MACHINERY INDUSTRY
SALES MARKETTING AND SERVICES OF A INDUSTRY
DESIGN AND DEVOLOPEMENT OF THE MACHINES INVOLVED

PS-I station: Agromach Engineering Pvt. Ltd. , Faridabad

Student

Name: JAY MITTAL .(2021B4A41356P)

Student Write-up:

PS-I Project Title: Streamline the working of industry

Short Summary of work done: I have worked in each department of industry like finance, HR , production, sales and services etc. collobrate with other department for smooth and time saving fuctioning of each department

Objectives of the project: Learn more about the industry and working of each and every department of a industry and how each department coordinate with each other

Tool used: S/w like tallyprime, solid works

Details of Papers/patents:N/A

Brief description of the working environment: The working environment is quite good, but the location is creating issues like no transport available, etc. We expect the company to have great infrastructure and be in a good loacation.

Academic courses relevant to the project: N/A

Learning Outcome: I have learnt about company and its sturacture and how each and every department work together and learnt more about the finance and specifically production and how company manager its production rate and learnt more about the sales and marketing of a company and how from scratch company find its customers and built a brand value by providing its regular and proper services nad learnt more about the management and overall functioning of a company.

PS-I station: Agromach Engineering Pvt. Ltd. , Faridabad

Student

Name: ANCHIT MAHESHWARI(2021B5A42452G)

Student Write-up:

PS-I Project Title: Streamlining the process of manufacturing machines

Short Summary of work done: I was assigned to get comfortable with erp software and resources were provided for the same. Increasing the efficiency of the company was highly depeneded on the incoming of resources like metal sheets and spare parts. As destoner was one of the major machinery manufactured by the company, we were required to get complete understanding of it, afterwards we had to test and trail with raw paddy

Objectives of the project: 1.Increasing efficiency of the company with the use of enterprise resource planning (ERP) software. 2. Designe machines on solid works software . 3.Complete understanding of important rice milling machines

Tool used: Enterprise resource planning , solidworks.

Details of Papers/patents:NA

Brief description of the working environment: The working environment was decent although the place where the cutting , welding , shot casting processes takes place were a bit messy. One can expect the complete understanding of what a core mechanical company and how a machine is manufactured from a sheet of metal to a fully functional complex machine.

Academic courses relevant to the project: Workshop

Learning Outcome: 1. ERP software
2. Destoner machine
3. Husk aspirator
4. Solid works [basic]

PS-I station: Alucoat , Jaipur

Student

Name: JAYESH JOSHI .(2021A4PS0885P)

Student Write-up:

PS-I Project Title: Aluminum extrusion designs using AutoCAD

Short Summary of work done: We were first told to search about some topics regarding Aluminum industry. Then, we were taught to design aluminum extrusion designs using autocad.

Objectives of the project: To design aluminum extrusion, doors and windows more efficiently using AutoCAD.

Tool used: AutoCAD

Details of Papers/patents: None

Brief description of the working environment: We had to go to work only 1 or 2 days in a week and the employees at the company were very friendly and supportive.

Academic courses relevant to the project: Material Science(2-1) and Manufacturing Processes(2-2).

Learning Outcome: Revision of many concepts of courses like manufacturing processes and material science. Also, you get a good grasp over AutoCAD.

PS-I station: Alucoat , Jaipur

Student

Name: AVI SANCHETI .(2021A4PS1702P)

Student Write-up:

PS-I Project Title: Aluminium designs using AutoCad LT

Short Summary of work done: We learned how a industry works, even had hands on experience on how a window is made and assembled and how glass is fit into it, we also designed various designs using AutoCad LT consisting of sliding windows, doors S,R series and much more.

Objectives of the project: Designing the profiles that are available on their website and subsequently modifying them accordingly as per needs using AutoCad

Tool used: Autocad and mechanical courses basics

Details of Papers/patents: None

Brief description of the working environment: We learned how a industry works, even had hands on experience on how a window is made and assembled and how glass is fit into it, we also designed various designs using AutoCad LT consisting of sliding windows, doors S,R series and much more. All in all working environment was pretty good, even we had lunch with the owner and also he was pretty keen to teach each and every detail

Academic courses relevant to the project: Material Science, MNM, Mechanics of solid, Engineering Graphics

Learning Outcome: We learned how a industry works, even had hands on experience on how a window is made and assembled and how glass is fit into it, we also designed various designs using AutoCad LT consisting of sliding windows, doors S,R series and much more.

PS-I station: Alucoat , Jaipur

Student

Name: DEVANSH DHADDA .(2021A4PS3209H)

Student Write-up:

PS-I Project Title: Aluminium Extrusion and AutoCAD

Short Summary of work done: We made 40 mm sliding windows, learnt in depth about the processes involving fabrication and we also made profiles for customers. Designing was very crucial and we made it through using AutoCAD

Objectives of the project: We learnt about fabrication and made extrusion designs using AutoCAD

Tool used: AutoCAD

Details of Papers/patents: No

Brief description of the working environment: The environment was really nice and welcoming, our mentor was one of the best people I have met. He encouraged us a lot to learn new things and allowed us to slide between projects as you like. I wanted to know about Aluminium business and learnt a lot.

Academic courses relevant to the project: Engineering Graphics, a little bit of Mechanisms and Machines

Learning Outcome: Learnt about Aluminium Industry, extrusion, fabrication and designing

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PS-I station: Alucoat , Jaipur

Student

Name: HARSH JAIN(2021B4A42868G)

Student Write-up:

PS-I Project Title: Secession on Aluminium door and window hardware and accesories materials and their practical usages

Short Summary of work done: I learned about aluminium. We fabricated a 44mm series sliding window ourselves. Also we learnt how to design it ourselves using autocad. Also the pros and cons of the procedure. We also learnt about the use cases and the installations.

Objectives of the project: The objective was to learn how aluminium windows and doors are designed, and fabricated along with all the supplementary details .

Tool used: Powder Coating Booth, Autocad software, Milling machine, Heating oven

Details of Papers/patents: No

Brief description of the working environment: The work environment was safe, efficient, and environmentally responsible. There was advanced technology, promoted employee well-being, encouraged teamwork and communication, and offered opportunities for growth while complying with regulations.
The expectation were not that high, but when I saw the premises, they were awesome. A huge factory with more than 50 workers. They all were friendly and helpful.

Academic courses relevant to the project: Engineering Graphics and Workshop

Learning Outcome: Fabrication and Designing of various aluminium profiles of doors and windows.

PS-I station: Apollo Tyres Ltd. , Gurugram

Student

Name: SHIVAM AGARWAL(2021A4PS2883G)

Student Write-up:

PS-I Project Title: Footfall at Brand Retail Outlets (BROs)

Short Summary of work done: To prepare a service strategy, we first understood the operations and the services provided at the Apollo Tyres BROs. During our project, we visited 35+ retail outlets, including of Apollo Tyres & other competitor brands like MRF, Bridgestone, Continental Tyres,etc. Understood the pain points of the dealer as well as customers by asking through a framed questionnaire. Finally prepared a service based strategy incorporating upcoming technologies and learnings from different tyre brands. The strategy also involved the learnings from companies of different sectors like from Starbucks. In conclusion, our project involved market research, competitor analysis, suggesting new services & improvisation of the existing ones.

Objectives of the project: To prepare services based strategy for increasing the footfall at the Brand Retail Outlets of Apollo Tyres.

Tool used: Canva, Microsoft office

Details of Papers/patents: Nothing

Brief description of the working environment: The working environment was not very good as we had to visit the retail outlets in the summer. We were called by the company to their head office only twice during the project. Also, we expected that the company can at least reimburse our travel expenses but we didn't get anything like that.

Academic courses relevant to the project: Nothing

Learning Outcome: (1) Learned about different services offered by Apollo Tyres at their BROs.
(2) Identified the pain points of the dealer and the customers in the existing services.
(3) Learned how to do market research and prepare a optimal strategy for the organisation.

PS-I station: Apollo Tyres Ltd. , Gurugram

Student

Name: PRANSHUL GARG(2021A4PS2894G)

Student Write-up:

PS-I Project Title: Servicing strategy for premium two wheelers

Short Summary of work done: For the project on developing a comprehensive servicing strategy for premium two-wheelers, I conducted in-depth research and analysis to understand the challenges faced by owners in this exclusive segment. Through market research, customer surveys, and interviews, I gained valuable insights into the unique needs and expectations of premium two-wheeler owners. To address the identified problems effectively, I proposed a comprehensive loyalty program as the centerpiece of the strategy. The loyalty program offers various benefits, including specialized technician training, a diverse range of tire options, convenient pickup and delivery services, streamlined servicing processes, and collaborations with local bike clubs. I collaborated

with industry experts and explored competitor practices to identify best practices for servicing premium two-wheelers. The data collected through primary and secondary research allowed me to recommend tailored solutions to meet the specific demands of this niche market. The final conclusion emphasized the company's commitment to providing unparalleled service experiences, fostering lasting customer relationships, and establishing itself as the go-to service provider for premium two-wheelers. The project's outcomes are expected to set new benchmarks in the industry and elevate the ownership experience for esteemed clientele.

Objectives of the project: To make an extensive strategy to increase customers for servicing of premium 2 wheelers

Tool used: Excel, Google Sheets, Customer Feedback Software, Slack

Details of Papers/patents:-

Brief description of the working environment: During the PS-I (Professional Skills - I) internship, I had the opportunity to work in a dynamic and collaborative environment. The company fostered a culture of innovation, encouraging creativity and critical thinking. As an intern, I was expected to actively participate in various projects, taking on responsibilities and contributing to the team's success.
The company emphasized a strong focus on customer satisfaction and quality of service, aiming to exceed customer expectations. I learned the importance of attention to detail and delivering excellence in every aspect of the work.
Throughout the internship, I had the chance to develop essential professional skills, such as effective communication, time management, and problem-solving. Collaborating with experienced professionals provided valuable insights into industry practices and real-world challenges.
The internship not only enriched my knowledge and technical skills but also taught me the significance of teamwork and adaptability in a fast-paced work environment. I am grateful for the learning opportunities and practical experiences gained during the PS-I internship, which have significantly contributed to my personal and professional growth.

Academic courses relevant to the project: Principle of management, data structure, consulting

Learning Outcome: Consulting, market research, data collection.

PS-I station: Apollo Tyres Ltd. , Gurugram

Student

Name: AMAY BHATIA .(2021A4PS2918H)

Student Write-up:

PS-I Project Title: Passenger Vehicle Fleet Management

Short Summary of work done: We went to multiple fleet owners and discussed their various pain areas they have in their day to day work flow for example frequent puncturing of tyres, disloyal and uneducated drivers tending to steal their tyres etc. Then after enough survey is done, to come up with solutions to these problems like installing puncture repair dispensers near petrol pumps for self repair, conducting mandatory workshops for drivers etc.

Objectives of the project: To increase and preserve the tie up of the passenger vehicle fleets by learning of their major pain areas in their business and figuring out ways on how apollo tyres can help them

Tool used: NA

Details of Papers/patents:NA

Brief description of the working environment: We were stationed at the service center of apollo tyres to have a more on hand experience with the customers and to experience the issues faced in real time.

Academic courses relevant to the project: Few manufacturing and management courses may be deemed relevant but not necessary

Learning Outcome: Communication skills and consultation were major learnings along with the insides on a major tyre company

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PS-I station: Arics infrastructure , Ranchi

Student

Name: SHARAJ BIJOY BARUAH .(2021A3PS2934H)

Student Write-up:

PS-I Project Title: Supply chain management

Short Summary of work done: Week1 : Understanding of Supply Chain Operations. The students will be working on analyzing the existing supply chain operations at Arics Infrastructure and international practices being followed in Material Management. Week2-3 : Understanding of different modes of tendering and tendering procedure The students will have practical insight of the procurement process and provisions of tender documents in different modes of tendering. Risk analysis and its mitigation. Week 4- Understanding of stakeholder, cross functional team in an Organisation Roles of different deptt. and the cross functional role at and top corporate. Week 5-6 : Project Costing, budgeting and Cash Flow Week 7 : Negotiation Strategy, Contract Formation and Understanding of Contract Management for high value project Week 8 : Summary and PS learning report submission by student

Objectives of the project: To understand supply chain management

Tool used: Management skills

Details of Papers/patents:N/A

Brief description of the working environment: Very healthy working environment

Academic courses relevant to the project: Principles of Management

Learning Outcome: Understanding of supply chan management

PS-I station: Arics infrastructure , Ranchi

Student

Name: PRIYAM KUMAR PAUL .(2021A3PS2973H)

Student Write-up:

PS-I Project Title: Supply Chain Management

Short Summary of work done: Supply Chain Management: I have acquired in-depth knowledge and experience in supply chain management, learning how to plan, implement, and optimize the flow of goods and information effectively. My learning has focused on enhancing efficiency and cost reduction in supply chain operations. Indian Contract Act: I have studied and familiarized myself with the Indian Contract Act, 1872, understanding the intricacies of contract formation, enforcement, and the implications of contract breaches. This knowledge has enabled me to ensure fair and legally binding agreements. Incoterms: Through my learning journey, I have gained a thorough understanding of Incoterms and their significance in international trade. I now comprehend the responsibilities, cost allocation, and risk transfer between buyers and sellers during global transactions. EPC (Engineering, Procurement, and Construction): As part of my learning experience, I have delved into the world of EPC projects, grasping the fundamentals of project management from conception to completion. I have gained insights into designing, procurement strategies, and successful project execution.

Objectives of the project: EPC, Incoterms

Tool used: Excel, Python

Details of Papers/patents:NIL

Brief description of the working environment: Working Environment:
During my Professional Skills I (PS-I) experience, I found the working environment to be dynamic and challenging. The company fostered a collaborative atmosphere, encouraging open communication and teamwork.
Expectations from the Company:
As an intern during PS-I, I had certain expectations from the company. Firstly, I expected the company to provide me with a structured learning experience, where I could apply my theoretical knowledge to real-world situations. Additionally, I looked forward to receiving constructive feedback and guidance from mentors, enabling me to improve my skills and performance.
Learning during PS-I:
Throughout my PS-I, I gained invaluable learning experiences. I had the chance to apply theoretical concepts to practical projects, deepening my understanding of various subject areas. The exposure to real-world challenges helped me develop problem-solving skills and taught me how to work effectively as part of a team.

Academic courses relevant to the project: FOFA, DRM, POE

Learning Outcome: Understanding of various types of contracts.

PS-I station: Arics infrastructure , Ranchi

Student

Name: HEMANT SANJAY BADGUJAR(2021A4PS2486G)

Student Write-up:

PS-I Project Title: Renewable energy market analysis

Short Summary of work done: Making a report on wind and solar energy industry. Analysing maintenance and operation cost,major players ,Capex etc

Objectives of the project: Reasearch on wind and solar energy power plants

Tool used: MS Excel, MS word

Details of Papers/patents:NA

Brief description of the working environment: We had online meetings, learned the import and export business.
Learned about analysing the different industries

Academic courses relevant to the project: Principles of management

Learning Outcome: Analysing market of different industries

PS-I station: Arics infrastructure , Ranchi

Student

Name: ARCHIT ARYA(2021A4PS2981G)

Student Write-up:

PS-I Project Title: Supply chain management and market research on wind energy

Short Summary of work done: During my internship under the BITS Pilani Practice School Program at Arics Infrastructure, I had the opportunity to delve into diverse aspects of the industry. The internship commenced with essential teachings on Incoterms, equipping me with an understanding of international trade procedures and responsibilities. Next, I delved into marketing logics, where I learned to strategize and optimize promotional efforts to effectively reach target audiences. This knowledge proved invaluable as I progressed to studying Indian and international contract law, providing a solid foundation for dealing with legal aspects of business transactions. As my internship advanced, I had the privilege of coordinating research work focused on major wind energy manufacturing capabilities in India. This undertaking allowed me to gain firsthand insights into the booming renewable energy sector and its potential impact on sustainable development. Throughout the internship, I experienced the practical application of theoretical concepts, fostering a holistic understanding of the industry's dynamics. The experience broadened my perspective, emphasizing the significance of informed decision-making, ethical practices, and adaptability in a rapidly evolving business environment. The internship at Arics Infrastructure served as an enriching platform to learn, collaborate, and contribute to meaningful projects. It not only enhanced my technical expertise but also honed my interpersonal skills, allowing me to thrive in a professional setting. I am grateful for this invaluable opportunity, as it has shaped my aspirations and deepened my passion for the field, inspiring me to pursue a career that drives positive change in the world.

Objectives of the project: To understand the flow of materials through various processes and also the level of risk involved. Also an understanding of wind energy levels in india

Tool used: Powerpoint

Details of Papers/patents: None

Brief description of the working environment: The working conditions at Arics Infrastructure were conducive to learning and growth. The company emphasized a collaborative approach, encouraging open communication and teamwork. This fostered a positive atmosphere where I felt comfortable seeking guidance from mentors and colleagues.

Academic courses relevant to the project: None as this was related to practice school

Learning Outcome: Analyze material flow, assess risks, and understand wind energy levels in India to optimize processes and propose sustainable solutions

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PS-I station: Arics infrastructure , Ranchi

Student

Name: ARVIND AMAR INAMDAR(2021A8PS1553G)

Student Write-up:

PS-I Project Title: Wind Energy

Short Summary of work done: Current installations of wind energy

Objectives of the project: Current installations in india

Tool used: Word, ppt

Details of Papers/patents:-

Brief description of the working environment: Learnt to make good ppt and hence enhacing the communication skills.

Academic courses relevant to the project: None

Learning Outcome: Learnt presentations skills.

PS-I station: Arics infrastructure , Ranchi

Student

Name: CHAUHAN DHRUV SINGH MADAN PAL SINGH(2021A8PS2917G)

Student Write-up:

PS-I Project Title: Market Research on Renewable Energy Sources: Wind Energy

Short Summary of work done: In the beginning, we were made familiar with some important terms and concepts of the supply chain industry. Later on, we were assigned a project that required us to undertake a wind sector market analysis. I chose the subtopic - Government Subsidies for Establishing Wind Energy Projects. To work on this project, I had assistance from my peers and mentoring from Mr. Rajeev Prasad. I delved into researching various government incentives and financial aids available for wind energy initiatives. I collected and analysed data to understand the impact of these subsidies on the growth of the wind sector.

Objectives of the project: To collect and analyse data of different states and suggest places where setting up wind energy projects would be economically feasible

Tool used: Excel, Power Point

Details of Papers/patents:-

Brief description of the working environment: -

Academic courses relevant to the project: -

Learning Outcome: I learned how to acquire the most legitimate and relevant data from authentic sources such as research papers, etc. PS-1 also greatly improved my presentation skills; I learned how to present professionally.

PS-I station: Baga Cement Works , Solan (H.P.-

Student

Name: HANSAL SINGH .(2021A3PS2984H)

Student Write-up:

PS-I Project Title: Automation of traffic control using PLCs

Short Summary of work done: Imagine a groundbreaking project called "Smart Traffic Buddy: Your Traffic Guardian," where we're working tirelessly to make your daily commute smoother and safer. Our goal is to transform how traffic is managed using cool

technology like Programmable Logic Controllers (PLCs). We've all experienced frustrating traffic jams and delays, right? Well, with Smart Traffic Buddy, those days will soon be a distant memory. Our system uses PLCs to control traffic signals in real-time, allowing us to adjust signal timings on the fly based on the actual traffic conditions. Think about it - when the roads are jam-packed, our Smart Traffic Buddy algorithms will intelligently prioritize directions with higher traffic volume, ensuring everyone gets their turn without sitting in long queues. Safety is our utmost priority! We've designed the system to detect emergency vehicles like ambulances and fire trucks. When they're on the move, the Smart Traffic Buddy will automatically create a clear path for them, ensuring they reach their destination quickly and safely. You'll love how user-friendly our system is. We've created an intuitive interface, so you can easily configure traffic parameters, check out real-time traffic data, and monitor the system's performance. And hey, accidents happen, but we're prepared! With redundant PLCs and fail-safe mechanisms in place, you can rest assured that the traffic signals will keep working even in the rare event of a system glitch. Before launching Smart Traffic Buddy, we thoroughly test and simulate the system under different scenarios to make sure it's ready for the real world. We're excited to bring Smart Traffic Buddy to your city and transform your daily commute into a stress-free and enjoyable experience. Together, we'll make traffic woes a thing of the past and embrace a smarter, safer future on the roads!

Objectives of the project: To reduce man power

Tool used: OpenPLC software,excel

Details of Papers/patents:N/A

Brief description of the working environment: During the Practice School (PS-I), I had the privilege of being part of a dynamic and innovative working environment at Ultratech Cement. The company culture was characterized by a strong emphasis on teamwork, open communication, and a passion for pushing the boundaries of cement technology. As an intern, I was warmly welcomed into the team, and my mentors and colleagues were always approachable and eager to help me learn and grow.

From Ultratech Cement, I had high expectations of gaining practical experience and applying my theoretical knowledge to real-world projects in the cement industry. I was keen to work on tasks that challenged me and allowed me to contribute meaningfully to the company's goals of providing high-quality cement products and sustainable solutions. Fortunately, my expectations were not only met but exceeded.

During PS-I, I was assigned to work on various projects related to cement manufacturing processes, quality control, and sustainability initiatives. I had the opportunity to get hands-on experience with cement production techniques, quality testing methods, and the implementation of eco-friendly practices. Additionally, I actively participated in team meetings, site visits, and workshops, where I collaborated with experienced engineers and professionals, absorbing valuable insights and industry best practices.

Ultratech Cement provided a conducive learning environment, allowing me to explore the intricacies of cement manufacturing and research. I was encouraged to ask questions,

experiment with ideas, and seek feedback on my work, which significantly contributed to my growth as an aspiring engineer in the cement industry.
The PS-I period was an invaluable learning experience, both technically and professionally. I not only honed my technical skills related to cement production and quality control but also developed essential soft skills like effective communication, problem-solving, and project management. As a result, I gained a deeper understanding of the cement industry's complexities and the importance of sustainable practices.
Overall, my time at Ultratech Cement during PS-I was enriching, inspiring, and formative. I am grateful for the supportive working environment, the opportunities to learn and contribute to the cement industry, and the lasting connections made with my colleagues. I am now more confident and excited to continue my journey in the field of cement technology, equipped with practical experience and a determination to make a positive impact in the cement manufacturing sector.

Academic courses relevant to the project: MPI,Control systems, microcontrollers

Learning Outcome: Communication skills,PLC Programming, Project Planning

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PS-I station: Baga Cement Works , Solan (H.P.-

Student

Name: AMISH SHARMA(2021B2A13013G)

Student Write-up:

PS-I Project Title: Quantitative and Qualitative Analysis of Raw materials in manufactured Cement

Short Summary of work done: I had to work closely with my mentor where he provided me with the knowledge and resources to work on my project. I collected data (numeric values of the parameters under study) by talking to different members of the organization belonging to different departments. After this, I had to analyze the data and give my suggestions on possibilities of improving the quality of cement and use sustainable alternatives to cement.

Objectives of the project: Performing a Qualitative and Quantitative Analysis of raw materials in manufactured cement., Optimising the quality of cement, suggesting sustainable alternatives of cement.

Tool used: NA

Details of Papers/patents:NA

Brief description of the working environment: The working environment was good . The employees were cooperative. My mentor was busy most of the times so I couldn't get time to speak with him everyday but still he made sure I had all the resources and knowledge to complete the project. I liked interacting with the people in different departments. It was a good experience overall.

Academic courses relevant to the project: Solod State Chemistry
Physical Chemistry-1
Inorganic Chemistry-1

Learning Outcome: Role of different chemical compounds in determing the quality of cement.
In depth knowledge about Cement Manufacturing process and the equipments used.
Data Analysis skills
Communication

PS-I station: Baga Cement Works , Solan (H.P.-

Student

Name: DEEPANSHU(2021B2TS2073P)

Student Write-up:

PS-I Project Title: Inventory Management and Cost Optimisation

Short Summary of work done: Highlighted supply chain efficiency and risk management. Emphasised customer satisfaction and financial impact. Discussed data analytics and software usage. Outline strategic planning and adaptability.

Objectives of the project: Meet customer demand promptly. Minimize stockouts and lost sales. Reduce holding costs and optimize cash flow. Minimize obsolescence risks. Improve production planning and efficiency.

Tool used: Na

Details of Papers/patents: Na

Brief description of the working environment: physical setting, organisational culture, management style, team dynamics, and communication practices . All were good.

Academic courses relevant to the project: Finance

Learning Outcome: Understanding inventory control and its impact on customer satisfaction.
Knowledge of cost optimization techniques for better financial outcomes.
Enhancing supply chain efficiency and risk management skills.
Strategic planning for adaptable and sustainable practices.

PS-I station: Baga Cement Works , Solan (H.P.-

Student

Name: PRAVEEN SINGH(2021B5TS2064P)

Student Write-up:

PS-I Project Title: Impact of (ASR) Alkali Sulphate Ratio on process and Clinker quality

Short Summary of work done: ASR Impact on Clinker User Impact of ASR Alkali Sulphate Ratio on process and Clinker quality objectives of the project ChatGPT In the context of cement production, ASR (Alkali-Sulphate Ratio) refers to the ratio of alkalis (mainly sodium and potassium oxides) to sulphates (mainly in the form of sulfur trioxide, SO3) in the raw materials used for clinker manufacturing. The ASR plays a significant role in the process and clinker quality objectives of a cement project. Let's explore its impact: Clinker Formation and Burning Process: The ASR can affect the formation of clinker during the burning process in the cement kiln. An inappropriate ASR can lead to the formation of unstable and undesirable clinker phases, which may result in decreased clinker quality and reduced cement strength. To achieve optimum clinker formation, it is essential to maintain a suitable ASR in the raw mix. Clinker Composition: The ASR influences the chemical composition of the clinker produced. High ASR can lead to the presence of excess alkalis in the clinker, which can result in various issues, such as delayed cement setting time and potential for alkali-aggregate reaction (AAR) in concrete.

AAR is a chemical reaction between the alkalis in cement and certain reactive minerals in aggregates, which can cause concrete expansion and cracking over time, compromising the concrete's durability. Clinker Grindability: The ASR can impact the grindability of the clinker. Clinker with a high ASR may be more challenging to grind, leading to higher energy consumption during cement milling. This, in turn, affects the overall energy efficiency of the cement production process. Clinker Cooler Operation: An inappropriate ASR can affect the performance of the clinker cooler, potentially leading to clinker agglomeration or coating on the cooler grate and negatively impacting the cooling efficiency. Proper cooling of clinker is crucial for achieving good quality cement. Cement Quality: The quality of clinker directly influences the quality of the final cement product. High ASR can result in cement with reduced strength, setting time issues, and potential durability concerns due to AAR. On the other hand, a balanced and appropriate ASR can help produce high-quality cement with desired strength and setting characteristics. Cement Performance in Concrete: The ASR of cement can impact its performance when used in concrete. High ASR cement may lead to concrete with reduced long-term durability and increased risk of cracking due to AAR. It can also affect the workability and setting time of concrete, making it crucial to control the ASR in cement to achieve the desired concrete properties. In conclusion, maintaining an optimal ASR is vital for achieving the process and clinker quality objectives of a cement project. Careful selection and blending of raw materials, along with appropriate process control measures, are essential to control the ASR and produce high-quality clinker and cement while ensuring long-term durability and performance of the final concrete product. User Impact of ASR Alkali Sulphate Ratio on process and Clinker quality Major Learning Outcomes ChatGPT The impact of ASR (Alkali-Sulphate Ratio) on the process and clinker quality in cement production can have several major learning outcomes, which are essential for optimizing the cement manufacturing process and ensuring the production of high-quality clinker. Here are the key learning outcomes: Optimal Raw Material Selection: One of the primary learnings is the importance of selecting and blending raw materials with appropriate alkali and sulphate content. Understanding the impact of ASR on the clinker formation process allows cement manufacturers to optimize their raw material choices to achieve the desired clinker quality. Process Control and Optimization: Cement production involves a complex set of chemical reactions. By comprehending the influence of ASR on the process, cement plant operators can implement better process control measures. This includes maintaining the right temperature profiles in the kiln, optimizing the residence time, and ensuring proper raw mix proportions to produce high-quality clinker consistently. Clinker Phase Composition: Knowledge of ASR's impact on clinker phase composition helps in controlling the presence of undesirable phases, such as free lime, belite, and excessive alkali-based compounds. This knowledge facilitates adjustments to the raw mix composition to achieve a more desirable clinker phase composition, resulting in improved cement performance. Clinker Grindability and Energy Efficiency: Understanding the correlation between ASR and clinker grindability empowers cement manufacturers to optimize their grinding processes. This leads to energy savings and improved overall energy efficiency in the cement production process. Clinker Cooler Management: Adequate knowledge of ASR's effect on clinker cooler performance helps operators maintain efficient cooling, preventing agglomeration and ensuring better quality clinker output. Quality Control and Assurance: By grasping the impact of ASR on clinker quality,

cement producers can implement robust quality control measures at different stages of the production process. This includes rigorous testing of raw materials, intermediate products, and final clinker to ensure compliance with quality standards. Cement Performance Prediction: Knowledge of the relationship between ASR and cement properties enables cement manufacturers to predict the performance of their cement in concrete mixes. This helps engineers and construction professionals make informed decisions about the suitability of cement for specific applications, taking into account factors like strength, setting time, and long-term durability. Sustainable Cement Production: Awareness of ASR's impact contributes to sustainable cement production by reducing the likelihood of AAR-related issues in concrete structures. This, in turn, enhances the long-term durability and service life of concrete constructions, resulting in reduced maintenance and repair costs. In conclusion, a deep understanding of the impact of ASR on the process and clinker quality in cement production yields valuable insights for optimizing the cement manufacturing process, improving clinker quality, and ensuring the production of high-performance cement that meets industry standards and satisfies customer requirements. These learning outcomes are crucial for the long-term success and sustainability of cement manufacturing operations. User Impact of ASR Alkali Sulphate Ratio on process and Clinker quality Short Summary of work done during PS-I (in not more than 250 words) ChatGPT During Phase-I (PS-I) of the project, the main focus was to investigate the impact of the ASR (Alkali-Sulphate Ratio) on the cement production process and clinker quality. The research work encompassed a comprehensive study involving raw material analysis, clinker formation kinetics, process control measures, and cement performance evaluation. The first step involved detailed raw material characterization, including the alkali and sulphate content in various raw material sources. This analysis allowed for the selection and blending of appropriate raw materials to achieve the desired ASR in the raw mix. The findings of PS-I contribute significantly to sustainable cement production, as the project emphasizes the prevention of alkali-aggregate reaction (AAR) by controlling the ASR in clinker and cement. This work lays a solid foundation for further research and optimization in Phase-II (PS-II) to advance cement production practices and ensure the consistent delivery of high-quality cement products..

Objectives of the project: In the context of cement production, ASR (Alkali-Sulphate Ratio) refers to the ratio of alkalis (mainly sodium and potassium oxides) to sulphates (mainly in the form of sulfur trioxide, SO3) in the raw materials used for clinker manufacturing. The ASR pla

Tool used: The impact of ASR (Alkali-Sulphate Ratio) on the cement production process and clinker quality involves the use of various development tools, both hardware and software, to conduct research, collect data, and analyze results. Below are the key tools used

Details of Papers/patents: As an AI language model, I don't have direct access to specific papers or patents. However, I can provide you with general information about the impact of ASR (Alkali-Sulphate Ratio) on the cement production process and clinker quality, as well as example

Brief description of the working environment:
ASR (Alkali Sulphate Ratio) has a significant impact on the cement manufacturing process and clinker quality. The alkalis in cement raw materials, such as sodium and potassium, can react with the sulphates present during the clinkerization process. This reaction forms compounds like alkali sulphates, which can lead to disruptive reactions during cement hydration, causing a phenomenon known as "alkali-aggregate reaction." This reaction can result in the expansion and cracking of concrete structures over time, compromising their integrity.
Maintaining an appropriate ASR is crucial to avoid such issues and ensure a high-quality clinker. A lower ASR is generally desired as it reduces the risk of alkali-aggregate reactions, thereby improving the long-term durability and strength of the concrete.
In the working environment of a cement manufacturing company, strict quality control measures are in place to monitor the ASR and other parameters that influence clinker quality. Continuous monitoring and adjustment of raw material composition, as well as the process parameters, are essential to achieve optimal ASR levels and produce consistent high-quality clinker.
During the Practical Training Phase-I (PS-I), a student or trainee is expected to gain practical experience in the cement production process. They should familiarize themselves with the various stages of cement manufacturing, including raw material preparation. Understanding the significance of factors like ASR, heat balance, and energy consumption during the process is crucial.

Academic courses relevant to the project: The ASR (Alkali Sulphate Ratio) plays a crucial role in the cement manufacturing process and significantly impacts clinker quality. Here's how it affects both aspects: Impact on the Process , Impact on Clinker Quality ,

Learning Outcome: The impact of ASR (Alkali-Sulphate Ratio) on the process and clinker quality in cement production can have several major learning outcomes, which are essential for optimizing the cement manufacturing process and ensuring the production of high-quality clinker. Here are the key learning outcomes: Optimal Raw Material Selection, Process Control and Optimization, Clinker Phase Composition.

PS-I station: Baga Cement Works , Solan (H.P.-

Student

Name: ARVIND SINGH(2021B5TS2070P)

Student Write-up:

PS-I Project Title: Studying About Cement Manufacturing Processes And Assessment Of Heat Balance In Cement Industry(Preheater, Kiln, Cooler) And Ways To Optimize Efficiency

Short Summary of work done: the cement manufacturing process is to produce high-quality cement efficiently and sustainably. This involves optimizing the heat balance, implementing energy-efficient measures, reducing costs, and minimizing environmental impact. The major outcomes include high-quality cement production, energy efficiency, cost reduction, environmental sustainability, operational excellence, and fostering continuous improvement and innovation within the industry.

Objectives of the project: Ensuring the proper chemical composition of the raw materials to produce cement with desired properties and strength. Minimizing environmental impact by reducing energy consumption, carbon emissions, and waste generation. Maintaining consistent and reliab

Tool used: Na

Details of Papers/patents: Na

Brief description of the working environment: During the PS-I, interns can expect a professional working environment where they are encouraged to learn and grow. Companies have expectations for interns to actively seek knowledge, demonstrate professionalism, contribute to projects, and show initiative in problem-solving. Interns are provided with opportunities for skill development, mentorship, and collaboration with team members.

Academic courses relevant to the project: Physics

Learning Outcome: High-Quality Cement Production: The primary outcome is the production of high-quality cement that meets the required standards and customer specifications. This includes achieving the desired chemical composition, fineness, and strength properties of the cement product.
Energy Efficiency and Cost Reduction: By optimizing the heat balance and implementing energy-efficient measures, the cement industry can reduce energy consumption, leading to cost savings in fuel and electricity expenses. Improved energy efficiency also contributes to the industry's overall sustainability and competitiveness.
Environmental Sustainability: The cement manufacturing process has a significant environmental impact, particularly in terms of carbon emissions. Optimizing efficiency helps reduce greenhouse gas emissions, mitigating the industry's carbon footprint and promoting environmental sustainability.
Waste Heat Recovery and Utilization: Implementing waste heat recovery systems allows the capture and utilization of heat generated during the manufacturing process. This

outcome leads to additional energy savings, reduced reliance on primary fuels, and potential power generation or other heating applications.
Operational Excellence: Optimizing the cement manufacturing process improves overall operational efficiency, including enhanced process control, reduced downtime, improved maintenance practices, and increased production capacity. These outcomes contribute to better productivity and cost-effectiveness.
Regulatory Compliance and Health & Safety: By adopting energy-efficient practices and optimizing the manufacturing process, cement manufacturers can meet regulatory requirements and ensure a safe working environment for employees.
Continuous Improvement and Innovation: The pursuit of efficiency optimization leads to a culture of continuous improvement within the cement industry. This outcome fosters innovation, research, and the adoption of advanced technologies to further enhance the manufacturin

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PS-I station: Baga Cement Works , Solan (H.P.-

Student

Name: TINKU SAINI(2021D2TS2083P)

Student Write-up:

PS-I Project Title: Impact of CSR on Turmeric and Organic Natural farming

Short Summary of work done: During PS-I, I conducted research to explore the impact of Corporate Social Responsibility (CSR) initiatives on turmeric cultivation and organic natural farming. The study aimed to understand how CSR practices could contribute to sustainable agricultural development and improve the livelihoods of farmers. Firstly, I analyzed various CSR programs implemented by companies in the agricultural sector, focusing on initiatives that specifically targeted turmeric cultivation and organic farming. These programs often involved providing farmers with access to improved seeds, organic fertilizers, and advanced farming techniques. Additionally, some companies offered training sessions and workshops to educate farmers on sustainable practices. Next, I conducted field visits to turmeric-growing regions and interviewed farmers who participated in CSR initiatives. The feedback from these farmers was overwhelmingly positive. They reported significant improvements in crop yield, quality, and income after implementing the sustainable practices promoted by CSR programs. The adoption of organic farming methods also led to reduced environmental impact, including decreased use of harmful chemicals and water conservation. Furthermore, the study assessed the

social and economic impact of CSR on farming communities. It was evident that CSR initiatives positively affected the livelihoods of farmers by enhancing their economic prospects and overall well-being. Increased income from improved crop yields allowed farmers to invest in education, healthcare, and better living conditions for their families. In conclusion, the research demonstrated that CSR initiatives have a significant and positive impact on turmeric cultivation and organic natural farming. These programs not only improve agricultural practices but also contribute to the economic and social development of farming communities. Encouraging more companies to engage in responsible and sustainable CSR practices can lead to a more sustainable and prosperous agricultural sector in the long term.

Objectives of the project: Measuring Crop Yield and Quality: One of the key objectives is to measure the quantitative and qualitative improvements in turmeric crop yield as a result of adopting sustainable practices encouraged by CSR programs. This data helps in assessing the econo

Tool used: However, when implementing CSR initiatives or projects related to turmeric and organic natural farming, certain development tools (hardware and software) might be used to support and facilitate these efforts. These tools can aid in data collection, analys

Details of Papers/patents: Community Development: CSR projects can focus on improving the livelihoods of turmeric farmers and their communities. By providing training, access to resources, and financial support, farmers can enhance their farming techniques and increase yields, lead

Brief description of the working environment: Expectations & Learning
Working Environment:
The working environment during your PS-I (Practical Semester I) is likely to be a combination of a professional and educational setting. As a PS-I student, you'll be joining the company as an intern or trainee, and you'll be exposed to the real-world application of your academic knowledge. The company will provide you with the necessary resources, guidance, and mentorship to enhance your skills and contribute to their projects. You'll be expected to work collaboratively with team members, adapt to the company's culture, and follow their policies and procedures.
Expectations from the Company:
The company will expect you to be enthusiastic, dedicated, and willing to learn during your PS-I. While you may not be an experienced professional, they will value your fresh perspective and ideas. Punctuality and regular attendance are essential, as you will be given real tasks and responsibilities that contribute to the company's objectives. Your supervisors and mentors will provide feedback on your performance, so it's crucial to be receptive to constructive criticism and demonstrate a willingness to improve.
PS-I: Expectations & Learning.
Working Environment:
The working environment during your PS-I (Practical Semester I) is likely to be a combination of a professional and educational setting. As a PS-I student, you'll be joining the company as an intern or trainee, and you'll be exposed to the real-world application

of your academic knowledge. The company will provide you with the necessary resources, guidance, and mentorship to enhance your skills and contribute to their projects. You'll be expected to work collaboratively with team members, adapt to the company's culture, and follow their policies and procedures.

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Learning during PS-I:

The primary goal of PS-I is to bridge the gap between theoretical knowledge acquired in classrooms and practical skills required in real-world scenarios. During this period, you'll have the opportunity to apply academic concepts to actual projects, gaining hands-on experience in your field of study. You will also develop essential professional skills like time management, communication, problem-solving, and teamwork.

Learning during PS-I is not limited to technical aspects; you will also gain insights into the company's work culture, industry practices, and business operations. Engaging with experienced professionals will expose you to new perspectives and best practices. Be proactive in seeking knowledge, asking questions, and taking on new challenges to maximize your learning experience.

In conclusion, the PS-I working environment offers a supportive and educational atmosphere where you can learn and grow both professionally and personally. Embrace the opportunities, be receptive to feedback, and make the most of your time to lay a strong foundation for your future career.

Working Environment:

The working environment during PS-I is a dynamic and diverse space, where you'll likely be surrounded by experienced professionals from various disciplines. It could be an office setting or a research facility, depending on the nature of the company or organization. You might also have the chance to work remotely, which would require self-discipline and effective communication.

Expectations from the Company:

The company will expect you to be proactive in taking on tasks, seeking guidance when needed, and showing a genuine interest in the work they do. Although you may not be as experienced as full-time employees, they will anticipate your commitment to contributing your best efforts and treating the internship as a valuable learning opportunity. Demonstrating a positive attitude, adaptability, and a willingness to collaborate will likely be highly regarded by the company.

Learning during

PS-I is an invaluable learning experience that allows you to apply theoretical knowledge to real-world challenges. You'll have the chance to work on practical projects, gaining insights into the complexities and intricacies of the industry. Throughout this period, you'll enhance your technical skills, problem-solving abilities, and project management capabilities.

Beyond technical skills, you'll also have the chance to develop crucial sof

Academic courses relevant to the project: Research and Innovation: CSR-backed research and development projects can drive innovation in organic natural farming techniques, crop varieties, and processing methods.

Learning Outcome: Economic Empowerment: CSR interventions in turmeric cultivation create opportunities for economic empowerment of farmers. By equipping them with knowledge, skills, and resources, farmers can improve their income levels, financial stability, and overall socio-economic status.

PS-I station: Bajaj Steel Industries Ltd. , Nagpur

Student

Name: MALLICK AMMAR AHMED ABSAR AHMED(2020A4PS1986G)

Student Write-up:

PS-I Project Title: Designing a fixture for a 161 Rib

Short Summary of work done: Learnt about the quality checking process that occurs in an industry in a step by step manner. I also learnt about designing a Jig and fixture for a workpiece machined in a cnc.

Objectives of the project: Design a jig and fixture for a 161 Rib

Tool used: AutoCAD, Fusion 360, Hand Drafting

Details of Papers/patents:nil

Brief description of the working environment: The working environment was quite friendly. Most of the work occurred on the Factory Floor and the Workers and Inspectors were really helpful in explaining everything. The company only expected us to do our work diligently and learn from the instructors.

Academic courses relevant to the project: Quality Controls, Engineering Graphics, Advanced Manufacturing Processes

Learning Outcome: Designing, Learning AutoCAD

PS-I station: Bajaj Steel Industries Ltd. , Nagpur

Student

Name: SHOUMIK NAG .(2021A4PS0042P)

Student Write-up:

PS-I Project Title: Welding and Inspection Systems

Short Summary of work done: I completed an internship at Bajaj Steel Industries, gaining comprehensive knowledge in core mechanical engineering. With a focus on welding and inspection systems, I completed projects that enhanced my skills. This report highlights my journey, including key learnings and achievements, showcasing the value of this transformative internship.

Objectives of the project: To understand and learn about welding and inspection systems and study their applications and industry

Tool used: Hardware - welding equipment

Details of Papers/patents:N/A

Brief description of the working environment: I completed an internship at Bajaj Steel Industries, gaining comprehensive knowledge in core mechanical engineering. With a focus on welding and inspection systems, I completed projects that enhanced my skills. This report highlights my journey, including key learnings and achievements, showcasing the value of this transformative internship.

Academic courses relevant to the project: Manufacturing Processes, Quality control

Learning Outcome: Leant about welding and inspection systems and their application in industry

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PS-I station: Bajaj Steel Industries Ltd. , Nagpur

Student

Name: AKSHAT MANDAVGANE(2021A4PS2187P)

Student Write-up:

PS-I Project Title: Recommending modifications in the design of fixtures for ribs

Short Summary of work done: Assigned the task to research and brainstorm new ideas for designing fixtures for ribs

Objectives of the project: To come up with a new customized fixture for ribs in CNC milling machine

Tool used: H/w - learned how to operate a 2 axes CNC lathe

Details of Papers/patents:-

Brief description of the working environment: To learn CNC part programming and resolve the real life problems faced by the company
To learn the basics of conveyor belts and their application in the industry

Academic courses relevant to the project: Manufacturing processes, Workshop

Learning Outcome: Learnt a lot about the ins and outs of this mechanical based industry

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PS-I station: Bajaj Steel Industries Ltd. , Nagpur

Student

Name: SAKSHAM BANSAL .(2021B4A42787P)

Student Write-up:

PS-I Project Title: Hydraulics

Short Summary of work done: I explored the significance of hydraulics in industrial applications, focusing on a bale press designed by Bajaj Steels for cotton processing and the hydraulic bench test machine. Hydraulics, based on fluid mechanics, offers efficient and reliable force generation through incompressible fluids like water and oil. The bale press compresses raw cotton into compact bales, optimizing storage and labour efficiency. The hydraulic bench test machine ensures reliable performance through testing and analysis. Integration of hydraulics in industries drives progress and innovation, revolutionizing processes and productivity.

Objectives of the project: Learning about hydraulics and its applications.

Tool used: vernier caliper, AutoCAD

Details of Papers/patents:.

Brief description of the working environment: Different teams, such as production, quality control, maintenance, and logistics, work together to achieve production goals and maintain product quality. Safety is of paramount importance. The adherence to safety protocols is strictly enforced to prevent accidents and injuries. Quality control measures are integrated into every step of the production process. Inspections, testing, and quality assurance procedures ensure that products meet the required standards before they are sent to the customer. Employees are encouraged to suggest ideas for enhancing processes, reducing waste, and increasing productivity. Regular reviews and feedback sessions contribute to a culture of constant growth and innovation.

Academic courses relevant to the project: Fluid mechanics, mechanism and machines, Engineering Graphics

Learning Outcome: Learnt about principles used in hydraulics and also designed hydraulic repair test bench

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PS-I station: Bharat Forge Ltd. , Pune

Student

Name: PIYUSH CHANDUKA .(2021A4PS1554P)

Student Write-up:

PS-I Project Title: Compressor Energy Saving

Short Summary of work done: my experience at Bharat Forge has been both insightful and rewarding. During my time there, I had the opportunity to observe and learn from real-world manufacturing processes, which complemented the theoretical knowledge I gained in college. I am grateful for the guidance and support provided by my mentor and colleagues at Bharat Forge. Their expertise and willingness to share knowledge greatly enhanced my understanding of the industry and enriched my learning experience. Additionally, I found the work environment to be collaborative and friendly, with all employees being helpful and encouraging throughout my tenure. Engaging in the implementation of my ideas, especially regarding compressor energy saving and the potential use of piezoelectric tiles, was a truly adventurous experience. It provided me with valuable insights into the challenges and opportunities faced by the industry in adopting sustainable technologies. Overall, my time at Bharat Forge has been a valuable learning journey. It has equipped me with practical skills, industry knowledge, and a deeper appreciation for the importance of energy conservation and innovation in the forging sector. I am confident that the experiences and lessons gained during this period will serve as a strong foundation for my future endeavors in the field of engineering and sustainability

Objectives of the project: Saving the energy consumed by air compressor used in the industry.

Tool used: internet, ChatGPT

Details of Papers/patents:NA

Brief description of the working environment: My experience at Bharat Forge has been both insightful and rewarding.
During my time there, I had the opportunity to observe and learn from real-world manufacturing processes, which complemented the theoretical knowledge I gained in college.
I am grateful for the guidance and support provided by my mentor and colleagues at Bharat Forge. Their expertise and willingness to share knowledge greatly enhanced my understanding of the industry and enriched my learning experience. Additionally, I found the work environment to be collaborative and friendly, with all employees being helpful and encouraging throughout my tenure.
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adopting sustainable technologies.
Overall, my time at Bharat Forge has been a valuable learning journey. It has equipped me with practical skills, industry knowledge, and a deeper appreciation for the importance of
of energy conservation and innovation in the forging sector. I am confident that the experiences and lessons gained during this period will serve as a strong foundation for my
future endeavors in the fields of engineering and sustainability.

Academic courses relevant to the project: Mechanics of machines, Heat Transfer

Learning Outcome: Study of compressors, Finding ways to conserve energy, reducing the use of compressed air,etc.

PS-I station: Bharat Forge Ltd. , Pune

Student

Name: PANDE VARUN VIJAY .(2021A4PS2191P)

Student Write-up:

PS-I Project Title: Implementation of Education and Training Modules

Short Summary of work done: Built training modules for workers of a manufacturing and machining division. These modules were about rules and about various parts of the machines. They covered identification of faults, symptoms and corrective actions.

Objectives of the project: To build and implement training resources

Tool used: Ms Excel, Ms PPT

Details of Papers/patents:N/A

Brief description of the working environment: Enriching work environment. Everybody at the workplace has time and was ready to reach me and answer my doubts. I expected the company to teach me the intricacies of working in a manufacturing plant and wanted to get more experience under my belt about the core industry.

Academic courses relevant to the project: Product management, material science, manufacturing processes, mechanics of solids

Learning Outcome: Working of the factory, importance of safety and quality

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PS-I station: Bharat Forge Ltd. , Pune

Student

Name: GAURAV DURAPHE .(2021A4PS2525H)

Student Write-up:

PS-I Project Title: Single Minute Exchange of Die

Short Summary of work done: Time study of die setup and die grinding process, SOP on inward inspection,

Objectives of the project: To reduce die setup time

Tool used: MS excel, Minitab

Details of Papers/patents:NA

Brief description of the working environment: Professional and helpful mentors, and a great place for learning. Attended many training sessions on die care, non-destructive testing, general safety, ISO standards and MSA. Learnt a lot about forging processes in an automated press setup. Hands-on experience with inward inspection using vernier callipers, micrometers and height gauge.

Academic courses relevant to the project: ME F219, ME F216

Learning Outcome: Forging and processing knowledge,

PS-I station: Bharat Forge Ltd. , Pune

Student

Name: ADITYA DIXIT .(2021A4PS2527P)

Student Write-up:

PS-I Project Title: Reducing Die Setup Time of Forge Modernisation Division III

Short Summary of work done: We learnt about the different activities of the die changeover procedure by interacting with machine operators and other employees, carefully observed the entire procedure several times by being on the shop floor, recorded a video of the entire procedure, noted down time taken for each activity after segregation and gave suggestions regarding its improvement

Objectives of the project: To observe activities taking place during die changeover and reducing the time taken to perform these activities

Tool used: -

Details of Papers/patents:-

Brief description of the working environment: It was a busy environment with continous production and processing taking place but the induction program and the interaction with employees helped gain valuable insight and understand the working of the organization better

Academic courses relevant to the project: Workshop Practice, Manufacturing Processes

Learning Outcome: Learnt about the structure of the organization, its different departments, production of various components through forging and understanding the process around observing a problem and coming up with useful suggestions

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PS-I station: Bharat Forge Ltd. , Pune

Student

Name: PREKSHA SHARMA .(2021ABPS2470P)

Student Write-up:

PS-I Project Title: INDUSTRY 4.0

Short Summary of work done: I created a training module on industry 4.0 for skill development of employees. I also attended other industry related trainings on many important concepts like 5'S, TPM, etc.

Objectives of the project: The goal was to make the employees more aware of the technological advancements in the worl that they do regularly. This was to spark some ideas for innovation.

Tool used: -

Details of Papers/patents:-

Brief description of the working environment: Great working and learning environment and everyone was really helpful.

Academic courses relevant to the project: Machines and mechanisms, Operations Management, Casting forming welding

Learning Outcome: Many technical industry related knowledge as well as soft skillls and interpersonal skills like communication, team work, and presentation skills.

PS-I station: Birla Furukawa Fibre Optics Pvt. Ltd. , Goa

Student

Name: SHREEJA GADA .(2021A4PS1530H)

Student Write-up:

PS-I Project Title: Study of Colour Measurement and Noise Reduction Techniques

Short Summary of work done: Our project was 'Study of colour measurements and Noise Reduction Techniques'. As part of this we learnt and researched about different colour scales that are generally used in industries to measure colour. The colouring process of fibre optics is important as it helps in identification of individual fibres and makes installation and maintenance easier and safer. We learnt about the colour coding standards that are used for optical fibres, according to their type and fucntion. We were also shown the working of the machinery that is used for colouring the fibres in the manufacturing plant. In order to exactly replicate the colour as per the customers' requirements, colour scales are used and our task was to study these scales in detail.

Objectives of the project: To understand various colour scales and how they are used in the fibre optics industry as part of the colouring process and to come up with techniques to reduce noise levels in the manufacturing plant

Tool used: Python

Details of Papers/patents:NA

Brief description of the working environment: .

Academic courses relevant to the project: -Technical report writing
-Manufacturing processes

Learning Outcome: - Various colour scales used in this industry for colour measurement
- Feasible techniques for reducing noise levels inside a manufacturing plant
- Corporate culture
- Daily functioning of a manufacturing company

PS-I station: Birla Furukawa Fibre Optics Pvt. Ltd. , Goa

Student

Name: ANISH HATUA .(2021A8PS0741P)

Student Write-up:

PS-I Project Title: Study of Colour Measurements and Noise Reduction Techniques

Short Summary of work done: We were divided into 2 teams of 3 members each. 2 projects were allocated to both teams and we were given almost 3 weeks duration to come up with solutions to the proposed problem statements and present them to the company officials. Our project topics were the Study of Colour Measurement Techniques and Munsell Scales, and Noise Reduction techniques. The first project aimed at educating the company workers about the colour scales and the techniques used to measure colour in industries, as they were mostly unaware of the science behind the technology they were using. We had to gain in-depth knowledge of colour theory, colour scales and models and the various scales being used in the industrial world. Additional information was needed to connect this information to the optical fibre industry and the application of colour in that context. The second project was intended for us to come up with solutions to the present high levels of noise in the post-draw section in the plant. Similar to the first project, we had to come up with ingenious solutions to this problem and present a detailed analysis considering costs, plant layout and equipment, and prerequisites present in the industry. We were provided various resources from the company itself for these projects including data from the equipment that they used and given various tours of the facility itself.

Objectives of the project: To make a detailed study to conduct and report the finding from basic knowledge of colour measurement techniques and to study some innovative ways to reduce industrial noise.

Tool used: Canva

Details of Papers/patents:US Patent: Process of colour coding an optical fibre

Brief description of the working environment: We observed the optical fibre manufacturing process, from start to finish, including the various stages like the draw process, post-draw stage and quality analysis. From this, we got to learn the entire supply chain of the optical fibre and cable industry. We also learnt in detail about optical fibres, their types, the science behind them and the technology used in making these optical fibres. It was also a great experience working at a manufacturing plant, getting detailed insights into its working and manufacturing processes in general. The company was very supportive through out the program and provided all the help that we needed, including taking us on multiple plant visits and setting up meetings with experts in their respective fields.

Academic courses relevant to the project: Business Communication, Technical Report Writing

Learning Outcome: Learnt about various colour scales and models and ways to reduce noise in industrial settings.

PS-I station: Birla Furukawa Fibre Optics Pvt. Ltd. , Goa

Student

Name: ANISHA JAYANT JOSHI(2021B1A43042G)

Student Write-up:

PS-I Project Title: Study of Product Cooling Methods

Short Summary of work done: During my PS1 course at Birla Furukawa Fiber Optics Pvt Ltd I had the opportunity to gain practical experience in the field of mechanical engineering. The course was a valuable learning experience, which allowed me to gain insight into the manufacturing processes and quality control of optical fibers. One of the most interesting aspects of my course was learning about the different types of optical fibers and their properties. I was able to observe the manufacturing process firsthand, from the initial stages of pulling the fiber to the final stages of testing the finished product for quality control. I was impressed by the precision and accuracy required in this process, and I gained a new appreciation for the complexities involved in the manufacturing of optical fibers. Working in a team environment was also a valuable experience for me. I had the opportunity to collaborate with professionals from diverse backgrounds, which allowed me to develop my interpersonal skills and learn from the vast knowledge and experience of my colleagues. I was also able to improve my technical abilities by using advanced equipment and software. Overall, my PS1 course at Birla Furukawa Fiber Optics Pvt Ltd was a rewarding experience. I gained valuable skills and knowledge that will be useful in my future career. I am grateful for the opportunity to have worked with such a talented and dedicated team, and I look forward to applying the skills and knowledge I gained during my course to future projects and endeavors.

Objectives of the project: To find a better, feasible, cost effective method to cool the optical fiber after being drawn from the raw material (Preform). And to find a way to reuse the gasses and reduce the wastage of gas in order to reduce cost of production.

Tool used: Canva, Excel, Google Docs, Draw.io

Details of Papers/patents:US Patent 5,470,716(1995):Cooling optical fibers using liquid coolants
US Patent 6,382,426(2002):Cooling optical fibers using gas coolants
US Patent 7,236,697(2007):Method for cooling optical fibers using phase-change materials

US Patent 8,504,208(2013):C

Brief description of the working environment: As a student pursuing an M.Sc Biological Sciences and B.E. Mechanical degree, I was excited to embark on the PS1 course in an optical fiber manufacturing unit. I have always been fascinated by the advancements in telecommunication technology, and I believed this course would allow me to gain practical experience in the industry.

During the duration of the course, I expected to learn about the different types of optical fibers, their properties, and their manufacturing processes. I was also keen to understand how optical fibers are used in communication networks, and how they are tested for quality control.

I anticipated working in a team environment, collaborating with professionals from diverse backgrounds. I wanted a chance to develop my interpersonal skills and learn from my colleagues' vast knowledge and experience.

Moreover, I was eager to improve my technical abilities by using advanced equipment and software. I understood that the manufacturing process involved precision and accuracy, and I looked forward to contributing to the success of the company by ensuring that the project made by us comes of their use and is at par with their expectations from us.

Overall, I was excited to gain hands-on experience in the optical fiber manufacturing industry, and I wanted it to provide me with valuable skills to help shape my future career in mechanical engineering.

Academic courses relevant to the project: Thermodynamics, Materials and Science, Optics, Principles of Economics, Electrical Sciences, General Chemistry.

Learning Outcome: The PS1 course provided a unique opportunity for me to gain practical industry experience and have first-hand experience in the corporate environment and the practices that are essential from a work point of view.

Following are a few takeaways from the PS1 course-

One of the primary takeaways from the course was gaining insight into the intricate process of optical fiber manufacturing. From understanding the initial raw material selection to the final quality control measures, I learned about the various stages involved in creating high-quality optical fibers. This exposure broadened my understanding of the industry and enhanced my technical knowledge.

During the course, I worked alongside skilled professionals and technicians who guided me through hands-on tasks. This practical experience allowed me to apply theoretical knowledge gained from textbooks to real-world scenarios. I became familiar with the operation of advanced machinery and equipment used in manufacturing, gaining proficiency in fiber drawing, coating, and testing.

Another valuable aspect of the course was the emphasis placed on quality assurance and control. I learned how meticulous testing procedures and stringent quality checks are implemented to ensure the production of flawless optical fibers. Witnessing these measures firsthand instilled in me a sense of commitment to delivering products of the highest quality.

Working in an optical fiber manufacturing unit also honed my teamwork and communication skills. Collaborating with professionals from diverse backgrounds and disciplines, I realized the significance of effective communication in achieving shared goals. I learned how to contribute as a team member, exchange ideas, and seek assistance when needed, fostering a collaborative work environment.

PS-I station: Birla Furukawa Fibre Optics Pvt. Ltd. , Goa

Student

Name: VASHISHTH PATEL(2021B4A82781G)

Student Write-up:

PS-I Project Title: Study of product cooling methods

Short Summary of work done: I worked on developing a cooling methods for optical fibres, developing a noise reduction system , gas recycling system

Objectives of the project: To find an alternative method to current cooling methods

Tool used: Spectrophotometer, Patent papers

Details of Papers/patents:US Patent 5,470,716: This patent, issued to John A. Bucaro in 1995
US Patent 6,382,426: This patent, issued to Paul J. Kucera in 2002
US Patent 7,236,697: This patent, issued to David S. Gemmell in 2007
US Patent 8,504,208: This patent, issued to Mark J.

Brief description of the working environment: Working environment was very friendly and was a brief introduction how the corporate world works. My expectations was about PS1 was to know more about careers in optical fibres industry and where would it lead which was well satisfied

Academic courses relevant to the project: Thermodynamics, Mechanical oscillation and waves

Learning Outcome: Corporate culture, knowledge about different divisions of an industry

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PS-I station: Birla Furukawa Fibre Optics Pvt. Ltd. , Goa

Student

Name: KRISHNA GOUR .(2021B5A43193H)

Student Write-up:

PS-I Project Title: Optical Fibre Cooling Methods

Short Summary of work done: I was involved in the "Cooling of Optical Fibre" project, which aimed to develop a method for efficiently cooling molten Silica Preform while minimizing the reliance on Helium, keeping in mind that minimum changes to existing system was a requisite. Additionally, after successfully completing my project, I dedicated my remaining time to voluntarily designing a recycling system for Deuterium gas, which was being wasted during the Gas Dozing process.

Objectives of the project: My project "Cooling of Optical" aimed towards developing a method for efficiently cooling molten Silica Preform while minimizing the reliance on Helium, keeping in mind that minimum changes to existing system was a requisite. Additionally, after successfu

Tool used: Internet

Details of Papers/patents: None

Brief description of the working environment: By the PS-1 station I expect to acquire a solid foundation of professional skills relevant to my field of education. This includes areas such as optics, thermodynamics, teamwork, time management, problem-solving, and critical thinking. The course focuses on developing practical skills that can be applied in real-world professional settings. This involves learning how to write effective emails, make presentations, conduct meetings, manage projects, or handle conflicts. PS-1 courses often encourage self-reflection and self-awareness to help one understand one's strengths, weaknesses, and areas for improvement. This might involve personality assessments, feedback sessions, or reflective exercises to enhance one’s professional development. The company might provide opportunities for networking, group projects,

or collaborative activities to help you develop these abilities. PS-1 course might cover topics such as professional ethics, integrity, and social responsibility to ensure you have a solid understanding of ethical standards in your chosen field. PS-1 course can help one boost one's confidence and personal growth by providing guidance on self-presentation, public speaking, and personal branding. It may also include activities to enhance your emotional intelligence and resilience. The course might focus on preparing you for your future career by providing guidance on job searching, resume writing, interview skills, and overall career planning.

Academic courses relevant to the project: PS-1 course equips students with foundational problem-solving techniques and strategies that can be applied across various disciplines. These skills can help me approach complex academic problems, assignments, and projects more effectively. PS-1 course em

Learning Outcome: During the PS-1 course, I have gained an understanding of the principles and techniques involved in cooling and manufacturing optical fiber. This includes studying heat transfer mechanisms, thermal management methods, and draw-tower processes etc. Through the course, I have gained hands-on experience in implementing cooling solutions for optical fiber. This involves designing and building cooling systems, working with relevant equipment and tools, and troubleshooting cooling-related issues. During the course, I have deepened my understanding of fiber optic communication systems, including the behavior of optical fibers under different temperature conditions and the impact of heat on signal transmission.

PS-I station: Birla Polyfibers , Harihar

Student

Name: CHIRAG R HARITHSA .(2021A1PS2585P)

Student Write-up:

PS-I Project Title: Effect of Temperature profile on the recovery of material at Lime Kiln

Short Summary of work done: We were given introduction to each part of the plant which helped us understand the working of the industry. We then interacted with professionals in the industry who guided us throughout the project. The project was involving a real world problem and we were able to provide possible solutions.

Objectives of the project: To study the issues in Lime Kiln and Improve the efficiency of the process

Tool used: Microsoft Word, PowerPoint, excel.

Details of Papers/patents: None.

Brief description of the working environment: Working environment was excellent. Company was very accommodating of our requests. Learning was related to working of the organization.

Academic courses relevant to the project: Chemical process calculation, heat transfer, fluid Mechanics, thermodynamics, engineering chemistry, material science.

Learning Outcome: With the completion of the project, we were able to put forward some solutions which may reduce the formation of rings in the lime kiln and hence reduce the environmental pollution associated with it.

PS-I station: Birla Science Centre , Pilani

Student

Name: GAURI SONAWANE(2021A3PS1386G)

Student Write-up:

PS-I Project Title: Design and deve of security systems

Short Summary of work done: Designed a smartsecurity system for the museum which includes various sensors like RFID card , keypad, fingerprint and ov7670 cam

Objectives of the project: Interactive learning

Tool used: Arduino mega , Arduino ide , fungerp,RFID card , kcd , uni

Details of Papers/patents: Ni

Brief description of the working environment: It was pleasant and very well organised

Academic courses relevant to the project: Digital design

Learning Outcome: Implementation and in person experience of using various hardware

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PS-I station: Birla Science Centre , Pilani

Student

Name: S SANJAY REDDY .(2021A8PS2952H)

Student Write-up:

PS-I Project Title: Design and Development of sentiment analyzer using AI

Short Summary of work done: We designed a binary classifier to classify the sentiment of statements

Objectives of the project: To predict sentiment of the sentences

Tool used: Google collaboratory

Details of Papers/patents:NA

Brief description of the working environment: Birla science centre is an innovation hub , environment was very friendly

Academic courses relevant to the project: Machine Learning

Learning Outcome: Natural language processing

PS-I station: Birla Science Centre , Pilani

Student

Name: HARSHWARDHAN BHARDWAJ .(2021A8PS3000H)

Student Write-up:

PS-I Project Title: Design and Development of Website for Birla Science Centre

Short Summary of work done: To create an interactive website for Birla Science Centre. The website serves as a platform to showcase the institution's mission, exhibits, and educational initiatives related to science, technology, art, and history.

Objectives of the project: To create an interactive website for Birla Science Centre

Tool used: HTML CSS javascript wix

Details of Papers/patents: None

Brief description of the working environment: The peer group and mentors along with the staff created a great learning environment for us.

Academic courses relevant to the project: CP

Learning Outcome: Had great first-hand experience on languages such as HTML, CSS, JS and even tools like Wix

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PS-I station: Birla Science Centre , Pilani

Student

Name: SHOUNAK ACHARYA .(2021AAPS2120H)

Student Write-up:

PS-I Project Title: Design and Development of an AI Based Sentiment Analysis and Prediction of Speech Input using NLP and Python

Short Summary of work done: We used open source speech-to-text AI models to convert an audio input into transcribed text. This text was then subsequently fed into a trained model that uses techniques of NLP to use plot the data against a logistic regression curve and classify it as a positive or negative sentiment. The prediction is then represented by a "happy" or "sad" face that is promptly displayed on the screen.

Objectives of the project: To take a speech input and predict the sentiment in it by classifying it as positive/negative

Tool used: Raspberry Pi 4B (4GB RAM), Python and its libraries

Details of Papers/patents:none

Brief description of the working environment: Being affiliated with a Science Gallery, the company expected us to brainstorm our own projects that allign with the functioning of the Science Centre. We were given the freedom to develop technology that enhance the museum experience for the visitor and/or the employees and were supported with the adequate funding/inventory to fulfill this. The working environment was good with all the basic facilities and more.

Academic courses relevant to the project: IOT, Natural Language Processing, Machine Learning

Learning Outcome: Being comfortable with using hardware modules like Raspberry Pi and Arduino. I also learnt to deploy python libraries and use NLP to train models

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PS-I station: Birla Science Centre , Pilani

Student

Name: PANCY SINGLA(2021B4A32166G)

Student Write-up:

PS-I Project Title: Design and development of Multipurpose Smart Safety Locker for Birla Museum using Arduino IDE

Short Summary of work done: We have designed a smart security locker including Keypad, RFID card sensor etc. The integration of a fingerprint scanner adds an extra

layer of security to the smart locker system. Authorized individuals can access their assigned locker with a simple touch, minimizing the risk of unauthorized entry and ensuring reliable protection of personal belongings. User passwords further reinforce the system's security, allowing only individuals with the correct password to access the locker. This additional authentication factor adds an extra barrier against unauthorized entry. RFID technology enables quick and contactless authentication by assigning users RFID cards or tags that can be easily scanned to grant access to their designated lockers. Face recognition technology accurately identifies authorized users by capturing and analyzing facial features.

Objectives of the project: Creating a smart locker for visitors in museum including: 1. Keypad (Password which changes every time) 2. RFID card sensor with cards and tags with unique ids 3. Fingerprint sensor 4. Face Recognition

Tool used: H/w: Arduino Mega ,LCD display, RFID card kit, Fingerprint sensor, Servo motor, Keypad, Jumper wires and breadboard. S/w: Arduino IDE

Details of Papers/patents:none

Brief description of the working environment: The centre offers a very comfortable and joyful working atmosphere. We had strict guidelines regarding attendance and timing. There, the mentors are incredibly helpful and friendly. They showed us how to proceed, gave us the resources we needed, and assisted us when we ran into challenges. I relished my time there, and I'm eager to return to an environment like that.

Academic courses relevant to the project: Microproc and interfacing

Learning Outcome: 1. Hardware knowledge: Arduino boards, sensors and other input and output devices
2. Software: Arduino IDE programming (it uses a language based on C++)
3. integration: integrating many small sensors together and compiling code

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PS-I station: Birla Science Centre , Pilani

Student

Name: A HARSHITH SARMA .(2021B4A32788P)

Student Write-up:

PS-I Project Title: Web development

Short Summary of work done: Learnt some new languages and made a website

Objectives of the project: My project was to make a website for bsc

Tool used: Html, css, java script, react js

Details of Papers/patents: No

Brief description of the working environment: It was nice

Academic courses relevant to the project: None

Learning Outcome: Learnt how to make a website, design aspects.

PS-I station: Comprotech Engineering Pvt. Ltd. , Hyderabad

Student

Name: SOUMILI BOSE .(2021A4PS2265H)

Student Write-up:

PS-I Project Title: A Comprehensive Study on Aerospace Manufacturing: Processes, Assemblies, and Components

Short Summary of work done: Visited Comprotech Engg. Pvt. Ltd. for orientation and tour of departments. Explored composite manufacturing, bonding processes, explosive bolts, and missile projects (MRSAM, LRSAM, etc.). Worked on Mid Semester presentation and report, analyzed process documents, and observed various components and their functions.

Objectives of the project: This report provides an in-depth analysis of the manufacturing processes employed in aerospace engineering.The report aims to provide insights into this aspect of the defense industry and help us understand its various processes and parts.

Tool used: Workshop tools, Canva

Details of Papers/patents: No

Brief description of the working environment: The people we met were friendly and ready to guide us. We expected a specified project but were not given due to the time constraint. But that didn't prohibit us from learning and observers from the technicians. I would recommend this PS station only if a specified project is alloted.

Academic courses relevant to the project: Manufacturing Processes, Workshop, Material Science

Learning Outcome: The report critically examines various manufacturing processes, including machining, composite manufacturing, precision casting,Assembly and Fitting, Quality Control. Each process is evaluated based on its strengths, limitations, and suitability for aerospace applications. This report also aims at explaining the thrust vector control assembly parts and the detonators associated with it along with the important quality assurance tests to ensure there is minimal margin of error.

PS-I station: Comprotech Engineering Pvt. Ltd. , Hyderabad

Student

Name: SIDDHARDHA SAMRUDH CHILLARA .(2021A4PS2461H)

Student Write-up:

PS-I Project Title: A Comprehensive Study on Aerospace Manufacturing: Processes, Assemblies, and Components

Short Summary of work done: Visited various departments on different days. Learnt the know-how of each department by respective in-charges.

Objectives of the project: To learn about the various manufacturing processes in the industry along with their application in the procurement of various products.

Tool used: No tools were used

Details of Papers/patents: Nil

Brief description of the working environment: Company provided a comfortable, air-conditioned environment. Company employees were friendly and helpful when approached regarding doubts.

Academic courses relevant to the project: Manufacturing Processes

Learning Outcome: Extensively learnt about various Manufacturing Processes.
Analyzed the assemblies of various components.
Developed a better understanding on the business aspect in the sale of products.

PS-I station: Comprotech Engineering Pvt. Ltd. , Hyderabad

Student

Name: JITEN JAJU .(2021A4PS2497H)

Student Write-up:

PS-I Project Title: A Comprehensive Study on Aerospace Manufacturing: Processes, Assemblies, and Components

Short Summary of work done: My work during the PS firstly involved keenly looking at the various manufacturing process and the machines used in them, go through the different shopfloors and get an idea about the company. Then we went on to learn about Composite parts manufacturing stages and the importance of it and what parts it is useful for. We also learned about the Thrust Vector Control and its assembly involving various parts. This TVC system is an integral part of the MRSAM and LRSAM missiles used by the Indian Army. We went on to documenting all these processes and creating a final report on the outcomes of what we have learnt. In all, this PS was a great learning experience in this domain of the industry.

Objectives of the project: To study the various processes and Components in the Aerospace and Defence industry

Tool used: None

Details of Papers/patents: None

Brief description of the working environment: The working environment involved various people working on different machines and tools on different shop floors to make a component from its raw material to the deliverable product. With deadlines to meet, all the employees in the company were completely focused on the work but at the same time they were polite enough to answer all our queries and guide us through and explain the process.We were not alloted a specific mentor, the CEO himself assigned various people to us on different days to guide us, this lead to a wide range of learning.

Academic courses relevant to the project: Manufacturing Processes
Material Science and Engineering

Learning Outcome: Got to learn the in depth assembly of the thrust vector control used in the LRSAM and MRSAM missiles, Composite component manufacturing, their quality control process and quality assurance tests along with studying different materials used.

PS-I station: Comprotech Engineering Pvt. Ltd. , Hyderabad

Student

Name: DWARAMPUDI SANTOSH REDDY .(2021B5A42800P)

Student Write-up:

PS-I Project Title: A Comprehensive Study on Aerospace Manufacturing: Processes, Assemblies, and Components

Short Summary of work done: We did an in-depth analysis of the manufacturing processes employed in aerospace engineering. The aerospace industry demands high-quality, reliable, and high-performance components and structures. We also had critically examined the various manufacturing processes, including machining, composite manufacturing, precision casting, Assembly and Fitting, Quality Control. Each process is evaluated based on its strengths, limitations, and suitability for aerospace applications.

Objectives of the project: To study the various processes and components in an aerospace manufacturing industry.

Tool used: Canva, Milling, Turning, Lathe, Composite, Autoclave, CMM, Resources

Details of Papers/patents:A Comprehensive Study on Aerospace Manufacturing: Processes, Assemblies, and Components

Brief description of the working environment: For the PS-1 we got to visit the company and research regarding various processes, assemblies and components manufactured in an Aerospace industry, mostly regarding the manufacturing of thrusting of a defensive missile. We got to see various assemblies like Thrust Vector Jet Vane Assembly, Booster Nozzle Assembly and many more.

Academic courses relevant to the project: Workshop, CMM

Learning Outcome: Got Familiar with the industrial environment of an Aerospace and Defense Industry. Learnt various processes as designing, and various processes included in manufacturing.

PS-I station: CPCL - Chennai Petroleum Corporation Ltd. , Chennai

Student

Name: DHRUVIT JAIN .(2021A4PS0916P)

Student Write-up:

PS-I Project Title: Basic pressure vessel design and heat transfer across pump seal

Short Summary of work done: Our work started off with exploration of different departments within CPCL, understand their way of working and the problems they solve. This took around the first 25 days of our PS-1, we visited many departments like Process, Inspection, Projects, Offsite Maintenance, Finance, Marketing etc. Having spend a day (or more) in these departments we were able to understand how they work and tackle problems. Post that we were allotted our projects which consumed the remaining part of our PS-1. We were divided in 3 groups, all the groups performing different

Objectives of the project: To learn and understand design of basic pressure vessel and analyse heat transfer for a heat exchanger use to cool a pump seal

Tool used: S/W - AutoCad Fusion 360

Details of Papers/patents: None

Brief description of the working environment: Working environment in CPCL(being a PSU) is rather stressful with some issue or the other coming up everyday but this indeed also provides us with an opportunity to see practical working and problem solving involved inside the industry.
For someone aiming to join a PSU in future CPCL is probably the best way to learn more about it. How to get a job in a psu and learning the way work is done here can be thoroughly understood.

Academic courses relevant to the project: Mechanics of Solids (3rd Semester)
Heat transfer (4th semester)

Learning Outcome: Understanding how pressure vessels are designed and what all challenges are faced by engineers. Also to perform the project I had to learn AutoCad Fusion 360
Heat transfer analysis rather helped us to understand the practical implementation of the subject done in the previous semesters

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PS-I station: Delhi Metro Rail Corporation , New delhi

Student

Name: ARYAN CHATURVEDI .(2021A2PS1578P)

Student Write-up:

PS-I Project Title: Construction and maintenance for track and Civil Structures.

Short Summary of work done: We thoroughly investigated several aspects of metro rail construction and maintenance throughout the course of the project. Week after week, we descended into the depths of track analysis, concentrating on tools like the Calipri machine that allowed us to properly measure and assess rail geometrical attributes. We also studied the techniques used to construct underground metro trains and stations, including the cut-and-cover method and the use of soldier pile technology for support and stability. We also learned about the oscillation monitoring system, which is crucial for ensuring passenger comfort and safety while also lengthening the lifespan of the

infrastructure. We learned about specialised tools used in the project, like rail jack machines and bolt tightening machines, which are all necessary.

Objectives of the project: To learn how tracks are maintained and inspected in a metro rail network, how civil structures are inspected for potential defects and how preventive and corrective maintenance is offered ; How a tunnel is constructed.

Tool used: Caliper machine Vibration monitoring system Railway crane machine Bolting machine Tunnel boring machine (TBM) Bi-reflex targets Molds or molds for melting ring segments Reinforcing materials (steel bars or fiber reinforcement) Pressure-based tunnel boring

Details of Papers/patents:N/A

Brief description of the working environment: The working atmosphere was dynamic and engaging during the project and provided practical visibility into various aspects of metro construction and maintenance. We were able to visit several areas, including warehouses, foundries, and tunnel construction sites, to see real-world applications of the technology and procedures we studied. Company expectations focused on active engagement, enthusiasm for learning and our ability to adapt to different work environments. We had to interact with the technologies and tools presented to us, ask questions and try to understand their practical impact on subway traffic. Attention to detail and adherence to safety regulations were critical when dealing with specialty equipment and supplies. As we often interacted with experts, engineers and maintenance staff during our travels, the organization respected our teamwork and communication skills. Accuracy and professionalism were important as we represented the company locally. Overall, the organization provided a suitable learning environment that required us to understand the intricacies of metro construction and maintenance and use our knowledge to effectively advance the field in the future.

Academic courses relevant to the project: Surveying , CPT, Mechanics of solids, Civil Engineering Materials , soil mechanics

Learning Outcome: Working of Tunnel Boring Machine, Construction methods of underground stations and elevated stations.

PS-I station: Delhi Metro Rail Corporation , New delhi

Student

Name: JOSHI ROHAN ABHIJIT .(2021A2PS2161P)

Student Write-up:

PS-I Project Title: Civil Engineering at DMRC

Short Summary of work done: Throughout the project, we conducted a thorough investigation of many areas of metro rail construction and maintenance. We dived into the complexities of track analysis week after week, focusing on technology such as the Calipri machine, which allowed us to precisely gauge and evaluate rail geometrical properties. We also looked into the procedures utilised to build subterranean metro rails and stations, such as the cut and cover approach and the usage of soldier pile technology for stability and support. We also heard about the oscillation monitoring system, which is essential for guaranteeing passenger safety and comfort while also increasing the infrastructure's lifespan. Throughout the project, we became acquainted with specialised instruments such as rail jack machines and bolt tightening machines, all of which are required for efficient and secure rail repair. In addition, we witnessed the casting process for tunnel ring segments, recognising its critical role in the construction of strong tunnel infrastructure. We realised the need of thorough planning, quality control, and cutting-edge technologies in the construction of various portions throughout our journey. Overall, the project gave us excellent insights into the multidimensional world of metro rail construction, maintenance, and safety measures, giving us with practical knowledge and abilities that will allow us to actively contribute to the development of dependable and efficient metro systems.

Objectives of the project: Learn about different technologies used for track analysis and their impact on metro movement, Understand the construction of underground metro tracks and stations using the cut and cover method, Explore soldier pile technology for building new undergroun

Tool used: Calipri machine Oscillation monitoring system Rail jack machine Bolt tightening machine Tunnel boring machine (TBM) Bi-reflex targets Formwork or moulds for casting ring segments Reinforcement materials (steel bars or fibre reinforcement) Pressure-based

Details of Papers/patents:NA

Brief description of the working environment: The working atmosphere was dynamic and immersive throughout the project, providing hands-on exposure in various facets of metro rail construction and maintenance. We were able to tour several areas, such as depots, casting yards, and tunnel construction sites, to see real-world applications of the technology and procedures we studied about.

The company's expectations were centred on our active engagement, enthusiasm to learn, and adaptability to various work contexts. We were supposed to interact with the technologies and tools that were provided to us, asking questions and attempting to comprehend their practical consequences in metro rail operations. In our encounters with specialised gear and equipment, attention to detail and adherence to safety regulations were critical.
Furthermore, because we frequently interacted with experts, engineers, and maintenance employees during our travels, the organisation respected our teamwork and communication skills. Punctuality and professionalism were essential because we were representing the company on-site.
Overall, the organisation provided a suitable learning atmosphere, expecting us to understand the complexities of metro rail construction and maintenance and utilise our knowledge to effectively contribute to the sector in the future.

Academic courses relevant to the project: Mechanics of solids, construction planning and technology, soil mechanics

Learning Outcome: Understanding the various technologies used for track analysis and their importance in ensuring rail safety and efficiency.
Gaining insights into the construction methods of underground metro tracks and stations, including the cut and cover technique and soldier pile technology.
Familiarity with the Calipri machine for precise rail profile scanning, allowing for early detection of defects and reducing maintenance costs.
Learning about the oscillation monitoring system's significance in maintaining passenger comfort and infrastructure integrity in metro rail networks.
Acquiring knowledge of specialized devices used during rail maintenance, such as rail jack machines and bolt tightening machines, for efficient and safe maintenance procedures.
Observing the casting process of tunnel ring segments and understanding the critical role they play in building reliable and durable tunnel infrastructure.

PS-I station: Delhi Metro Rail Corporation , New delhi

Student

Name: SARTHAK MENDIRATTA .(2021A4PS2528P)

Student Write-up:

PS-I Project Title: To increase non-fare revenue channels for DMRC

Short Summary of work done: In my 7 week PS duration I worked on the following- One of them was to identify strategic players and reach out to them for co-branding of metro stations, and convince these prospective bidders to put in their money. Second was the reserve price financial modelling of stations for the PB tender released by DMRC. And thirdly, I contributed in hosting a stakeholder interaction by reaching out to 70+ brokers and developers for the Property Development tender.

Objectives of the project: To identify channels of revenue other than metro fares and push these channels to increase revenue.

Tool used: Ms Excel, PPT , Google Sheets

Details of Papers/patents:-

Brief description of the working environment: I was working with Auctus advisors who were functioning as consultants to Delhi Metro Finance Department. The learning environment was extremely collaborative and helped me in my personal growth while giving a taste of corporate life at the same time.
My major learning through the PS was , how consultants and their clients collaborate over projects, and how each part of the problem is approached piecewise. Through my personal work I was able to hone my communication , logical and marketing skills and also get exposure to how bids work and how companies focus to increase their revenue.

Academic courses relevant to the project: NA

Learning Outcome: strategy consulting/financial model long/ contractual terms/marketing/pitching

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PS-I station: Delhi Metro Rail Corporation , New delhi

Student

Name: NAMAN JAIN .(2021A4PS2669P)

Student Write-up:

PS-I Project Title: 1. A STUDY ON BUILDING MANAGEMENT SYSTEM(BMS), TUNNEL VENTILIATION SYSTEM(TVS), ENVIRONMENT AND CONTROL SYSTEM (ECS) AND SOLAR SYSTEMS 2. A study on Electrical and Mechanical Systems on Airport Express line

Short Summary of work done: At DMRC, I was required to visit numerous metro stations and depots to gain knowledge of the various systems in use. I learned primarily about the generation and use of solar energy at stations and facilities, as well as the tunnel ventilation system and environmental control system required for any underground station. I also learned about the Building Management system, which is designed to reduce decision-making and implementation delays in the event of an emergency. I also learned that the Airport Express line utilises an exclusive Baggage Handling System. In addition, I gained knowledge of the fire detection and suppression system.

Objectives of the project: To learn about Solar Panels (Energy generation and distribution), Tunnel Ventilation System and Environmental Control Systems. Also to learn about peculiar systems such as Baggage Handling System on Airport Express line and Building Managment System used

Tool used: Hardware- NA, Software - Microsoft Excel and Word

Details of Papers/patents:NA

Brief description of the working environment: The work environment was very conducive. Many Assistant Managers, Junior Engineers, Section Engineers, and Assistant Section Engineers were involved. All of them were affable and taught me well about the complexities of the systems employed. They instructed both theoretically and practically. They were always amenable to questions and their resolution. They even provided me with instructions and operation manuals to aid in my learning process. I was required to create a PowerPoint presentation every week to provide the Assistant Manager with a weekly summary of my work and a monthly report of my work.

Academic courses relevant to the project: Heat Transfer, Mechanics of Solids, Mechanisms and Machines, Digital Design, Electrical Machines, and Advanced Mechanics of Solids

Learning Outcome: I gained knowledge on the solar policy of DMRC, the implementation of an efficient Tunnel Ventilation system, the provision of adequate ventilation for people in stations, the maintenance of the climate in underground stations, and the contribution towards a greener future as an organization. I also learned about the system responsible for reducing the decision making time during emergencies, known as Building Management System.

PS-I station: Delhi Metro Rail Corporation , New delhi

Student

Name: ABHINAV MISHRA .(2021A7PS0706P)

Student Write-up:

PS-I Project Title: Implementing a CRUD Web Application using Laravel Framework

Short Summary of work done: The objective of my project was to create a Web Application that gives a System Administrator a dashboard to interact with to Create/Read/Update/Delete entries of metro card holders. Any changes made by the System Admin are visible instantaneously on the Database. Furthermore, more than one SystemAdmin can log in to the application and make changes. The application also provides the functionality of running search queries on the database of metro card holders. All results are paginated. Pagination is also implemented when a system admin accesses the database. To log in as a system admin, the user either has to have a registered e-mail id and password or create a new one from the login page and be added to the list of accepted system admins. The System Admin accounts have a layer of security implemented through password protection. Laravel's inbuilt hashing mechanism hashes the password and stores them corresponding with the list of System Admins in the database. A separate database of metro card holders exists, which can be manipulated only by the logged-in System Admins.

Objectives of the project: Designing a CRUD web apllication using Laravel implementing search, pagination and other related features

Tool used: Laravel Composer, XAMPP Control Panel, VS Code

Details of Papers/patents: None

Brief description of the working environment: The working enviroment was very good and the mentor was very supportive. She guided me to help me understand the use of Laravel framework and gave constant input and suggestions to help me increase the functionality of the web application. She also walked me through the actual code base that DMRC uses on Laravel and helped me understand how the framework and its various functionalities are implemented. She helped me understand how the MVC architecture is leveraged for smooth operation.

The mentor gave me sufficient time and resources to first understand how php code is written, how the laravel framework is used and its related functionalities.

Academic courses relevant to the project: DBMS
DSA
OOPS

Learning Outcome: -MVC architecture of Laravel
-HTML CSS basics, desigining web pages
-Handling SQL which reflects the changes made to the database

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PS-I station: Delhi Metro Rail Corporation , New delhi

Student

Name: TUSHAR VIGH .(2021A8PS0288P)

Student Write-up:

PS-I Project Title: Increasing Non-Fare Box Revenue of DMRC

Short Summary of work done: During my 7-week PS duration, I accomplished the following: Identified strategic players and pitched co-branding opportunities for metro stations, persuading prospective bidders to invest. Conducted reserve price financial modeling for stations in the PB tender released by DMRC. Organized a stakeholder interaction by engaging 70+ brokers and developers for the Property Development tender.

Objectives of the project: To identify channels of revenue other than metro fares and push these channels to increase revenue.

Tool used: MS Excel , PPT

Details of Papers/patents:-

Brief description of the working environment: During my internship with Auctus Advisors, serving as consultants to Delhi Metro Finance Department, I experienced a highly collaborative learning environment that nurtured my personal growth and provided a glimpse into corporate life. Major takeaways included understanding the dynamic

collaboration between consultants and clients in project execution, and the systematic, piecewise approach to problem-solving. My personal work allowed me to refine communication, logical, and marketing skills, as well as gain valuable insights into the bidding process and revenue generation strategies employed by companies. The experience was invaluable in preparing me for a successful career in the consulting and business domain.

Academic courses relevant to the project: Fundamentals of Finance and Accounting

Learning Outcome: Strategic and Implementation consulting, financial model , contractual terms , market making

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PS-I station: Delhi Metro Rail Corporation , New delhi

Student

Name: YASH KUMAR .(2021B4A32330P)

Student Write-up:

PS-I Project Title: MTS connectivity in tunnel

Short Summary of work done: I was alloted telecom department(S&T), and my workplace consisted of all the substations from Udyog bhawan to Samaypur. Initially I had to learn about how various components in racks for different systems like CCTV,Radio,PAS etc. are connected. I got to know about error checking procedures in case of failure in various equipments. I did some research on how handover procedures can be optimized inside tunnels to ensure seamless connectivity.

Objectives of the project: To find an alternative solution for when there is a handhshake/handover problem in underground stations.

Tool used: MTS-4 (Motorola trunked system), Bi-directional amplifiers and co-axial cables.

Details of Papers/patents: None

Brief description of the working environment: The working environment was completely safe even though it was underground. I never had to go near OHE or any

other high voltage equipment . The engineers on site were very cooperative and they catered to my issues whenever they had a chance. DMRC offers great chance to learn about various systems in every core department. The learning experience is nice but interns are not allowed to work with actual equipment and the most you get to do is look at the equipment functioning and being repaired whenever necessary. But again I learn quite a lot about Radio connectivity on a very huge scale, and there are many such systems in telecom department to be explored.

Academic courses relevant to the project: Communication Systems(EEE F311)

Learning Outcome: TETRA connectivity, connection procedures, handshake and handover mechanisms.

PS-I station: Delhi Metro Rail Corporation , New delhi

Student

Name: ANIRUDH(2021B4A82579G)

Student Write-up:

PS-I Project Title: How to minimize time in AFC

Short Summary of work done: Applied queueing theory on fare collection system to minimize wait time of passengers at a general station with provided footfall

Objectives of the project: To apply queueing theory on AFC

Tool used: AFC gates,TVM,TOM,EFO

Details of Papers/patents: None

Brief description of the working environment: Working environment was safe and friendly.I was allowed to look at the components functioning and being repaired in real time. however I was not allowed to work with some of the sensitive equipment. I learnt some real life application of Queueing theory.

Academic courses relevant to the project: Queueing theory(operation research)

Learning Outcome: Application of queueing theory

PS-I station: Disha Auto Components , Aurangabad

Student

Name: YASH SHIRISH MUNDADA .(2021A7PS0001P)

Student Write-up:

PS-I Project Title: Creating an Attendance/ Employee Management/ Human Resources Management Software portal for DISHA Auto Components Pvt Ltd.

Short Summary of work done: This project focuses on developing an HR Management System (HRMS) using the MERN stack. The HRMS aims to streamline and automate various HR processes, including employee management, leave management, and training tracking. By leveraging technologies such as React, Node.js, Express, and MongoDB, the system offers a user-friendly interface, robust data storage, and seamless integration. The HRMS addresses the specific needs of the organization, including support for blue-collar employees and efficient document management. Through this project, the goal is to enhance HR operations and optimize administrative tasks.

Objectives of the project: To create an Attendance/ Employee Management/ Human Resources Management Software portal

Tool used: Figma, MongoDB, NodeJS, ExpressJS, ReactJS, Postman, Git

Details of Papers/patents:NA

Brief description of the working environment: Working environment was good, company had high expectations from BITS students, learning was also good because of this.

Academic courses relevant to the project: OOPS, DBMS

Learning Outcome: Learnt development to a good extent

PS-I station: Disha Auto Components , Aurangabad

Student

Name: SAKSHAM GUPTA(2021B4A42778G)

Student Write-up:

PS-I Project Title: Improvement in Tool Life of Cutting Blade and CNC inserts

Short Summary of work done: The primary focus of our project was on bringing improvement in tool life of blade used in cutting machines and inserts used in CNC machines. To achieve this, we first properly observed each and every production process carried out at the work floor which involved these tools. While observing, we took note of all the factors which could be increasing tool wear and affecting tool life and hence, decreasing the number of jobs currently being achieved by the tool. We also tried to suggest possible solutions for the same. Next, we recorded the complete set of data about the parameters being set and results achieved by multiple machines for several days and compiled it all together in sets. Then, we analysed this compiled data using graphs and charts to find how tool life is being affected by each factor and by how much. This helped us to finally draw conclusions about current state of tool life and the ways by which we can further improve it, resulting in increased efficiency and, hence, productivity of the company.

Objectives of the project: 1) Get a proper idea of the current status of all the data involving manually set parameters as well as results being achieved on shop floor. 2) To find ways to bring an overall improvement in tool life of the cutting blade and inserts used in CNC Machine

Tool used: Hardware - Company Tally sheet, Software - MS Excel, MS Powerpoint, MS Word

Details of Papers/patents:1) Pradeep Kumar, “Study on Tool Life and its Failiure Mechanisms”, Sep 2015
2) Luo, S. Y., Liao, Y. S. and Tsai, Y. Y. 1999, Wear Characteristic in Turning High Hardness Alloy Steel by Ceramic and CBN Tools, Elsevier Journal of Materials Processing Techn

Brief description of the working environment: The main workplace for the execution of our project was the workshop floor of the company. We mainly worked in the cutting and CNC sections of the company as our project domain was related to these sections.
Cutting section consisted of 5 bandsaw blade cutting machines and 1 saw cutter (mainly used for tubings). On our project, we focused on “Cutting Machines- 11 and 12”.
CNC Section consisted of 17 CNC machines of different varieties and purpose out of which we mainly focused on “CNC Machine-18 and 19” (Casing section) and “CNC Machine- 7 and 10” (Tubing section) for the purpose of our project.
It was a great learning experience of being allowed to look at real time industrial processes up close and a closer look at problems and solutions faced in large scale industrial operations.
I came out with a lot of great insights and soft skills that were required hand in hand throughout the project work as well as course evaluations. My communication and interpersonal skills also got better as I had to communicate with supervisors, industry mentors and workers on a daily basis. And I also improved my critical thinking & decision making skills as I observed and gained new knowledge from their experiences.

Academic courses relevant to the project: Important topics relevant to project -
1) Tool Life & Factors affecting it
2) Methods to measure Tool Life
3) Tool wear & its effects
4) Feed Rate & Depth of Cut

Learning Outcome: 1) Observational skills through regular visit to shop floor and watching processes.
2) A learning experience of being able to look at real time industrial processes up close and an insight on problems and solutions faced in large scale industrial operations.
3) Technical, interpersonal and communication skills through communicating with machine operator, supervisors, and industry professionals.
4) Problem solving and critical thinking skills.
5) Analysing and Documentation skills by keeping a regular record of data (parameters and results) and then trying to find patterns by comparing them.

PS-I station: Disha Auto Components , Aurangabad

Student

Name: SUDHANSHU SHIRISH BONDRE(2021B5A40898G)

Student Write-up:

PS-I Project Title: Improvement in tool life of cutting blades and CNC inserts

Short Summary of work done: During the PS period, we were asked to collect data and were told to try and find out reasons for tool failure. Based on this, we were told to try and find solutions to solve the issue of reduced tool life.

Objectives of the project: To improve tool life of tools and try to find reasons for tool failure

Tool used: Software- Excel, word, Hardware- Company tally sheets

Details of Papers/patents: No papers or patents application

Brief description of the working environment: The working environment was good and the employees were very friendly and helpful. They explained the processes and also cleared our doubts, which proved to be very helpful for us.
The company expected us to resent concrete data on the tools which they used and also wanted us to try and find out the reasons for tool failure in the machines. This helped us in understanding in how tool life affects a manufacturing industry and also its processes. PS helped learn about the importance of planning, communication and execution as these are some of the determining factors on which an industry runs.

Academic courses relevant to the project: Workshop

Learning Outcome: Throughout the project, I understood how an industry works and the work flow in an industry. The project helped me learn new things like material science, and was comparable to the workshop course which we had studied. It also helped me understand how even small changes or modifications to a process can lead to many consequences. It helped me improve my non-technical skills like communication, writing formal content and team coordination.

PS-I station: DRDO - Advanced Systems Lab , Hyderabad

Student

Name: SEHAJPAL SINGH SARAO .(2021A4PS1360P)

Student Write-up:

PS-I Project Title: Application of CFRP composites in aerospace industry

Short Summary of work done: Acquired knowledge of composites using books and other literature reviews. Then we learnt the basics of an aerospace launch vehicle, its parts and propulsion techniques. We wrote a detailed report on application of CFRP composites in aerospace industry. Then we performed statistical analysis of composite properties using excel and wrote our finding about the same.

Objectives of the project: Write a detailed report on Composites and statistical analysis of mechanical properties of composites

Tool used: Excel

Details of Papers/patents:NA

Brief description of the working environment: The working environment was nice. The scientist and his team were very cooperating and taught us a lot of things. We went to labs and witnessed some tests. No electronic items are allowed inside due to security reasons. But overall, the learning environment is very conducive, and you will get to witness the assembly and manufacturing of weapon systems in DRDO.

Academic courses relevant to the project: Probability and Statistics, Mechanics of Solids

Learning Outcome: Learnt about composites, their properties, manufacturing techniques and basics of an aerospace vehicle.

PS-I station: DRDO - Advanced Systems Lab , Hyderabad

Student

Name: SMRITHI L SERAMALANNA(2021A4PS2018G)

Student Write-up:

PS-I Project Title: Application of CFRP Composites for Aerospace Applications

Short Summary of work done: Report Writing, Research Paper Writing, Testing, Manufacturing

Objectives of the project: To complete a technical report outlining concepts of Aerospace vehicles, Rocket Propulsion Systems, Rocket Motor Casings, Introduction to Composites, Polymeric Composites and Methods of Experimental Characterization of these properties through rigorous li

Tool used: Microsoft Office

Details of Papers/patents: Statistical Analysis on Material Properties of Carbon/Epoxy Composites through Experimental Characterization (not published) The paper aimed to predict a model fitting the data and predict property values for desired relibility. The reliability analysis d

Brief description of the working environment: The work environment was welcoming with multiple opportunities to learn and people willing to teach. It gave a good introduction to the line of research and development in the field of defence. It also gave me exposure to work ethics, principle of punctuality and the chain of command followed in the industry. I got a chance to meet likeminded interns from different NITs, working on various domains which led to a knowledge transfer and opened up pathways for future collaborations .The internship improved my communication skills, broadened my network and also taught me how to approach senior members of an organisation. Personally, it introduced me to a new domain of Aerospace, which broadened my view on potential research topics, further internship opportunities and job domains.

Academic courses relevant to the project: Advanced Mechanics of Solids, Applied Statistical Methods, Probability and Statistics

Learning Outcome: 1. Technical Report Writing based off of literature review on the concepts outlined above.
2. Approach to writing a research paper from the states of literature review to compiling references.
3. Lab and testing site visits to understand practical application of concepts studied in literature.

PS-I station: Forbes Marshall Pvt Ltd , Pune

Student

Name: ISHAAN SHANKAR .(2021A4PS0788P)

Student Write-up:

PS-I Project Title: Value Stream Mapping in FGS Packing area.

Short Summary of work done: This Value Stream Mapping (VSM) project aims to analyze and optimize the value stream within FGS Packing Area. This project encompasses both the current state VSM and the future state VSM to identify opportunities for improvement and enhance our overall operational efficiency. By examining our existing processes and envisioning an optimized future state, we can strategically align our operations with customer needs, streamline workflows, and eliminate waste. This introduction sets the stage for a comprehensive and forward-thinking project that will drive continuous improvement and enable us to deliver better solutions to improve efficiency of FGS Packing Area.

Objectives of the project: Value stream mapping of Current and future state to increase efficiency

Tool used: Value Stream Mapping and it's mapping softwares, AutoCAD, Lean management Tools.

Details of Papers/patents:SMID (Single Minute Identification of Delivery) concept

Brief description of the working environment: The working environment during PS-I (Practice School-I) is a crucial phase for students as it provides them with their first exposure to a professional setting. Typically, PS-I is a mandatory part of various educational programs, where students undergo practical training in a real-world workplace. At Forbes Marshall, our environment is characterized by a blend of academic learning and practical implementation. We as students are expected to adapt to the professional culture, collaborate with colleagues, and follow workplace norms.
PS-I is an invaluable opportunity for students to bridge the gap between theoretical knowledge acquired in classrooms and real-world application. It allowed us to learn Lean management principles like Value stream mapping, Kaizen, Kanban, real time problem-solving, teamwork etc. By working alongside experienced professionals, we observed and learn from their expertise.
The PS-I experience provided me with the insight into potential career paths, helping me make informed decisions about their future. It offered me a chance to explore different roles within the company, enhancing career clarity.

Academic courses relevant to the project: Lean Manufacturing (DEL of mechanical)

Learning Outcome: Lean management principles such as Value stream mapping, Kaizen , KanBan , Gemba, etc.

PS-I station: Forbes Marshall Pvt Ltd , Pune

Student

Name: KARTIK PARAMESHWARAN(2021A4PS2901H)

Student Write-up:

PS-I Project Title: AUTOMATION OF ALERTS TO BE GENERATED IN A BOILER WHEN A PARAMETER IS OFF THE ALLOWABLE LIMIT

Short Summary of work done: To find out the advantages, disadvantages, importance of the parameters that effect the efficiency of the boiler overall and then to automate the alert generation when these parameters go off limit

Objectives of the project: To automate the alert generation in a boiler house when a parameter is off limit

Tool used: Grafana, Prometheus, SQL, Microsoft Excel

Details of Papers/patents:NA

Brief description of the working environment: The working environment of the company was really good. They promoted the interns to ask questions while working and they made sure that we didn't have any by the end of the internship. The main learning during PS-I was the understanding of the working of a MNC of as well as the working of a boiler house and the people behind its proper functioning.

Academic courses relevant to the project: Applied Thermodynamics, Heat Transfer

Learning Outcome: Understanding MS Excel for Data Analysis, Intermediate level SQL learning , Grafana and Prometheus basics

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PS-I station: Forbes Marshall Pvt Ltd , Pune

Student

Name: RAVITEJ SUDHIR GHALSASI(2021A4PS2904G)

Student Write-up:

PS-I Project Title: Value Stream Mapping of Finished Goods Store Packing Area

Short Summary of work done: In conclusion, the value stream mapping (VSM) project has been a transformative and empowering journey that led us towards operational excellence and process optimization. By systematically analyzing our current state processes, identifying wasteful elements, and gaining invaluable insights into our value streams, we have discovered numerous opportunities for improvement. The comprehensive approach, addressing the eight types of waste, has resulted in streamlined processes, reduced lead times, minimized costs, and improved customer satisfaction. This remarkable achievement was made possible through the collaborative efforts of cross-functional teams, which have fostered a culture of continuous improvement, tapping into the creativity and skills of our employees. Moving ahead, we remain committed to monitoring and measuring our performance, implementing further enhancements to ensure sustained success. The value stream mapping project has given us the tools to drive positive change, eradicate waste, and unleash the full potential of our organization's value streams. Armed with this newfound knowledge and an unwavering dedication to continuous improvement, we are well-positioned to thrive in an increasingly competitive business landscape.

Objectives of the project: We aim to optimize the existing operations to be simpler, more transparent, error-free, and more time and cost-efficient. This can be achieved by implementing Lean Principles. Thus, our first goal is to create existing and ideal VSM of the FGS processes.

Tool used: Value Stream Mapping softwares, Root cause analysis diagrams, Autocad

Details of Papers/patents:SMID (Single Minute Identification of Delivery) concept

Brief description of the working environment: Our mentor, Mr. Richard sir, has been highly supportive and has personally looked to it so that we get industry exposure and domain knowledge. He also organized a three-day training session with an external consultant, which greatly enhanced my understanding.

While completing the project, we interacted with many people in the organization, all very helpful.
We also learned the Lean Manufacturing concepts and tools like Valve Stream Mapping. Overall the PS station was very understanding & I am happy that we could create a positive impact there.

Academic courses relevant to the project: Lean Manufacturing

Learning Outcome: Lean Framework, Value Stream Mapping, Root cause Analysis, ERP systems

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PS-I station: Funskool (india) Ltd. , Goa

Student

Name: HASAN ZAREB .(2021B3A80951P)

Student Write-up:

PS-I Project Title: Product development

Short Summary of work done: During my time as a product development intern at Funskool, I worked with the technical team which is repsonsible to bring the product from ideation stage to production stage. During my time, I worked on developing several toys but the major ones were - Juicer, Washing Machine, Doctor's Kit and Steam Iron. The major work in the line of developing these toys that i had to do was help the team in their assembly and completing the electric connections in the toys and modifying their function according to the requirements.

Objectives of the project: To design and develop new toys for the company. The basic idea of the toy is provided by the Headquarters in Chennai, we have to turn that idea into a functional product.

Tool used: Soldering Iron, silicon glue gun, Premium grade adhesives, Lathe and milling tools, multimeter motors etc

Details of Papers/patents:NA

Brief description of the working environment: I had to face the problem of language barrier in the organization as almost everyone here speaks and converse in Konkani. The only time they used English or Hindi was when they addressed directly to me. So whenever i went with a team senior to different departments to talk to them about something or ask for their help, the learning chances to obtain new knowledge about their department and how they worked substantially reduced.
I had expected that a company as famous as funskool would be better organized and would involve me more or give me role suitable to me and my field.

Academic courses relevant to the project: Mechanical Workshop, Technical Report Writing, Organization psychology, Principles of management. (Mechanical branch courses which teach CAD software would be really relevant to the project as that was half the work, they do in the designing stage.

Learning Outcome: Knowledge about 3D printing and different types of materials used in 3D printing and toy making, Knowledge about different types of processes required in the development of a toy like molding, sonic welding, spray painting, dough making, and Assembly. I also learned how different departments in an organization work together to achieve one common goal.

PS-I station: G.M.R. Varalakshmi Foundation , Hyderabad

Student

Name: ANIRUDH NARLA .(2020A3PS1351H)

Student Write-up:

PS-I Project Title: Student counselling for lower gifted children

Short Summary of work done: The work which has been assigned to me was the counselling of students from 1st to 10th grade.The students are referred as "lower gifted children" since they are young students in their schooling age.All of the students in this institution come from very underprivileged families with no idea about education and it is the goal of this institution(GMR varalaxmi foundation) to provide free education to the underprivileged families who especially lost their lands in shamshabad for the construction of the airport.Me and my teammate sai tejeshwar have conducted various surveys by travelling to 3 different villages in the vicinity of the airport and interacted with the students and their families to get to know the problems they were facing.Throught he

surveys and meeting we have observed many things. For instance a few students face the problem of smartphone addiction.The excessive use of smartphones has hindered their ability to study and has distracted them from their education. Some students show much more interest in co curricular activites than studies which has also distected them. Some students face a much more sever problem of lack of fundamentals. This was the result of not listening and concentrating in the online classes during the pandemic over the span of 2 years which has severely affected them in their higher grade.This is a large problem for the students of higher grade like 9th and 10th. But we have proposed a few solutions to these issues.We have advised the parents to keep their students under strict monitoring and we have proposed evening tuitions with special attention to the slow learners.We have also got to know that there was a bit of bad influence and the students should be punished accordingly.We have even taught subject material like maths and science to students of 9th and 10th grade and took various sessions for different topics in both subjects.We even taugth students basic english and trained them for their upcoming conference where they have to introduce themselves and tell about their lives.We have also worked in collaboration with a non profit organisation called aizebra and conduct tuitions for students.We have done 4 student surveys after visiting 4 villages in proximity to the airport.

Objectives of the project: To find solutions to the issues of underprivileged students of GMR varalaxmi foundation

Tool used: none of the above. Unfortuantely i have not been assigned any work related to my field but more of a social service

Details of Papers/patents: None

Brief description of the working environment: The working environment was very similar to that of an institute where different teachers/mentors are assigned their respective roles. Me and my partner Sai Tejeshwar were assigned the role of counselling the students from class 1 to 10th. The institute expected me to propose solutions for the low performing students which constitute about 25% of the student body. The goal was for the majority of the students to score atleast 60% from the next year onwards. To achieve this, me and my partner conducted many surveys in 4 villages in proximity to the airport to find out the issues these students were facing. I have mentioned all the activities I have done in the previous question(short summary question).Unfortunately the institute could not offer any work related to my field(Electrical and Electronics) which is why I had to choose counselling for the project. The work has benefitted me both professionally and personally. Professionally, this project has given me an outlook of how an institute works and gave me significant job experience in the field of counselling which is very prevalent in many organisations and institutions. Personally, this has given me enough knowledge to help my friends and family who are in desperate need of career advice and school related advice.

Academic courses relevant to the project: None

Learning Outcome: Experience in the profession of teaching and counselling young children/college students with mental and skill issues. Also, expertise in dealing and forming schedules and study plans for vulnerable students to cope with their current situation and education. Also got a brief outlook of how an organisation works. It not only helped me professionally but also personally so i could even use the knowledge gained here to help other family members who possibly need career guidance and student advice since this is a very prevalent issue in modern society. It is also a very charitable thing to help underprivileged students who face a wide variety of issues so it is a thing to be proud of.

PS-I station: G.M.R. Varalakshmi Foundation , Hyderabad

Student

Name: SAI RIDDI S .(2021A1PS3047H)

Student Write-up:

PS-I Project Title: Making short videos for "assistant electrician course"

Short Summary of work done: We recorded video lectures in the light board studio room and post that, we edited the videos using Microsoft Clipchamp and added elements such as relevant images, illustrations and animations to make the videos more engaging.

Objectives of the project: To make engaging and interactive video lectures for the students enrolled in assistant electrician course at GMR Varalakshmi foundation to help them in their learning process

Tool used: OBS Studio, Lightboard, Microsoft Clipchamp

Details of Papers/patents: None

Brief description of the working environment: The mentors were able to guide us effectively and give us constructive feedback from time to time. we were able to show them our work on a weekly basis to take reviews on how we can improve.

Academic courses relevant to the project: EEE F111 Electrical Sciences

Learning Outcome: Majorly, we learnt how to work in a corporate environment as a team, how to communicate effectively and take and give constructive feedback. In the video editing process, we learnt how to edit videos, add images animations and other such elements to make the videos appealing

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PS-I station: G.M.R. Varalakshmi Foundation , Hyderabad

Student

Name: GAUTHAM R .(2021A4PS2773H)

Student Write-up:

PS-I Project Title: Making short videos on Two-Wheeler Technician Course

Short Summary of work done: At GMR Varalakshmi Foundation, I had the privilege of embarking on an internship focused on editing educational videos, where I was exposed to the fascinating world of multimedia production and its profound impact on enhancing learning experiences. We tasked to make 20 short educational videos on the topic two-wheeler technician course within the alloted time.

Objectives of the project: Making short videos on Two-Wheeler Technician Course

Tool used: H/w- Light Board S/w-Clipchamp,Canva

Details of Papers/patents: Nil

Brief description of the working environment: The working environment at GMR Varalakshmi Foundation was very professional and working and interacting with different people at GMRVF was a very valuable experience for me.

Academic courses relevant to the project: A Basic understanding of 2-wheeler parts and engines is useful

Learning Outcome: Improved time-management skills, Learned more about 2-wheeler engines

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PS-I station: G.M.R. Varalakshmi Foundation , Hyderabad

Student

Name: VIVEKA MANDALA .(2021AAPS1531H)

Student Write-up:

PS-I Project Title: Making short videos on "Refrigeration and Air Conditioning"

Short Summary of work done: My project is to make short videos on "Refrigeration and Air Conditioning" for school and college dropouts. For this, we had to shoot short videos using Lightboard software. While shooting, we have to set up and operate cameras, lighting equipment, and audio devices. After shooting a video, we import and organise the raw footage in an editing software called DaVinci Resolve. After editing the video, we show the edited video to the trainer in charge, ask for feedback, make the necessary changes to the edited video, and upload the final edited video to the drive.

Objectives of the project: The project aims to produce a series of educational short videos on the topics of refrigeration and air conditioning as part of a three-month training program provided by GMRVF for unemployed youth , school and college dropouts in rural communities.

Tool used: H/w: Lightboard technology: is a tool used for creating engaging and visually appealing presentations or instructional videos. S/w: DaVinci Resolve: is a professional video editing and colour grading software developed by Blackmagic Design. OBS Studio: i

Details of Papers/patents:none

Brief description of the working environment: During my internship at GMR Varalakshmi Foundation, the working environment was fun and collaborative, fostering a positive and supportive atmosphere for growth and learning. They were encouraging and motivated us to complete assigned tasks on time.
Throughout the internship, I had ample opportunities for learning and skill development. I was exposed to various aspects of video production, starting from filming, editing, and post-production techniques. GMR Varalakshmi Foundation provided us with access to Lightboard software, allowing us to gain hands-on experience in video production. The internship instilled valuable time management skills as you juggle multiple tasks and deadlines in the video production process.

Academic courses relevant to the project: Refrigeration and Air Conditioning

Learning Outcome: Working on video production has been an enriching experience. I learnt how to plan, script, shoot, and edit videos, ensuring that the content is informative, engaging, and visually appealing. I also gained insights into the importance of effective communication and storytelling through visual media, as well as the technical aspects of videography.

PS-I station: G.M.R. Varalakshmi Foundation , Hyderabad

Student

Name: BHUVANA SINGAMANENI .(2021AAPS2271H)

Student Write-up:

PS-I Project Title: Making short videos on the topic Refrigeration and Air Conditioning

Short Summary of work done: As a part of the 3 month free practical training giving to unemployed youth and college dropouts given by GMR Varalakshmi Foundation , we were asked to make short videos that would be helpful for the students to understand the concepts better. We learnt video shooting and video editing and incorporated animations and motion graphics to make the content more engaging.

Objectives of the project: Shooting and editing short videos

Tool used: OBS Studio, Light board technology, Davinci resolve for editing

Details of Papers/patents:none

Brief description of the working environment: It was a friendly and welcoming work environment. All the mentors and trainers in charge were really helpful and made learning easier for us.

Academic courses relevant to the project: EEE

Learning Outcome: Video editing,Video shooting

PS-I station: G.M.R. Varalakshmi Foundation , Hyderabad

Student

Name: SAI VINEETH K .(2021B1A12435H)

Student Write-up:

PS-I Project Title: Model Village Development and Mobile Medical Unit

Short Summary of work done: We visited few villages for MMU and interviewed the beneficiaries about the care they were receiving and whether the medicines were having a positive effect or not. We suggested some improvements that could be made for this function better like a digital database of the beneficiaries. For model village, we were asked to make a plan to develop Airport colony into a model village. We were given some indicators that are required for a model village. Based on that we made action plans and budget plans.

Objectives of the project: To help in developing a village into a model village. To learn the inner workings of MMU and suggest improvements

Tool used: None

Details of Papers/patents: None

Brief description of the working environment: The working environment was healthy, comfortable and fun. The expectations of the company were above average. They didn't keep high expectations over us. I learnt how to work as a team. PS-I helped me develop my communication skills.

Academic courses relevant to the project: None

Learning Outcome: Communication skills, Teamwork

PS-I station: G.M.R. Varalakshmi Foundation , Hyderabad

Student

Name: Srikrishna Vardhan Muttangi(2021B2A43140H)

Student Write-up:

PS-I Project Title: Higher Gifted Children

Short Summary of work done: Contacted students from villages around Shamshabad and found their problems in pursuing their education, and helped find solutions so that they may excel in their endeavours.

Objectives of the project: To help students from villages surrounding Shamshabad in their educational pursuits.

Tool used: MS Word, Excel and PowerPoint.

Details of Papers/patents: None.

Brief description of the working environment: The working environment was professional and conducive to diligent efforts from the students.

Academic courses relevant to the project: None.

Learning Outcome: Learnt how to deal with and navigate through a variety of new social settings which equips one the ability to better understand people and their perspectives.

PS-I station: G.M.R. Varalakshmi Foundation , Hyderabad

Student

Name: CHAKINALA ASHISH .(2021B2A83139H)

Student Write-up:

PS-I Project Title: counseling of gifted students

Short Summary of work done: counseling of gifted students

Objectives of the project: counseling

Tool used: excel

Details of Papers/patents:no

Brief description of the working environment: we have me meet the gifted children and counsel them

Academic courses relevant to the project: no

Learning Outcome: communication

PS-I station: G.M.R. Varalakshmi Foundation , Hyderabad

Student

Name: NAKSHATRA POTULA .(2021B3A80673H)

Student Write-up:

PS-I Project Title: Making video lectures for 'Assistant Electrician Course'

Short Summary of work done: Preparing video lectures for Assistant Electrician Course offered by the institute

Objectives of the project: To serve the students of the foundation with engaging and interactive video lectures

Tool used: Microsoft Clipchamp

Details of Papers/patents:none

Brief description of the working environment: The working environment was supportive, motivated by the work of the company and their motive was inspiring

Academic courses relevant to the project: Electrical courses

Learning Outcome: Teamwork, Collaboration and Professionalism

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PS-I station: G.M.R. Varalakshmi Foundation , Hyderabad

Student

Name: HARSHAVARDHAN SUBHAS HAJERI(2021B5A42402G)

Student Write-up:

PS-I Project Title: Model Village and Mobile Medical Unit

Short Summary of work done: We were assigned to look into the functioning of Mobile Medical Unit operating under GMR Varalakshmi Foundation and were assigned to find out how we could improve the total impact on the community by the Mobile Medical Unit. We were able to suggest many improvements which include digitalization of records to increase efficiency and reliability. And we suggested methods to improve beneficiary interaction. In the second project, the foundation aims to develop a small locality called Airport Colony into a Model Village, and we suggested plans to achieve the indicators that we provided to us by the foundation.

Objectives of the project: To study Model Villages and Mobile Medical Unita

Tool used: Microsoft Office

Details of Papers/patents: None

Brief description of the working environment: Mentors from the station were very supportive and we freely interacted with other staff members of the foundation. Any form of communication was encouraged from the staff. I have had the opportunity to go to the field where GMR Varalakshmi foundation runs its community welfare efforts and interact with the locals, increasing my exposure to different culture of people.

Academic courses relevant to the project: NONE

Learning Outcome: Improved interpersonal communication skills

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PS-I station: G.M.R. Varalakshmi Foundation , Hyderabad

Student

Name: ADITYA NARAYAN SINGH .(2021B5A43182H)

Student Write-up:

PS-I Project Title: Model Village Development and Mobile Medical Units

Short Summary of work done: During my internship at GMR Varalakshmi Foundation (GVF), I had the opportunity to work on diverse projects related to education and community development. I collaborated with a dedicated team, conducting research, analyzing data, and contributing to the implementation of social development programs. Engaging with local communities, I witnessed firsthand the positive impact of GVF's initiatives. The internship provided valuable learning opportunities, including workshops and mentorship programs, which enhanced my communication, teamwork, and problem-solving skills. Overall, the experience at GMRVF was enriching, expanding my knowledge and instilling a deep sense of purpose to make a meaningful contribution to society.

Objectives of the project: To develop a model village and improve the working of Mobile Medical Units

Tool used: Excel, MS word

Details of Papers/patents: Nill

Brief description of the working environment: During my two-month internship at GMR Varalakshmi Foundation (GVF), I have had a truly enriching and transformative experience. From the very beginning, I was warmly welcomed into a supportive and collaborative working environment.
Throughout my time at GVF, I had the opportunity to work on various projects related to education and community development. I collaborated closely with a dedicated and passionate team, learning from their expertise and commitment to creating sustainable change. I was encouraged to contribute my ideas and perspectives, and my opinions were valued and respected.

One of the highlights of my internship was the hands-on experience I gained in project management. I was entrusted with tasks such as conducting research, analyzing data, and assisting in the implementation of social development programs. This allowed me to develop practical skills and a deeper understanding of the challenges and opportunities in the field of corporate social responsibility.

Additionally, I had the privilege of engaging directly with local communities, witnessing the positive impact GVF has on people's lives. The Foundation's commitment to empowerment and inclusivity was evident in all the initiatives undertaken, and it was inspiring to see the transformative effects of their work firsthand.

Furthermore, GVF provided ample learning opportunities through workshops, training sessions, and mentorship programs. These initiatives equipped me with valuable skills in communication, teamwork, and problem-solving, which will undoubtedly benefit me in my future endeavors.

In conclusion, my internship at GMR Varalakshmi Foundation has been a truly rewarding experience. It has not only expanded my knowledge and skills but also instilled in me a deep sense of purpose and the desire to contribute meaningfully to society. I am immensely grateful for the opportunity to be part of such a remarkable organization.

Academic courses relevant to the project: Management

Learning Outcome: During my internship at GMR Varalakshmi Foundation (GVF), I gained several significant learning outcomes:

Understanding of Corporate Social Responsibility (CSR): Working at GVF provided me with a comprehensive understanding of CSR and its role in sustainable development. I learned about the importance of businesses engaging in social initiatives and how they can positively impact communities.

Project Management Skills: GVF offered me hands-on experience in project management. I learned how to plan, execute, and monitor projects effectively, ensuring their successful implementation. This included tasks such as setting objectives, creating timelines, and coordinating resources.

Research and Analysis: I developed strong research and analytical skills during my internship. I learned how to gather and interpret data, conduct needs assessments, and evaluate the impact of social development programs. These skills are crucial for evidence-based decision-making.

Community Engagement: Engaging directly with local communities was a significant part of my internship. I learned how to build relationships, understand community needs, and design initiatives that empower and uplift marginalized populations. This experience enhanced my empathy and cross-cultural communication skills.

Collaboration and Teamwork: Working with the dedicated team at GVF taught me the importance of collaboration and teamwork in achieving shared goals. I learned how to effectively communicate, delegate tasks, and leverage the strengths of team members to maximize productivity and outcomes.

Social Impact Assessment: GVF emphasized the measurement and evaluation of social impact. I gained knowledge of various methodologies and tools used to assess the effectiveness and sustainability of social development interventions. This understanding will be invaluable in future endeavors.

Personal Growth and Empathy: The internship experience at GVF had a profound impact on my personal growth. Witnessing the positive change brought about by GVF's initiatives deepened my empathy and reinforced the importance of social responsibility. It inspired me to continue working towards making a meaningful difference in society.
Overall, my internship at GMR Varalakshmi Foundation provided me with a holistic learning experience encompassing CSR, project management, research, community engagement, teamwork, social impact assessment, and personal growth. These valuable skills and insights will undoubtedly shape my future endeavors and contribute to my professional development.

PS-I station: Gates India Pvt. Ltd., , Faridabad

Student

Name: ARUSH BOSE(2021A8PS2849G)

Student Write-up:

PS-I Project Title: Unveiling Machine Marvels by Decoding Instrumentation, Control, and Machine Design of Machinery and Components

Short Summary of work done: My tenure as an intern in Gates India was truly memorable. My mentor, Mr. Rahul Dev, was very helpful and he motivated me to understand the practical aspect of the courses that I have studied. PS-1 was really helpful for me to understand and develop a career plan for my future.

Objectives of the project: The report provides an overview of the two types of machines, their functions, and their importance in creating secure and reliable hose connections. It discusses the components, working principles, and control systems of the machines.

Tool used: Hardware

Details of Papers/patents:NA

Brief description of the working environment: Working Environment- A corporate office on the first floor, and the shop floor on Ground Floor.
Learning- Practical use of the theory that I studied in my courses.

Academic courses relevant to the project: EM

Learning Outcome: Instrumentation & Control

PS-I station: Gates India Pvt. Ltd., , Faridabad

Student

Name: SHREYA PANDEY .(2021A8PS3118H)

Student Write-up:

PS-I Project Title: UNVEILING MACHINE MARVELS BY DECODING INSTRUMENTATION, CONTROL, AND MACHINE DESIGN OF MACHINERY AND COMPONENTS

Short Summary of work done: Studied various types of machines used in the hose industry and their functions and workings. Correlated the learning in courses and their practical application.

Objectives of the project: To Study about the different machines used in the industry and their working

Tool used: Prepared the presentation on ms word and presentation on canva

Details of Papers/patents:1. Kaithari, D. K. ., Krishnan, P. K. ., & Al Burtamani, H. M. S. . (2021). Recent Design and Development of Hydraulic Jack Operated Hose Crimping Machine. Advanced Aspects of Engineering Research Vol. 16, 95–113. https://doi.org/10.9734/bpi/aaer/v16/1739

Brief description of the working environment: The office staff was extremely helpful and supportive. The working environment was extremely nurturing. Our mentor helped us at every step and was there in case of any difficulty. The HR head was approachable and agreeable. The company allowed space to grow and made me realize my future in corporate. I was fortunate to experience a highly conducive and vibrant working environment. The company culture emphasized teamwork and collaboration, fostering a sense of camaraderie among colleagues. Everyone was approachable, and the open-door policy encouraged open communication, enabling me to seek guidance and support whenever needed.

The company had clear expectations from interns, including a strong commitment to professionalism, punctuality, and a proactive approach to tasks. They encouraged us to take ownership of our projects and provided the necessary resources to excel. Meeting deadlines and delivering high-quality work were essential, and I embraced these expectations as opportunities for personal growth and development. The hands-on experience exposed me to real-world scenarios and allowed me to apply theoretical concepts from my academic studies. My mentors and supervisors provided valuable feedback and constructive criticism, which motivated me to continually improve and refine my abilities.

Academic courses relevant to the project: Electrical Machines, Control Systems and Control Systems lab.

Learning Outcome: Learning about various types machines in use in the hose industry, their uses and working.

PS-I station: Goa Shipyard Ltd. , Goa

Student

Name: HARSH RAKESH BHANUSHALI(2021A3PS1154G)

Student Write-up:

PS-I Project Title: Data acquisition using Masibus 5006 RN and SCADA Software

Short Summary of work done: We worked on making a mechanical pressure monitor into a software alternative. we made SCADA implementation / GUI for the same. We made graphs and tables to represent and analyze data.

Objectives of the project: instead of using mechanical gauges convert it into a digital format for monitoring and storing

Tool used: H/w- masibus 5006rn + 4-20mA signal generator + rs485-usb convertor + laptop s/w- QmodbusMaster + Ellipse E3

Details of Papers/patents:we were working with manual of masibus and various transducers

Brief description of the working environment: everyone was very helpful. We learnt how to talk to senior management of a company. When we did not have a project we had to talk to a lot of company managers to get ourselves a good project. I believe this would have helped by gain experience and bettered our soft skills

Academic courses relevant to the project: Digital design , Control Systems

Learning Outcome: various protocols used in transducers / Pinouts of controllers & PLC / communication protocols and SCADA software / graphing the data fetched

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PS-I station: Goa Shipyard Ltd. , Goa

Student

Name: GIRIK ROUTARAY .(2021A4PS0120P)

Student Write-up:

PS-I Project Title: Study on Ship Design and Optimising Pipe Bending Properties

Short Summary of work done: We learntabout the mechanism of ships and the parts involved - engines, shafts and propellers. We read about each of their functioning and purpose. We also learnt about seatubes and sea piping system meant for cooling and water supply. We observed and experimented at pipe bending workshop were pipes are bent and connections are made. With the help of previous research and certain approximations, we also obtained and confirmed an approximate equation to predict springback angle.

Objectives of the project: To study designing of ships and solve the problem of springback in manual pipe bending

Tool used: Excel for data and plots, Semi Automatic Pipe bending machine at GSL

Details of Papers/patents:(Reeds Marine Engineering and Technology 08) Paul A Russell_ Leslie Jackson_ Thomas D Morton - Reeds Marine Engineering and Technology Volume 8_ General Engineering Knowledge For Marine Engineers-Bloo

Brief description of the working environment: It was a nice experience, different from a typical industry since its govt. and lot of the information we gain is classified. The

mentors in GSL taught us thoroughly and the learning process was smooth. We had guidance whenever we needed it.

Academic courses relevant to the project: Mechanics of Solids, Manufacturing Processes

Learning Outcome: We learn the processes involved in pipe fitting in ships and the difficulties faced in semi automatic pipe bending

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PS-I station: Goa Shipyard Ltd. , Goa

Student

Name: MEGH KUMAR BAFNA(2021A8PS2992G)

Student Write-up:

PS-I Project Title: Data Aquisition System using SCADA and Masibus 5006rn

Short Summary of work done: Configuring, Calibrating and Using the Masibus 5006rn. Learning various communication protocols. Making the basic Circuit used. Learning and using Elipse E3 SCADA Software and its various functionalities.

Objectives of the project: Making a Simple Data Aquisition system using Masibus 5006rn Controller and using SCADA software to make a usable GUI for the user and to log data received from the sensor

Tool used: Masibus 5006rn, Elipse E3 Software

Details of Papers/patents:N.A

Brief description of the working environment: -> Visiting various ships (Coast Guard, LPG Carriers, INS Shardul).
-> Learning the basic structure and parts of a ship.
-> Learning of the functioning and work of the Instrumentation Dept of GSL.
-> Configuring, calibrating and using the Masibus 5006rn Module.
-> Learning and using Elipse E3 SCADA Software and its various functionalitites.

Academic courses relevant to the project: Electrical Machines, Digital Design

Learning Outcome: Configuring Masibus 5006rn, using SCADA software

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PS-I station: Goa Shipyard Ltd. , Goa

Student

Name: PUSHKAR .(2021B3A33035H)

Student Write-up:

PS-I Project Title: Data acquisition using Masibus 5006 RN and SCADA Software

Short Summary of work done: We worked on Masibus and SCADA. The Masibus 5006 RN offers accurate and real-time monitoring of critical parameters such as temperature, pressure, level, and flow. It can seamlessly integrate with various ship systems, allowing operators to closely monitor essential processes.

Objectives of the project: Data acquisition using Masibus 5006 RN and SCADA Software

Tool used: Masibus 5006RN and SCADA

Details of Papers/patents: Masibus 5006RN User Manual

Brief description of the working environment: The PS I provided a dynamic and collaborative working environment. It catered to our professional growth and learning. The company valued open communication and encouraged us to go an extra mile to do our project, the staff was helpful and were willing to provide us any information we needed.

Academic courses relevant to the project: EEE courses

Learning Outcome: Introduction to Design/ Technical Services Department. Data Acquisition using SCADA and Masibus 5006 RN, Configuring and using SCADA Software, Working of various Monitoring and Control System on Ship. Designing Electrical Circuit, Configuring MASIBUS, using various communication protocols.

PS-I station: Goa Shipyard Ltd. , Goa

Student

Name: SHAIK MOHAMMED IKHLAS(2021B4A32515G)

Student Write-up:

PS-I Project Title: Data acquisition using Masibus 5006 RN and SCADA Software

Short Summary of work done: used a 4-20mA sensor to simulate pressure data , sent it to the the masibus 5006 rn that was connected to the laptop via RS484 to VGA PORT where the data was read and displayed in a gui which we built using Elipse e3(scada software) to graph the pressure readings after linearizing it .we were simulating how pressure in a ship tank would vary.

Objectives of the project: Acquiring Data on pressure inside a tank using the masibus software

Tool used: masibus 5006rn,rs485 , ellipse e3

Details of Papers/patents:ellipse e3 manual, masibus manual

Brief description of the working environment: GSL being a govt. organization worked like a govt. organization, expecting a good learning environment which they provided and learnt about ships and the machines used in ships

Academic courses relevant to the project: em LAB , digital design

Learning Outcome: scada, circuitry,ellipse e3

PS-I station: Grasim Industries - Staple Fibre Division , Nagda

Student

Name: OM SHIV SHARMA .(2021A1PS2139P)

Student Write-up:

PS-I Project Title: THEORETICAL WATER REQUIREMENT IN EACH PROCESS OF VISCOSE FIBRE MANUFACTURING

Short Summary of work done: It was about calculating water required in each of the processes of viscose fibre manufacturing . We had talks with the industry mentor and learnt a lot through the PS. We gave power point presentations which enhanced our presentation skills. We also had group discuss

Objectives of the project: STUDY THEORETICAL WATER REQUIREMENT IN EACH PROCESS OF VISCOSE FIBRE MANUFACTURING

Tool used: Powerpoint, Google Docs, Canva

Details of Papers/patents:no

Brief description of the working environment: It was overall nice experience and as it was online so we were not able communicate to more workers.

Academic courses relevant to the project: Chemical

Learning Outcome: Calculated theoretical water requirement in process of manufacturing viscose fibre.
Insights about future possible water requirements.
Ideas about water quality management.

PS-I station: Grasim Industries - Staple Fibre Division , Nagda

Student

Name: SHIVIKA DHANDHANIA .(2021A1PS2140P)

Student Write-up:

PS-I Project Title: Theoretical water requirement in viscose fiber manufacturing

Short Summary of work done: Go through different research papers and understand viscose manufacturing process. Procure data from the plant and make suitable calculations

Objectives of the project: Calculate and find out avenues of water requirement in viscose manufacturing

Tool used: None

Details of Papers/patents: None

Brief description of the working environment: Work environment was good. Learnt a lot about chemical processes calculation

Academic courses relevant to the project: CPC, SP-1

Learning Outcome: Learnt about a functioning of a chemical plant

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PS-I station: Grasim Industries - Staple Fibre Division , Nagda

Student

Name: GARV GUPTA .(2021A1PS2371P)

Student Write-up:

PS-I Project Title: Theoretical water requirement in each process of viscose fiber manufacturing.

Short Summary of work done: As a team, we wrote reports on the industry, the processes involved in the manufacturing and the water required in viscose fiber production.

Objectives of the project: To calculate the water required in the production of the viscose fibre from raw materials.

Tool used: Various development tools were used.

Details of Papers/patents:NA

Brief description of the working environment: The PS was online. But I did learn how the industry works and what are the things that are required to smoothly run a process and then access its data to plan further actions.

Academic courses relevant to the project: Grasp on the concepts of Chemistry is very relevant to the project.

Learning Outcome: I learnt how the industry works and what are the things that are required to smoothly run a process and then access its data to plan further actions.

PS-I station: Grasim Industries - Staple Fibre Division , Nagda

Student

Name: ANSHUL ANKIT .(2021A1PS2578P)

Student Write-up:

PS-I Project Title: Calculation of theoretical steam requirement in the production of sodium sulphate

Short Summary of work done: The industry's data was provided which included the capacity of tanks and tubes, diamteres and on their basis had to do the calculation

Objectives of the project: To calculate steam required in the production of glauber's salt

Tool used: MS Excel

Details of Papers/patents: No

Brief description of the working environment: It was online so couldn't learn a lot of things.

Academic courses relevant to the project: CPC

Learning Outcome: Applied the concepts of chemical engg cdc like cpc to calculate feed steam required

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PS-I station: Grasim Industries - Staple Fibre Division , Nagda

Student

Name: PALASH GAGRANI .(2021A1PS2580P)

Student Write-up:

PS-I Project Title: Study theoretical water requirement in each process of viscose manufacturing

Short Summary of work done: We started our work with Arun sir by first preparing the seminar 1 report on grasim industries history and its role in viscose manufacturing. Then we had our 1st group discussion with mentor with Amit Sir on topic of India's Defence exports. Next we prepared our midsem report in which I prepared the topics of industrial discipline and future water requirements. Then we had our GD2 in which we discussed about use of electric vehicles in India. Next we contacted our industry mentor for meeting and he provided us data of water requirement. At last we prepared our end semester report incorporating all the collected information and presented the report to Arun Sir in seminar 2.

Objectives of the project: Objectives include to know about grasim nagda plant and the manufacturing of viscose fiber. We had to calculate the water requirement for viscose manufacturing by using data provided by industry mentor.

Tool used: Microsoft Excel, Powerpoint, Google Docs, Microsoft Word

Details of Papers/patents: No

Brief description of the working environment: Our PS1 was in hybrid mode so most of us opted for online mode but still the working environment from what we learned was through professional and resourceful. Our expectations from the company was to give a suitable environment for completion of our PS and it provided us with all resources necessary. Our faculty and industry mentor were in constant touch with us and provided us with all the necessary resources and help. We learned about viscose fiber

manufacturing and water quality in great detail and even calculated the water required for its making by using the data provided to us.

Academic courses relevant to the project: Mainly chemical courses of 2nd year

Learning Outcome: We learned about the functioning of grasim plant and it's role in viscose fiber manufacturing. We also learned to work in group whose goal was to prepare project reports, present seminars and study about various processes of viscose fibre manufacturing.

PS-I station: Grasim Industries - Staple Fibre Division , Nagda

Student

Name: TRIDIBES GUHA .(2021A1PS2582P)

Student Write-up:

PS-I Project Title: Study Theoretical Steam Requirement in Sodium Sulphate Manufacturing

Short Summary of work done: I did primary research on various structures of Grasim industries and their financial summary for 2021-2022. Then we were divided into multiple groups and given a project to work on. In the scope of research, I covered various processes that go into the manufacturing of viscose rayon and sodium sulphate, including the spin bath regeneration process. In addition to this, I tried to gauge the company's product to get an insight into the functioning of the company. Furthermore, I learnt about the triple effect evaporators and how various heaters and vapour heads are designed and what function they serve.

Objectives of the project: To calculate the amount of steam in the manufacturing of sodium sulphate

Tool used: Excel

Details of Papers/patents:https://ebner-co.de/en/portfolio/regeneration-of-spin-bath-and-production-of
-sodium-sulfate-in-the-viscose-fiber-industry/
http://iwma.in/pdf/Mr.%20N%20Prakash.pdf

https://www.textilecoach.net/amp/manufacturing-process-of-viscose-rayon

Brief description of the working environment: The internship at Grasim Industries Ltd., Nagda, has provided me with
adequate knowledge of the Viscose staple fibre industry and the process of manufacturing and the chemistry involved. Despite being online, it was a useful experience to learn how a chemical engineering plant operates. Apart from the knowledge I gathered, I was able to gain industry experience through my interactions with my PS faculty, Arun Maity, and plant mentor, Pradeep Tripathi.

Academic courses relevant to the project: Seperation Process 1 and Chemical Process Calculations

Learning Outcome: We learnt various manufacturing processes of viscose rayon with the help of a flow diagram indicating all the processes involved. We specifically chose to study the properties of viscose plants as Grasim Industries is the world's first viscose plant to have a zero-liquid discharge plant. Following this, we learned about the spin bath regeneration Process and how it is used to produce sodium sulphate. The spin bath is a solution that becomes exhausted and diluted during the spinning
process, where sodium sulfate and water are formed from soda lye and sulfuric acid
One commonly employed method for sodium sulphate production involves
using anhydrous evaporators, particularly the triple effect evaporator. For that, we had to learn more about the working of evaporators, heaters, and vapour heads.
To conclude, our analysis of the multi-effect evaporator system confirms its functionality and effectiveness, contingent on proper management of factors like flow rate, temperature variance, and heat transfer area. It also highlighted the benefits of energy conservation through the re-use of steam from one stage to heat the subsequent stage

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PS-I station: Grasim Industries - Staple Fibre Division , Nagda

Student

Name: SIDDHESH MANDHANA(2021A1PS2800G)

Student Write-up:

PS-I Project Title: Study theoretical steam requirement in sodium sulphate manufacturing

Short Summary of work done: We basically had to determine the amount of steam used to manufacture sodium sulphate. So woth the help of our mentor and our faculty we obtained a flowsheet of the plant and evaluated its steam consumption from the data provided

Objectives of the project: To study theoretical steam requirement in sodium sulphate manufacturing

Tool used: Excel, word, canva, powerpoint

Details of Papers/patents:NA

Brief description of the working environment: It was very friendly, Arun sir and Pradeep sir were very helpful and I learned a lot about the chemical industry

Academic courses relevant to the project: Fluid mechanics, heat transfer, chemical process calculations, separation processes

Learning Outcome: Process calculations, separation processes, heat transfer operations

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PS-I station: Grasim Industries - Staple Fibre Division , Nagda

Student

Name: DURGA SIVANI KUNAPAREDDY .(2021A1PS3051H)

Student Write-up:

PS-I Project Title: Study theoretical steam requirements for sodium sulphate

Short Summary of work done: Collected information regarding Nagda plant, Manufacturing of Viscous Rayon, Sustainable Waste Management in textile idustries, Triple Effect Anhydrous Evaporator

Objectives of the project: Manufcturing

Tool used: In manufacturing of viscous fiber different type of spinners are used, for winding winder is used

Details of Papers/patents: Had done a report which contain the introduction of Grasim industries including various fields in nagda plant. second we came across various steps involved during manufacturing of viscous fiber and Sustainable Waste Management Practices in the Textile Ind

Brief description of the working environment: Done PS-I through online mode and have conducted meetings through online-mode, Faculty as well as mentor held meetings regularly, Group discussion were conducted through google meet were each member of the group were involved in addition to this presentation of our reports was also done by each group and followed by few questions which were asked by the faculty during our final report.

Academic courses relevant to the project: Separation Process

Learning Outcome: Manufacturing of Viscous Fiber

PS-I station: Grasim Industries - Staple Fibre Division , Nagda

Student

Name: SIDDHANT SHARMA .(2021A2PS2133P)

Student Write-up:

PS-I Project Title: Optimization of Biological Effluent Treatment Plant

Short Summary of work done: Understood the working of an ETP from our Mentor and what are the constraints used for the proper working.

Objectives of the project: To Understand the working of ETP

Tool used: Excel

Details of Papers/patents:N/A

Brief description of the working environment: It was an Online station. I think I improved my communication skills significantly apart from learning the working of ETP

Academic courses relevant to the project: N/A

Learning Outcome: COD and Bod Levels in the Grasim Industries ETP and how much Influent comes in daily.

PS-I station: Grasim Industries - Staple Fibre Division , Nagda

Student

Name: YATIN KUMAR SAHU .(2021A2PS2607P)

Student Write-up:

PS-I Project Title: OPTIMIZATION OF BIOLOGICAL EFFLUENT TREATMENT PLANT

Short Summary of work done: it may be said that since the wastewater is treated before being discharged so as to retain any useful or reusable substances it may contain, and primarily to minimise the negative environmental consequences its discharge may have, it is required to implement cost-effective methods to assess the potential ill-effects and take appropriate measures to manage them. Doing so ensures sustainable development and ensures that no living organism is harmed, considering that the lifeforms are interdependent for sustenance. The choice of method rests on several considerations, such as the nature of the industry, its scale of production, its location, and the associated expenses, etc.

Objectives of the project: Wastewater Chemical Treatme

Tool used: NO

Details of Papers/patents:A REPORT
ON
OPTIMIZATION OF BIOLOGICAL EFFLUENT TREATMENT
PLAN

Brief description of the working environment: We have allotted Online practice school station.. we got to learn everything as per our expectations everything was perfect we enjoyed a lot..with Arun sir

Academic courses relevant to the project: Yes

Learning Outcome: strategies for increasing the efficacy of wastewater treatment

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PS-I station: Grasim Industries - Staple Fibre Division , Nagda

Student

Name: KRISHNANGSHU BHATTACHARJEE(2021A3PS2823G)

Student Write-up:

PS-I Project Title: Power Plant Automation using PLC Technology

Short Summary of work done: Our PS station operated in hybrid mode - i.e. both online and onsite. I had chosen the online mode. On the first day an online orientation session was conducted in which our Faculty Mentor introduced himself and we gave our introduction. He gave an overview of the tasks we were supposed to do in the next one and a half months. A few days later, groups were made as per the branch of the students and Projects were allotted accordingly. My group had six members - including me. An industry mentor was assigned to each group. Project specific Whatsapp groups were made and we started our work for the project. We were supposed to carry out an extensive research on our project topic by gaining all necessary information from our Industry Mentor. We had virtual meets with our Industry Mentor who helped us a lot in our project. The meets were kept short yet effective. All the evaluation components like Quiz, Group Discussion (GD), Seminar and Report Submission were conducted as per the timeline given by PS Division. Apart from this we were supposed to maintain a regular Diary wherein we had to write about our daily work during the PS tenure. Whenever we had any doubt, we could approach our Faculty Mentor and he was very quick to respond to our queries.

Objectives of the project: The objective of my project was to study the Power Plant Automation using PLC Technology at Grasim Industries Limited, Nagda

Tool used: MS Word, MS Powerpoint, and various resources from the internet

Details of Papers/patents:-

Brief description of the working environment: The working environment was very good. I had wonderful team members who were really co-operative and helpful. Eventually, all of us - from different campuses - bacame good friends. We were given work which was relevant to our project topic. We were not asked to do any irrelevant or unnecessary work and there was proper work-life balance. My expectations from the company were mostly fulfilled. Only if I had opted for the onsite mode, I feel, I could have got the opportunity to visit the power plant. Other than that it was a wonderful experience to work in a huge organization like Grasim Industries.

Academic courses relevant to the project: Control Systems, Microprocessors and Interfacing, Principles of Management

Learning Outcome: I learnt about the processes involved in the working of the huge Power Plants at Grasim Industries, Nagda. I got to know about the use of Programmable Logic Controller (PLC) for the automation of these power plants. I, also, learnt about other important plant technologies like Distributed Control System (DCS), Wagon Tipplers, and Internet of Things (IoT).
My presentation skills have enhanced and I got to learn about time management, work-life balance and team work.

PS-I station: Grasim Industries - Staple Fibre Division , Nagda

Student

Name: MANISH INDORIA .(2021A4PS0137P)

Student Write-up:

PS-I Project Title: Advances in Design of Modern Thermal Power Palnt

Short Summary of work done: I worked on Thermal Cycle efficiency and suggested some advanced combined vapor cycle to increase overall efficiency.

Objectives of the project: To increase efficiency of Thermal Cycles and find better fuel alternative

Tool used: Google scholar

Details of Papers/patents: Rice Husk as fuel alternative

Brief description of the working environment: Overall working environment is good and i enjoyed overall work.

Academic courses relevant to the project: Thermodynamics

Learning Outcome: Thermal Cycles

PS-I station: Grasim Industries - Staple Fibre Division , Nagda

Student

Name: KANIKA JAIN .(2021A4PS2898H)

Student Write-up:

PS-I Project Title: Advancement in thermal power plant

Short Summary of work done: Learned thermodynamic cycles involved in production of viscous fiber plant. Learned about how industry work.

Objectives of the project: To see what advancements in thermal power plant like IOT, CCS, ZLD etc..

Tool used: google

Details of Papers/patents:-

Brief description of the working environment: -

Academic courses relevant to the project: thermodynamics

Learning Outcome: 1. Alternate products that can be used in place of coal.
2. Learned about css, zld, iot etc.

PS-I station: Grasim Industries - Staple Fibre Division , Nagda

Student

Name: RAHUL YADAV .(2021A7PS0535P)

Student Write-up:

PS-I Project Title: market study of viscose fiber and scope of digital marketing

Short Summary of work done: i have learnt many things like how an industry can work and what are the major requirements of an industry and how can we improve it and about its market and its scopes in digital marketing

Objectives of the project: study of market and increase the scope and growth of industry in digital marketing

Tool used: tools used like online platforms

Details of Papers/patents:research on market study an scope in digital marketing for viscose fiber

Brief description of the working environment: all things are good

Academic courses relevant to the project: POM , DAS

Learning Outcome: learn about market study and scopes in digital marketing for an industry

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PS-I station: Grasim Industries - Staple Fibre Division , Nagda

Student

Name: SONAKSHI SHARAN .(2021A8PS3002H)

Student Write-up:

PS-I Project Title: Power Automation using PLC technology

Short Summary of work done: I was the project lead .My project was POWER AUTOMATION USING PLC TECHNOLOGY .We made ppt of topics like PLCs, DCS, IOT and wagon tippler

Objectives of the project: Uses of PLCs(Programmable Logic controllers) , DCS(Distributed control system), IOT(Internet of things) used in Power Plant

Tool used: We made ppt on each topic

Details of Papers/patents:no

Brief description of the working environment: Our mentor is very cooperative and answered our doubts calmly. I had online PS at Grasim Industries.

Academic courses relevant to the project: Control Systems , Internet of Things

Learning Outcome: How PLCs are used in power plant ,work of dcs in plant and use of Iot technology in automation

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PS-I station: Grasim Industries - Staple Fibre Division , Nagda

Student

Name: MS. ANISHA(2021B1TS2072P)

Student Write-up:

PS-I Project Title: Optimization of biology effluent treatment plant

Short Summary of work done: Optimization of Biological Effluent Treatment Plant is the plant were the whole industry waste come and collect in different different ponds and after they try to get water clean for re use

Objectives of the project: How we recycle the waste water for reuse

Tool used: Different different tools are used like strainer,grid chamber, biological reactor, primary clarified

Details of Papers/patents: Na

Brief description of the working environment: Na

Academic courses relevant to the project: Na

Learning Outcome: How the water clean for reuse

PS-I station: Grasim Industries - Staple Fibre Division , Nagda

Student

Name: JATIN RAWAT(2021B1TS2080P)

Student Write-up:

PS-I Project Title: Optimization of Biological effluent treatment plant

Short Summary of work done: So on 30 may i have reached nagda so initially they provide us the basic pp kit which includes helmet shoes and eyeglasses so on 5 they tell us the safety measurements and how we can be safe in the industry the environmental of that industry is Higley corrosive although agter safety measurements we go to plant visit like etp wtp finance staple fiber we visted these plant separately like we have hybrid ps station so my main work is that i have to collect the information like cod bod level so that our group can make report i also learn how etp work

Objectives of the project: How etp works and how we can optimize the waste water

Tool used: To be honest sir is not provide the information about tool mainly there are some chemical level which they have to manage

Details of Papers/patents: No

Brief description of the working environment: So if we talk about work experience they not really responsiveness although if we go there they treat us nicely so my expectations of these ps was like i will get some industry knowledge and i was quite good experience for me.

Academic courses relevant to the project: Not really

Learning Outcome: How nagda station purify water for domestic use as well as the industrial use and wastages of water is nil there.

PS-I station: Grasim Industries - Staple Fibre Division , Nagda

Student

Name: TASHI(2021B2A12270P)

Student Write-up:

PS-I Project Title: Water requirement in viscose fiber processes

Short Summary of work done: It was basically about calculating theoretical water required in each process of viscose fiber manufacturing. We had talks with the industry mentor and learnt a lot through the PS. We gave power.points presentation which enhanced our.presentation skills. We also had group discussion which helped in communication skills and working with our group helped us in team work. It was good.

Objectives of the project: To calculate water requirement

Tool used: Powerpoint, Google docs, canva

Details of Papers/patents: Not as such

Brief description of the working environment: There could have been more communication with the industry workers to say about that since my station was online.

Academic courses relevant to the project: Chemical

Learning Outcome: How water requirement calculation are practically done in chemical industries

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PS-I station: Grasim Industries - Staple Fibre Division , Nagda

Student

Name: HARSHITA YADAV(2021B2TS2053P)

Student Write-up:

PS-I Project Title: Theoretical calculation of water requirement in each process of viscose fibre manufacturing

Short Summary of work done: The PS-1 at Grasim started with a brief overview of the industry as well a sproject allocation. We leanrt about the various subsidiaries of Grasim and its main products. Later on we had multiple meetings (online format) witg our industry mentor who shared crucial happenings and details from the plant. We recieved real time data for the theoretical calculation aspect of our project. Throughout the project we had a few group discussions and seminars all of which helped in building soft skills. Our project also focused on improving the water quality, for which we studied about the various parameteres and their impacts. The entire project required for us to go through mulitple resources available on the internet and as shared by our PS faculty. Thr project was a blend of understthe theory behind every manufacturing process and the practical aspect of the water requirement in each of those.

Objectives of the project: The project focused on calculating the water requirement for every process involved in viscose manufacturing. Grasim is one of the leading manufacturers of viscose staple fibre. The objective was to understand various manufacturing processes involved and

Tool used: Google Suite

Details of Papers/patents: No.

Brief description of the working environment: The work environment was very comfortable. Both our PS faculty and industry mentor were available throughout the course of the project which helped us immensely.

Academic courses relevant to the project: Separation Processes (Chemical Course)

Learning Outcome: We learned how vuscose is manufactured and how the Nagda plant reuses water for various non critical processes. We also learnt about the significance of using soft water for the manufacturing processes. We gathered a deep understanding of the water quality aspect as well as understanding the industry parameters. Additionally, as it was a group project we learnt the foundations of team work which has proven to be extremely helpful.

PS-I station: Grasim Industries - Staple Fibre Division , Nagda

Student

Name: VINEETA BUGALIA(2021B3TS2058P)

Student Write-up:

PS-I Project Title: Financial performance analysis of Grasim Industries ltd

Short Summary of work done: Visit to different plants of Grasim like Viscose,auxiliary, spinning,ETP,power plant etc. Get to know about the working of Accounts and Finance department as it is divided into different departments like Store,billing,payment,GST and taxation,balance sheet formation,Management information system etc.Get to know about the working behind the SAP app used here. Worken on the project of analyzing the financial performance of it using ratio and graph analysis drom the data of annual financial report

Objectives of the project: Financial Analysis of Grasim using Ratio Analysis

Tool used: Excel,PowerPoint, word, PPE kit(helmet,shoes,jacket)

Details of Papers/patents:NA

Brief description of the working environment: Working environment is highly corrosive in terms of plant activity.Safety is their main priority which is taken very seriously and all the rules are followed properly.
Visit to different plants of Grasim like Viscose,auxiliary, spinning,ETP,power plant etc. Get to know about the working of Accounts and Finance department as it is divided into different departments like Store,billing,payment,GST and taxation,balance sheet formation,Management information system etc.Get to know about the working behind the

SAP app used here. Working on the projectof analyzing the financial performance of it using ratio and graph analysis from the data of annual financial report.

Academic courses relevant to the project: Fundamentals og accounts and finance,Mathematical and statistical method

Learning Outcome: Official interaction with Industry employees, Working of industry,exposure to industrial environment, working of Finance and Accounts department.

PS-I station: Grasim Industries - Staple Fibre Division , Nagda

Student

Name: MANISHA SAINI(2021B3TS2060P)

Student Write-up:

PS-I Project Title: Financial performance analysis of grasim industries limited

Short Summary of work done: During the PS-1 we visited different plants in grasim industry like ETP, energy center, WTP. Also visited different sections in finance and accounts department including billing, payment(domestic and foreign payments), stores. Also learned the calculations of financial ratios of industry and comparing them with previous years. We got to know many financial terms related to balance sheet.

Objectives of the project: Financial analysis of grasim industries

Tool used: Excel and ppe kit

Details of Papers/patents: Na

Brief description of the working environment: Working environment of the industry was good .we were provided with the conference room to work in. We also got to know the working of finance and accounts department as it was related to our project. Additionally we visited the different plants in the industry.

Academic courses relevant to the project: Fundamentals of finance and accounts, mathematical statistical methods

Learning Outcome: Financial analysis, comparing ratios and working of finance department

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PS-I station: Grasim Industries - Staple Fibre Division , Nagda

Student

Name: MS. SANJANA SAINI(2021B4TS2063P)

Student Write-up:

PS-I Project Title: Financial analysis of grasim industries

Short Summary of work done: As I done my ps-1 onsite,I used to visited the plant daily me and my team survey the whole plant and tried to get the knowledge as much as we can we visited the particular department of finance separately and collected the data and also try to understand their work.

Objectives of the project: To know about financial statements and financial status of Grasim VSF

Tool used: In finance field

Details of Papers/patents: By report

Brief description of the working environment: Environment is good and the working staff and employees in the company are also very helping in nature they help us during our whole ps -1 and we spend a great time there.

Academic courses relevant to the project: Finance

Learning Outcome: We get to know about major ratios financial statements of the company

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PS-I station: Grasim Industries - Staple Fibre Division , Nagda

Student

Name: AISHWARYA SINGH(2021B4TS2089P)

Student Write-up:

PS-I Project Title: Financial performance analysis of grasim industries ltd.

Short Summary of work done: Ratio analysis and how to calculate different financial ratios, visited different plants such as ETP, WTP, VISCOSE, SPINNING AND AFTER TREATMENT, CHEMICAL AND TEXTILE LAB, etc learned about chemical and textile lab as a learning of special project.

Objectives of the project: Financial analysis of Grasim industries through Ratio Analysis

Tool used: Microsoft excel, Microsoft PowerPoint, Microsoft word etc

Details of Papers/patents:N/A

Brief description of the working environment: Working environment is good, everyone at the company is friendly and very cordial, Willing help and mentor provides data as per request.
People at the factory take extra precautions and safety is their top priority.
As due to manpower shortage people are very busy and does not have much time to provide for the interns so more time from the people at the company would be good, and providing conveyance to the factory. Ratio analysis of Financial ratios, usage of software SAP in different account departments of Grasim.

Academic courses relevant to the project: Fundamentals of finance and account, Optimization

Learning Outcome: Ratio analysis of Financial ratios, usage of software SAP in different account departments of Grasim.

PS-I station: Grasim Industries - Staple Fibre Division , Nagda

Student

Name: AAYUSHMAN SINGH .(2021B5A11315P)

Student Write-up:

PS-I Project Title: Optimization Biological effluent treatment plant

Short Summary of work done: Analysed various processes involved in Biological effluent treatment plant .Also analysed various variius reserch paper to get a deeper understanding of the field and industial processes.We all did some research on how AI and ML can contribute in these feilds.We were in constant touch with Arun Sir and our project mentor Anil Vijay sir who provided us with various dat related to Etp of grasim industry Nagda.

Objectives of the project: To get an understanding of ETP of grasim industry and industrues in general.

Tool used: Excel and apart from that just reading and googling about a lot of thing.

Details of Papers/patents:NA

Brief description of the working environment: The environment of of the station was very comforting .We were given a lot of freedom in terms what we want to explore .Also the the deadline for different task were always decided after discussing with the group leaders . If we got things wrong we were always given a chance to do it again .Also the feedback given was very encouraging. All the mentor provided us proper guidance and were expert in their fields. You will be having great time learning if you intern at Grasim Industry ,Nagda.

Academic courses relevant to the project: As I have not yet started my B.E(chemical) so I am not I a position to to comment on that.

Learning Outcome: Got to learn about new things related to ETP .

PS-I station: Grasim Industries - Staple Fibre Division , Nagda

Student

Name: SUNAYANA S MORO(2021B5A82185G)

Student Write-up:

PS-I Project Title: PLC integration in powerplant

Short Summary of work done: During PS-I, I was involved in the design and implementation of a PLC-based automation system for the wagon tippler at the power plant. I collaborated with the automation team to research and test the control logic for precise gripper engagement, rotation control, and material discharge coordination. Additionally, I researched integrated safety interlocks and emergency stop systems to enhance operational safety. I also contributed to studying the development of the fault detection and diagnostics module, which enabled real-time monitoring of critical parameters and provided alerts for maintenance personnel. The project focused on optimizing the unloading process, enhancing safety measures, and improving data management for data-driven decision-making.

Objectives of the project: The primary objectives of the project were: a) To implement a PLC-based automation system for optimizing the wagon tippler operations at the power plant, aiming to increase efficiency and reduce unloading times. b) To enhance safety measures through the

Tool used: a) Programmable Logic Controllers (PLCs) for automation and control of the wagon tippler operations. b) Supervisory Control and Data Acquisition (SCADA) systems for remote monitoring and control. c) Data analysis software for analyzing performance paramet

Details of Papers/patents: The working environment during PS-I was highly dynamic and challenging. I was part of a collaborative team of experienced engineers and professionals. The company provided a supportive and conducive learning environment, encouraging active participation a

Brief description of the working environment: The working environment during PS-I was highly dynamic and challenging. I was part of a collaborative team of experienced engineers and professionals. The company provided a supportive and conducive learning environment, encouraging active participation and innovation.
The expectations from the company were to contribute actively to the project, demonstrate a keen interest in learning, and adhere to safety and quality standards. I was also expected to communicate effectively, seek guidance when needed, and show adaptability in dealing with technical challenges.
During PS-I, I learned the significance of practical applications of engineering concepts, gained proficiency in PLC programming and automation, and developed a deeper

understanding of project management and data-driven decision-making. The exposure to real-world industrial projects enabled me to grasp the importance of teamwork, adaptability, and continuous learning in the engineering domain. I look forward to building on these experiences and contributing meaningfully to future projects in the field of automation and control systems.

Academic courses relevant to the project: Electronics courses

Learning Outcome: During PS-I, I gained valuable insights and knowledge in the following areas:
a) Practical Application of Theoretical Concepts: I had the opportunity to apply the theoretical knowledge gained during academic studies to real-world projects and challenges. This experience enhanced my understanding of engineering principles and their practical implications.
b) Project Management: I learned the importance of effective project management, including planning, scheduling, and resource allocation. It helped me develop skills in coordinating tasks and meeting deadlines.
c) Hands-on Experience with Tools and Technologies: Working on the project allowed me to gain hands-on experience with various tools and technologies commonly used in the industry, such as PLC programming, SCADA systems, and data analysis software.
d) Team Collaboration: Collaborating with multidisciplinary teams exposed me to diverse perspectives and improved my teamwork and communication skills. I learned to work cohesively with colleagues from different backgrounds to achieve common project goals.

PS-I station: Grasim Industries - Staple Fibre Division , Nagda

Student

Name: MANUSHREE SAINI(2021B5TS2065P)

Student Write-up:

PS-I Project Title: ADVANCES IN DESIGN OF MODERN THERMAL POWER PLANT

Short Summary of work done: I analysed the potential applications of IoT technology in thermal power facilities and how we can benefit from this innovative approach. I have examined the challenges and future directions of modern thermal power plant design. Through my investigation, I have identified several key areas that require attention.

Objectives of the project: The purpose of this paper is to investigate potential strategies for upgrading the Grasim Power Plant's Thermal Power Plant in order to raise productivity while decreasing overhead expenses. We analyze the plant in detail to find out what may be changed t

Tool used: Power point, Microsoft word

Details of Papers/patents:none

Brief description of the working environment: This transformative and enlightening learning experience was truly unparalleled. I am absolutely thrilled and filled with immense gratitude to have been granted an incredible chance to start on a transformative journey as an esteemed intern within the prestigious realm of a Core Chemical Engineering Company. The Simulation Work seamlessly integrated the fundamental principles with cutting-edge technology, resulting in a truly innovative effort. Challenging at first, yet unquestionably rewarding in the end. I got an overview of how industries work in real life.

Academic courses relevant to the project: none

Learning Outcome: I learned the significance of certain processes and the necessity of various divisions in a manufacturing facility. I discovered how IOT can be utilized in thermal power plants. In addition to developing technical skills, I also improved my communication and public speaking abilities. Working in a group also allowed me to develop skills in collaboration and cooperation.

PS-I station: Grasim Industries - Staple Fibre Division , Nagda

Student

Name: MS. JAYSHREE(2021D2TS2075P)

Student Write-up:

PS-I Project Title: Market study of Viscous fiber and scope of digital marketing

Short Summary of work done: Making 2 reports mid sem and comprey and 1gd , 2quiz is happened.

Objectives of the project: To understand more about garism industry

Tool used: Nothing

Details of Papers/patents: Nothing

Brief description of the working environment: To gain practical knowledge and Industrial exposure to enhancing my skills whatever I learned during degree course. This PS training will also helps me to build professional career in corporate world.

Academic courses relevant to the project: Don't know

Learning Outcome: To understand more about garism industry.

PS-I station: Grasim Industries - Staple Fibre Division , Nagda

Student

Name: DEPIKA(2021D2TS2087P)

Student Write-up:

PS-I Project Title: Market study of viscous fibre and scope for digital marketing

Short Summary of work done: In the orientation report, I conducted a study on Grasim Industries, providing essential details about the company and its operations. The report covered key aspects, such as their products, market presence, and financial performance. Additionally, I conducted a market study on viscous fiber, focusing on understanding its demand and applications. The study included an analysis of consumer behavior, market trends, and the potential scope for digital marketing strategies. Overall, the orientation report highlighted vital information about Grasim Industries, while the market study explored the opportunities and challenges in the viscous fiber industry, providing valuable insights for business decision-making.

Objectives of the project: The objective of this market study on viscous fiber and its scope for digital marketing is to gain a deep understanding of the viscous fiber industry, consumer behavior, and market trends. The study aims to explore the potential opportunities and challeng

Tool used: Secondary Research, Web Analytics, Competitive Analysis Tools, Content Management Systems, Data Analytics, Market Research, Data Visualization Tools

Details of Papers/patents: Na

Brief description of the working environment: I'm off-campus for the ps1. So I don't have any knowledge about the company's working environment.

Academic courses relevant to the project: Na

Learning Outcome: The study provides a comprehensive understanding of the viscous fiber industry, consumer insights, and digital marketing strategies. It assesses the potential for digital marketing, identifies growth opportunities, and offers actionable recommendations for enhanced competitiveness and adaptability.

PS-I station: Grasim Industries - Staple Fibre Division , Nagda

Student

Name: RAHUL(2021D2TS2088P)

Student Write-up:

PS-I Project Title: Market Study Of Viscose Fiber/Scope For Digital Marketing

Short Summary of work done: I've collected data of market of vsf by mentor at and

Objectives of the project: The VSF Industry, Vsf scope , Scope for digital Marketing

Tool used: Plant is having 2 Mills and having 11 machines to work efficiently and has a capacity of 230 ton per day.

Details of Papers/patents: Nan

Brief description of the working environment: Working environment of the station is good as well as loop holes majorly the environment was corrosive and it is found that the

overall demand of vsf of grasim Industries is loosing it's trend comparatively to the previous years.

Academic courses relevant to the project: Market Study of Vsf
Sap of Vsf of over-all units of grasim
Demand And Supply Chain

Learning Outcome: The Demand And Supply Of Vsf By Grasim, Vsf Major Industries Marketting Startegies, Process Of Making Fiber From Viscose . Indian Vsf Industries and their market .

PS-I station: Gujarat Environment Management Institute , Gandhinagar

Student

Name: DEVENDRA NIMBALKAR .(2021A1PS2364P)

Student Write-up:

PS-I Project Title: APPLICATION OF STATISTICAL ANALYSIS FOR THE IDENTIFICATION OF POLLUTED STRETCHES OF THE SABARMATI RIVER

Short Summary of work done: First study of Downstream Locations of Sabarmati river , Then study of water quality and parameters involved in it ,then study of water quality index , calculation of WQI , studying trends and variousions in WQI value and its parameters , learned Different methods of statistical analysis, such as Pearson correlation coefficient ,Multiple Linear Regression (MLR), Standardized coefficients, Multicollinearity statistics. made an MLR model for the designated locations for the given time period of data sampling. Learned the variations in the water paramerters and WQI values, identified polluted locations based on WQI value

Objectives of the project: ● To study the water quality data of one year for 9 locations (middle stream and downstream) of Sabarmati. ● To identify the polluted locations of river Sabarmati based on WQI. ● To study the methods of statistical data analysis ● Establish multiple li

Tool used: excel,word Doc,XLstate,researchpapaer

Details of Papers/patents:IS10500(2012):Drinking water
http://cgwb.gov.in/documents/wq-standards.pdf
Designated Best Use Water Quality Criteria
https://cpcb.nic.in/wqm/Designated_Best_Use_Water_Quality_Criteria.pdf
Comprehensive Report of Sabarmati River.
https://gemi.gujarat.gov.

Brief description of the working environment: The work environment was very good, helpful, motivating, and comfortable. Everyone helped me a lot; my queries were answered very quickly, and with effort, they guided me with the project at every step.
All my expectations were fulfilled at the centre. The first thing was the study of the Sabarmati River with reports and articles , then the study of the downstream locations of the river,then the study of water quality ,parameters and WQI index . learned Different methods of statistical analysis, such as Pearson correlation coefficient ,Multiple Linear Regression (MLR),
Standardized coefficients, Multicollinearity statistics.
made an MLR model for the designated locations for the given time period of data sampling.
Learned the variations in the water paramerters and WQI values,
identified polluted locations based on WQI value .

Academic courses relevant to the project: learning about Water Quality parameters , Statistical methods, and there calculations

Learning Outcome: 1)water quality
2)water quality index
3)Significance of the parameters use in WQI
4) Statistical Analysis
5) Pearson correlation coefficient
6) Multiple Linear Regression (MLR)
7) Standardized coefficients
8) Multicollinearity statistics
9) downstream study of Sabarmati River
10) trends in the parameters and WQI value of downstream locations
11)Significance of the locations
12)identification of polluted river locations

PS-I station: Gujarat Environment Management Institute , Gandhinagar

Student

Name: SRUSHTI THAKKER .(2021B1A10785P)

Student Write-up:

PS-I Project Title: Plastic degradation using Pseudomonas sp.

Short Summary of work done: I checked the Plastic degrading activity of Pseudomonas by using Gas chromatography and FTIR analysis. Cultured and did tests on the bacteria to confirm its species. Checked the plastic degrading activities through weight loss, GC and ftir.

Objectives of the project: 1) Culturing of Pseudomonas. 2) Checking it's plastic degrading activity through GC-MS-MS and FTIR. 3) Comparing the pre weight and post weight of the plastic granules before and after introducing them to the bacteria.

Tool used: It was a Wet Lab project. No software was used as such.

Details of Papers/patents:nil

Brief description of the working environment: It was a great experience working here. I got to do the project of my interest and domain. The staff and my mentor were very supportive in every aspect. I got a lot of hands on experience, and now I can handle most of the laboratory apparatus on my own. I also got to work on sophisticated instruments like GC and FTIR.

Academic courses relevant to the project: Biological Sciences

Learning Outcome: Performing culturing of the bacteria and many biochemical tests on it, Working on Gas Chromatography and FTIR machines, and understanding their mechanism, got used to reading a lot of research papers and learnt many more valuable lessons.

PS-I station: Hindustan Colas , Mumbai

Student

Name: KHATRI HARSHVARDHAN KESHAV .(2021B2A22160P)

Student Write-up:

PS-I Project Title: Emulsion treated bases

Short Summary of work done: We had to make moulds everyday and do tests on them .

Objectives of the project: Reducing testing time of Emulsion treated bases from 4 days to 1 day

Tool used: Various testing apparatus

Details of Papers/patents: None

Brief description of the working environment: It was very good. The staff and employees were easy to talk to and interesting. They guided us and shared their experience

Academic courses relevant to the project: Civil courses, mostly highway engineering

Learning Outcome: About emulsions, aggregate, highway construction

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PS-I station: Hindustan Colas , Mumbai

Student

Name: KUSHAGRA SANDEEP SINGHVI .(2021B4A21718P)

Student Write-up:

PS-I Project Title: REDUCTION OF SETTING TIME OF EMULSION TREATED BASE FROM 4 DAYS TO 1 DAY

Short Summary of work done: t is concluded that the setting time of the emulsion stabilised base(ETB) can be reduced by conditioning in an oven at 115C for 3 hours and then in vacuum at 60C for 3 hours. The strength of the moulds depends on factors like

pre-conditioning time and temperature, conditioning time and temperature, cooling time and uniformity of composition of mould.We are working on the base layer of the pavement. First we started with the testing of the aggregates. Aggregate tests like sieve analysis, flakiness and elongation index, impact value test,los angeles abrasion test, sand equivalent test, specific gravity test were performed

Objectives of the project: The aim of our project is to reduce the settling time of emulsion treated base (ETB) from 4 days to one day

Tool used: NA

Details of Papers/patents:NA

Brief description of the working environment: The company environment was pretty good, helpful ,supportive and at the same time very approachable so we were given attention and also assisted in need
the company expectations were to fulfill the project

Academic courses relevant to the project: Not yet started with my dual degree

Learning Outcome: Learned to perform various tests on aggregates,Went through a series of tests and well defined procedures.
learnt a lot regarding how road is made
so overall was a great learning experience

PS-I station: HPCL , Vishkhapatnam

Student

Name: PRIYANSHU JHA .(2021A4PS2199P)

Student Write-up:

PS-I Project Title: Study of Rotary Maintenance

Short Summary of work done: Throughout the duration of PS, we had learnt about all the different kinds of equipments used in a refinery. Being an intern in the maintenance department, our goals was to cover both rotary maintenance for the first 4 weeks and then static maintenance for the 2 remaining weeks. During our time in rotary division, we

had learnt about compressors, pumps, fans, blowers, gas turbines and condition monitoring of these equipment. Coming to equipments, we had understood their internal working, their types and the tools required for their maintenance. We had also learnt about different types of bearings and valves used in an equipment. During our time in static maintenance, we had learnt about heat exchangers, boilers, heaters and surge drums. For our final project, we were supposed to make a report on all the observations we had made during our time in rotary division.

Objectives of the project: Understanding rotary equipments used in a petrochemical refinery.

Tool used: Helmet and safety shoes.

Details of Papers/patents: None.

Brief description of the working environment: The working environment was really employee-friendly although the work life balance was a little unsatisfactory. The working staff really made time for us as well as provided us with manuals when they were busy. We were supposed to report at 9 am and leave by 4:30 pm.
We weren't allowed to use to phone storage facility as phones inside the refinery wasn't allowed, due to which we couldn't bring our phones. We were taken to the field during our first half where we were supposed to make our observations while the equipment was running, and in our second half we were either taken to workshop to learn about the internal working of the equipments where they were shut down and getting repaired upon, or given manuals to read. It was a very fun experience as we were able to corelate the theory taught in our class and the practical knowledge provided to us at the refinery.

Academic courses relevant to the project: Heat transfer, Mechanisms and machines, Fluid mechanics, Mechanics of solids, Manufacturing processes.

Learning Outcome: Gained experience on the internal working of an industry as well as learnt about rotary equipments used for mechanical maintenance.

PS-I station: HPCL , Vishkhapatnam

Student

Name: ABHIROOP VENKATA VADDIPARTHI .(2021A4PS2378H)

Student Write-up:

PS-I Project Title: Study Of Rotary Equipment

Short Summary of work done: I have gone around observing various rotary equipment within a petroleum refinery and was taught in detail on how to plan each of them in case making our own equipment and various safety allowances that were taken to prevent any failure in any of the equipment. I also went on to study about various various predictive maintence techniques used in the refinery and how each of them are analyzed by an anlyst to maintain the integrity of the refinery.

Objectives of the project: Study the various kinds of rotary equipment found in a petrochemical refining industry .

Tool used: ---None---

Details of Papers/patents:---None---

Brief description of the working environment: Every morning starts with reporting to new engineers, after which they brief them about specific equipment that we are going to ponder studying and discussing.
This was followed by visiting the field and visually learning about the equipment and getting much more knowledge about it's past history, it's failures, and why it actually failed. The field visit usually extends until the afternoon, when a lunch break is provided.
After this followed a deep conversation with each of the engineers, which was mostly knowledge transfer about the equipment that we observed and any kind of doubt clarification if required.

Academic courses relevant to the project: Primemovers and Fluid Machinery (PMFM), Vibrational Analysis.

Learning Outcome: 1. I learned about how various rotary equipment like pumps, compressors, fans, blowers, and various others work.
2. I've also learned on how to troubleshoot them in case of any emergency within the equipment.
3. I've also been shown in detail the various processes that occur in a petroleum refinery.

PS-I station: HPCL , Vishkhapatnam

Student

Name: ABHISYANTH REDDY DONTHI REDDY .(2021A4PS2772H)

Student Write-up:

PS-I Project Title: Study Of Heat Exchangers

Short Summary of work done: Everyday we Report to an Assigned Engineer who would then take us to the unit he is assigned to where we would enquire about the various equipment present there along with their Design,Function and Maintainance Measures taken in order to keep the unit running at perfect condition and After that Visit the Maintenance Workshop present where we would interact with the various technicians present and learn about the finer details of the equipment Present there

Objectives of the project: To Study About the design and maintenance of heat exchangers

Tool used: --

Details of Papers/patents:-

Brief description of the working environment: The Working Environment is typical of that of a Government office, where there is a clear Hierarchy present. The Engineers issued Permits for Various Tasks to the technicians, who then carried out the work required. Above the Engineers were Managers of Different Grades who ensured that their Respective departments were functioning smoothly.
The Company Expected us to learn about the types of Exchangers in Each unit and how those different units come together to keep the production of Various products running smoothly.
I have greatly benefited from this Experience as I learned how to visualize the various concepts that I have learned in different courses. The Infrastructure is extremely impressive and makes us wonder about the various marvels of Engineering. For Someone Intrested in Pursuing a field in the Oil and Gas Industry or Just Manufacturing Industry in General this would be a Priceless Experience

Academic courses relevant to the project: Heat Transfer, Manufacturing Process, and Applied Thermodynamics

Learning Outcome: Some key Learning Outcomes that can be Expected are:
1. Understanding of Heat Transfer Principles
2. Knowledge of Heat Exchanger Types and Applications:
3.Real-world Applications
4. Heat Exchanger Maintenance and Fouling

5. Heat Exchanger Design Considerations
6.Safety Considerations
7. Visualization of Concepts Taught in Courses like Heat Transfer

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PS-I station: HPCL , Vishkhapatnam

Student

Name: SUJAL KUMAR KAR .(2021A4PS2991H)

Student Write-up:

PS-I Project Title: Study on Heat Exchangers

Short Summary of work done: Everyday we report to an assigned engineer who would then take us to the unit he is assigned to, where we would enquire us about the various equipment present there along with their design, function and maintenance measures taken in order to keep the unit running at perfect condition. After that we visit the maintenance workshop present, where we would interact with various technicians present there and learn about the finer details of the equipment present in the refinery.

Objectives of the project: To study About the design and maintenance of Heat Exchangers

Tool used: N/A

Details of Papers/patents:N/A

Brief description of the working environment: The Working Environment is typical of that of a Government office, where there is a clear Hierarchy present. The Engineers issued Permits for Various Tasks to the technicians, who then carried out the work required. Above the Engineers were Managers of Different Grades who ensured that their respective departments were functioning smoothly.
The Company Expected us to learn about the types of Exchangers in Each unit and how those different units come together to keep the production of Various products running smoothly.
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impressive and makes us wonder about the various marvels of Engineering. For Someone Intrested in Pursuing a field in the Oil and Gas Industry or Just Manufacturing Industry in General this would be a Priceless Experience.

Academic courses relevant to the project: Heat Transfer, Manufacturing Process, and Applied Thermodynamics

Learning Outcome: Some key Learning Outcomes that can be Expected are:
1. Understanding of Heat Transfer Principles
2. Knowledge of Heat Exchanger Types and Applications:
3.Real-world Applications
4. Heat Exchanger Maintenance and Fouling
5. Heat Exchanger Design Considerations
6.Safety Considerations
7. Visualization of Concepts Taught in Courses like Heat Transfer

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PS-I station: I care Foundation A.C , Mexico

Student

Name: YASH TIWARI .(2021A8PS2705P)

Student Write-up:

PS-I Project Title: Fund raising

Short Summary of work done: Approached organisation and individuals for raising the fund. Edited videos and made posters for the campaign.

Objectives of the project: To raise funds for the raices community.

Tool used: S/w- adobe photoshop, Adobe premiere Pro, danadora

Details of Papers/patents: Nothing

Brief description of the working environment: Working environment was very decent and met many amazing people.

Academic courses relevant to the project: None as such.

Learning Outcome: 1. I learnt a lot about fund raising like different techniques and stuff. 2. Improved my skills in video editing.

PS-I station: I care Foundation A.C , Mexico

Student

Name: SHASHANK REDDY GADE .(2021AAPS0482H)

Student Write-up:

PS-I Project Title: Harvesting Hope

Short Summary of work done: Hosted crowdfunding campaing in donadora website. Then, researched and formulated social media plan. Implemented the plan and started publicizing the campaign. Recieved feedback and resolved queries regarding payments. Recieved donations.

Objectives of the project: Raise funds for canpaign to build orchard with greenhouse and rainwater harvesting system to help support 80 middle school students

Tool used: Canva, animaker, google docs, Microsoft designer

Details of Papers/patents:N/A

Brief description of the working environment: Online working environment with meet every week regarding progress and status of project. The company expects us to achieve objectives communicated by us in weekly meets. PS 1 helped us in gaining experience in management, project ideation, marketing plan, team management and effective communication.

Academic courses relevant to the project: NA

Learning Outcome: Project management, team communication, ideation

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PS-I station: I care Foundation A.C , Mexico

Student

Name: DAKSH PRATAP SINGH(2021AAPS2774G)

Student Write-up:

PS-I Project Title: Feeding Dreams

Short Summary of work done: Was successfully able to raise 750 mexican pesos with the help of my team and combined efforts with Icare foundation officials.

Objectives of the project: Raising money for a community called Raices in Mexico inorder to build huertos

Tool used: Canva, Adobe Aftereffects, Adobe Photoshop, Powerpoint

Details of Papers/patents:N/A

Brief description of the working environment: Gave clear and firm instructions and helped us all the way through the campaign. Also encouraged us to work in teams thereby increasing the sense of teamwork among students and helping us understand how professional presentations and meetings are conducted and what is expected of the employees/interns

Academic courses relevant to the project: Technical Report Writing

Learning Outcome: Networking, Advertising, Fundraising, Presentation making and presenting, Conferencing and group discussions

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PS-I station: I care Foundation A.C , Mexico

Student

Name: RAGHAV .(2021B1AA1688P)

Student Write-up:

PS-I Project Title: Crowdfunding event for raices community

Short Summary of work done: publicising the campaign to raise more and more funds

Objectives of the project: to raise funds for raices community students

Tool used: designing apps like canva , adobe and video editing tools

Details of Papers/patents:no

Brief description of the working environment: the working environment was very nice. The PS was online but the response from the team was very good

Academic courses relevant to the project: no

Learning Outcome: softskills and how to reach out to people

PS-I station: I care Foundation A.C , Mexico

Student

Name: PRATYAKSHA GUPTA .(2021B1AB2740P)

Student Write-up:

PS-I Project Title: TO RAISE FUNDS FOR INTSALLATION OF GREENHOUSE IN MEXICAN MIDDLE SCHOOL.

Short Summary of work done: Our task was to create a campaign that would help collect funds for installation of greenhouse. we started with researching about the raices community of mexico and how could we help them. then with necessary details we launched the campaign and used to design two. posters per week for publicity work. we collected funds of about 776 mexican dollars.

Objectives of the project: TO RAISE FUNDS AS WELL AS TO EDUCATE PEOPLE ABOUT IMPORTANCE OF GREENHOUSE AND SELF SUFFICIENCY

Tool used: CANVAS FOR POSTER DESIGNING

Details of Papers/patents:NO

Brief description of the working environment: Working enviourment is good . mafer maam tries to explain us every aspect we could do.

Academic courses relevant to the project: TECHRE , EPS

Learning Outcome: CROWDFUNDING WAYS , POSTER DESIGNING , OUTREACH, COMMUNICATION SKILLS, VIDEO MAKING

PS-I station: I care Foundation A.C , Mexico

Student

Name: SOHAM MANGLE .(2021B3A71068H)

Student Write-up:

PS-I Project Title: Seeds Of Change

Short Summary of work done: We run over campaign on donadora which is website for fund raising campaigns. We had to create videos, short videos, gifs, posts etc. to promote our campaigns. For promotion we had to reach out via email, through social media, etc. Our team managed to raise around 200 dollors

Objectives of the project: To collect donations to provide one greenhouse with huerto

Tool used: Excel, Donadora, Hunter.io

Details of Papers/patents: None

Brief description of the working environment: The working environment is great. They are very friendly and responsive. Over all I enjoyed working with I-Care and got to learn about raising funds

Academic courses relevant to the project: Trw

Learning Outcome: Collecting donations , raising funds

PS-I station: I care Foundation A.C , Mexico

Student

Name: RUCHIKA MAHAJAN(2021B4A32532G)

Student Write-up:

PS-I Project Title: Sustainability and education

Short Summary of work done: For the project given I and my team(total 4 members in team) created a webpage, designed variuos posters(total 2 both in english in spanish), also made videos(in total 5). Along with this we contacted many influencers and potential donors to collect money for the noble cause I care foundation is visioning for

Objectives of the project: To provide one huerto with greenhouse and rainwater harvesting system to the Middle School of Raíces community to support more than 80 students through the production of organic food for their own consumption.

Tool used: For poster we used canva, for video editing ised VideoProc and canva, for webpage wr used framer.ai

Details of Papers/patents: We didn't refer any research papers as such but went through the documents provided by the ps station

Brief description of the working environment: The working environment was motivating and enthusiastic. All the team members and ps faculty, mentors were supportive and ever ready for clarifying the doubts.It was fun working on the project. In this short span of time, I've gained invaluable knowledge and experiences that I'll forever be grateful for.

Academic courses relevant to the project: As such no particular courses required. I knew canva and video editing a little bit which helped me in the project and also enhanced these skills in this span of time

Learning Outcome: Enhanced my skills of communication, video editing. Also designed a web page with the team which I have not done before

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PS-I station: I care Foundation A.C , Mexico

Student

Name: LOVENYA JAIN .(2021B4AA1732P)

Student Write-up:

PS-I Project Title: Manos Amigas (Grow, Harvest, Sustain: Green Homes for Raices)

Short Summary of work done: Designed a webpage, implemented it using no-code tool Framer. Made 2 PPTs as Midsem and endsem reports, contacted people, influencers, youtubers, instagrammers, and others, for outreach, donations, and posting about our campaign on their social handles.

Objectives of the project: To provide one Huerto with a greenhouse and rainwater harvesting system to the Middle School of Raíces community to support 80 students. And for that, we were to raise $970 USD/$16700 MX pesos. It was essentially a crowdfunding campaign and we had to co

Tool used: Framer, Microsoft PPT

Details of Papers/patents:N/A

Brief description of the working environment: Learned webpage designing, marketing, outreach, how to conduct a successful crowdfunding campaign.

Academic courses relevant to the project: N/A

Learning Outcome: Marketing, Webpage designing, Outreach

PS-I station: IFP-Petro Products (P) Ltd , Ghaziabad

Student

Name: RAJVARDHAN SINGH .(2021A1PS2158P)

Student Write-up:

PS-I Project Title: A Comprehensive Study on Used Cooking Oil to Biodiesel Conversion, Base Oil Quantification, and Challenges in Re-Refining Synthetic Oils

Short Summary of work done: During my internship at IFP PETRO PRODUCTS Ltd. in the quality assurance department, I worked on a project focused on converting used cooking oil into biodiesel. Throughout the internship, I gained valuable experience in understanding the biodiesel production process, implementing quality control measures, analyzing biodiesel samples, and adhering to safety protocols. The project aimed to produce a renewable and eco-friendly alternative to conventional diesel fuel, emphasizing the importance of environmental sustainability. Additionally, working with a team improved my communication and problem-solving skills, contributing to a well-rounded learning experience in the field of chemical engineering.

Objectives of the project: The combination of these projects offers valuable insights into the potential of UCO as a renewable fuel source, the precise analysis of base oil composition, and the challenges faced in the re-refining of synthetic oils. use of resources.

Tool used: Websites for reading research papers like scihub , research gate

Details of Papers/patents:NA

Brief description of the working environment: The working environment during my internship at IFP PETRO PRODUCTS Ltd. was professional and collaborative. The company provided a well-equipped laboratory for conducting experiments and research related to the biodiesel conversion project. The team members, including supervisors and colleagues, were supportive and encouraged knowledge sharing and learning
Expectations from the company were to actively participate in the project and contribute to its success. As an intern, I was expected to demonstrate a strong work ethic, show eagerness to learn, and take responsibility for assigned tasks. Timely completion of the project milestones and adherence to safety protocols were paramount.

Academic courses relevant to the project: Organic Chemistry,Chemical Reaction Engineering, Separation Processes

Learning Outcome: During my internship at IFP PETRO PRODUCTS Ltd. under the quality assurance department, the major learning outcomes I gained from working on the project on used cooking oil to biodiesel conversion are:
Understanding the biodiesel production process: I learned the entire process of converting used cooking oil into biodiesel, including the various chemical reactions involved and the necessary equipment and steps for the conversion.

PS-I station: IFP-Petro Products (P) Ltd , Ghaziabad

Student

Name: YASH PANDEY .(2021A8PS3194H)

Student Write-up:

PS-I Project Title: COMPREHENSIVE PROFILING OF THE USED OIL INDUSTRY & STRATEGY ANALYSIS IN A RE-REFINERY

Short Summary of work done: We had several objectives. Market Research of Textile and Coal industry, regarding the volume of used oil consumed , to find market potential, then Complete company profiling of Kabadiwalla Connect, an rising innovator in the circular economy of the developing world. then Company profiling of Castrol,, then Hosting a MERN stack website on CPanel, Draft out the requirements in a Reverse Vending Machine for Used Oil Collection. Make a flowchart for the monthly GST payment schedule of the company., finalle Computing energy consumed in VI department in plant.

Objectives of the project: Market Research of Textile and Coal industry, Complete company profiling of Kabadiwalla , Hosting a MERN stack website on CPanelDraft out the requirements in a Reverse Vending Machine for Used Oil Collection. and Computing energy consumed in VI departme

Tool used: Plesk, CPanel

Details of Papers/patents:none

Brief description of the working environment: Very diligent environment, very encouraging mentors and seniors in the company, only wish was that if the task given to

us could be done in online mode, then the company should give students the option to so their tasks from home.

Academic courses relevant to the project: Principles of Economics

Learning Outcome: Learned about economics of a circular economy company, learnt to host a website using Plesk annd CPanel

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PS-I station: Indian Red Cross Society - IT , Mumbai

Student

Name: HARSHAL CHANDRAKANT GUTHE(2021AAPS2926G)

Student Write-up:

PS-I Project Title: Increasing Awareness through Social Media Marketing

Short Summary of work done: To let people know about the facilities provided at Adams Wylie Physio Rehab Centre and the campaigns carried our by IRCS Mumbai.

Objectives of the project: To let people know about the facilities provided at Adams Wylie Physio Rehab Centre and the campaigns carried our by IRCS Mumbai.

Tool used: Excel, Social Media Apps, Extensions on chrome, PowerPoint

Details of Papers/patents: No

Brief description of the working environment: Already mentioned in report

Academic courses relevant to the project: None

Learning Outcome: Social Media Marketing

PS-I station: Indian Red Cross Society Gujarat State Branch , Ahmedabad

Student

Name: NILESH GUPTA .(2021A3PS1699H)

Student Write-up:

PS-I Project Title: Emergency Training Management and Designing

Short Summary of work done: Automation of the ground work ongoing there and digital marketing by desiging posters, banners for them. I also helped them in ground level management.

Objectives of the project: Healthcare management and designing

Tool used: Excel, Photoshop, Canva

Details of Papers/patents: None

Brief description of the working environment: Working environment was decent, all the facilities were provided to us. I learnt the day to day working in the corporate sector and in technical skills i learnt coding and video editing.

Academic courses relevant to the project: None

Learning Outcome: Ground level management and tech

PS-I station: Indian Red Cross Society Gujarat State Branch , Ahmedabad

Student

Name: GAURAV KHURDIA .(2021A4PS2552H)

Student Write-up:

PS-I Project Title: First aid training managment and Designing

Short Summary of work done: Manage the first aid training program and did various Designing work

Objectives of the project: Management and designing

Tool used: Excel and Canva

Details of Papers/patents: None

Brief description of the working environment: Good working environment

Academic courses relevant to the project: None

Learning Outcome: Management

PS-I station: Institute of Seismological Research (Onsite) , Gandhinagar

Student

Name: PRAKHAR KUMAR .(2020A2PS1739P)

Student Write-up:

PS-I Project Title: Prediction of SRGC using machine learning

Short Summary of work done: The PS-1 project involved acquainting myself with the field of seismology. I read the introduction manuals to various aspects of seismology i.e. microzonation, hazard assesment, geophysics & gravometery. After developing this overarching understanding of the field, I started working on Radon precursory research. This required me to go through research papers relevant to this area. During this literature review, I understood the context, background and state of the art of this precursory research area. The next step involved implementing a special machine learning technique on the Institute's Dataset. I acquainted myself with the libraries and conducted preliminary data analysis on the 2 year multi-sensor dataset. This involved data cleaning, summarising and checking for erroneous readings and patterns. The processed data was then passed through an implementation of a delegated regression model (xgboost +

svmr). I trained this model on various combinations of training and testing data and with different hyperparameters. The predictions made by the model were then analyzed for accuracy and other factors. The results obtained are up to par with the current state of this field and promising enough to continue further research.

Objectives of the project: Analyze radon time series database, Implement delegated regression on the multivariate timeseries database, analyze the results for earthquake precursory significance

Tool used: R, tidyverse, xgboost, e701, linux

Details of Papers/patents:NA

Brief description of the working environment: The institute is an amalgam of multidisciplinary approaches to seismology. There are civil engineers, geophysicists, geologists and other specialists who work in various labs. Interacting with this diverse team has helped me broaden my understanding of various subject matters and have a more holistic outlook of this field.
The research environment is comfortable, where one can oneself decide the pace of their work.

Academic courses relevant to the project: BITS F464 - Machine Learning, Data Science minor

Learning Outcome: Implementing machine learning in a real world scenario
Data analysis and preprocessing for non-linear natural datasets

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PS-I station: Institute of Seismological Research (Onsite) , Gandhinagar

Student

Name: GUNNAM JAYARAM CHOWDARY .(2021A2PS3056H)

Student Write-up:

PS-I Project Title: Seismic micro zonation

Short Summary of work done: Our mentor provided 43 samples of soil belonging to different places of kanpur City . We have performed moisture content, specific gravity, seive analysis, hygrometer analysis. We learnt the objectives of seismic micro zonation , hazard level and how India is divided into 4 zones , which area falls under a specific zone. Hazard level of a particular area.

Objectives of the project: Specific Gravity test is used to know the load bearing capacity of soil. Moisture content is used to know the liquefaction factor. Grain size analysis is to determine the different particles of soil and their reactions and about Seismic micro zonation.

Tool used: None (our work is completely based on (Geo technical)soil lab

Details of Papers/patents: None

Brief description of the working environment: The working environment is very well, they have provided all necessary materials and work place with good internet connectivity. Their teaching and doubt clarification made us learn deeply. Most of our project work is based on lab tests and they have assisted us well

Academic courses relevant to the project: Soil mechanics

Learning Outcome: I got to know what is seismic micro zonation, Hazard level of a particular city and Index properties of soil .

PS-I station: Institute of Seismological Research (Onsite) , Gandhinagar

Student

Name: MAHIN PATEL .(2021B3A41086P)

Student Write-up:

PS-I Project Title: GUI for Gujarat seismic activity

Short Summary of work done: Made a website for showing the output of the data

Objectives of the project: To make a GUI for compiling Gujarat seismic datas.

Tool used: YouTube, Chat GPT

Details of Papers/patents:NA

Brief description of the working environment: Working environment was not conducive.

Academic courses relevant to the project: My project was not related to my field as I am Eco + Mech graduate. But the project was related to coding background.

Learning Outcome: Web development for presenting data.

PS-I station: Institute of Seismological Research (Onsite) , Gandhinagar

Student

Name: MANAN GUPTA .(2021B4A41689P)

Student Write-up:

PS-I Project Title: Coulom stress variation of the 2011 earthquake epicentral region

Short Summary of work done: learned about coulomb stress and its role in seismic hazard assesment. Learned Coulomb 3d application integrated with MATLAB. Learned about and worked on Generic mapping tool.

Objectives of the project: Learn about coulomb stress variation and plot maps using plotting tools like GMT

Tool used: Pygmt: a python library, Coulomb 3d, matlab, gmt

Details of Papers/patents: No such paper/patent filed

Brief description of the working environment: Working environment was good and satisfactory, mentor and institute was of helping nature. Learnt many new thing described above.

Academic courses relevant to the project: No such courses

Learning Outcome: Working under a guide, time constrained environment, learnt about earthquake and it's properties, coulomb stress, and plotting tools like GMT

PS-I station: Institute of Seismological Research (Onsite) , Gandhinagar

Student

Name: HEMANK JAIN .(2021B5A12505P)

Student Write-up:

PS-I Project Title: Detection of earthquake precursors and radon time series using AI/ML

Short Summary of work done: After completing my first year of learning C language, I embarked on a new coding journey with Python. This experience has been incredibly rewarding, as it has not only enhanced my programming skills but also kindled a passion for seismology and earthquakes. Python's diverse libraries, such as scikit-learn, Numpy, and pandas, have opened up a world of possibilities for data analysis. I now feel confident in exploring other datasets independently in the future. Delving into data science and machine learning, I have gained valuable insights into using various models effectively. Throughout this project, I extensively worked with stacking and AAIF techniques, further solidifying my understanding of their importance. Additionally, I have learned the significance of statistical analysis techniques in the context of Machine Learning, comprehending their strengths and limitations when handling data. This project's impact extends beyond just machine learning; it has provided valuable insights into how organizations operate. I am excited about the prospects of applying my newfound skills in future projects, as they will undoubtedly be invaluable assets for problem-solving and decision-making. The journey has been enriching, equipping me with a versatile skill set that I envision will shape my future endeavors across various domains.

Objectives of the project: to predict radon concentration

Tool used: python

Details of Papers/patents:"Anomaly Classification for Earthquake Prediction in Radon Time Series Data Using Stacking and Automatic Anomaly Indication Function"
~ ADIL ASLAM MIR,1,2 FATIH VEHBI C¸ELEBI,1 MUHAMMAD RAFIQUE,3 M. R. I. FARUQUE,4 MAYEEN UDDIN KHANDAKER,5 KIMBERLEE J

Brief description of the working environment: During my internship at my station, I thrived in a supportive and welcoming environment. Transitioning from C to Python coding, I explored libraries like scikit-learn, Numpy, and pandas, honing my data analysis skills. Delving into machine learning, I grasped various models and specialized in stacking and AAIF techniques. Moreover, I grasped the significance of statistical analysis in Machine Learning. The experience not only enriched my technical abilities but also taught me effective teamwork and organizational dynamics, setting a strong foundation for my future projects and aspirations. Grateful for this transformative opportunity, I look forward to making meaningful contributions in data science and beyond.

Academic courses relevant to the project: Computer programming

Learning Outcome: Machine learnig, Python, earthquakes, corporate behaviours

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PS-I station: Institute of Seismological Research (Onsite) , Gandhinagar

Student

Name: SHREYAS ASHISH KESKAR(2021B5A13034G)

Student Write-up:

PS-I Project Title: Detection of Earthquake precursors in Radon time series using AI/ML

Short Summary of work done: We explored the use of multiple linear regression and artificial neural networks in analyzing soil radon data for seismic precursory studies.

Objectives of the project: Detection of Earthquake precursors in Radon time series using AI/ML

Tool used: Pyrhin, Jupiter Notebook etc

Details of Papers/patents:NA

Brief description of the working environment: As an ML intern at ISR Gandhinagar, I was expected to actively participate in a research project and contribute to the development and implementation of machine learning models for seismological data

analysis. My responsibilities included data preprocessing, feature engineering, model selection, and evaluation etc.
The working environment was great, and mentors were very supportive. I also made connections with some MSc/PhD students in Geophysics and Seismology at the institute.

Academic courses relevant to the project: CP

Learning Outcome: Multiple Linear Regression, ANN, Python, Numpy, Pandas etc

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PS-I station: Institute of Seismological Research (Onsite) , Gandhinagar

Student

Name: TAVISH NATH SHARMA(2021B5A43005G)

Student Write-up:

PS-I Project Title: Predicting Radon Concentration

Short Summary of work done: As an ML intern at ISR Gandhinagar, the company would expect you to actively participate in research projects and contribute to the development and implementation of machine learning models for seismological data analysis. Your responsibilities might include data preprocessing, feature engineering, model selection, and evaluation.

Objectives of the project: Predicting Radon Concentration

Tool used: python, R

Details of Papers/patents:none

Brief description of the working environment: Working environment was healthy and supportive

Academic courses relevant to the project: computer programming

Learning Outcome: learned python

PS-I station: Institute of Seismological Research (Onsite) , Gandhinagar

Student

Name: OJAL AGRAWAL(2021B5TS2054P)

Student Write-up:

PS-I Project Title: Building a GUI for seismic activities

Short Summary of work done: We started working on creating a prototype for web page and then worked on adding different forms of specifications and worked on creating a GUI and plotting more thand 100s of stations in Gujrat. On the course of whole P.S. we had little to no coding experience but in the end we successfully presented a web page to our mentor with Gujarat's map plotted all stations and made it researchers job of extracting data a little easier.

Objectives of the project: To build a web page with all the seismic earthquake data and station data for any research to extract the required data.

Tool used: Soft ware of python different libraries

Details of Papers/patents: No

Brief description of the working environment: We were expected to learn organisational skills, learning practical skills and gaining in hands experience. The environment was professional and we were able to work on project beyond our theoretical scope.

Academic courses relevant to the project: CP

Learning Outcome: Python, GUI, HTML, Source code

PS-I station: IOCL - Information Services , Mumbai

Student

Name: AAYUSH RANJAN JHA .(2021B5A70895P)

Student Write-up:

PS-I Project Title: Revamping the IOCLintranet Site

Short Summary of work done: During my PS-1 internship at IOCL Mumbai, I gained practical experience in Information Services through assignments in UI/UX design, web development, auditing, and reporting. Highlights of my work include: Developed a PowerPoint presentation on "101 Principles of UI/UX," showcasing my research and presentation skills while deepening my understanding of user-centered design. Built a user-friendly survey website using Django, django-survey forms, and crispy-forms with Bootstrap, allowing IOCL employees to provide feedback on the intranet site. Conducted an audit of the developing intranet site, proposing improvements such as color gradients aligned with the IOCL brand and voice search functionality. Performed a comprehensive Lighthouse analysis to assess the site's performance, accessibility, best practices, and SEO. Integrated survey findings and Lighthouse analysis results into a detailed audit report, highlighting flaws and providing recommendations for optimization.

Objectives of the project: To revamp the intranet site of IOCL mumbai

Tool used: Figma: You utilized Figma for UI/UX design tasks. Figma is a collaborative design tool that allows for creating, prototyping, and collaborating on designs in real-time. Django: You employed Django, a high-level Python web framework, for developing the su

Details of Papers/patents:-

Brief description of the working environment: During my PS-1 internship at IOCL Mumbai, the working environment was professional and focused on learning and growth. We were asked to work in a hybrid mode, which involved going to the office twice a week to meet with our mentor. These meetings provided an opportunity to discuss our work, receive feedback, and evaluate our progress. The company had high expectations, encouraging active participation and timely delivery of high-quality work. I had the opportunity to apply theoretical concepts in real-world projects, gaining practical exposure in UI/UX design, web development, and auditing. Collaborating with industry experts and receiving feedback during our in-person meetings enhanced my skills and understanding. The internship fostered a culture of collaboration, effective communication, and teamwork, allowing me to expand my knowledge and strengthen interpersonal skills. Overall, the internship provided a valuable learning experience in a professional setting, exposing me to industry-standard tools and technologies, while the hybrid mode allowed for regular in-person interaction and evaluation with my mentor.

Academic courses relevant to the project: User Interface Design: This course covers the principles and techniques of designing effective user interfaces, focusing on usability, interaction design, and user-centered design methodologies. It would provide a solid foundation for understanding UI/UX

Learning Outcome: 1)Understanding of UI/UX Principles: Gained a solid understanding of UI/UX principles and their practical application in design projects.
2)Proficiency in Tools and Technologies: Developed proficiency in tools like Figma and Django, as well as conducting a Lighthouse analysis.
3)Collaboration and Communication Skills: Enhanced collaboration and communication skills through working with industry experts and participating in meetings and discussions.
4)Attention to Detail and Analytical Thinking: Sharpened attention to detail and analytical thinking abilities through tasks such as conducting audits and analyzing data.
5)Continuous Learning and Adaptability: Embraced a mindset of continuous learning, staying updated with industry trends, and adapting to new practices.
6)Professional Skill Development: Developed a range of professional skills, including project management, problem-solving, time management, and organization.
7)Practical Application of Academic Knowledge: Successfully applied theoretical knowledge gained during academic studies to real-world projects and problem-solving.
8)Hands-on Experience: Acquired practical experience by working on assignments and projects related to UI/UX design and information services.
9)Integration of Feedback: Learned to incorporate feedback from industry experts and make improvements based on their suggestions and insights.
10)Valuable Networking: Established professional connections and networks with industry experts, providing opportunities for mentorship and future collaborations.

PS-I station: IOCL - Operation LPG Operation , Delhi

Student

Name: ANUBHAV SINHA(2021A4PS2893G)

Student Write-up:

PS-I Project Title: Study of HVLRs and safety operations in tt loading/unloading

Short Summary of work done: firstly we were given a tour of the terminal and instructed about how the work is done here.since it is a terminal is responsible for the storage and transportation of hazardous products especially ATF.

Objectives of the project: To understand the importance of safety while filling tanker trucks and also about the importance of HVLRs.

Tool used: different types of valves

Details of Papers/patents:NA

Brief description of the working environment: positive work culture and my mentors were a constant source of motivation.

Academic courses relevant to the project: Fluid Mechanics, applied thermodynamics

Learning Outcome: industry experience , hands on experiences

PS-I station: IOCL - Operation LPG Operation , Delhi

Student

Name: VAIBHAV JAIN .(2021B1A42475P)

Student Write-up:

PS-I Project Title: Firefighting water engines

Short Summary of work done: PS at Indian Oil Corporation Limited, Delhi has been a wonderful experience. I have met many new people, gained new skills, and learned about professional practices in offices. This internship has been incredibly educational, offering me the chance to learn various aspects of thermal and fluid engineering and understand the importance of safety in the oil and gas industry.

Objectives of the project: To learn the core mechanical pumps and engines used in the industries.

Tool used: H/w

Details of Papers/patents:NA

Brief description of the working environment: PS at Indian Oil Corporation Limited, Delhi has been a wonderful experience. I have met many new people, gained new skills, and learned about professional practices in offices. This internship has been incredibly educational, offering me the chance to learn various aspects of thermal and fluid engineering and understand the importance of safety in the oil and gas industry.
As a mechanical engineering student, it was a good experience with some engines, pumps, and some core engineering tools required in the industries. And I thoroughly enjoyed the all-over experience here.
I firmly believe that the skills I have acquired during this period will undoubtedly prove beneficial in my future endeavors.

Academic courses relevant to the project: core mechanical courses like fluid mechanics,etc

Learning Outcome: Leaned about the different types of mechanical pumps used in the oil industries and also about the safety measures to be taken care in such industries.

PS-I station: IOCL - Operation/Maintenance , Delhi

Student

Name: KUSHAGRA SINGH .(2021A3PS2393P)

Student Write-up:

PS-I Project Title: Understand SCADA System operation in Pipeline

Short Summary of work done: The station in itself is very lite and does not involve a lot of work.The station incharges are really friendly though and try to help us out in any way.During the course of the project I learnt how modern control systems are used in industry and the interplay of different branches of engineering in modern industries.

Objectives of the project: The objective of the report was to understand how SCADA is used as a modern control system to help in pipeline operations

Tool used: Microsoft Office Suite

Details of Papers/patents:.

Brief description of the working environment: The company does not put a lot of pressure on the students and asks us to go forward with the project based on our understanding.

Academic courses relevant to the project: Control System

Learning Outcome: Learnt about Modern Control System and how they are used

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PS-I station: IOCL - Operation/Maintenance , Delhi

Student

Name: MRITYUNJAY GOSWAMI .(2021A4PS2291H)

Student Write-up:

PS-I Project Title: Study and Design of Centrifugal Pumps using CAD

Short Summary of work done: I undertook the project of designing Centrifugal Pump using SolidWorks and deriving performance reports using ANSYS Fluent. This also included studying the various parts of the pump and important performance curves.

Objectives of the project: To learn about turbomachinery used in pipelines operations at IOCL.

Tool used: SolidWorks, MATLAB, ANSYS

Details of Papers/patents:N/A

Brief description of the working environment: The environment is very friendly, warm and welcoming. IOCL is a fortune 500 and Maharatna company of India which gives you a once in a lifetime experience to work with professionals who strive hard to make sure that energy requirements of this nation are met and no pump stations go dry due to lack of fuel at any time of the day.

Academic courses relevant to the project: PMFM (Taught in 3rd Year Mechanical CDC)

Learning Outcome: - Proficiency in using CAD Software like SolidWorks and ANSYS
- MATLAB Fundamentals
- Operations of a big and key player in Energy Sector of India

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PS-I station: IOCL - Operation/Maintenance , Delhi

Student

Name: BHASKAR .(2021B2A42761P)

Student Write-up:

PS-I Project Title: Operations and Maintenance

Short Summary of work done: During my internship at Indian Oil Corporation Limited (IOCL), I had the opportunity to gain valuable experience and insights into the petroleum industry. The internship lasted for a period of six months, and I was assigned to the Refining and Marketing division. Throughout the internship, I was actively involved in various projects and tasks related to the refining processes and marketing strategies.Working closely with experienced professionals at IOCL, I learned how to navigate the complexities of a large-scale organization and witnessed the importance of teamwork and effective communication in achieving organizational goals. The internship also provided me with exposure to real-world challenges and problem-solving opportunities, enhancing my analytical and critical thinking skills.

Objectives of the project: Understanding Operations at IOCL

Tool used: -

Details of Papers/patents:-

Brief description of the working environment: During my internship at OpenAI, the working environment was both dynamic and supportive, providing an invaluable learning experience. From the moment I joined the team, I was welcomed with open arms, and my colleagues were always eager to help and share their knowledge.

The company fostered a culture of continuous learning and encouraging interns and employees.

Academic courses relevant to the project: -

Learning Outcome: In-depth understanding of the petroleum industry's operations, processes, and challenges.
Knowledge of refining technologies and their impact on production efficiency and environmental sustainability.
Exposure to the regulatory framework and safety protocols in the oil and gas industry.
Understanding of corporate social responsibility and the positive impact businesses can have on communities.
Strengthened analytical and critical thinking skills through problem-solving opportunities.

PS-I station: IOCL - Operation/Maintenance , Panipat

Student

Name: DEVANSH MAHESHWARI .(2021AAPS0984P)

Student Write-up:

PS-I Project Title: Pipelines,PLC and Intrusion Detection System at an Oil Refinery

Short Summary of work done: Saw the Intrusion detection system working live as well as the R-LNG Pipelines which had a gas chromatograph installed. Also learned the basics and ideas behind PLC programming in an ESD situation.

Objectives of the project: Making a PLC logic

Tool used: PLC

Details of Papers/patents: None

Brief description of the working environment: IOCL is a very good company to work at, The staff there is very intelligent and proactive in taking initiatives to teach you

Academic courses relevant to the project: Control Systems

Learning Outcome: PLC programming, Intrusion Detection System , Overall working of the Pipelines division

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PS-I station: IOCL - Operation/Maintenance , Panipat

Student

Name: ROHIT BIND .(2021ABPS2719P)

Student Write-up:

PS-I Project Title: Operation and maintenance of pump&valves

Short Summary of work done: We had regular visit of site, where we learned how does centrifugal,booster, reciprocating pump works,and the various defect that occur in these pumps and how to takle these defect or in short maintainece of pump

Objectives of the project: I understood how does the maintenance of pump and valves work ,and also i was interested to understand how does supply chain of iocl works ,so somewhat i understood supply chain also

Tool used: We saw working of mainline pump (centrifugal pump) and also reciprocating pump and the various working of valves and their importance for example gas over oil actuator (GOV),HAND OPERATED valve (globe valve) mainline block valve and flow control valve

Details of Papers/patents: None

Brief description of the working environment: We had regular visit of site ,where mohit sir used to explain the working of pump (mainline pump and reciprocating pump) and booster pump which play very important role for mainline pump since mainline pump work on specific pressure energy of fluid other their is more chances of getting defects in pump,so booster pump used to provide required presure energy in fluid .
Overall the working environment was very good ,every one was very helpful their.

Academic courses relevant to the project: Material science and machanism and machine

Learning Outcome: I learn to work with team ,

This station help me to acquire a special skill that is networking with senior and other ,
I learned how centrifugal and reciprocating are playing their major role in pumping of fluid
,
And what are the major defects that occur in pump

PS-I station: IOCL - Operation/Maintenance , Vadodara

Student

Name: RUDRA MATHUR .(2021A1PS2112P)

Student Write-up:

PS-I Project Title: Overview of the Gujarat Refinery(Production Department

Short Summary of work done: We had to make report of all the units that we visited during our PS-1 which includes all the details of the plant, mass balance of the incoming and outgoing streams as well as a basic plant layout.

Objectives of the project: To understand and analyze different units of the refinery and how crude is processed to different petrochemical products for the market

Tool used: MS Excel, communication skills, Ms power point presentation

Details of Papers/patents: Not really, we used to read manuals inside the refinery.

Brief description of the working environment: The working environment can be really relaxed to being too hectic, it depends on the position of the employee. For the contract workers who do all the jobs inside the refinery, it’s a hectic and a dangerous environment but for the engineers and diploma holders, it’s very relaxing, they just need to sit in the DCS room and monitor the plant.

Academic courses relevant to the project: Thermodynamics, Fluid Mechanics, Heat Transfer, Process Calculations, Mass transfer

Learning Outcome: Refinery operations, communication skills

PS-I station: IOCL - Operation/Maintenance , Vadodara

Student

Name: ABHAY KUMAR CHOUDHARY .(2021A1PS2595P)

Student Write-up:

PS-I Project Title: Operations and Maintenance

Short Summary of work done: Work summary: We had to make report of all the units that we visited during our PS-1 which includes all the details of the plant, mass balance of the incoming and outgoing streams , furnace efficiency as well as a basic plant layout.

Objectives of the project: To study the Operations and Maintenance of the different primary and secondary units of the IOCL VADODARA Refinery

Tool used: Ms Excel, Ms PowerPoint, Communication skills

Details of Papers/patents: None

Brief description of the working environment: The working condition was quite fascinating we were allocated Conference hall to study the Unit manual and an engineer was allocated to us who would take us to the field for showing practical applications . The entire unit was very supportive and nice to us trainees.

Academic courses relevant to the project: Heat Transfer, Fluid Mechanics, Process Calculation, Heat Transfer

Learning Outcome: Learned the Practical application of the Mass balance , Heat exchangers , Pumps Refinery operations, communication skills

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PS-I station: IOCL - Operation/Maintenance , Vadodara

Student

Name: ABHIGYA KHANDELWAL(2021A1PS2969G)

Student Write-up:

PS-I Project Title: Overview of Gujarat Refinery

Short Summary of work done: We had to make report of all the units that we visited during our PS-1 which includes all the details of the plant, mass balance of the incoming and outgoing streams as well as a basic plant layout.

Objectives of the project: To understand the refinery operations, application of college therory and to have an understanding of the petroleum industry in genral

Tool used: MS Excel, communication skills, good physique (you will need to walk a lot inside of a refinery), Ms power point presentation

Details of Papers/patents: Not really, we used to read manuals inside the refinery.

Brief description of the working environment: The working environment can be really relaxed to being too hectic, it depends on the position of the employee. For the contract workers who do all the jobs inside the refinery, it's a hectic and a dangerous environment but for the engineers and diploma holders, it's very relaxing, they just need to sit in the DCS room and monitor the plant.

Academic courses relevant to the project: Thermodynamics, Fluid Mechanics, Heat Transfer, Process Calculations, Mass transfer

Learning Outcome: Refinery operations, communication skills

PS-I station: IOCL - Operation/Maintenance , Vadodara

Student

Name: SHREYANSH SINGH .(2021A2PS2519P)

Student Write-up:

PS-I Project Title: Structure Maintenance and Civil Methodologies in Process Industry.

Short Summary of work done: I was in Civil Maintenance Department. During PS-I have learned about various methodologies and processes in the Refinery. I have visited the Thermal Power Station, Cooling towers, coke yard etc. I have also visited construction sites in and around the Refinery. I have seen the various practical applications of Engineering concepts in Industry. I also went to a new IOCL refinery construction site. I primarily learned about the maintenance and repair activities of structures in the Refinery.

Objectives of the project: To get an overview of the various civil works in a Process industry. The project was focused on maintenance activities of various structures in a Refinery.

Tool used: N/A

Details of Papers/patents:N/A

Brief description of the working environment: The working environment in the station is very uplifting. The employees are friendly and the peer relations are very cordial. There is an organized work structure. The work pressure is not too great. The work hours are moderate. It is a premier organization of the country. Overall, it was a great experience.

Academic courses relevant to the project: CE F213 - SURVEYING,
CE F230 - CIVIL ENGINEERING MATERIALS

Learning Outcome: The project provided a number of learning opportunities.
1. Applications of Civil Engineering concepts in the Industry, e.g. using surveying techniques to estimate volume of Coke Stacks.
2. Completed a study on an OISD document dealing with guidelines for maintenance and repair activities for Civil structures.
3. Learned about Retrofitting aspects in relation to Oil and Gas Industry.
4. Garnered insight into the working of cooling systems in the Refinery and Heat exchangers.

PS-I station: IOCL - Operation/Maintenance , Vadodara

Student

Name: SHIVAM HEMANT PACHCHIGAR(2021A3PS2228G)

Student Write-up:

PS-I Project Title: Interference Mitigation: Cathodic Protection Solutions

Short Summary of work done: The project titled "Interference Mitigation: Cathodic Protection Solutions" focuses on addressing the issue of excessive charge accumulation on a pipeline within the framework of IOCL (Indian Oil Corporation Limited). The aim of the project was to develop a solution for the problem of corrosion of IOCL pipelines due to DC interference. We took a comprehensive explanation of cathodic protection (CP) techniques and their benefits in preventing pipeline corrosion. We specifically analyzed the issue caused by the proximity of an anode bed belonging to an undisclosed company referred to as XYZ, which has led to an excessive charge build-up on the IOCL pipeline. The project highlights the importance of swiftly identifying and implementing an effective mitigation method to prevent further damage to the pipeline's primary coating and ensure the continued protection and integrity of the infrastructure. It emphasizes the need for careful evaluation and selection of appropriate solutions based on the specific circumstances and extent of the interference. Overall, the project aims to provide an efficient and cost-effective solution to mitigate the interference and ensure the long-term integrity of the IOCL pipelines.

Objectives of the project: Develop a solution for the problem of Corrosion of IOCL Pipelines due to DC Interference

Tool used: NA

Details of Papers/patents: Peabody's control of pipeline corrosion, 2nd edition; NACE (National Authority of Corrosion Engineering) papers

Brief description of the working environment: -

Academic courses relevant to the project: Electrical machines

Learning Outcome: Learn how corrosion is prevented in pipelines and learn about various mitigation methods

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PS-I station: IOCL - Operation/Maintenance , Vadodara

Student

Name: VRAJ MAHESHBHAI SANARIYA(2021A3PS2658G)

Student Write-up:

PS-I Project Title: Interference Mitigation: Cathodic Protection Solutions

Short Summary of work done: The project titled "Interference Mitigation: Cathodic Protection Solutions" focuses on addressing the issue of excessive charge accumulation on a pipeline within the framework of IOCL (Indian Oil Corporation Limited). The aim of the project was to develop a solution for the problem of corrosion of IOCL pipelines due to DC interference. We took a comprehensive explanation of cathodic protection (CP) techniques and their benefits in preventing pipeline corrosion. We specifically analyzed the issue caused by the proximity of an anode bed belonging to an undisclosed company referred to as XYZ, which has led to an excessive charge build-up on the IOCL pipeline. The project highlights the importance of swiftly identifying and implementing an effective mitigation method to prevent further damage to the pipeline's primary coating and ensure the continued protection and integrity of the infrastructure. It emphasizes the need for careful evaluation and selection of appropriate solutions based on the specific circumstances and extent of the interference. Overall, the project aims to provide an efficient and cost-effective solution to mitigate the interference and ensure the long-term integrity of the IOCL pipelines.

Objectives of the project: Develop a solution for the problem of Corrosion of IOCL Pipelines due to DC Interference

Tool used: NA

Details of Papers/patents: Peabody's control of pipeline corrosion, 2nd edition; NACE (National Authority of Corrosion Engineering) papers, CP3

Brief description of the working environment: Good Working Environment, Helpful and friendly Mentors

Academic courses relevant to the project: Electrical Machines(EM), Electrical Sciences(ES)

Learning Outcome: Learn how corrosion is prevented in pipelines and learn about various mitigation methods

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PS-I station: IOCL - Operation/Maintenance , Vadodara

Student

Name: PARTH POL(2021A4PS1556P)

Student Write-up:

PS-I Project Title: On-line retrieval of Corrosion Coupon

Short Summary of work done: In PS-1, I understood and got exposed to the working environment and how further preparations need to be made in order to be ready for placements. I learned how to implement previously learnt knowledge and how to interact with colleagues and seniors.

Objectives of the project: To reduce the economic loss to IOCL caused due to shutdowns while installing and retrieving the coupons

Tool used: Corrosion coupon, SCADA software, Spanners, and Allan keys.

Details of Papers/patents: None

Brief description of the working environment: The expectations from the company were to provide a learning environment and healthy interactions. The PS-1 company treated us with hospitality and provided an excellent working environment. Experienced and knowledged PS instructors were appointed who guided us throughout our project and helped us learn the technical aspects required to land a job in the particular sector

Academic courses relevant to the project: Corrosion, Cathodic Protection, Hot tapping in pipelines, Centrifugal pumps

Learning Outcome: Better understanding of corrosion, Learning the workings of Centrifugal pump, Watching real life workings in a company and professional interactions in a workplace, performing logistics on how loss of material is measured in monetary terms

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PS-I station: IOCL - Operation/Maintenance , Vadodara

Student

Name: VARUN MANOHAR CHITLANGE(2021B1A12733P)

Student Write-up:

PS-I Project Title: Operations/Maintenance

Short Summary of work done: Studied about overall functioning of different units in the refinery. We were able to study about different types of furnaces, distillation column and further equipments of the refinery. We did mass balances of units like AU-I CDU, DHDS, DHDT, different units of GHC, different units of LAB complex, MSQU, and FCC unit. We calculated furnace furnace efficiencies of MSQU and DHDS unit. We got to know about "Net zero by 2046" project of IOCL. Also, we were able to know about industrial standard safety norms. Thus, we were able to get a overview of the Petroleum refinery.

Objectives of the project: To study the overall operations going on in the refinery

Tool used: None

Details of Papers/patents:-

Brief description of the working environment: Overall working environment of the refinery is very friendly and supporting. What we expected from this training is that we should be able to understand "How petroleum Refining industry works?" And company was successful to fulfil our expectations.
And we were also able to achieve our motive of this vocational training

Academic courses relevant to the project: None

Learning Outcome: Learning about different equipments in the refinery.
Overall operations going on in the refinery

PS-I station: IOCL - Operation/Maintenance , Vadodara

Student

Name: DAMERLA TARUN SAI(2021B1A12741P)

Student Write-up:

PS-I Project Title: Overview of primary and secondary processing units in the refinery.

Short Summary of work done: We had to make report of all the units that we visited during our PS-1 which includes all the details of the plant, mass balance of the incoming and outgoing streams as well as a basic plant layout.

Objectives of the project: To understand the refinery operations, application of college therory and to have an understanding of the petroleum industry in genral.

Tool used: MS Excel, communication skills, good physique (you will need to walk a lot inside of a refinery), Ms power point presentation

Details of Papers/patents: Not really, we used to read manuals inside the refinery.

Brief description of the working environment: Working conditions: The working environment can be really relaxed to being too hectic, it depends on the position of the employee. For the contract workers who do all the jobs inside the refinery, it's a hectic and a dangerous environment but for the engineers and diploma holders, it's very relaxing, they just need to sit in the DCS room and monitor the plant.

Academic courses relevant to the project: Thermodynamics, Fluid Mechanics, Heat Transfer, Process Calculations, Mass transfer

Learning Outcome: Refinery operations, communication skills

PS-I station: IOCL - Operation/Maintenance , Vadodara

Student

Name: SIDHANT BHATTACHARJEE .(2021B2A12745P)

Student Write-up:

PS-I Project Title: Operation and maintenance

Short Summary of work done: We had to make report of all the units that we visited during our PS-1 which includes all the details of the plant, mass balance of the incoming and outgoing streams as well as a basic plant layout.

Objectives of the project: To study the operation and maintenance of different units present at refinery

Tool used: Ms excel ,Ms ppt , communication skills

Details of Papers/patents: Nil

Brief description of the working environment: Desk job during panel study and field work during plant excavation

Academic courses relevant to the project: Heat transfer, mass transfer, thermodynamics, fluid dynamics

Learning Outcome: Understand

PS-I station: IOCL - Operations , Mumbai

Student

Name: VINAYAK BHATTACHARYA .(2021B3A42570P)

Student Write-up:

PS-I Project Title: EV market entry strategy for Oil Marketing Companies (OMCs)

Short Summary of work done: My work mostly focused on research. The topics I researched on were increasing revenue for ROs, corporate media optimisation and market entry strategy.

Objectives of the project: The main objective was to collate data, analyse it and predict a holistic approach that OMCs can take when venturing into the EV market.

Tool used: Excel, Canva

Details of Papers/patents:NA

Brief description of the working environment: The working environment was very conducive to learning, and interns were given a lot of freedom regarding the kind of project that they wanted to pursue. Mentors were very approachable, which allowed us to clear the numerous doubts we had efficiently. IOCL, more than anything, expected us to learn about the industry, which I think we successfully did.

Academic courses relevant to the project: Fundamentals of Finance and Accounting

Learning Outcome: - Understood how OMCs function
- Understood how the Indian and international oil markets function
- Understood how the EV market functions

PS-I station: IOCL - Operations/Maintenance , Mathura

Student

Name: VINAYAK GUPTA .(2021A1PS2365P)

Student Write-up:

PS-I Project Title: In-depth Study of Continuous Catalytic Reforming Unit and detailed simulation of Naphtha splitter column its processes variables along with its efficiency increased

Short Summary of work done: Main work include deep and insightful study of the operating manual of various plants, along with live onsite experience of equipment used in the refinery and how they work. Detailed information for running of simulations and process variable and condition of the Naphtha Splitter Unit had been learned. Learned the way how to professionally interact with people working in the organizations.

Objectives of the project: To Deeply understand the process of refining Naphtha to produce Motor Spirit blending stock with high RON, and to under stand DWSIM software to design Process flow diagram of Naphtha Splitter Unit of CCRU in order to compare its newly installed divided wa

Tool used: DWSIM

Details of Papers/patents:none

Brief description of the working environment: Working environment here was more or less flexible for interns as they were not only bound to what they were allotted, everyone here was friendly and was eager to teach the Vocational Trainee as according to them "we were the future employee so we need better understanding of refinery". Company's expect only learning and understanding behaviour of the trainee along with the basic Discipline, Punctual, Diligent, Observational and Patience.
I had learned how to be Punctual, Observational, curious and calm during working or learning inside Refinery as its the place with some dangers and thus proper training on this is also given in start of the training by fire and safety department here. Most important learning is that how to work in corporate world and how to maintain a good communication in order to work something out successfully.

Academic courses relevant to the project: Kinetic reactor design, Separation Processes I and II, Chemical Thermodynamics, Heat transfer, and Fluid Dynamics.

Learning Outcome: Had a hand on Experience of DWSIM, a simulator use to design and solve Process flow diagram which in directly depicts how the refinery mechanism/ units are installed and run on a daily basis. Got to know the process behind the refining of crude and the economic industrial reaction on which this refinery works in order to refine its crude and produce valuable, market sellable products. Whole processing of the crude into its multiple component is learned and experienced.

PS-I station: IOCL - Operations/Maintenance , Mathura

Student

Name: HEMANTI BISWAS .(2021A1PS2406H)

Student Write-up:

PS-I Project Title: Calculation of efficiency of Reactor Feed Furnace in OHCU

Short Summary of work done: To find the efficiency of the reactor feed furnace inOHCU and to see if any requirements are required in the design of the furnace in order to increase the efficiency

Objectives of the project: To find the efficiency of the reactor feed furnace in OHCU and to see if any requirements are required in the design of the furnace in order to increase the efficiency.

Tool used: H/w

Details of Papers/patents: None

Brief description of the working environment: Everyone was very enthusiastic and always mindful of the work they were doing. The ecological environment was always kept under care. They followed all the laws and protocols of the government. They were very disciplined in terms of safety.

Academic courses relevant to the project: Fluid Mechanics, Thermodynamics, Separation Process, Heat Transfer, Numerical Method

Learning Outcome: Observation skills, how to solve a problem set, collecting the proper data's and learning the laws applied in the equipments.

PS-I station: IOCL - Operations/Maintenance , Mathura

Student

Name: SAKSHAM CHABA .(2021A1PS2592P)

Student Write-up:

PS-I Project Title: In-depth study of MSQU and Prime-G units for the production of motor-spirit (petrol) in compliance with the latest BS-VI norms

Short Summary of work done: My work was to understand and learn practically through site visits that how is the final motor-spirit (petrol) made from different techniques used in the refinery. My main focus was to understand two units mostly which were MSQU (Motor Spirit Quality Upgradation) Unit and Prime-G Unit. A short brief about the working of MSQU and Prime-G unit is that we do isomerization reaction in MSQU Unit to increase the octane number of the feed so as to take it to the BS-VI limit whereas in Prime-G units we, majorly do hydrodesulfurization to decrease the sulfur content in feed and simultaneously avoiding octane number loss since its octane number is already very high. Finally my work was to understand how is the blending of different quality of MS from

different units is done and in what variable ratios they are mixed to form the final motor-spirit (MS) for sale in market.

Objectives of the project: 1)To understand how is the motor spirit made from light naphtha with the help of different units in refinery such as MSQU and Prime-G. 2)To understand the material and energy balance of MSQU and Prime-G unit to analyze its input and output stream data.

Tool used: ASPEN HYSYS, CHEMCAD – for modelling chemical processes; HTRI – for designing of Heat Exchangers; MATLAB – to perform mathematical calculations related to unit processes.

Details of Papers/patents:I completed a report on - "In-depth study of MSQU and Prime-G units for the production of motor spirit (petrol) in compliance with the latest BS-VI norms." under the guidance of my mentor Mr. K.K Tiwari, Training and Learning Department and my PS-1 instru

Brief description of the working environment: The working environment was very positive and motivating even though the day to day jobs that are performed in the refinery are very risky. The safety norms and rules & regulations were strictly followed and extreme discipline inside the refinery was a major highlight.
There was also a strong push for innovation and development of new technologies which could be seen from the fact that IOCL has developed its own patented technique of producing motor-spirit that has very high octane number of 120. The employees there were very cooperative and helped us in every possible way to understand the working of the refinery.
The motivation and eagerness to learn more among the other trainees and apprentices made me even more motivated and disciplined towards my job and my PS journey joyful.

Academic courses relevant to the project: Separation Processes, Heat Transfer, Fluid Mechanics, Process Design & Principles, Reactor Design

Learning Outcome: 1)learnt the history of refinery's development and composition of petroleum. 2)learnt about the refinery products, test methods and petroleum properties. 3)learnt about various petroleum refinery processes such as distillation, combustion, cracking etc.

PS-I station: IOCL - Operations/Maintenance , Mathura

Student

Name: JYOTIRADITYA SIKDER .(2021B2A12762P)

Student Write-up:

PS-I Project Title: Corroboration of Heat Exchanger using CFD and In-depth Study of Sulphur Recovery Unit

Short Summary of work done: I have acquired significant expertise through Operations, Maintenance, and Technical Services CFD intern at the Oil Refinery of Indian Oil Corporation Limited in Mathura. In the aforementioned position, my primary responsibility entailed the validation of data obtained from heat exchangers through the utilisation of Ansys Fluent computational fluid dynamics (CFD) analysis. The present study involved acquiring proficiency in the application of ASTM D86, as well as gaining knowledge regarding regime flow dynamics. The software tool DWSim was employed to determine the fluid and vapour velocities at the outlet, along with the calculation of the Reynolds number. Furthermore, I have acquired valuable knowledge pertaining to the operational and maintenance protocols, process flow diagrams, chemical reactions, including control panels associated with the Sulphur Recovery Unit. In addition, I had the privilege of acquainting myself with the operational aspects and process flow diagrams including field visits pertaining to several other units, namely AVU (Atmospheric Vacuum Unit), VBU (Viscosity Breaker Unit), CCRU (Continuous Catalytic Reforming Unit), FCCU (Fluid Catalytic Cracking Unit), DHDT (Diesel Hydrotreating Unit), DHDS (Diesel Hydrodesulfurization Unit), LPG (Liquefied Petroleum Gas), ETP (Effluent Treatment Plant) and MSQU (Motor Spirit Quality Unit).

Objectives of the project: A detailed overview about the operations and process engineering in the Sulphur Recovery Unit and to validate and predict fouling in shell tube Heat Exchanger of MSQ unit using Computational Fluid Dynamics Simulation.

Tool used: Indian Oil Control Panel Operations of different units, Fusion 360, Ansys Fluent, and DWSim.

Details of Papers/patents: Project report submission to IOCL. As per terms of the company, no publication or patent is allowed in this Vocational Training Program.

Brief description of the working environment: The professionals within the company exhibit a high degree of affability and promote an environment advantageous to knowledge acquisition for individuals seeking such opportunities. In order to acquire knowledge, it is imperative to engage with individuals from diverse units and actively pursue opportunities to gather insights from their expertise. Due to the nature of heavy-duty industries, it is imperative that field visits be conducted under the supervision of an accredited professional. This precautionary measure is necessary due to the presence of

numerous toxic gases within the environment, which pose significant health risks to individuals.
Carrying of electronic devices is prohibited within the premises as a precautionary measure to address concerns pertaining to privacy and safety.
Heat exchanger data was validated using Ansys Fluent CFD. This study taught regime flow dynamics and ASTM D86. DWSim computed output fluid and vapour velocities and Reynolds number. SRU, AVU, VBU, FCCU, CCRU, DHDT, DHDS, OHCU, LPG, ETP, MSQU, OCTAMAX, PrimeG and BBU's operations, maintenance, process flow diagrams, chemical reactions, and control panels were also covered. I have been offered the opportunity to engage in extensive learning at a highly esteemed PS Station, recognised as a prominent entity within the field of Chemical Engineering.

Academic courses relevant to the project: Chemical Process Calculation; Fluid Mechanics; Advanced Fluid Mechanics; Numerical Methods of Chemical Calculation; Heat Transfer.

Learning Outcome: This study aims to provide a comprehensive overview of the process and production engineering involved in the production of various petroleum products, namely LPG, Naptha, Gasoline, Diesel, ATF, SKO, Bitumen, and Sulphur, derived from crude oil with high levels of impurities and viscosity. The production of these petroleum products requires a series of intricate processes and engineering techniques to effectively refine the crude oil.
The present study involves acquiring proficiency in utilising the ASTM D86 method and comprehending the concept of regime flow. These skills will be applied in conjunction with the DWSim software to determine the fluid and vapour velocities at the outlet, as well as calculate the Reynolds number. Fouling phenomena occur within heat exchangers, specifically in various configurations such as U-tube, kettle, and other shell-and-tube heat exchangers. Understanding the safety considerations and implementation of Total Production Maintenance (TPM) in heavy-duty industries, such as refineries, is of paramount importance.

PS-I station: IOCL - Operations/Maintenance , Mathura

Student

Name: PULKIT TRIPATHI .(2021B4A42779P)

Student Write-up:

PS-I Project Title: Pumps and Valves

Short Summary of work done: Main work is to get an experience of how an actual chemical plant works and what are the limitations of the real world.Learning how to approach an engineering problem practically not theoretically.

Objectives of the project: to gain deep learning about how pumps and valves are used in the real large scale refineries.

Tool used: no

Details of Papers/patents:no

Brief description of the working environment: nice working environment but also depends on your IOCL mentor. Not too much work load .Main learning is the experience that you get by working with real engineers.

Academic courses relevant to the project: Thermodynamics

Learning Outcome: Experiential knowledge of industry and deep knowledge of centrifugal pumps

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PS-I station: IOCL - Retail Automation , Mumbai

Student

Name: TANISH NEVATIA .(2021AAPS0313P)

Student Write-up:

PS-I Project Title: UI/UX Design

Short Summary of work done: During my internship at IOCL, I focused on UI/UX development for the intranet site. I gained valuable insights into UI/UX principles and best practices, utilizing design tools like Figma to create engaging interfaces. Conducting thorough audits and user surveys, I collected valuable feedback to drive data-driven improvements. Working collaboratively with my mentor, Mr. Arun Verma Sir, and fellow interns, we successfully implemented user-driven design changes, elevating the overall user experience. This enriching journey at IOCL has equipped me with practical skills and

a deeper understanding of UI/UX, fostering my passion for creating intuitive and user-centric digital platforms.

Objectives of the project: These objectives included gaining knowledge and expertise in UI/UX principles, conducting an audit of the intranet site, creating a user survey, and providing recommendations for improving the site's UI/UX experience.

Tool used: Figma, Lighthouse, Canva

Details of Papers/patents: Report named 'UI/UX Design'

Brief description of the working environment: During my PS-I at IOCL, the working environment was dynamic and collaborative, fostering professional growth. Clear expectations were set, encouraging dedication and proactive participation. I gained valuable insights and technical skills through mentorship and engaging with experienced professionals. IOCL's commitment to continuous learning was evident through regular training sessions. Overall, the experience was transformative, equipping me with the knowledge and confidence to excel in my responsibilities and embrace new challenges. Grateful for the opportunity to contribute to such a prestigious organization.

Academic courses relevant to the project: Technical Report Writing

Learning Outcome: Acquired proficiency in using Figma and its various features for designing engaging user interfaces.
Developed training and presentation skills, effectively delivering complex concepts to a diverse audience.
Strengthened knowledge of design tools used in the industry and their applications.

PS-I station: IRCS - Non Tech , Delhi

Student

Name: THAREJA DHRITI(2021A8PS1659G)

Student Write-up:

PS-I Project Title: 1. E-LEARNING MODULE 2. AUDIO-VISUAL PROJECT FOR BLOOD DONATION 3. FINANCIAL ANALYSIS OF IN-KIND DONATIONS

Short Summary of work done: Worked with the group on three projects. First, we developed E-Learning Modules as per the manual provided by IRCS for their First-Aid course. Second, we created a 4-minute video about the myths of blood donation. Third, we made a ppt recommending the appropriate recording of in- kind donations in their book of accounts.

Objectives of the project: 1. Creating ppt slides following the content of the First-Aid manual of IRCS's First Aid Course 2. Creating a video about the myths of blood donation 3. Analysing how in-kind donations can be recorded in the book of accounts

Tool used: Canva, Animaker, Narakeet

Details of Papers/patents:NA

Brief description of the working environment: Everyone at IRCS was welcoming and assured our comfort at all times. They assisted us whenever required. We had frequent contact with our mentor. We were also introduced to someone from the finance department to help us with the finance project.

Academic courses relevant to the project: NA

Learning Outcome: Graphic design, video editing, animation, financial analysis, content writing

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PS-I station: IRCS - Non Tech , Delhi

Student

Name: SAMAY GOYAL .(2021ABPS2486P)

Student Write-up:

PS-I Project Title: 1. E-LEARNING MODULE 2. AUDIO-VISUAL PROJECT FOR BLOOD DONATION 3. FINANCIAL ANALYSIS OF IN-KIND DONATIONS

Short Summary of work done: Worked with the group on three projects required by the organisation. Firstly, developed E-Learning Modules as per the manual provided by the Indian Red Cross Society for their First-Aid course. Secondly, created a 4-minute video highlighting and dispelling prevalent blood donation myths. Lastly, was responsible for

providing recommendations to the organisation on the appropriate recording of in- kind donations in their book of accounts.

Objectives of the project: 1. to devise a short course for students who give the exam for jobs like conductors etc. 2. devise an awareness campaign using video editing skills for blood donation . 3 . present suggestions of how the organisation should manage their finances when it c

Tool used: Canva, Animaker , Prezi

Details of Papers/patents:N/A

Brief description of the working environment: The working environment is cool enough. The CA was provided by the organisation only and was very helpful . We were guided by the mentor regularly and had cordial relations with her. Learning was majorly self learning.

Academic courses relevant to the project: Not any

Learning Outcome: Learnt the skills of video editing , interpersonal communication skills , how to do financial research using balance sheet

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PS-I station: IRCS - Non Tech , Delhi

Student

Name: YASH GOYAL .(2021B2A12764P)

Student Write-up:

PS-I Project Title: 1. E-LEARNING MODULE 2. AUDIO-VISUAL PROJECT FOR BLOOD DONATION 3. FINANCIAL ANALYSIS OF IN-KIND DONATIONS

Short Summary of work done: Worked with the group on three projects required by the organisation. Firstly, developed E-Learning Modules as per the manual provided by the Indian Red Cross Society for their First-Aid course. Secondly, created a 4-minute video highlighting and dispelling prevalent blood donation myths. Lastly, was responsible for providing recommendations to the organisation on the appropriate recording of in- kind donations in their book of accounts.

Objectives of the project: 1. to devise a short course for students who give the exam for jobs like conductors etc. 2. devise an awareness campaign using video editing skills for blood donation . 3 . present suggestions of how the organisation should manage their finances when it c

Tool used: Canva, Animaker , Prezi

Details of Papers/patents:NA

Brief description of the working environment: The working environment is cool enough. The CA was provided by the organisation only and was very helpful . We were guided by the mentor regularly and had cordial relations with her. Learning was majorly self learning.

Academic courses relevant to the project: not as such

Learning Outcome: Learnt the skills of video editing , interpersonal communication skills , how to do financial research using balance sheet

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PS-I station: IRCS - Non Tech , Delhi

Student

Name: VANSH .(2021B4A10841P)

Student Write-up:

PS-I Project Title: 1. E-LEARNING MODULE 2. AUDIO-VISUAL PROJECT FOR BLOOD DONATION 3. FINANCIAL ANALYSIS OF IN-KIND DONATIONS

Short Summary of work done: course. Secondly, created a 4-minute video highlighting and dispelling prevalent blood donation myths. Lastly, was responsible for providing recommendations to the organisation on the appropriate recording of in- kind donations in their book of accounts.

Objectives of the project: 1. to devise a short course for students who give the exam for jobs like conductors etc. 2. devise an awareness campaign using video editing skills

for blood donation . 3 . present suggestions of how the organisation should manage their finances when it c

Tool used: Canva, Animaker , Prezi

Details of Papers/patents:NA

Brief description of the working environment: The working environment is cool enough. The CA was provided by the organisation only and was very helpful . We were guided by the mentor regularly and had cordial relations with her. Learning was majorly self learning

Academic courses relevant to the project: not as such

Learning Outcome: Learnt the skills of video editing , interpersonal communication skills , how to do financial research using balance sheet

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PS-I station: IRCS - Non Tech , Delhi

Student

Name: ANMOL MURTI .(2021B4A82539H)

Student Write-up:

PS-I Project Title: E modules, blood donation, financial analysis

Short Summary of work done: Make informational presentations, video about blood donation

Objectives of the project: Visual learning, myths of blood donation

Tool used: Video editing, script writing, PowerPoint

Details of Papers/patents: No

Brief description of the working environment: Good working environment

Academic courses relevant to the project: No

Learning Outcome: Humanitarian aid, do's and don't of blood donation

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PS-I station: IRCS - Nontech , Bhopal

Student

Name: AARYAN SINHA .(2021A3PS0098P)

Student Write-up:

PS-I Project Title: First aid training web development

Short Summary of work done: The creation of the first aid training registration website involved several key steps and components: 1. **Planning and Requirements Gathering:** The process began with comprehensive planning and discussions to identify the project's goals and requirements. Stakeholders outlined the objectives of the website, such as providing easy access to first aid training courses, enabling online registration, and managing participant data efficiently. 2. **Design and User Interface:** A user-friendly and intuitive interface was designed to ensure a smooth user experience. The website's layout, color scheme, and navigation were carefully crafted to facilitate easy registration and course browsing. 3. **Front-End Development:** Skilled front-end developers implemented the approved design, using HTML, CSS, and JavaScript. They ensured that the website was responsive, compatible with various devices and browsers, and visually appealing. 4. **Back-End Development:** The back-end development team worked on the server-side functionality of the website. They used programming languages such as Python, PHP, or Ruby, and a database system (e.g., MySQL or PostgreSQL) to store and manage registration data, course details, and user accounts. 5. **Registration and Payment Processing:** Integration with secure payment gateways was implemented to facilitate online payments for course registrations. This enabled users to select their preferred training course, provide necessary details, and make payments seamlessly. 6. **User Authentication and Security:** To protect users' personal information and ensure privacy, robust user authentication and security measures were incorporated. Encryption methods were used to safeguard sensitive data. 7. **Course Management System:** An admin panel was created to manage training courses, including scheduling, updating course content, and handling instructor details. 8. **Notifications and Communication:** The website incorporated an automated email notification system to send registration confirmations, course reminders, and updates to participants. 9. **Testing and Quality Assurance:** Rigorous testing was performed to

identify and fix any bugs, usability issues, or security vulnerabilities. This included both manual testing by QA teams and automated testing where feasible. 10. **Deployment and Launch:** Once the website passed all tests and met the project's requirements, it was deployed to a production server, making it accessible to the public. 11. **Marketing and Promotion:** To attract users and increase visibility, marketing efforts such as search engine optimization (SEO), social media campaigns, and online advertisements were undertaken. 12. **Maintenance and Updates:** After the website's launch, regular maintenance and updates were performed to ensure its smooth operation, fix any issues that arose, and add new features or courses as needed. Overall, the first aid training registration website aimed to make the process of signing up for first aid courses easy, efficient, and accessible to a wide audience, contributing to the promotion of first aid knowledge and preparedness in the community.

Objectives of the project: to make a registraion website for ircs bhopal for their first aid program and including apis

Tool used: VS code

Details of Papers/patents: None

Brief description of the working environment: IRCS Bhopal operates with a clear mission to alleviate human suffering, promote health and well-being, and empower vulnerable communities. The organization's employees and volunteers are driven by a shared sense of purpose and dedication to making a positive impact on people's lives.

Academic courses relevant to the project: None

Learning Outcome: web languages like html css java appscript and api

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PS-I station: IRCS - Nontech , Bhopal

Student

Name: BHAVESH MAURYA .(2021A4PS2899H)

Student Write-up:

PS-I Project Title: Web Development

Short Summary of work done: Developed a data-entry website which could store all the data of funds that was recieved to IRCS by different schools with a user friendly interface.

Objectives of the project: Developing a data entry website

Tool used: HTML, CSS, Google Sheets

Details of Papers/patents:N/A

Brief description of the working environment: he company fosters a positive and inclusive working environment, where collaboration, innovation, and teamwork are highly valued.During your internship, the company expects you to be committed, dedicated, and eager to learn. They value your unique perspective and fresh ideas, and they encourage you to actively contribute to projects and initiatives.Throughout your internship, you will have the chance to engage in hands-on learning and gain practical experience in your chosen field.

Academic courses relevant to the project: None

Learning Outcome: HTML, CSS, Google Sheets

PS-I station: IRCS - Nontech , Bhopal

Student

Name: KARTIKEYA GUPTA(2021A4PS3043G)

Student Write-up:

PS-I Project Title: CSR Project

Short Summary of work done: Collected data and documents and compiled then to create our CSR project

Objectives of the project: To create a CSR for a company

Tool used: Google Docs, Google Presentation

Details of Papers/patents: None

Brief description of the working environment: The working environment was really good and I enjoyed working here

Academic courses relevant to the project: None

Learning Outcome: CSR Filing

PS-I station: IRCS - Nontech , Bhopal

Student

Name: SHIVANI SAINI .(2021B1A10825P)

Student Write-up:

PS-I Project Title: First aid training program website development

Short Summary of work done: I designed the website for first aid training programme and also designed portal for it

Objectives of the project: To develop a website for training individuals in first aid

Tool used: Visual studio code, word press, sublime text for HTML and CSS

Details of Papers/patents: No

Brief description of the working environment: The working environment was growth oriented that helped in developing new skills and transparency was maintained

Academic courses relevant to the project: Computer programming taught in first year served as a base for this project.

Learning Outcome: Website designing, portal designing

PS-I station: IRCS - Nontech , Bhopal

Student

Name: ARINJAY SAXENA .(2021B1A12274P)

Student Write-up:

PS-I Project Title: First aid portal

Short Summary of work done: During this project, I embarked on the journey of learning how to develop a data entry portal using Google Apps Script and Google Sheets, and connecting it to a website hosted on GoDaddy. This experience has been incredibly valuable for my personal growth and skill development in the fields of scripting languages, web development, and database management. As I delved into the functionalities of Google Apps Script, I discovered its versatility in creating custom functions and automating tasks within the Google Workspace environment. Through hands-on experimentation with code, I gained a deeper understanding of its potential for practical applications, and this newfound knowledge has opened up exciting possibilities for future projects. Working with Google Sheets as the database for the data entry portal was eye-opening. I learned essential concepts of data management, data validation, and real-time collaboration. Designing a user-friendly interface for data input and implementing data validation rules were particularly valuable skills that I honed during the process. Understanding how to ensure data security while facilitating multiple users to work simultaneously added to the complexity of the project but also deepened my knowledge of data handling. Integrating the data entry portal with a website hosted on GoDaddy introduced me to the world of web development practices. I explored front-end design and focused on creating a seamless user experience. Additionally, understanding the intricacies of back-end integration provided valuable insights into how different components work together to form a cohesive system. Beyond technical aspects, I also gained valuable insights into project management, client communication, and time management. Working on a real-world project from start to finish allowed me to experience the challenges that may arise during implementation and helped me develop problem-solving and critical thinking skills. This learning experience has significantly broadened my skill set and opened up exciting career opportunities in web development, software engineering, and data management. Whether pursuing personal growth or aiming for a career change, the knowledge and skills acquired during this project have set the foundation for my success in the tech industry. I am eager to apply these learnings in future endeavors and continue expanding my expertise in this ever-evolving field.

Objectives of the project: Developing a portal to enter data of first aid and deploy it on ircs mp website

Tool used: Visual code studio, GoDaddy

Details of Papers/patents: Not applicable

Brief description of the working environment: The company was amazing the mentors were very supportive and helping.

Academic courses relevant to the project: learned team coordination and communication. Learned basics of web development

Learning Outcome: Learned how to use APIs and Javascript

PS-I station: IRCS - Nontech , Bhopal

Student

Name: RACHIT CHOUDHARY(2021B3A42147G)

Student Write-up:

PS-I Project Title: Corporate Social Responsibility (CSR)

Short Summary of work done: I was assigned to draw a project report as the CSR proposal for IRCS Bhopal. The proposal was to get the equipment and machines for the X-Ray and Dental unit for Red Cross Hospital, Bhopal. I interacted with the hospital to understand their requirements. I interacted with staff from administration and accounts to get the official documents.

Objectives of the project: The primary objective of this project was to get the equipment and machines for the X-Ray and Dental unit for the Red Cross Hospital, Bhopal.

Tool used: Google Docs

Details of Papers/patents: None

Brief description of the working environment: IRCS Bhopal has a remarkable reputation for fostering a friendly and inclusive working environment. With a team of dedicated staff and volunteers, the atmosphere is characterized by warmth, respect, and

collaboration. The organization encourages open communication and values the input of every team member, irrespective of their position or tenure. The organization recognizes the importance of work-life balance and offers flexible working hours.

Academic courses relevant to the project: None

Learning Outcome: Working on a CSR project helped me understand hospital management and the procedure and requirements of the CSR project. It also helped improve project management and team-building skills by giving the industry exposure.

PS-I station: IRCS - Nontech , Bhopal

Student

Name: VAIBHAV SHARMA .(2021B4A12328P)

Student Write-up:

PS-I Project Title: First aid portal

Short Summary of work done: Created a web portal using js, typescript and html . Integrated google translate API

Objectives of the project: To create a first aid portal

Tool used: Google translate API tool, typescript,html,javascript

Details of Papers/patents: No

Brief description of the working environment: Working environment was good

Academic courses relevant to the project: No

Learning Outcome: Learnt web development

PS-I station: IRCS - Nontech , Chandigarh

Student

Name: AMAN PHOGAT .(2021A7PS0582P)

Student Write-up:

PS-I Project Title: App development

Short Summary of work done: I made an app for the Punjab State Red Cross which will aid them in collecting blood and donations easily through the app and our challenge was also to make the app user friendly.

Objectives of the project: Developing an app for the Punjab State Red Cross

Tool used: Java, flutter, kotlin

Details of Papers/patents: No

Brief description of the working environment: The organisation had a healthy working environment, staff was eager to help us and assist us in the project, the company readily provided all the resources we needed.
I learned how to work in teams and improved my presentation skills.

Academic courses relevant to the project: Object oriented programming

Learning Outcome: Teamwork and Team leading skills

PS-I station: IRCS - Nontech , Chandigarh

Student

Name: PRIYANSHU VERMA(2021B1TS2081P)

Student Write-up:

PS-I Project Title: website development

Short Summary of work done: As part of the talented team at IRCS Company, I was entrusted with the exciting opportunity to develop their dynamic website. Combining my skills in HTML, Java, and WordPress, we crafted a versatile and engaging online platform that represents the company's values and offerings. With HTML as the foundation, we created a user-friendly interface and intuitive navigation, providing a seamless browsing experience. Integrating Java elements, I added interactive features, animated content, and personalized user interactions, enhancing visitor engagement. Leveraging the power of WordPress, we ensured easy content management, enabling swift updates and adaptability to market trends. The result is a visually captivating, responsive, and dynamic website that reflects IRCS Company's vision. Our team's collaborative effort has contributed to IRCS Company's digital success, positioning them as a leading player in their industry

Objectives of the project: To build a website for IRCS , learning backend ,front end , different langauges

Tool used: WORDPRESS , HTML , JAVA , CSS

Details of Papers/patents:NONE

Brief description of the working environment: The working environment at IRCS Company fosters a culture of collaboration, creativity, and growth. Team members are encouraged to communicate openly, share ideas, and work together to achieve common goals. The company values diversity and inclusivity, creating a supportive atmosphere where everyone's contributions are recognized and appreciated. IRCS Company encourages employees to take ownership of their projects and initiatives, fostering a sense of responsibility and empowerment. Learning is a core pillar at IRCS Company, with ample opportunities for professional development. Employees are encouraged to expand their skill sets through workshops, training programs, and access to cutting-edge resources. The company values continuous improvement and actively supports employees in achieving their career aspirations. Overall, IRCS Company's working environment is conducive to growth, both professionally and personally. The company's expectations drive individuals to reach their full potential, while the emphasis on learning ensures that employees stay at the forefront of their respective fields, contributing to the company's ongoing success.

Academic courses relevant to the project: Learning of figma , html , css

Learning Outcome: Learning backend , frontend different languages ex. PHP, JAVASCRIPT , HTML, website designing using figma , WordPress .

PS-I station: IRCS - Nontech , Chandigarh

Student

Name: YAKSHAYAM VATS .(2021B2A12746P)

Student Write-up:

PS-I Project Title: Non Tech

Short Summary of work done: I was responsible for designing the website of IRCS,Punjab. I used a software called figma and canva to design the website

Objectives of the project: To redesign the preexisting website of IRCS

Tool used: Figma, Canva and Wordpress

Details of Papers/patents: None

Brief description of the working environment: The working environment was great and all the people were extremely supportive.

Academic courses relevant to the project: Computer Programming

Learning Outcome: To learn the basics of using Figma and UI/UX Design

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PS-I station: IRCS - Nontech , Chandigarh

Student

Name: ARPITA(2021B3TS2051P)

Student Write-up:

PS-I Project Title: Web development

Short Summary of work done: At our PS, we had firstly learnt about various departments and working of IRCS and afterwards we were assigned two projects i.e. to make an app and website for punjab red cross. Before midsem, I worked on the UI/UX designing and designed a website for IRCS. Afterwards, I worked on the frontend part.

Objectives of the project: To make a working website for IRCS.

Tool used: Figma, Wordpress

Details of Papers/patents: Nil

Brief description of the working environment: The working environment of the Indian Red Cross Society in Punjab is characterized by a dedicated team of staff and volunteers who engage in humanitarian activities. They respond to disasters, provide medical services, organize blood donation drives, implement community health programs, and conduct awareness campaigns.
From the company we learned a lot about humantarian activities, managing events and camps, app and web development.

Academic courses relevant to the project: Computer programming

Learning Outcome: UI/UX designing and knowledge about the languages such as html, css and javascript.

PS-I station: IRCS - Nontech , Chandigarh

Student

Name: PARTH KHANNA .(2021B5A12794P)

Student Write-up:

PS-I Project Title: Web Development

Short Summary of work done: I started from scratch learning web development, in which I learned 7 different programming language and few other things like colour theory and Wordpress.

Objectives of the project: To redesign the website

Tool used: WordPress, VS Code, Python, HTML, CSS, Django, PHP, JavaScript.

Details of Papers/patents:none

Brief description of the working environment: Very Good, Friendly, Helping, Professional.

Academic courses relevant to the project: Computer Programming,WordPress, VS Code, Python, HTML, CSS, Django, PHP, JavaScript.

Learning Outcome: Learned web development

PS-I station: IRCS - Nontech , Chandigarh

Student

Name: NISHA SATYAPAL(2021D2TS2091P)

Student Write-up:

PS-I Project Title: Website

Short Summary of work done: In my ps-1 i learn lot of thing . we have to re design the website of ircs in this i was working on content writing and verification of the information

Objectives of the project: Redesign the website of ircs punjab

Tool used: Canva

Details of Papers/patents: None

Brief description of the working environment: The working environment, expectation from the company, learning during PS-1 is good

Academic courses relevant to the project: Nil

Learning Outcome: Major learning of this internship is content writing, communication ,coding and many more thing

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PS-I station: IRCS - Nontech , Lucknow

Student

Name: VIKHYAT GIRI .(2021A4PS2670P)

Student Write-up:

PS-I Project Title: Web development for Ircs

Short Summary of work done: I made a webite with the help of html, css and javascript for the ircs. It was a blood donation website

Objectives of the project: Built a website as a front-end developer particularly blood donation website

Tool used: html, css and js

Details of Papers/patents:none

Brief description of the working environment: it was a good working experience but no formal sitting place was given to the students in the company.

Academic courses relevant to the project: None

Learning Outcome: Learnt front-end developer

PS-I station: IRCS - Nontech , Lucknow

Student

Name: SHUBHASHISH SINHA .(2021B2A12760P)

Student Write-up:

PS-I Project Title: Junior and Youth proposal for IRCS

Short Summary of work done: Not a good station that should be continued

Objectives of the project: Creating proposal for youth and junior to help them understand red cross

Tool used: Canva

Details of Papers/patents: None

Brief description of the working environment: No expectations left

Academic courses relevant to the project: None

Learning Outcome: Nothing

PS-I station: IRCS - Tech , Bhopal

Student

Name: BHAVYA GARHWAL .(2021A3PS2111H)

Student Write-up:

PS-I Project Title: Fund Management System

Short Summary of work done: We have to built a short fund management website for the red cross so that they can manage their funds properly.

Objectives of the project: We have to build a website which manages all the funds of JRC and YRC.

Tool used: HTML, CSS, JAVASCRIPT

Details of Papers/patents:none

Brief description of the working environment: They had a good working environment, flexible hours of working and no pressure from the staff.

Academic courses relevant to the project: None till now.

Learning Outcome: Front end and back end development

PS-I station: IRCS - Tech , Bhopal

Student

Name: SATYAM KUMAR(2021B1A42813G)

Student Write-up:

PS-I Project Title: Fund Management System

Short Summary of work done: During my internship at the Indian Red Cross Society in Bhopal, I had the opportunity to develop a fund management system using HTML, CSS, and Google Sheets. Additionally, I successfully hosted the system using GitHub. This project aimed to streamline and improve the organization`s financial management processes, enabling efficient tracking and utilization of funds. The fund management system I created was designed to be user-friendly and intuitive, ensuring the organization's staff could easily navigate and operate it. By utilizing HTML and CSS, I implemented a visually appealing and responsive user interface that facilitated seamless interaction. To ensure real-time data synchronization and accessibility, I integrated Google Sheets into the system. This feature enabled authorized personnel to access and modify financial information from various devices and locations, promoting collaboration and enhancing efficiency. Moreover, Google Sheets provides robust data storage and

security, mitigating the risk of data loss or unauthorized access. Hosting the fund management system on GitHub further enhanced its accessibility and convenience. This facilitated easy adoption and reduced dependency on specific devices or software. Overall, my internship project at the Indian Red Cross Society involved the development of a fund management system using HTML, CSS, and Google Sheets. This system aimed to streamline financial operations, improve record-keeping, and enhance collaboration within the organization. By leveraging GitHub for hosting, I ensured easy access and increased usability. Through this project, I gained valuable experience in web development, data management, and teamwork, contributing to the organization's mission of serving the community more effectively.

Objectives of the project: The need for an efficient fund management system became apparent due to the increasing volume of donations received by the Indian Red Cross Society Bhopal. As the organization grew, the existing manual processes for recording and tracking donations became

Tool used: Google Sheets,Github

Details of Papers/patents: Not Any

Brief description of the working environment: During my (PS-I) at the Indian Red Cross Society in Bhopal, I was immersed in a dynamic and supportive working environment.
From the company, I had high expectations of gaining practical experience in the field of fund management and honing my skills in web development. I aimed to contribute meaningfully to the organization's objectives while learning from experienced professionals. Additionally, I sought opportunities to improve my project management and communication skills through interactions with colleagues and supervisors.
Throughout my PS-I, I had the privilege of working on developing a fund management system using HTML, CSS, and Google Sheets. This experience provided invaluable insights into web development, database management, and user interface design.
Moreover, during my time at the Indian Red Cross Society allowed me to develop essential soft skills. I improved my ability to work in a team, communicate effectively, and adapt to changing project requirements.
Overall, my PS-I experience at the Indian Red Cross Society provided a stimulating and enriching learning environment. The internship not only met my expectations but also exceeded them, offering me a solid foundation for my future career endeavors.

Academic courses relevant to the project: CS F111,POE

Learning Outcome: I had learned about Time Management, Communication Skills, Web Development Technologies like HTML,CSS,JS. Additionally, I had also learned about Google Sheet Integration into the website & used hosting platform like github.

PS-I station: IRCS - Tech , Chandigarh

Student

Name: KAVISH .(2021A7PS0559P)

Student Write-up:

PS-I Project Title: App Development

Short Summary of work done: During PS-1 we worked as a team to develop a mobile application for Indian Red Cross Society. The application has many features which are helpful for the Society and keep relevant information about the work done and works to be done by the Society in future.

Objectives of the project: To Develop a user friendly app for the organisation

Tool used: Flutter, Java, Kotlin, node.js

Details of Papers/patents: No

Brief description of the working environment: As a computer science intern at IRCS, I experienced a dynamic working environment that combines technical expertise with a humanitarian focus. Working alongside a team of dedicated teammates , I had the opportunity to collaborate and innovate, contributing to meaningful projects.

Academic courses relevant to the project: Object Oriented Programming

Learning Outcome: Team Management, leadership, team work

PS-I station: IRCS - Tech , Chandigarh

Student

Name: LAKSHAY GARG(2021A7PS2056G)

Student Write-up:

PS-I Project Title: Development of app for IRCS

Short Summary of work done: Gaining valuable experience and skills in the field, I had the opportunity to work on app development projects during my internship. Here's a short summary of the accomplished work - I worked closely with the app development team and actively contributed to the project. Utilizing programming languages and frameworks specific to app development, I have acquired knowledge in Kotlin language for Android. In order to me-et project goals and deadline-s, I actively interacted with my group of team membe-rs. This resulted in effective- teamwork. Overall, my internship was a great experience.

Objectives of the project: Objective of the project is the development of an android application that helps IRCS in it's functionality and along with that it should be easy to interact and learn for the end users.

Tool used: Kotlin, Flutter, Android Studio

Details of Papers/patents:NIL

Brief description of the working environment: The work environment during my internship program was very good. I am part of a team that supports open communication, where everyone is invited to share ideas and implement those on the projects. The workplace creates an atmosphere for learning and growth.
The company expected us to make an app based on the suggestions that were provided by them. The app should contain all the information about different work done by them and also services provided by them.
I learnt various development processes. I also gained some experience in working with languages such kotlin and also android studio while learning how to build the app. Also working in a team environment allowed me to develop better communication skills.
Overall, my internship helped me in learning various new things.

Academic courses relevant to the project: CodeLab for Android Kotlin Fundamentals
https://www.youtube.com/@flutterdev
https://www.udemy.com/course/android-studio-masterclass/

Learning Outcome: Major Learning outcomes are understanding the development process, learning different programming languages, management of the project and adapting to different changes overtime.

PS-I station: IRCS - Tech , Chandigarh

Student

Name: VILAKSHAN GUPTA(2021A7PS2075G)

Student Write-up:

PS-I Project Title: Web Development

Short Summary of work done: Designed the website of the Punjab Red Cross Society in wordpress.

Objectives of the project: To design a website for the organization

Tool used: WordPress (S/W)

Details of Papers/patents:-

Brief description of the working environment: Here at Punjab Red Cross Society, the working environment was very good, the staff was very cooperative and helped the students in whatever they could. Everyone worked together as a team for the project assigned to us.

Academic courses relevant to the project: Web Development

Learning Outcome: Learnt HTML,CSS , JavaScript

PS-I station: IRCS - Tech , Delhi

Student

Name: JOY SINHA .(2021A8PS1606H)

Student Write-up:

PS-I Project Title: FDRS form portal

Short Summary of work done: The work was fairly related to the domain which I had applied for. The work was entirely web development which helped us gain first hand experience for a real world project.

Objectives of the project: The objective was to create a data entry form which would collect data from the user and store it in the existing database of IRCS.

Tool used: React for frontend, nodejs for backend.

Details of Papers/patents: None

Brief description of the working environment: The working environment was mediocre at best. There is no water filling station so you're on your own for that. The washroom is fairly clean. For lunch, you either need to bring your own food or go have it from nearby canteen or you can use the metro to go wherever to eat if you can manage the time. There is no dress code and the working hours are not very strict either. Overall a fine experience.

Academic courses relevant to the project: None

Learning Outcome: Web development

PS-I station: IRCS - Tech , Delhi

Student

Name: MAYANK KUMAR SINHA(2021B4TS2078P)

Student Write-up:

PS-I Project Title: Quality Assurance and Bug Reporting (JioEngage Features)

Short Summary of work done: During our internship, we accomplished various tasks that enhanced our skills and knowledge. Firstly, we created user flow reports for the PLAY&WIN section of the MYJIO app and the BOURNVITA TJK engagement, providing detailed insights into the user journey. We also formulated queries for the TAP2WIN

feature and test cases for the BOURNVITA TJK engagement, ensuring comprehensive testing coverage. Additionally, we generated queries, test cases, and test scenarios for the FUNZONE section of PLAY&WIN, enabling thorough testing of its functionalities. We documented a list of bugs encountered during testing and even created a hypothetical bug list to understand the distinction between priority and severity. Furthermore, we formulated queries and test cases for the Scratch Card widget, contributing to its development and testing process. As part of the DevOps platform, we added features, user stories, test cases, and bugs for both the FUNZONE section and the Scratch Card widget, promoting collaboration between developers and testers. Through these tasks, we gained practical experience in user flow analysis, query formulation, test case development, and effective use of the DevOps platform.

Objectives of the project: To check the features that are released in JIO ENGAGE for any bugs and report any bugs found.

Tool used: MS EXCEL, MS WORD, FIGMA, AZURE DEVOPS

Details of Papers/patents:NONE

Brief description of the working environment: Over the course of the past 35-40 days, our PS has been a rich learning experience. We started by immersing ourselves in the Jio Engage platform and understanding its functionalities. We then delved into the world of software development, discovering the crucial roles played by teams like CSM, product management, and QA in ensuring the smooth and timely delivery of bug-free products and features. Learning about the Software Development Life Cycle (SDLC) provided us with a structured approach, covering planning, product definition, design, development, testing, and deployment. A particular focus was placed on the responsibilities of the QA team, involving understanding the product, creating meticulous test cases, and striving for minimal bugs. We comprehended the testing life cycle and honed our skills in documenting user flow reports, generating queries, and formally documenting test cases and bugs. Moreover, we explored business concepts underlying engagements on the MYJIO app and gained insights into admin settings, bug life cycles, severity vs. priority in bug reporting, and the creation and documentation of different testing scenarios. Finally, we were introduced to the DEVOPS platform, where we learned how to collaboratively work on features by adding user stories, test cases, and bugs. Overall, our internship has provided us with a well-rounded understanding of software development processes, testing methodologies, and effective documentation techniques.

Academic courses relevant to the project: NONE

Learning Outcome: Throughout our internship and the completion of various tasks, we have gained significant learning outcomes:
1.User Flow Analysis: We have acquired the ability to analyze and document user flows, understanding the sequence of steps and interactions involved in different app sections and engagements.

2.Query Formulation: We have developed skills in formulating clear and concise queries based on design specifications, ensuring a thorough understanding of requirements and facilitating effective communication with the development team.
3.Test Case Development: We have honed our skills in creating comprehensive test cases, defining specific steps, and expected outcomes for thorough testing coverage.
4.Bug Identification and Documentation: We have learned how to identify, document, and report bugs encountered during testing, ensuring that they are appropriately logged for resolution.
5.Understanding Severity and Priority: Through the creation of a hypothetical bug list, we have gained a clear understanding of the distinction between severity and priority in bug reporting, allowing us to prioritize issues effectively.
6.Collaboration on DevOps Platform: By working with the DevOps platform, we have learned how to add features, user stories, test cases, and bugs, facilitating seamless collaboration between developers and testers and enhancing overall project management.
7.Practical Application of SDLC: Our tasks have provided us with hands-on experience in applying the Software Development Life Cycle (SDLC), particularly in the areas of planning, design, development, testing, and deployment.
8.Comprehensive Testing Approach: Through the creation of test scenarios, we have learned how to develop diverse testing scenarios, considering different user interactions and potential scenarios to ensure thorough testing coverage.
9.Documentation Skills: We have strengthened our ability to formally document various aspects, including user flow reports, queries, test cases, and bug reports, ensuring clear and organized project documentation.
These learning outcomes have equipped us with valuable skills and knowledge in software development, testing methodologies, bug tracking, and effective communication within a development team.

PS-I station: IRCS - Tech , Delhi

Student

Name: VISHNU GOYAL(2021B5TS2068P)

Student Write-up:

PS-I Project Title: FDRS project, Website Wireframing, Digital Blood Donor form with E-signature

Short Summary of work done: Basically we have completed 3 projects as part of the tech team.In the website wireframing project we made then a website on figma with more user friendly interface, attracting features,etc

Objectives of the project: 1. To build a database system layer wise for the districts then state and finally the NHQ. 2.To build a website in Figma Suggesting changes from the existing website. 3. Converting the Blood donor Form to a Digital Form with an E-Sign.

Tool used: Figma

Details of Papers/patents: Nothing

Brief description of the working environment: The working environment was fine. Could Have been better. The Company met our expectations.

Academic courses relevant to the project: Computer Programming

Learning Outcome: Learning Outcomes included How to use Figma in Detail, Full back-end and front end development.

PS-I station: IRCS - Tech , Delhi

Student

Name: TUSHAR SAINI(2021D2TS2086P)

Student Write-up:

PS-I Project Title: 1. Designing Wireframes for the IRCS website, 2. Integration of eRaktkosh with an online blood, 3. FDRS data collection form-portal

Short Summary of work done: During PS-I, the work involved designing wireframes for the IRCS website, integrating eRaktkosh with an online blood bank, and creating a web-based portal for data collection in the FDRS scheme. The projects focused on improving user interface design, enhancing blood inventory management, and streamlining data collection for better scheme administration.

Objectives of the project: 1. Designing Wireframes for the IRCS website: The objective is to create visual representations of the IRCS website's layout and design, helps to

understand its structure and user interface before development. 2. Integration of eRaktkosh with an online bl

Tool used: Figma, MySQL, MongoDB, Node.js, Django, HTML, CSS, JavaScript, React, Angular

Details of Papers/patents:N/A

Brief description of the working environment: The working environment during PS-I was dynamic and collaborative. The team members likely collaborated on projects, sharing ideas and expertise. Regular meetings and discussions are conducted to ensure smooth progress and address any challenges. The atmosphere was conducive to learning and skill development.
The IRCS organization likely expected us to actively contribute to the assigned projects. They were expected to deliver high-quality work within the given timeframes, demonstrating a strong work ethic and professionalism. Companies also anticipated us to adapt to new challenges.
During PS-I, we as interns had the opportunity to gain practical experience in our chosen field. Learned valuable skills related to website design, system integration, and web-based form development. Enhancing our technical and problem-solving abilities.

Academic courses relevant to the project: Non, self taught, I'll be taking related courses to go on advanced level in my 3rd year as electives.

Learning Outcome: 1. Designing Wireframes: Understanding user interface design principles, information architecture, and effective communication of website layouts.
2. eRaktkosh Integration: Gaining expertise in system integration, real-time data exchange, and efficient blood inventory management.
3. FDRS Data Collection Portal: Developing skills in web-based form development, data validation, and secure data management for improved scheme administration.

PS-I station: IRCS - Tech , Lucknow

Student

Name: DHRUV VERMA .(2021A8PS2711P)

Student Write-up:

PS-I Project Title: Blood donation website

Short Summary of work done: Build a website for IRCS Lucknow

Objectives of the project: To make a user friendly blood donation website for donars and receivers

Tool used: Html css javascript react js php

Details of Papers/patents: Blood donation repository in GitHub

Brief description of the working environment: Build a blood donation website using html css java script and php

Academic courses relevant to the project: ENI branch

Learning Outcome: Frontend Backend Designer

PS-I station: IRCS , Dehradun

Student

Name: SHIRASKAR SOHAM VIJAY.(2020A4PS2108G)

Student Write-up:

PS-I Project Title: App development

Short Summary of work done: Made app, PDFs, questionnaire and speeches for the organization.

Objectives of the project: To create an informative app for increasing reach of the app.

Tool used: Flutter, other mobile apps

Details of Papers/patents:na

Brief description of the working environment: it was all good.

Academic courses relevant to the project: NA

Learning Outcome: Learnt fluttr , figma and dart for app dev.

PS-I station: IRCS , Dehradun

Student

Name: HARSH MALIK .(2021A1PS1661P)

Student Write-up:

PS-I Project Title: Web Development Project

Short Summary of work done: I worked closely with IRCS to develop their website

Objectives of the project: To develop the website for IRCS

Tool used: Django, Figma

Details of Papers/patents: No

Brief description of the working environment: It was a good learning curve

Academic courses relevant to the project: Copywriting

Learning Outcome: Learned about website development

PS-I station: IRCS , Dehradun

Student

Name: ISHIKA BHANDARI(2021A1PS2279G)

Student Write-up:

PS-I Project Title: Social media and app development

Short Summary of work done: Designed app. Held some events. Also conducted quizzes in the uttarakhand state and have out the certificates and prizes. Had first aid training workshop.

Objectives of the project: App development and social media handling

Tool used: -

Details of Papers/patents:-

Brief description of the working environment: Working environment is really good and the organization is very supportive. Learnt a lot of things. The organization was helpful more than I expected.

Academic courses relevant to the project: App development

Learning Outcome: Learnt UI designing

PS-I station: IRCS , Dehradun

Student

Name: JUGRAAJ SINGH .(2021A1PS2589P)

Student Write-up:

PS-I Project Title: App Development

Short Summary of work done: App Development , managing all the social media accounts and activity

Objectives of the project: to work on the app for the IRCS app

Tool used: VS Code

Details of Papers/patents:none

Brief description of the working environment: It is very effective

Academic courses relevant to the project: DSA ,

Learning Outcome: Learned to work in a team , and working on how is the app made.

PS-I station: IRCS , Dehradun

Student

Name: PURUJIT PADHY .(2021A2PS2629P)

Student Write-up:

PS-I Project Title: App development

Short Summary of work done: Designed the complete UI/UX for the app using Canva anf figma software. Figma was taught to me by a senior member of the Technical team. I then worked on improving the overall performance and feel of the app. I also worked on fraud analysis of various companies that applied for first aid training for thier employees. Some companies were applying for certificates without even undergoing any training. I have also helped in digitalization of important data that is to be stored for futher proceses.

Objectives of the project: To create a fully functional informative app that shows various data of services provided by redcross in a district wise fashion.

Tool used: Canva, Figma, Excel

Details of Papers/patents: No

Brief description of the working environment: The working environment was quite healthy and supportive. The mentor, Mr. Harish Chandra Sharma guided us in most of the work and helped us achieve better than what we thought we coud. All the team

members showed enough enthusiasm towards helping each other. Various soft skills were also focused during the PS which helped me uplift my skill levels.

Academic courses relevant to the project: None.

Learning Outcome: UI/UX Design of application, API calling, major functionalities of an app, Figma , Canva.

PS-I station: IRCS , Dehradun

Student

Name: YASHVARDHAN SHARMA .(2021A3PS2645P)

Student Write-up:

PS-I Project Title: Development of Red Cross Uttarakhand App

Short Summary of work done: The work involved creating an app for showing the working of Red Cross, Uttarakhand. The app was made in Flutter (frontend) & Django (Backend).

Objectives of the project: To develop an informative app for Red Cross Uttarakhand

Tool used: Flutter, Django, Dart

Details of Papers/patents: None

Brief description of the working environment: The expectation of the station were quite high, as the app had to be inaugurated by the Governor of Uttarakhand. Yet at the same time, the feature list was manageable according to the time. The support was great, we had professional help whenever we needed, inspite of the station not being a technical one. Our mentor was very helpful and considerate of our needs.

Academic courses relevant to the project: None

Learning Outcome: App Development

PS-I station: IRCS , Dehradun

Student

Name: GURJOT SINGH BINDRA .(2021A4PS2373H)

Student Write-up:

PS-I Project Title: Social Media , app and web dev

Short Summary of work done: Na

Objectives of the project: make app for the company and increase there social media presence

Tool used: na

Details of Papers/patents:none

Brief description of the working environment: not good and boring

Academic courses relevant to the project: nothin

Learning Outcome: corperation and work life balance

PS-I station: IRCS , Dehradun

Student

Name: SUGAM CHAUHAN(2021A4PS2405G)

Student Write-up:

PS-I Project Title: IT/Management/healthcare

Short Summary of work done: very good, helpful employee, very kind

Objectives of the project: Website upgradation, app development

Tool used: HTML, CSS, EXCEL

Details of Papers/patents:none

Brief description of the working environment: very good; had high expectation and lived up to the mark

Academic courses relevant to the project: HTML, CSS

Learning Outcome: website making(understood frontend in detail), social media influencing and handling

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PS-I station: IRCS , Dehradun

Student

Name: ADITYA PANT(2021A4PS2901G)

Student Write-up:

PS-I Project Title: Social Media management

Short Summary of work done: I had the honour of spearheading the Indian Red Cross Society's social media operations and organising statewide online quizzes during my 50-day term there. My primary goal as a responsible student managing their social media platforms was to engage the audience, promote awareness about the society's projects, and educate the public about vital healthcare problems and their possible solutions. I created a series of interactive online quizzes covering a wide range of topics, including first aid, environment day, Youth Day, and blood donation, etc. with thorough planning and implementation. These questions were created to both entertain and educate players, establishing a sense of community and information exchange. I used several social media platforms and tailored advertising methods to maximise the quiz's visibility and assure a successful statewide reach. We used linkedin, Instagram, youtube and twitter to publicize

our work. Furthermore, I worked together with other Indian Red Cross Society team members to obtain important information, generate interesting content, and maintain a consistent brand voice across all social media channels. This joint approach guaranteed that the quizzes were consistent with the society's aims and effectively communicated vital health and safety lessons.

Objectives of the project: 1. Ensure that all the social media handles of IRCS-Uttarakhand are up-to-date. 2. Curate and hold quizzes on various occassions for the IRCS. 3. Design and deliver certificates to the winners and participants of these quizzes

Tool used: Figma, canva

Details of Papers/patents:none

Brief description of the working environment: The environment was very supportive. The mentor was supportive throughout the course of the PS and was diligent with work, and expected good work from us as well. This helped us grow into responsible workers. The company can be expected to be quite reliant. They are very supportive and don't just think of what work they can expect from us, but also were very keen on giving back to us, and as a result had us complete a training on first aid.
There was a lot of practical learning from the PS. I leant skills like team leadership, communication skills, etc.

Academic courses relevant to the project: Business Analysis and Valuation

Learning Outcome: 1. How various social media channels are managed
2. How to conduct online quizzes on a mass scale
3. Team management and leadership.

PS-I station: IRCS , Dehradun

Student

Name: AKHOURI SHAKALYA .(2021ABPS2721P)

Student Write-up:

PS-I Project Title: Web Dev

Short Summary of work done: Helped in creating a website ,official work regarding digitization of official documents,finanacing and accounting.

Objectives of the project: To create a website for Ircs Dehradun

Tool used: HTML,CSS,JAVASCRIPT,Bootstrap,figma

Details of Papers/patents:.

Brief description of the working environment: Working evnvironment was quite good and healthy ,they were very flexible with whatever technology we wanted to work with.

Academic courses relevant to the project: Computer programming

Learning Outcome: Got to learn about various technologies used in web dev ,also learned about finance m

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PS-I station: IRCS , Dehradun

Student

Name: SANYA AGARWAL .(2021B1A32314P)

Student Write-up:

PS-I Project Title: Social media management

Short Summary of work done: Designing certificates for quizzes and organizing events related to important days such as environment day , blood donor day etc.

Objectives of the project: Renewing the social media channels and updating them

Tool used: Canva

Details of Papers/patents: None

Brief description of the working environment: Nice working experience

Academic courses relevant to the project: None

Learning Outcome: Designing

PS-I station: IRCS , Dehradun

Student

Name: ABHISHEK BISHT .(2021B2A12485P)

Student Write-up:

PS-I Project Title: Web dev

Short Summary of work done: Learnt pyhtin, django, bootstrap framework. Made website of ircs, did some office work, learnt to make effective presentations, learnt cpr and first aid techniques

Objectives of the project: To make website of ircs

Tool used: Django, bootstrap, html , css, javascript,

Details of Papers/patents: No

Brief description of the working environment: Very good working environment, learnt web development

Academic courses relevant to the project: C programming

Learning Outcome: Learnt web dev

PS-I station: IRCS , Dehradun

Student

Name: MUDDAM SANJITH .(2021B4PS1112H)

Student Write-up:

PS-I Project Title: Create a website, app

Short Summary of work done: We have developed annapp for the organisation which would help them in registering new users and volunteers. It would also be of great help during the disasters

Objectives of the project: To create a website and app for the organisation

Tool used: Flutter, dart

Details of Papers/patents: None

Brief description of the working environment: It is a small organisation with 4 employees. It is a ngo which is government funded and runs parallel to the government which specialises on the services like disaster management and blood donation

Academic courses relevant to the project: C programming

Learning Outcome: Learnt about frontend development, html , css and flutter

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PS-I station: IRCS , Dehradun

Student

Name: VISHAL THANGEDA(2021B5A42954G)

Student Write-up:

PS-I Project Title: Social media and app development

Short Summary of work done: Designed app. Held some events. Also conducted quizzes in the uttarakhand state and have out the certificates and prizes. Had first aid training workshop.

Objectives of the project: App development and social media handling

Tool used: -

Details of Papers/patents:-

Brief description of the working environment: Working environment is really good and the organization is very supportive. Learnt a lot of things. The organization was helpful more than I expected.

Academic courses relevant to the project: -

Learning Outcome: Learnt UI designing

PS-I station: Johnson Controls (I) Pvt. Ltd. , Pune

Student

Name: RAMANUJ PRALHAD SOMANI(2021A4PS0311G)

Student Write-up:

PS-I Project Title: Cost cutting in sheet metal fabrication

Short Summary of work done: My role was to understand the principles of VAVE (value added value engineered) and to learn about various sheet metal operations. Look into the design guidelines for various operations that are performed on sheet metal. Read and document about various interlocking mechanisms and where they can be used to join sheet metal components.

Objectives of the project: Learn and understand the principles ov value engineering

Tool used: Power Point

Details of Papers/patents: None

Brief description of the working environment: Work environment was focused and relaxed, learnt various principles that goes into product development. Major learnings however were understanding the organization structure various levels of managment, process followed in product development, importance of integrated documentaion softwares, understanding PLM(product life-cycle managment). Understood the role of project manager and various techniques deployed by them to improve efficiency of development.

Academic courses relevant to the project: Manufacturing Process

Learning Outcome: Understanding principles of design for manufacturing and value engineering

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PS-I station: JSW Steel , Vijaynagar

Student

Name: PRAMESH MJ .(2021A1PS3050H)

Student Write-up:

PS-I Project Title: Identification of action learning projects for another set of high potential candidates

Short Summary of work done: Our project is to identify action learning projects for another set of high potential candidates.Senior manager and team created a mail id for the summer intern at JSW Steel and every detail about the project will be communicated through that mail.We received mail from senior manager about the details of employees and managers..We were asked to create a google form to collect review about the 3 workshops from the employees.The review helps them to conduct the upcoming workshop as a part of executive development program.We created a google form and it was floated by senior manager to the 21 employees.The responses from the employees was explaining more about the previous 3 workshops,they also mentioned how was the workshop helpful for the self development.They gave suggestion for the upcoming workshop as well.We were asked to create a powerpoint presentation by senior manager with the responses we received.The template for the google form as well as powerpoint

presentation was given by senior manager.The powerpoint presentation contain all the question, responses from the employees and details about the employees.We collected the comments about the ongoing action learning and suggestion about the action learning project for the well being of company through google form from the managers.

Objectives of the project: To identify action learning project for high potential candiates

Tool used: Software

Details of Papers/patents:1)https://bluepointleadership.com/resource/10-best-practices-action-learning-projects/
2)https://practera.com/action-learning-projects/#:~:text=What%20is%20action%20learning%3F,relevant%20to%20their%20future%20goals.
3)https://business.rice.edu/action-le

Brief description of the working environment: The working environment was excellent.I expected the company to train us to gain more knowledge about the action learning project and its execution.The learning from the PS-1 is how to execute action learning project and collect reviews about the action learning project.The action learning helps to develop interpersonal and professional growth.

Academic courses relevant to the project: Principles of management

Learning Outcome: The major learning outcome is about action learning project and it is executed in a company.

PS-I station: JSW Steel , Vijaynagar

Student

Name: AKSHITA RAJ PANDITA .(2021A4PS0151P)

Student Write-up:

PS-I Project Title: Job analysis using competency Mapping

Short Summary of work done: The project involved identifying and defining key competencies required for successful job performance within the organization. My group

followed a structured methodology that included a comprehensive job analysis through a well-designed survey administered to 168 JSW employees across various departments and levels. The survey responses were meticulously analyzed to gain insights into job responsibilities, required skills, and other attributes. Based on the data analysis, we created competency-based job descriptions, providing clarity to potential candidates, existing employees, and management about roles, responsibilities, and performance expectations. The project aimed to optimize talent management practices, improve performance evaluation, and align individual capabilities with organizational goals. Challenges faced during the project included a lack of enthusiasm from some employees to participate in the survey and limited outreach to potential respondents due to incomplete contact details. We recommended highlighting the survey's importance, using different communication channels, providing feedback to participants, and using multilingual surveys to improve the process.

Objectives of the project: 1. Identify technical job roles within JSW Steel. 2. Define key technical competencies required for each role. 3. Consult human resource business partners (HRBP's) for input and insights. 4. Create a competency framework with clear definitions and profici

Tool used: NA

Details of Papers/patents:NA

Brief description of the working environment: During our internship, we had an online PS station which resulted in limited interaction with my fellow batchmates. To enhance collaboration, our department(HR) was divided into three subgroups, each comprising three members. Our sub-group worked on the same project and shared the responsibility of creating a unified report, which facilitated some level of interaction. However, the workload was relatively light, which offered limited opportunities for socializing. The online meetings with our mentor took place once a week, during which he assigned us manageable tasks. He was understanding of our status as second-year students and ensured we were not burdened with excessive work. His support and guidance throughout the internship made the experience highly rewarding and enjoyable. Working in this company has been an overall positive and enriching experience for all of us.

Academic courses relevant to the project: No courses as such

Learning Outcome: During my internship, I had the opportunity to gain valuable insights into the functioning of an HR department. I delved into the concept of competency mapping, learning about its various types and the overall process involved. Engaging in one-on-one interactions with JSW employees allowed me to understand their experiences and perspectives better. Additionally, I conducted a survey, meticulously analyzed the data, and as a result, developed a comprehensive understanding of the core aspects of different companies. This enriching experience has provided me with practical knowledge and a deeper appreciation for the complexities of HR practices in real-world settings.

PS-I station: JSW Steel , Vijaynagar

Student

Name: SAMARTH JAISWAL .(2021A4PS1352P)

Student Write-up:

PS-I Project Title: Cold Rolling mill

Short Summary of work done: I was working in a team of 4 on Cold Rolling mill of Jsw plant . We were given two different projects. We divided the team into two parts and worked on the individual projectin a team of two.First we learned about Cold Rolling Mill , then the shear blade and how it works with different pressure. We then learned about the ways of wear and tear of the blade and deformitiesformed, Then we worked on the solution of the problem.

Objectives of the project: Shear Blade life Enhancement for high Strength

Tool used: Research Papers on shear blades And material shared by mentor.

Details of Papers/patents:N.A.

Brief description of the working environment: The PS was online so we didnt get the industrial environment the one we expected.But our mentors were really helpful as they showed the industry through videos.The project given was very exciting too.The mentors were very considerate and reachable.

Academic courses relevant to the project: Material Science, Mechanics of solids

Learning Outcome: I learned how a steel plant works ,the shear blades materials used And hw to enhance it.

PS-I station: JSW Steel , Vijaynagar

Student

Name: ADITYA GATTANI .(2021A4PS1510P)

Student Write-up:

PS-I Project Title: Hydraulic Circuit Analysis of Walking Mill in Hot Strip Mill Processes

Short Summary of work done: Our team was deeply engrossed in tackling the challenge of drift in a walking mill, a critical issue that had been impeding production efficiency. Through diligent research and collaborative efforts, we devised innovative solutions to rectify the drift problem, implementing advanced control systems and precision adjustments. Additionally, we focused on optimizing production accuracy by fine-tuning the manufacturing processes, streamlining workflows, and implementing quality checks. The outcome of our endeavors was a significant improvement in production output and accuracy, elevating the overall operational performance and success of the company.

Objectives of the project: Optimizing production accuracy and control hydraulic drift

Tool used: Excel, Docs, Powerpoint

Details of Papers/patents: None

Brief description of the working environment: The work environment at the company was remarkably friendly and conducive to productivity. Colleagues were supportive and approachable, fostering a strong sense of camaraderie among the team members. Collaboration and open communication were encouraged, leading to the seamless exchange of ideas and innovative solutions. The company's commitment to fulfilling all our expectations was evident in its dedication to providing excellent resources, professional growth opportunities, and a healthy work-life balance. This nurturing environment allowed us to thrive personally and professionally, resulting in a motivated and engaged workforce. Overall, our experiences within the organization were highly positive and left a lasting impression of appreciation and contentment.

Academic courses relevant to the project: Fluid Mechanics

Learning Outcome: Reheating Furnaces,Walking beam,Circuit Analysis.

PS-I station: JSW Steel , Vijaynagar

Student

Name: ARJUN RAGHURAMAN .(2021A4PS1648H)

Student Write-up:

PS-I Project Title: Job Analysis using Competency Model

Short Summary of work done: Aim of the project was to conduct job analysis in order to formulate job descriptions. This was done by making a questionnaire and floating it to 100 employees. Results of the questionnaire were evaluated to come up with brief job descriptions.

Objectives of the project: To conduct job analysis by finding technical competencies of employees to formulate a job description.

Tool used: MS Excel, MS Powerpoint

Details of Papers/patents:none

Brief description of the working environment: Had 1-3 meetings per week in online mode to update work done and to be assigned new tasks.

Academic courses relevant to the project: none

Learning Outcome: Leadership, communication, presentation skills

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PS-I station: JSW Steel , Vijaynagar

Student

Name: SHANTANU SUSANTA PANDA(2021A4PS1727G)

Student Write-up:

PS-I Project Title: OPTIMIZING PRODUCTION ACCURACY: HYDRAULIC CIRCUIT ANALYSIS OF WALKING MILL IN HOT STRIP MILL PROCESSES

Short Summary of work done: the relevant topics were studied, following which the circuit diagram of the walking beam was analysed and potential pitfalls were pointed out along with their solutions

Objectives of the project: to study hydraulic circuit and suggest changes to prevent cylinder drift and resulting loss in production time

Tool used: none

Details of Papers/patents:none

Brief description of the working environment: the mentors assigned to us were hard to get a hold off. they have never replied to mails, picked up call, replied to texts etc initially. we never received courteous replies stating that they were busy occupied etc from them either, just radio silence. our fic asked us to wait for their reply as they might be busy with end of month loads, however other groups had recieved adequate material and started regular work, whereas we didn't get a project assigned until late week 2. even after this, our new mentor was slow in passing out materials and instruction/clarification, and never met with us, only made conference calls. he was extremely reluctant to do that even, infact he didnt join a single conference call after midsems, which made our final submission extremely tricky. i would reccomend this to all students, as it would help them grapple the reality of how actually "professional conduct and communication" occur in a real workplace, where people will not acknowledge you unless they themselves require you.

Academic courses relevant to the project: Fluid dynamics, thermodynamics, material science, manufacturing processes

Learning Outcome: learning about the various types of valves etc in a hydraulic system, learning about hydraulic circuit diagrams

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PS-I station: JSW Steel , Vijaynagar

Student

Name: PRADHYUMN SINGH RATHORE(2021A4PS2828G)

Student Write-up:

PS-I Project Title: Project management dashboard

Short Summary of work done: The team was provided with industry data that contained information about the projects going on in the company, with relevant additional information about the same. We were supposed to clean this data and using excel, prepare a dashboard which would ease out the analysis of this data.

Objectives of the project: To clean the data provided by the station and prepare a dashboard for easy analysis

Tool used: Ms excel

Details of Papers/patents: None

Brief description of the working environment: Our mentor was very helpful throughout the project tenure. Not only he helped us with the project, but he would also provide insights about his industry and the working manner in a professional setting.

Academic courses relevant to the project: Data analytics

Learning Outcome: Ms excel, data analysis

PS-I station: JSW Steel , Vijaynagar

Student

Name: M C VRINDA(2021A4PS3077H)

Student Write-up:

PS-I Project Title: Digitalisation project management dashboard

Short Summary of work done: It was a good experience as it helped me in knowing about a part of the working world. I was given a very good team and mentor to work with.Overall it was a very good experience.

Objectives of the project: To create a dashboard to overview the ongoing projects in the company

Tool used: Excel, Power BI

Details of Papers/patents:A report was made regarding the dashboard created

Brief description of the working environment: The working environment was nice as I had a good team and mentor. The company has reached my expectation in showing the overall of its working.

Academic courses relevant to the project: None

Learning Outcome: Learnt how to make a dashboard from scratch

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PS-I station: JSW Steel , Vijaynagar

Student

Name: MINAL SHARMA .(2021A4PS3088H)

Student Write-up:

PS-I Project Title: Hydraulic Circuit Analysis of Walking Mill in Hot Strip Mill Processes

Short Summary of work done: We worked on solving the problem of drift in a walking mill and optimizing production accuracy

Objectives of the project: Optimizing production accuracy and control hydraulic drift

Tool used: Excel

Details of Papers/patents:nil

Brief description of the working environment: It was a very friendly and productive work environment.The company fulfilled all our expectations.

Academic courses relevant to the project: Fluid Mechanics

Learning Outcome: Reheating Furnaces,Walking beam,Circuit Analysis.

PS-I station: JSW Steel , Vijaynagar

Student

Name: KUNAL AGGARWAL .(2021AAPS1476P)

Student Write-up:

PS-I Project Title: Digitalization Project Management Dashboard

Short Summary of work done: We made a Project Management Dashboard to monitor the 600 or so Digitalizion projects being conducted in the department. They included metrics given in tabular form which we organised, sorted and created pivot tables for. after this we created great dynamic visualizations using slicers to incorporate into a dashboard for the upper management to monitor these projects.

Objectives of the project: Create Dashboard to monitor data and find good Insights from it.

Tool used: Microsoft Excel, PowerPoint, Word

Details of Papers/patents:N/A

Brief description of the working environment: Expectations were met, the environment was supportive and professional and we learnt a lot; like how to use Excel, how to handle data and how to work in a professional environment.

Academic courses relevant to the project: Workshop Practice.

Learning Outcome: Tools like Excel, PowerPoint and how to work in a professional environment

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PS-I station: JSW Steel , Vijaynagar

Student

Name: SHIVAM GAUTAM .(2021AAPS3018H)

Student Write-up:

PS-I Project Title: Action learning projects

Short Summary of work done: We worked in a team to establish an effective way of assigning action learning projects to high potential employees.

Objectives of the project: To establish an effective way of assigning action learning projects to high potential employees.

Tool used: Worked with excel and google forms

Details of Papers/patents: None.

Brief description of the working environment: The working environment was positive. The company helped mould myself in the corporate setting. My biggest takeaway would be the importance of team work and how helpful it is.

Academic courses relevant to the project: None.

Learning Outcome: Importance of Action learning projects and importance of teamwork

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PS-I station: JSW Steel , Vijaynagar

Student

Name: HARSHVARDHAN KEJRIWAL .(2021B2A41713P)

Student Write-up:

PS-I Project Title: Optimisation of pickling time for IFHS steel and Sheer Blade life enhancement for high strength steel

Short Summary of work done: My major task was to conduct research on the topic and then apply that knowledge to answer the stated problem statement. As we were unable to visit the PS, I was unable to set up the experiment and collect the necessary data for the study, so I had to rely on research publications to learn more about it. At the conclusion, we had to show our project to our mentors and PS faculty.

Objectives of the project: It was a research project to optimise the pickling time of IFHS steel in a tandem cold mill.

Tool used: No tools were required for the project

Details of Papers/patents:-

Brief description of the working environment: As it was an online station, we were unable to visit it. The PS faculty and project supervisors were very easily accessible and resolved our questions.

Academic courses relevant to the project: None

Learning Outcome: While working on this project, I learned a variety of concepts. I learnt about the steel manufacturing process, how tandem cold mills operate, the many portions of tandem cold mills, and how this manufactured steel is used in various industries.

PS-I station: Kalpataru Power Transmission Ltd. , Noida

Student

Name: AARUSH TEWARI(2021A2PS2163P)

Student Write-up:

PS-I Project Title: Medanta hospital site sector 50

Short Summary of work done: Visisting the site, working with various departments such as P &m, execution,quality ,admin and store. Weekly visits to the batching plant.

Objectives of the project: Analysis of construction processes and machinery

Tool used: Autocad (not mandatory)

Details of Papers/patents: None

Brief description of the working environment: The expectation was to get office work but we were alloted the construction site. The working environment of a site differs from that of the office. But that being said the experience was valuable as we got to learn new things. The site was safe as all the safety measures were in place.

Academic courses relevant to the project: Construction Planning and technology, (not mandatory)

Learning Outcome: Information about on site processes and machinery

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PS-I station: KEC International Limited , Jabalpur

Student

Name: PRANAV KUMAR MALAIYA .(2021A4PS1059P)

Student Write-up:

PS-I Project Title: How to reduce long cycle time in manufacturing?

Short Summary of work done: I was assigned to reduce cycle time in manufacturing. My mentor guided me and instructed to make a report of modern manufacturing methods to incorporate sustainable and lean manufacturing. I came across the Industry 4.0 methods which incorporate AI/ML, Big data analytics, cloud and other modern tools to reduce resource consumption and increase the overall productivity of the facility. I had to prepare a report on how I can incorporate these to KEC's current capacity and what will be the financial implications of the whole renovation. I was told to visit the factory floor

and survey the workers about their problems and what they'd like to be implemented. After this, I interviewed senior officials in the company, various department heads to get a better idea of what might be feasible. I also prepared a new factory layout to improve upon the existing structure, to improve the assembly line.

Objectives of the project: To reduce the time taken for production of finished goods from raw materials through incorporation of various modern manufacturing methodologies.

Tool used: S/w - Autodesk, Smartdraw, Excel; H/w - Manufacturing machines for various manufacturing process including notching, drilling, shaping, welding etc.

Details of Papers/patents:na

Brief description of the working environment: Working environment was good, the mentors and all the people working were really cooperative. The major problem with this station was the limited amount of exposure to current manufacturing scenario of the industry. The tools and processes being carried out are really basic and they don't add any value to the product. The limited resources and exposure was a major disappointment. Other than these factors, the working environment was professional and basic learnings can be expected such as cooperation and teamwork.

Academic courses relevant to the project: Manufacturing Processes, Advanced Mechanics of Solids, Fundamentals of Finance and Accounting, Principles of Economics

Learning Outcome: Learned about Industry 4.0, Applications of modern manufacturing methods to current manufacturing facilities, designing factory layout, ecnomics of running a manufacturing facility and financial planning for implementation of plans.

PS-I station: KEC International Limited , Jabalpur

Student

Name: SRIJAN KOSHTI .(2021ABPS1678P)

Student Write-up:

PS-I Project Title: Reduction in Steel Wastage

Short Summary of work done: Worked with the planning department and learned about steel wastage minimization using optimization and on site work flow analysis to provide suggestions to improve work efficiency.

Objectives of the project: Research about the material planning and coming up with optimization plans to reduce steel wastage.

Tool used: Excel

Details of Papers/patents:NA

Brief description of the working environment: Very comfortable and admitting, they taught us with great patience and cross-checked our work regularly, along with making valuable suggestions for constant improvement.

Academic courses relevant to the project: Metrology and Quality Assurance, Operations Management

Learning Outcome: Optimization via Excel, Planning

PS-I station: KEC International Limited , Jabalpur

Student

Name: ANIKET PANDEY .(2021B2AB2768P)

Student Write-up:

PS-I Project Title: Reduction of steel wastage

Short Summary of work done: I have learnt to make MRP manually. MRP, or Material Requirements Planning, is a crucial inventory management system used in manufacturing. It involves work in optimizing production processes and reducing costs. Engineers and software developers collaborate to design MRP software, allowing companies to plan, schedule, and control material requirements efficiently. MRP ensures that materials are available when needed, avoiding excess inventory or stockouts. The system factors in demand, lead times, and production schedules to streamline operations. Continuous advancements in AI and data analytics have led to enhanced MRP capabilities, aiding businesses in maintaining optimal inventory levels, improving supply

chain efficiency, and enhancing overall productivity. I have also prepared reports for optimising steel wastage by optimising machine and labour.To optimize steel wastage in a scaffolding workshop, a comprehensive approach involving both machines and labor is essential. Implementing advanced cutting machines with precision controls can minimize material waste during the fabrication process. Automation can streamline production, reducing human errors and optimizing resource utilization. Training the workforce in efficient cutting techniques and promoting waste-conscious practices further enhances material efficiency. Embracing lean manufacturing principles can lead to better material planning and utilization, maximizing output while minimizing scrap. Continuous monitoring and data analysis of wastage patterns allow for continuous improvement, ultimately resulting in a more sustainable and cost-effective scaffolding production process.

Objectives of the project: Objective of the project was to reduce the steel wastage at every stage of manufacturing.

Tool used: Some of the softwares used were- Excel , PowerPoint presentation and MS word.H/w-

Details of Papers/patents: Not applicable

Brief description of the working environment: During the PS-I , the working environment for a student or trainee is a blend of learning and practical application. The trainee is typically immersed in a professional setting, where they get hands-on experience and exposure to real-world scenarios related to their field of study.
Expectations from the company during PS-I revolve around providing a conducive learning environment and relevant projects that align with the trainee's educational background. The company is expected to assign mentors or supervisors who guide and support the trainee throughout their internship. This mentorship helps the trainee develop both technical skills and soft skills, enhancing their overall professional growth.
During PS-I, the trainee is encouraged to actively participate in projects, collaborate with teams, and contribute valuable insights. The learning process involves understanding the company's work culture, processes, and industry standards. Trainees are often exposed to various tools, technologies, and methodologies that are prevalent in the industry.
Overall, the PS-I experience aims to facilitate a seamless transition from theoretical knowledge to practical application. Trainees are encouraged to ask questions, seek guidance, and continuously improve their skills. The internship offers a valuable opportunity to bridge the gap between academic learning and industry requirements, fostering a holistic and well-rounded development.
In summary, PS-I provides a dynamic learning environment where the company supports the trainee's growth by offering relevant projects, mentorship, and exposure to real-world challenges, thereby preparing them for future professional endeavors.

Academic courses relevant to the project: Not applicable

Learning Outcome:

Some of the major learning outcomes were-
1.Learnt about manufacturing process
2. Learnt about how corporate works
3.Learnt to collaborate with people on projects
4. Learnt about planning of raw material and scrap management

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PS-I station: KEC International Limited , Jabalpur

Student

Name: PRANAV PANDEY .(2021B4A40794P)

Student Write-up:

PS-I Project Title: Reduction of Steel wastage

Short Summary of work done: During the 2-month practice school at KEC International Jabalpur, our project focused on reducing steel wastage in the manufacturing process. Our primary objective was to identify areas where steel was being wasted and implement effective solutions. We began by analyzing the Material Requirements Planning (MRP) and Scrap Reports to pinpoint patterns of excess steel usage and wastage. Through this analysis, we identified several key factors contributing to wastage, such as over-ordering, improper handling, and inefficient production practices. With these insights, we proceeded to conduct a workshop involving various stakeholders, including engineers, operators, and supervisors. During the workshop, we explored different possibilities and brainstormed potential solutions to reduce steel wastage. Some of the key solutions involved optimizing cutting processes, implementing better inventory management strategies, and enhancing training programs for employees to handle steel more efficiently. Overall, the practice school experience at KEC International Jabalpur provided valuable learnings on the importance of data-driven decision-making, collaboration among different teams, and the integration of technology to achieve significant reductions in steel wastage during the manufacturing process.

Objectives of the project: Minimize steel wastage through efficient resource utilization and process optimization in order to enhance sustainability and cost-effectiveness.

Tool used: MS Excel , MS PowerPoint

Details of Papers/patents:NA

Brief description of the working environment: During the practice school at KEC International Jabalpur, the working environment was engaging and informative. The induction process gave us an overview of various departments, and due to our interest in planning, we chose to work in that department on a project focused on reducing steel wastage.
The Planning Department's primary responsibilities included ordering raw materials based on requirements, providing fabrication plans to the workshop, and managing scrap. The work culture in the company was highly professional, and we were encouraged to take initiative and actively participate in the project.
Throughout the practice school, we gained valuable insights into the functioning of a manufacturing workshop. Working closely with the planning team and other stakeholders, we learned about the intricacies of material management, fabrication processes, and the challenges involved in reducing steel wastage.
The experience provided us with practical knowledge and hands-on experience in tackling real-world industrial issues. We also developed skills in data analysis, process optimization, and collaboration. It was a great learning experience that expanded our understanding of manufacturing operations and the importance of efficient planning in minimizing wastage.

Academic courses relevant to the project: Supply chain Management

Learning Outcome: 1. Identification of key areas contributing to steel wastage.
2. Implementation of measures to minimize scrap generation and optimize steel utilization.
3. Improvement of production processes to reduce rework and reject rates.

PS-I station: KEC International Ltd , Jaipur

Student

Name: ABHISHEK KOCHAR .(2021A4PS2495H)

Student Write-up:

PS-I Project Title: Helmet detection Ai

Short Summary of work done: Our time at kec has been great

Objectives of the project: Devlop a helmet detection ai

Tool used: Python,labelimg,google colab

Details of Papers/patents: None

Brief description of the working environment: Great environment

Academic courses relevant to the project: Fdsa

Learning Outcome: Learnt about ml and ai models

PS-I station: KEC International Ltd , Jaipur

Student

Name: GAURAV AGARWAL .(2021B4A41715P)

Student Write-up:

PS-I Project Title: Financial analysis of KEC international Ltd

Short Summary of work done: Done financial analysis of company for the past and predicted the future also in terms of finance

Objectives of the project: Become a financial analyst

Tool used: Excel, trading view, NSE

Details of Papers/patents: None

Brief description of the working environment: It's pretty good there,we learnt a lot and our mentors are very helpful

Academic courses relevant to the project: Fundamental analysis of finance and accounting, derivatives and risk management

Learning Outcome: Knowing all in and out about finance

PS-I station: KEC International Ltd , Jaipur

Student

Name: SHREYANSHI GARG .(2021B5A41891H)

Student Write-up:

PS-I Project Title: Analyzing KEC's Financial Position

Short Summary of work done: We were provided with the accounts of the company. Then we were tasked to find the financial ratios for the past 6 years, which were based on a number of parameters. We also did a technical analysis to find the stock trends and tried predicting the companies position in the future.

Objectives of the project: Analyzing the financial trend of the company in the past 6 years.

Tool used: Excel

Details of Papers/patents: There weren't any papers or patents. We just submitted a brief project report at the end.

Brief description of the working environment: The mentors at KEC are really helpful and helped us with any of the doubts and problems that we had. You can expect to find really helpful, accommodating mentors who excel in their respective fields. This is the reason why the company has excelled in the past few years. We gained the knowledge of looking into and analyzing the company's accounts.

Academic courses relevant to the project: Fundamental Analysis of Finance and Accounting
Derivatives of Risk and Management

Learning Outcome: Knowledge about stocks and financial ratios.

PS-I station: Kerala Infrastructure and Technogy for Education , Thiruvananthapuram

Student

Name: GARVIT JAIN(2021A7PS2840G)

Student Write-up:

PS-I Project Title: Creating a tool to transfer data

Short Summary of work done: To create scripts (in python, tried in C++ as well) that connects with MySQL server in KITE and OpenEMIS platform. Then transfer the data from KITE database to OpenEMIS (40 million entries in total) while sorting out any relations or constraints.

Objectives of the project: Create a digital product/script to migrate data from KITE official database to OpenEMIS platform

Tool used: Git, Docker, Docker Compose, Python, eDrawSoft, VScode or Pycharm Path intelliSense, MySQL, MySQL Workbench, OpenEMIS, phpMyAdmin, MariaDB, MS Excel, Python flask, csv and mysql.connector module.

Details of Papers/patents:NA

Brief description of the working environment: We only had 2 people in the team in which only I knew coding and MySQl, they were over expecting (way beyond my technical capabilities). They were working on a project that was not very well documented so we were stuck in all the steps and they were guiding us out of it. They were also pressurizing our co-instructor and were not willing to establish a connection with us. They want even more extension to PS, extending till sip preparation came.
Learning: I had my PS station changed from online to here offline (but it got converted to online on request). So they were expecting us like offline students. Also, if person with you don't know coding and you do, it is better for you to do work and give others to verify rather than dividing the work equally.

Academic courses relevant to the project: DBS- Database Systems, Computer Programming

Learning Outcome: Communication Skills and management skills development, stress management, learning about git, docker, flask, phpMyAdmin and python in details.

PS-I station: Kerala Infrastructure and Technogy for Education , Thiruvananthapuram

Student

Name: LAVANYA SABHARWAL .(2021B4AA2263H)

Student Write-up:

PS-I Project Title: Creating a tool to transfer data

Short Summary of work done: Firstly, We also worked on creating a mapping of terms from Sampoorna to OpenEMIS which would help facilitate transfer of data. Then, We worked on creating a script that transfers data from the source database to the target database. We also carried out detailed verification to ensure that script runs perfectly. We used Objects in code as it simplified the mapping process and allow specific fields to remain unmapped. We also suggested a plan for KITE Auditors to perform IT audits in a smart way which would improve efficiency, accuracy and save time. These included addition of features like Feedback forms, Data collection forms, Probabilistic approach to IT auditing etc. We also created UML diagrams to represent the finalized features list. We also worked with applications of databases, Docker compose and YAML file.

Objectives of the project: We were given the important task of creating a tool that will help move educational data from the KITE database (Sampoorna) to the OpenEMIS platform. KITE was initially using Sampoorna database to store all of their educational data and they wanted to tra

Tool used: Python, python modules like mysql.connector, OpenEMIS, Docker, Docker compose, MySQL, phpMyAdmin, Git, CSV files

Details of Papers/patents:NA

Brief description of the working environment: PS-1 at KITE was overall a great experience. We had daily stand-up meetings (morning and evening) where we were given tasks for the day and discussed the plan of action. I learned new technical skills as well as improvised problem-solving skills. I also learned how to manage time properly and meet deadlines. The team was also very helpful.

Academic courses relevant to the project: Computer programming

Learning Outcome: I learned many new technical skills. I have also made significant progress in efficiently working with MySQL. I learned about applications of databases, the usage of Docker and also acquired knowledge of Python and its modules. I also learned about OpenEMIS UI and the configuration of YAML file. I also experienced growth in my soft skills which included communication and teamwork. During PS-1 at KITE, I also had the opportunity to develop problem solving and time management skills.

PS-I station: Kotak Education Foundation , Mumbai

Student

Name: JAIN DIYAN VISHAL .(2021A1PS2121P)

Student Write-up:

PS-I Project Title: Data Analysis

Short Summary of work done: I was working on analysing different states in India to help expand KEF's operations and reach out the students that need educational support. I was focusing on expanding in Punjab and Odisha and need data related to the participation of the schools in that state and the performance of students. The data was extracted from National Academic Survey (NAS)'s reports that are uploaded on their website. This data had to be cleaned and analyzed to make certain decisions which will help KEF reach out the students, who need academic support, in Punjab and Odisha. I made a dashboard using Tableau to present the data I had extracted. Apart from this I had other tasks to perform such as comparing different project management software that KEF could use. Also, I was given the task to call underprivileged students and ask if they need a scholarship and guide them to complete the forms.

Objectives of the project: Collect data from website, input in spreadsheet and make a dashboard

Tool used: Excel, Tableau

Details of Papers/patents:NA

Brief description of the working environment: The working environment was great. the people were helpful and friendly and helped us gel in well. I was a bit disappointed when I was alloted the task because as per the sheet given by PS division the projects

were based on IT and coding but I was given the task of working on excel and making dashboards. They said there was a miscommunication between BITS and KEF which is why I was allotted some other department. I learnt how to make dashboards and became more prolific at using excel and working with large amounts of data. I learnt the importance of good communication and to discipline myself.

Academic courses relevant to the project: NA

Learning Outcome: Understood how to analyse data and create inferences from it as per the specific needs

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PS-I station: Kotak Education Foundation , Mumbai

Student

Name: VEDANT BHATT .(2021A4PS3080H)

Student Write-up:

PS-I Project Title: Work done at DLS Intervention

Short Summary of work done: During the first week of the internship at Digital Learning Solutions (DLS) at Kotak Education Foundation (KEF), the focus was on learning about product development and its implementation at the organization. The main objective of KEF is to provide quality education to the underprivileged sections of society, with a particular emphasis on the (M) ward of Mumbai. The analysis of the Oteach program, which is a teacher training program conducted by KEF, revealed a continuous drop in teacher attendance across sessions and below-average assignment submission rates. The teachers face challenges as they come from resource-constrained schools and are already burdened with additional responsibilities. To address these issues, several solutions were proposed, including spreading the sessions over longer durations, implementing a comprehensive evaluation approach, giving weightage to attendance in the scoring policy, and introducing rewards for regularity and performance. Another task involved understanding and analyzing the SurveyCTO platform, identifying unused features, and implementing them to strengthen the database and reduce manual work. The internship also involved analyzing data from the Digital Learning Solutions team, specifically focusing on the Oteach program. The dataset was unorganized and lacked proper linkage, so efforts were made to categorize and organize the data for future use. Furthermore, the intern worked on presenting the data in an attractive manner using

Lookerstudio software, creating a dashboard to showcase the DLS Oteach Teacher Performance Data 2021-22. The process involved learning about user interface design principles such as Gestalt Principles and Color Theory. Overall, the internship provided hands-on experience in product development, data analysis, and data visualization, contributing to the goal of improving education for the underprivileged through KEF's initiatives.

Objectives of the project: Data analytics, consulting and dashbboard creation

Tool used: Excel,Google sheets,surveyCTO,Tableau,App Script and basic Python

Details of Papers/patents:NA

Brief description of the working environment: 1. Working environment: Very good, doesnot pressurise you for anything. Just need to be regular.
2. Expectations: To come when called and complete the projects on time.

Academic courses relevant to the project: Data Analytics and Product Management

Learning Outcome: SurveyCTO, advanced MS Excel and Google Sheets, Google Looker Studio, Tableau

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PS-I station: Kotak Education Foundation , Mumbai

Student

Name: RISHI KETAN SHAH(2021A7PS2496G)

Student Write-up:

PS-I Project Title: Learnig Networks,Cloud Computing and Hardware auditing

Short Summary of work done: During my internship with the IT team, I was exposed to a diverse range of tasks that allowed me to develop valuable skills and knowledge in various areas. One of my primary focuses was on Linux commands and the Linux operating system, where I gained hands-on experience in navigating the terminal, managing files, and troubleshooting common issues. I also delved into computer networks, learning about subnetting and IP addressing, which provided me with a solid foundation in understanding how data is transmitted across networks efficiently and

securely. Additionally, I had the opportunity to work with an Active Directory Domain Services (ADDS) server, where I gained insights into user management, group policies, and network authentication protocols. One of the highlights of my internship was being involved in cloud computing projects. I collaborated on tasks such as setting up and configuring virtual machines, exploring different cloud service models, and implementing cloud-based solutions for various business needs. Furthermore, I actively participated in hardware auditing initiatives, where I honed my skills in inspecting, analyzing, and documenting hardware components and their configurations to ensure compliance with industry standards. Throughout my internship, I demonstrated adaptability, problem-solving, and teamwork skills, contributing to the success of various IT projects and gaining invaluable experience in the process

Objectives of the project: To get hand-on experience on Linux-OS. learn about vitual machines. how cloud computing is used in KEF

Tool used: 1)- Foritgate firewall 2)- google sheets 3)-AWS

Details of Papers/patents:NA

Brief description of the working environment: During my time at KEF, an NGO dedicated to educating underprivileged children, I had the opportunity to be part of the IT team. It was an incredibly enriching experience. The work environment at KEF was exceptional, fostering a positive and collaborative atmosphere.
The team at KEF was like a big family. We supported and encouraged each other, making the workplace feel warm and welcoming. I felt valued and appreciated for my contributions, which motivated me to do my best every day.
One thing that stood out was the open communication and respect among team members. Everyone's opinions were heard and valued, creating a space where ideas could freely flow. This allowed us to find creative solutions to IT challenges and improve our systems continually.
I also appreciated the emphasis on work-life balance at KEF. The organization understood the importance of personal time and provided flexibility when needed, which made me feel more balanced and focused during work hours.
Moreover, KEF's commitment to its mission of educating disadvantaged children was inspiring. Knowing that our IT efforts directly contributed to making a difference in these kids' lives gave a deeper sense of purpose to our work.
Overall, my experience with KEF's IT team was fantastic. The positive work environment, supportive colleagues, and meaningful mission made it a truly fulfilling place to work

Academic courses relevant to the project: Computer Networks
Network programming

Learning Outcome: 1)- Learning everything about how linux works and how is it different than other operating systems
2)- learning about ip addressing, subnetting firewalls will help me in my coursework

3)- learning about AD-DS server gave me exposure of authentication factors in case of centralized databases

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PS-I station: Kotak Education Foundation , Mumbai

Student

Name: KHANDELWAL AVICHAL PANKAJ .(2021B3A42317P)

Student Write-up:

PS-I Project Title: Centralised assessment planning(CAP)

Short Summary of work done: Research intern: making a centralised assessment plan for the company to evaluate it's ongoing projects

Objectives of the project: Set up guidelines for CAP

Tool used: Chatgpt, Google scholar

Details of Papers/patents: Nil

Brief description of the working environment: Everyone around is sweet and supportive. It's not a corporate environment. Interaction is semi formal.

Academic courses relevant to the project: Nil

Learning Outcome: Learning office etiquettes and researching in a professional setting

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PS-I station: Kotak Education Foundation , Mumbai

Student

Name: VIVEK RAMCHANDRA YEDGE(2021B3A82149G)

Student Write-up:

PS-I Project Title: Dls

Short Summary of work done: It was quite an experience

Objectives of the project: To make a website

Tool used: Html css Java nodejs

Details of Papers/patents: Not finding

Brief description of the working environment: Very good

Academic courses relevant to the project: Economics related

Learning Outcome: Html,css, sales

PS-I station: L&T Defence , Hazira

Student

Name: ARIN MANGAL .(2021A2PS2153P)

Student Write-up:

PS-I Project Title: ASC Canteen Extension Project

Short Summary of work done: 1) Designed floor layout for Gate 14 building, 2) Designed pantry for functions, 3) Observed construction of canteen, 4) Observed construction of flagpole, 5) Learnt about safety in big plants

Objectives of the project: Work on canteen extension in ASC

Tool used: AutoCAD, MS Power Point

Details of Papers/patents:M/A

Brief description of the working environment: During my Summer Internship at Larsen & Toubro Defense, the working environment was dynamic and inspiring. The company fostered a culture of innovation, collaboration, and excellence, which provided a supportive atmosphere for interns like me to thrive.

Academic courses relevant to the project: EG

Learning Outcome: Design skills, knowledge about defense

PS-I station: L&T Defence , Hazira

Student

Name: VIJAY KUMAR UDHANI .(2021B5AB1697P)

Student Write-up:

PS-I Project Title: 46-M MODULAR BRIDGE , LOAD TESTING AND WELDING TECHNIQUES

Short Summary of work done: Major work of my was to observe how they are doing such a big project and the process they follow which includes machining, grinding and welding and how important is to put together correct no. Of parts in their jobs and how do they maintain data in their computers and how to read actual blue print of different components of the bridge and maintain it drawing no. Plate no. Revisions no. And copy no. On the excel

Objectives of the project: To learn about difference manufacturing and fabrication processes

Tool used: MS, EXCEL ,AUTOCAD ,ERP

Details of Papers/patents: Na

Brief description of the working environment: Working environment is very supportive and healthy . From shop workers to top management everyone here is good enough to

help us wherever we need them and they trust us by sharing there confidential data with us approved by DRDO.

Academic courses relevant to the project: WORKSHOP AND ENGINEERING GRAPHICS

Learning Outcome: DIFFERENT WELDING TECHNIQUES , HOW. DOES EVERYTHING WORKS IN SUCH A BIG DEFENCE ORGANIZATION

PS-I station: L&T Defence , Powai

Student

Name: VIKRAMADITYA BHATTACHARJEE(2021A4PS0736P)

Student Write-up:

PS-I Project Title: PROJECT MANAGEMENT : GUNS SYSTEM BUSINESS

Short Summary of work done: Learnt about working and functioning of gun. Read about material properties such as stainless steel. Created purchase orders for various items. Prepared excel sheets to maintain data. Verified the quantity of items required are correct for ordering. Created dashboard for risk management. Made presentations of my work and presented the ppt to my mentor.

Objectives of the project: Learn about how a project is managed in the defence sector, from planning and execution till delivery of final product.

Tool used: MS Excel ,MS project, Product Lifecycle Management software, ERP, PowerBI , Concerto

Details of Papers/patents:N/A

Brief description of the working environment: Working environment is good. 9-5 working hours , includes breakfast and lunch time. Mentors and supervisors are helpful, friendly and insist you to learn. Can have a tea/coffee break too.
Expect mostly an office job in this L&T campus, as it is majorly responsible for management.

Learned to work in a team, manage a project from start to the end and understanding that it is not an individual contribution but the team works as a whole.
Learned to prepare presentations and presented my work to my mentor. Improved my soft skills.

Academic courses relevant to the project: Engineering Graphics, Workshop Practice, Manufacturing Process, Material Science

Learning Outcome: 1) Coordinating with team members and other departments to execute a task
2) Managing a project as a team
3)Learning about the working and technicalities of a gun
4)Improved communication skills
5)Ability to handle critical tasks within given time limit

PS-I station: L&T Defence , Powai

Student

Name: YASH MANOHAR AGARWAL .(2021A8PS3120H)

Student Write-up:

PS-I Project Title: Applications of Machine Learning in Underwater Sonar Technology

Short Summary of work done: My work was very self-paced. I had to go, sit in front of a PC and read as many research papers as I wanted to. At first I went through sonar and acoustic signals in underwater uses. Then I slowly shifted to learning about machine learning applications in them. In the end I deployed my own model to such a dataset. OpenCV was used for the same.

Objectives of the project: To learn about sonar and acoustic signals, ML algorithms and then applying ML algorithms on underwater acoustic signals.

Tool used: Hardware- nothing specific. Software- Colab, Python DL libraries, Scholar

Details of Papers/patents: None

Brief description of the working environment: The entirity of my work and the working environment was extremely chill. There were no compulsions from the company. A target was set on the first day and goal was to hopefully achieve it till the end of PS-1. No daily targets to meet. Self-paced. You can expected to be in a very stimulated environment involving lots of secret projects with the government but you wont get to work on them directly. The environment is friendly, and the employees are fun to have group discussions with, although the productivity aspect was extremely low overall, in my experience. Larsen and Toubro is a good company to have on your resume though. Try to communicate your interest in area of work to the HR assigned to you before they allot you something.

Academic courses relevant to the project: Machine Learning, Signals and Systems

Learning Outcome: Knowledge of underwater sonar and acoustic signal and processing methods ML and DL models and applications of the same.

PS-I station: L&T Defence , Powai

Student

Name: ANUJ MANOJ SHIKARKHANE(2021B5A71145G)

Student Write-up:

PS-I Project Title: Object Detection and Tracking

Short Summary of work done: During the project on object detection using YOLO (You Only Look Once) with a custom dataset, I implemented a real-time computer vision model for detecting multiple objects in images and videos. YOLO's efficiency and accuracy made it a prime choice for this task. To streamline development and deployment, Docker containers with GPU support were utilized, enabling seamless scalability and resource-efficient training and inference. This resulted in faster processing times and improved model performance. Creating a custom dataset was a primary focus, involving data collection, annotation, and preprocessing. This ensured the model's ability to accurately identify target objects in various contexts. Training the YOLO model on the custom dataset was accelerated using GPU support in Docker, reducing training times and optimizing performance. Fine-tuning and hyperparameter tuning further enhanced the model's accuracy and generalization. The project yielded remarkable results in object detection across diverse scenarios, showcasing the potential of YOLO and the benefits

of using Docker containers with GPU support for deep learning tasks. The custom YOLO model opens up possibilities for applications such as surveillance systems and autonomous vehicles.

Objectives of the project: 1. To train various state of the art object detection models on custom datasets. 2. Get acquainted with industry standard techniques to train and deploy computer vision models.

Tool used: Docker, NVIDIA GeForce RTX 3080 Ti GPU, Google Colab, Anaconda Distribution

Details of Papers/patents: Not allowed to disclose.

Brief description of the working environment: I was fortunate to be a part of a diverse team that encouraged creativity, critical thinking, and open discussions. A positive and inclusive work culture that fosters collaboration helped me bring out my best potential. Access to necessary resources, modern technologies, and training opportunities helped me in staying up-to-date with the latest industry practices and trends. Overall, it was a wonderful learning opportunity. I gained fundamental level knowledge about the field of computer vision and its immense applications.

Academic courses relevant to the project: Computer Vision, Deep Learning

Learning Outcome: 1. Learn the significance of the field of computer vision (object detection and tracking) and its various applications.
2. Learn how to train and deploy real time object detection models on large custom datasets.
3. Learn the various performance metrics that are used to evaluate the performance of a model on a dataset.
4. Learn how to prepare a custom dataset by understanding about concepts like segmentation, annotation, labelling, etc.
3. Develop the required skills and work ethics to work like a professional in this field.

PS-I station: L&T Defence , Talegaon

Student

Name: ADITYA LALIT SHRIPAD(2021A4PS2711G)

Student Write-up:

PS-I Project Title: Defence Project-B

Short Summary of work done: Full-kitting , categorizing the different parts required for a project according to their types and usage. Observing the project being commenced and correlating with the drawing sheets which I had read initially

Objectives of the project: Segregate and categorize different assemblies and learn various mechanical operations and mechanisms

Tool used: Excel

Details of Papers/patents: Not any

Brief description of the working environment: Friendly working environment, employees are kind and helpful. They helped me solve my doubts.

Academic courses relevant to the project: Mechanics of Solids , Advanced Mechanics of Solids, Material Science

Learning Outcome: Applications of theoretical concepts into practical real world projects.

PS-I station: L&T Defence , Talegaon

Student

Name: ARYAN SONTHALIA .(2021A4PS3103H)

Student Write-up:

PS-I Project Title: ARUDHRA ASSEMLY AND MODULAR BRIDGING SYSTEM DEVELOPMENT

Short Summary of work done: Worked on radar assembly and composite based modular bridge launching/ de-launching concept. Had to work on softwares related to Siemens NX, AutoCAD, Solidworks and Ansys. Also had to do shopfloor visits where weapons systems were majorly made.

Objectives of the project: Mainly to develop a concept of the bridging system

Tool used: Solidworks, Ansys, Siemens NX, AutoCAD, Excel for all calculation purposes

Details of Papers/patents: None

Brief description of the working environment: It was very tedious. Reached there by 8 am in the morning and left by 5:45 pm. If you stay in Pune, you'll have to wake up before 6 am to reach Talegaon and reach home only after 7 pm.
They are very strict with respect to electronics also. Mobile phones and laptops are not allowed inside the campus. They provide computers but that does not internet surfing access. Only certain sites like Gmail, meet, etc are whitelisted which are also based on intranet and all the internet data is scanned and then routed to intranet.
However, you gain access to all the design and development files there. If you have had experience upon any skill earlier, may it be CAD, FEA, even your academic skills if you studied the courses properly, it shall be temple of knowledge for you.
It is mostly based on how you interact with the people there using your communication skills as they are already very busy, but what you'll get to learn in a core-based company of defense field shall not be found anywhere else.

Academic courses relevant to the project: Mechanics of Solid (MOS)
Adanced Mechanics of Solid (AMOS)
Design of Machine Elements (DOME)
Machines and Mechanisms (M&M)
Material Science

Learning Outcome: Real world applications of the academic courses listed ahead

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PS-I station: L&T Defence , Talegaon

Student

Name: SOHAM ANAND KULKARNI(2021A8PS2756G)

Student Write-up:

PS-I Project Title: Pre production planning and design of CIWS

Short Summary of work done: The purpose of this report is to summarise the work that I have done over the past 7-8 weeks in my PS-1 internship at L&T defence, Talegaon. I have been working at the Electronics Shop department at L&T defence Talegaon. Over the course of my internship, I have worked on projects related to the gun system and firing system.. The report is broken down into 4 steps of my project . The report also highlights the important components of an engineering drawing , documents and testing. Electrical testing is an essential part of the production process, during the internship time was also spent learning different inspection of units , testing and card testing, the report also covers these techniques in brief. The report also covers, in brief, my understanding of Process and Methods Engineering, over the time I have worked at L&T defence Talegaon. The purpose of this report is to summarise the work that I have done over the past 7-8 weeks in my PS-1 internship at L&T defence, Talegaon. I have been working at the Electronics Shop department at L&T defence Talegaon. Over the course of my internship, I have worked on projects related to the gun system and firing system.. The report is broken down into 4 steps of my project . The report also highlights the important components of an engineering drawing , documents and testing. Electrical testing is an essential part of the production process, during the internship time was also spent learning different inspection of units , testing and card testing, the report also covers these techniques in brief. The report also covers, in brief, my understanding of Process and Methods Engineering, over the time I have worked at L&T defence Talegaon.

Objectives of the project: Pre production planning and prototype unit testing

Tool used: H/w

Details of Papers/patents: No

Brief description of the working environment: Very good environment, helpful seniors and colleagues are always ready to help you .

Academic courses relevant to the project: Digital Design , electrical machines , microprocessor and interfacing.

Learning Outcome: Electrical testing, vendor communication, Technical development

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PS-I station: L&T Defence , Talegaon

Student

Name: VISHNU VINAY(2021A8PS2989G)

Student Write-up:

PS-I Project Title: ESD technical gap assessment, development & implementation.

Short Summary of work done: I have made significant contributions in the development of an enhanced Electrostatic Discharge (ESD) protection system. Through my diligent research and innovative approach, I have successfully designed various improvements to the existing ESD protection mechanisms. By analyzing the vulnerabilities and limitations of conventional systems, I have devised novel strategies to enhance the robustness and efficiency of ESD protection. Finally created an excel sheet, after market research, consolidating all the data and possible implementations, with all the specifications, prices and dimensions.

Objectives of the project: To develop a new and improved ESD protection system for the electronics department at L&T.

Tool used: AutoCAD, Excel

Details of Papers/patents: None

Brief description of the working environment: The working environment was very supportive of us interns and the seniors always helped us out whenever we were in doubt about anything.

Academic courses relevant to the project: Engineering Graphics, Electronic Devices.

Learning Outcome: Learnt about the importance of ESD(Electrostatic discharge) protection in Industries.

PS-I station: L&T Defence , Talegaon

Student

Name: LOKHANDE SALONI .(2021ABPS2561P)

Student Write-up:

PS-I Project Title: Total Cycle Time Optimization of a Defence Project

Short Summary of work done: This project aimed to comprehend the assembly process flow and preassembly operations of a Defence project. This study involved identifying processes, calculating cycle times, and sequencing them. The critical path was determined, and constraints within the critical activities were acknowledged. These constraints were: 1. manufacturing problems; 2. a lack of skilled workforce; 3. issues with the inventory management system (resulting in high inventory holding costs); 4. insufficient manpower; and 5. a lack of quality checks with the vendor. The report also mentions the suggestions presented by me to the AIT 3 team for future improvements.

Objectives of the project: 1.To study the different roles and responsibilities pertaining to internal operations. 2.To study the manufacturing processes getting executed on the shop floor. 3.To understand the project sequence and study the project scheduling via MS Project 4.To stu

Tool used: MS Project

Details of Papers/patents:none

Brief description of the working environment: The policies are strict, but the culture of L&T Defence is just
as friendly. Ever since my arrival, everyone has been very open and
welcoming, addressing my doubts with the utmost patience and
helping me achieve my goals to the fullest. Imparting and sharing
knowledge are two of the values that form the core of this
company.
My expectations from the company were that I'd get to learn and implement the theoritical knowledge I had gained into practical learning.
During my internship, I was able to acquire a variety of skills and
put into practice the theoretical knowledge I had acquired over the
course of two years. It was gratifying to work at L&T Defence and
gain exposure to theories such as:
○ Gemba Kaizen,
○ Just-In-Time,
○ Zero Defects,
○ Total Quality Management, and
○ Total Productive Maintenance,
● Implementing theories like the theory of constraints and Inventory
Management Systems was extremely informative and engaging.

Academic courses relevant to the project: Operations Management, Metrology and Quality assurance

Learning Outcome: During my internship, I was able to acquire a variety of skills and

put into practice the theoretical knowledge I had acquired over the course of two years. It was gratifying to work at L&T Defence and gain exposure to theories such as:
○ Gemba Kaizen,
○ Just-In-Time,
○ Zero Defects,
○ Total Quality Management, and
○ Total Productive Maintenance,
● Implementing theories like the theory of constraints and Inventory Management Systems was extremely informative and engaging.
● Through my internship, I was able to hone my skills and expand my understanding of the field.
● I have also acquired experience with MS Project.
○ Gaining a deeper understanding of:
○ Critical Paths
○ Gantt Charts
○ Network Diagrams
○ Resource Allocation
○ Project Sequencing
○ Project scheduling, and much more.
● A remarkable outcome of my project is that I can successfully identify and list the various components and activities of the Defence Project and the scientific concepts behind them, as well as identify similar components in other mechanical products that one may come across.Being able to comprehend and apply one's knowledge in real life is a source of immense pride, and I am extremely grateful to be one of those individuals.
● In addition, I was able to gain an understanding of L&T Defence's testing procedures and requirements. This helped me comprehend how L&T Defence is able to manufacture products of such high quality, accuracy, and precision.
● Through my project,I have streamlined and organized the flow of Defence Project assembly-related activities.
● In my opinion, the most intriguing result of my project was the controlled Inventory Holding Cost, which was achieved using the Gantt Chart extracted from my MS Project file.
● This will result in significant time, resource, energy, and storage space savings

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PS-I station: Lakshyam , New delhi

Student

Name: AKASH JAIN .(2021AAPS2049H)

Student Write-up:

PS-I Project Title: Butterfly

Short Summary of work done: he work accomplished during my internship encompasses diverse contributions in social media campaigns, such as mobilizing support for the impactful "Hands to Heart" campaign and curating reference shots for photography. Additionally, I authored an article highlighting the significance of social internships and conducted research and outreach in various areas, including special schools, veterinary colleges, affordable insurance, and collaborations with college CSR departments. My involvement extended to marketing efforts, donor engagement, and skillful event coordination. Valuable hands-on experiences in education, mentorship, and animal welfare were gained through field visits to Kirby Place slums and Tuk Gilhari farm.

Objectives of the project: Child holistic development

Tool used: MS-Excel

Details of Papers/patents: None

Brief description of the working environment: Our expectations from the company were to be given work relevant to our academic curriculum and skill sets. In spite of not being given so, we learnt skills of community outreach during PS-1. We were required to visit a number of sites as part of assigned tasks, and that provided us with exposure to community service and animal welfare. The working environment was conducive to our working, and were supportive.

Academic courses relevant to the project: None

Learning Outcome: Community Service, Marketing Activities

PS-I station: Lakshyam , New delhi

Student

Name: RAIYAN KATARIA(2021AAPS2556G)

Student Write-up:

PS-I Project Title: Empowering Lives - A Journey of Interning at an NGO focused on helping children and women

Short Summary of work done: The work can be split into two sections - Onsite tasks and office based tasks. On-site tasks mainly involved going at different locations and centres of Lakshyam and directly helping out the beneficiaries by interacting with them, conducting workshops, donating goods and upkeeping the centres. Office tasks mainly involved contributing to work which helped the NGO run instead of directly helping the beneficiaries. These tasks mainly involved doing research work, communicating with other companies, schools and colleges, compiling data, and database management.

Objectives of the project: Gain hands-on experience. Learn about the NGO's mission and values. Contribute to meaningful projects utilizing our skillset Develop professional skills. Expand knowledge about social issues. Gain cross-cultural and diversity experience. Understand the

Tool used: Excel, Powerpoint

Details of Papers/patents:-

Brief description of the working environment: The working environment was a conductive one where the employees and staff I interacted with were both kind and helpful. They guided me throughout the internship. I expected to learn the workings of an organization and put my skills to use in a practical environment.
I learned a lot during my time interning here. Some of those learnings being-
1. Importance of Empowerment
2. Workings of an Organization
3. Collaboration and Partnerships
4. The power of awareness and education
5. Resilience of the human spirit
6. Continuous learning and adaptation
7. Marketing skills

Academic courses relevant to the project: M.Sc. Bio, B.E. Computer Science

Learning Outcome: 1. Community service
2. Importance of empowerment
3. Workings of organization

4. Collaboration and partnerships
5. The power and awareness and education
6. Marketing
7. Research
8. Database management

PS-I station: Log9 Materials Scientific Pvt. Ltd. , Bengaluru

Student

Name: PARAMESWARAN SARAVANAN .(2021A1PS2371H)

Student Write-up:

PS-I Project Title: Understanding and quantifying the degradation in lithium iron phosphate (LFP) cells

Short Summary of work done: I have learnt about the different chemistries of cells and their degradation factors. Mainly focused on cathode specific degradation modes of LFP chemistry. I have analysed the IC curve of the experimental data from labs. I performed peak analysis using features of interest (FOIs) as a basis to qualitatively predict the mode of degradation in LFP cells.

Objectives of the project: Understanding IC curves and observing the change in Features of Interest (FOI) thereby qualitatively analysing of IC curve.

Tool used: MS Excel

Details of Papers/patents: Research papers used:
1) Solvent free lithium iron phosphate cathode fabrication with fibrillation of polytetrafluoroethylene.
2) Generalised diagnostic framework for rapid battery degradation quantification with deep learning.
3) Lithium-ion battery SoH

Brief description of the working environment: The working environment was very productive and has a good work-life balance. I have familiarised myself with qualitative analysis of IC curves, detailed understanding on functioning of various lithium-ion

chemistries and their respective degradation factors. I would prefer better project allotments from the company in near future.

Academic courses relevant to the project: CHEM F111 , EEE F111 , CHE F214

Learning Outcome: •Qualitative analysis of experimental data.
•Clear understanding of degradation map of Gr/LFP cell.
•knowledge of cathode specific degradation modes of LFP cells.
•Detailed analysis of IC curves and respective peak analysis.

PS-I station: Log9 Materials Scientific Pvt. Ltd. , Bengaluru

Student

Name: GARG ACHINTYA JITENDRA .(2021A1PS2373P)

Student Write-up:

PS-I Project Title: Battery Design

Short Summary of work done: My work started with me learning SolidWorks from a youtube tutorial series. Once that was over I had to read and learn the basics about battery pack architecture and cooling systems used in batteries. After that I got a problem statement, to formulate a way to calculate the dimensions of a busbars using the value of resistivity of copper which was available for 20 degree celsius battery temperature. Since this was related to the current carrying capacity (J), I tried to use the basic formula of E = ρJ but then had to go down the challenging path to find out the electric field generated in plate conductors. So I tried to find out J using drift velocity. But that too needed me to find a way to calculate drift velocity in some other way which again entailed the need to find E. Thus, finally I used the equation P = I2R where P is the heat dissipated by the busbar and formulated an equation which gave the area of cross section of busbar using the value of length and made a calculator on excel for he same.

Objectives of the project: Formulate a way to calculate the dimensions of busbars used in electric vehicles

Tool used: SolidWorks

Details of Papers/patents:N/A

Brief description of the working environment: The working environment of Log9 was free. Here, at Log9 you are free to work, i.e., there is no one hovering above you constantly. But at the same time there are stand ups everyday, where the entire team discusses with each other, the work they did yesterday and the work they are going to do today, thus keeping everyone updated. This encourages everyone to be sincere without being forced and helps build an environment for trust and provides space for self growth. Everyone here is friendly, supporting and willing to help and there are many activities available to cool down after a stressful stint. My expectations from this PS was clarity whether I want to pursue core chemical or not. I learned SolidWorks, MATLAB and a lot about batteries and their cooling systems, something that I feel might come in handy in the near future.

Academic courses relevant to the project: Heat Transfer

Learning Outcome: SolidWorks, Basics about battery pack architecture, MATLAB

PS-I station: Log9 Materials Scientific Pvt. Ltd. , Bengaluru

Student

Name: ATHARVA SHRIRAM GHAISAS(2021A1PS2802G)

Student Write-up:

PS-I Project Title: Thermal Management of Aluminum Fuel Cell

Short Summary of work done: I figured the sources of thermal dump in an AFC along with the % contribution of each source.I was also able to extract hydrogen from the AFC which is a potential selling by-product.

Objectives of the project: Quantitative Analysis Sources of Thermal Dump in an AFC

Tool used: Research papers, Gas chromatography, Titration,Hydrogen extraction setup

Details of Papers/patents:NA

Brief description of the working environment: Peers were supportive and the company encouraged asking questions.Everybody maintained the right level of professionalism and helped me out at every step of my project

Academic courses relevant to the project: Heat transfer, general chemical kinetics and reactions

Learning Outcome: Got lab experience and learned about AFC by reading research papers.

PS-I station: Log9 Materials Scientific Pvt. Ltd. , Bengaluru

Student

Name: VISHNU RAJAGOPAL(2021A3PS1537G)

Student Write-up:

PS-I Project Title: Physics Based Model for Li ion cell in simulink

Short Summary of work done: I had to learn about cell modelling and state determining algorithms of Li ion battery cells. Then I had to make a physics based model of a Li ion cell in simulink

Objectives of the project: Converting python code to simulink model

Tool used: Simulink, matlab

Details of Papers/patents: State of charge estimation of lithium ion battery based on a novel reduced order electrochemical model

Brief description of the working environment: A very nice and welcoming place. It has a nice layout conducive to not only rest when needed but also work accordingly. Did not have to stay more than we had to, which meant we could leave as long as we finished our work for the day. Coming in late for various reasons, was allowed and the team was understanding and helpful when needed.

Academic courses relevant to the project: Control systems, electrical sciences

Learning Outcome: Call modelling, simulink

PS-I station: Log9 Materials Scientific Pvt. Ltd. , Bengaluru

Student

Name: SRIVENKAT VARMA NANDIMANDALAM .(2021A3PS2932H)

Student Write-up:

PS-I Project Title: GUI Application Development of Diagnostic Tool for Electric Vehicles

Short Summary of work done: As part of my project, I was introduced to various new technical concepts and technologies.A GUI Application for Diagnostic tool for Electric Vehicles with real-time periodicity analysis and message decoding capabilities was successfully developed. The project provided avenues to learn not just technical skills but also non-technical aspects of the corporate setting. Apart from the main project I also got the opportunity to build smaller applications like a recorded PCAN trace scenario simulator. The internship helped me realise my capabilities and enhanced my skills.

Objectives of the project: Build a GUI Application to record, analyse periodicity, decode CAN messages defined under the GB\T Charging Standard in real-time and generate decoded message traces.

Tool used: Python, PCAN-Basic API, cantools, PCAN-View, KVASER Database Editor, Notepad++

Details of Papers/patents: None

Brief description of the working environment: The people of the company are highly motivated and extremely approachable. There is a decent work-life balance with no typical bureaucratic rules enforced. You are allowed to explore, learn, relax and question freely, while focussing on your project and timelines. The mentors are very supportive and ready to guide you through your project and learning stints. You are only expected to be passionate, enthusiastic to learn and be inquisitive.

Academic courses relevant to the project: CS F111: Computer Programming

Learning Outcome: Multi-threaded programming, GUI Development and File Handling through Python, CAN Messages, Charging Standards, PCAN-Basic API, DBC files

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PS-I station: Log9 Materials Scientific Pvt. Ltd. , Bengaluru

Student

Name: ATHREYA UPADHYA(2021A3PS3045G)

Student Write-up:

PS-I Project Title: Electrical Test Setup Standardization for AFC

Short Summary of work done: Made a working web server from scratch. Raspberry Pi is the host for the web server and it is connected to multiple PCBs,which hold sensors, which feed the Pi data through a CAN network. Then that information is posted in real time on a local server. This allows monitoring of experimental data remotely.

Objectives of the project: Making a web server on a Raspberry Pi for real time data monitoring

Tool used: Python, HTML, JavaScript

Details of Papers/patents:-

Brief description of the working environment: Working environment was very nice. Everyone at the company was very helpful and encouraging. Learnt how to make a web server by going in with zero experience. Company expectations were high considering the fact that a PCB was designed even before the software was made.

Academic courses relevant to the project: Microprocessors and interfacing

Learning Outcome: Making a web server, communication network using CAN, hardware-software integration

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PS-I station: M/s Mosdorfer India Pvt. Ltd. , Nashik

Student

Name: PRIYANKA SATHEESAN NAMBOODRI .(2020B1AB1901P)

Student Write-up:

PS-I Project Title: Optimization and in-depth analysis of the supply chain for consumables

Short Summary of work done: We categorized the consumables according to the landed value cost to the company by ABC analysis. Category A consumables consisted of 70% of the landed value and less than 1% of the total number of items, so we restricted our focus to these items only. We analyzed consumption patterns, measured relevant KPIs, studied inventory data and understood the various procedures in place for the procurement of consumables. We identified weak points in the processes and gave relevant suggestions to improve the same.

Objectives of the project: □ Understand existing supply chain management process followed at MIPL for consumables □ In depth study of sub-procedures in practice for planning, procurement and safety stock qtys. of consumables, so that production proceeds seamlessly. □ Acquiring past

Tool used: Office 365 (Excel, Word, PowerPoint)

Details of Papers/patents:-

Brief description of the working environment: Mosdorfer provided a very supportive working environment, with immediate response to any of our needs. Everyone was very friendly and approachable. Working under the Vice President was a great learning opportunity as we were able to draw on from his expertise. We were also able to network with the Austrian management on their board visit and derive an international perspective of our field of work.

Academic courses relevant to the project: Operations Management, Supply Chain Management

Learning Outcome: Opportunity for students to study various inventory control techniques used in various renowned
industries to ensure optimum stock while minimizing excess inventory and associated costs.

 Students will gain insights into production planning and scheduling methodologies, such as ERP, MRP
capacity planning and production sequencing.
 Students will become familiar with complete supply chain management - handling supply chain data and
apply analytical tools and methods to identify areas for improvement by analyzing KPIs, conducting root
cause analysis, and implementing optimization strategies to enhance supply chain efficiency.
 Students will learn to critically think through supply chain issues in industry and acquire data-based
decision-making skills that will help them come up with feasible solutions.
 Students will gain unparalleled industry exposure, specifically in supply chain management of
consumables.
 Student will get exposure to professional work experience in a safe and structured environment with
help of experienced executives.
 Gaining industry experience helps student to practice and improve their industrial skills

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PS-I station: M/s Mosdorfer India Pvt. Ltd. , Nashik

Student

Name: ISHWARI SANTOSH BIDWAI(2021A4PS2903G)

Student Write-up:

PS-I Project Title: Aluminium Bronze casting

Short Summary of work done: I basically worked under production department. I took the readings of casting(die opening,pouring,part removing,finishing) for various RL components.We did graphical analysis for the same.We also briefly studied the stimulation reports.

Objectives of the project: Detail studying of aluminum bronze die casting

Tool used: Hardware: casting tools,temperature gun,cutting machine,belt grinding machine,finishing machines Software:autocad,magma

Details of Papers/patents: No

Brief description of the working environment: Working environment was healthy,although the environment in gdc was very suffocating.I learned many things during my tenure such as time management,teamwork. Overall it was a good experience.

Academic courses relevant to the project: Manufacturing processes

Learning Outcome: Industrial exposure, time handling ,developing communication skills,information on gdc

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PS-I station: M/s Mosdorfer India Pvt. Ltd. , Nashik

Student

Name: VISHAL DALAL .(2021A4PS3099H)

Student Write-up:

PS-I Project Title: SCM OF CONSUMABLES

Short Summary of work done: analysis of previous years data to create a strategy to improve inventory management for next year

Objectives of the project: optimize the processes involved in SCM of consumables

Tool used: Excel, VBA, ppt, word

Details of Papers/patents: Na

Brief description of the working environment: Supportive working environment.

Academic courses relevant to the project: Supply chain management, operations management

Learning Outcome: SCM, inventory management, operations management

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PS-I station: M/s Mosdorfer India Pvt. Ltd. , Nashik

Student

Name: ARYAN SANTOSH SOLANKE .(2021ABPS2722P)

Student Write-up:

PS-I Project Title: Industrial training of Non- ferrous (Alu. Alloy and Alu. Bronze) foundry.

Short Summary of work done: Take readings of cycle time by recording a video of the process and recording the time on a stopwatch while watching the video. Write time required for various parts of the process in excel sheet. 15 times a day

Objectives of the project: 1. Evaluate the impact of various factors on the casting quality and to propose recommendations for process optimization. 2. Understand type of casting defects, finding out defect wise root causes, and implement remedial actions and analyze the effe

Tool used: Phone, Temperature gun, MS excel

Details of Papers/patents:no

Brief description of the working environment: Mosdorfer India Private Limited is a subsidiary of Mosdorfer GmbH, a global company specializing in the design, development, and manufacturing of infrastructure solutions for the energy sector. Mosdorfer India operates in the field of power transmission and distribution, providing a wide range of products and services to utility companies and electrical infrastructure projects.
The company offers innovative solutions for overhead line fittings, including conductor accessories, insulators, suspension clamps, tension clamps, and hardware fittings. These products are designed to ensure reliable and efficient power transmission while maintaining high safety standards.
Mosdorfer India also provides engineering services and technical support to its customers, assisting in the design, installation, and maintenance of electrical networks. The company's expertise in power transmission and distribution enables them to offer tailored solutions and reliable project execution.

Academic courses relevant to the project: casting-forming-welding, manufacturing processes, operations management

Learning Outcome: 1. Study of material properties and understanding process details helps enhance the knowledge base and offer opportunity to implement theoretical knowledge in practical working.
2. Get exposure to professional work experience in a safe and structured environment with help of experienced engineers.

PS-I station: Mahindra Research Valley , Chennai

Student

Name: SHREERANG SUNIL GUPTE(2021A4PS0254G)

Student Write-up:

PS-I Project Title: Modulization of Agricultural Implements

Short Summary of work done: Read about various types of agricultural implements, types of tractors and different power settings they can operate at, mechanical design of the implements, variation of the implements with the soil type, final design of a modular implement that can be used for various field sizes.

Objectives of the project: To make modular agricultural implements that can be used on vehicles of varying horsepower

Tool used: -

Details of Papers/patents: Mahindra Patented Designs

Brief description of the working environment: Since it was online, the working environment was not much of a factor. The mentor and overall team was very professional, and asked for regular updates on the work. The company expectations changed with the level of understanding we had with the project.

Academic courses relevant to the project: Engineering Graphics was slightly relevant for the final sketch, but most of the work was simply reading and more research, which wasn't really taught in any course specifically.

Learning Outcome: Research on various agricultural implements, tractors, soil types etc, optimisation of mechanical design.

PS-I station: Mahindra Research Valley , Chennai

Student

Name: NAMAN SETH(2021A4PS0722P)

Student Write-up:

PS-I Project Title: Designing of Born EVs for maximising Efficiency and Range

Short Summary of work done: Exploring aerodynamic body features, CAD modelling a cara and performing simulations.

Objectives of the project: Conceptualising an EV by exploring aerodynamic features.

Tool used: Autodesk Fusion 360, Autodesk CFD Ultimate

Details of Papers/patents:NA

Brief description of the working environment: The working environment was very helpful. Everyone at Mahindra was always receptive to clearing questions and doubts. Being an online station, Mahindra was beyond my expectations.
I learned about CAD and simulation softwares.

Academic courses relevant to the project: Fluid Mechanics, Mechanics of Solids

Learning Outcome: Knowledge of numerous aerodynamic features, CAD and basic Simulations

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PS-I station: Mahindra Research Valley , Chennai

Student

Name: ADITYA BALASUBRAMANIAN .(2021A8PS1475H)

Student Write-up:

PS-I Project Title: Software-Defined Vehicles

Short Summary of work done: My goal was to design a software defined vehicle architecture. In other words, keeping the underlying hardware essentially unchanged, I needed to provide access to new hidden features by issuing software updates through the cloud To the computer governing the working of the hardware.

Objectives of the project: To design and attempt to implement a simple Software-Defined Vehicle Architecture

Tool used: H/W - Basic Circuit building using Arduino, Tinkercad, NodeMCU. S/W - Using the Blynk Clouding Platform to issue OTA updates

Details of Papers/patents: Nil

Brief description of the working environment: It was a rather enriching learning experience for me, working at Mahindra Research Valley. All the mentors from MRV and BITS were really helpful. They helped guide through every step of the process. They expected me a provide a final presentation of a simulation or a working model which I successfully did. This experience has widened my technical and philosophical perspective on how the world works.

Academic courses relevant to the project: Control Systems

Learning Outcome: Arduino, Tinkercad, Blynk Clouding Platform

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PS-I station: Maker & Asylum , Goa

Student

Name: ANIMESH KRISHNA TIWARI(2021B2A12300P)

Student Write-up:

PS-I Project Title: Project Riverwatch : Water Quality In Goa

Short Summary of work done: We tested the present water quality in Goa. Then using image segmentation we mapped out the sampling locations and collected a huge sample space of data for researchers , govt bodies to have access to.

Objectives of the project: To conduct tests across Goa's water bodies and map out the the sampling areas and publish the data on an online repository.

Tool used: 1. TDS meter 2.DO meter 3. pH meter , Trello

Details of Papers/patents: Nothing such published

Brief description of the working environment: Maker's Asylum was very different from a typical corporate office. It was a small building and had all sorts of modern tools related to creation of different things. The work environment was very healthy and our mentor , Mr Praneeth was very supportive. Our professor Mr Arfat Sofi was very helpful and made sure our PS was smoothly carried out. The company expected us to visit the river waters and talk and make friends with unknown fishermen , travel on their boat to the centre of rivers and collect samples. This was not considered safe by me and my peers. We developed negotiation skills , teamwork and presentation skills during our time at PS1. We got accustomed to meeting deadlines.

Academic courses relevant to the project: None out of the ones taken until now

Learning Outcome: 1. Teamwork
2. Presentation skills
3. Water Testing Protocols

PS-I station: Manodayam Pvt. Ltd. , Lucknow

Student

Name: AKSHAT MUDGAL(2021A8PS2912G)

Student Write-up:

PS-I Project Title: Mental Health diagnosis using Vocal acoustic feature

Short Summary of work done: It started with reading few research papers relating depression and vocal acoustic features. Later worked on python using the library Librosa, pandas etc . We used them to remove the background white noise using the noise reduction algorithm, plotting the spectrogram and extracting the MFCCs of the audio signals. These were converted into csv and later to be used in the machine learning model.

Objectives of the project: To use the voice samples and extract its features to implement in a model using for diagnosing mental health conditions

Tool used: Python

Details of Papers/patents: Papers about to research studies conducted on individuals with and without depression to use their voice samples and extract the acoustic features to identify the relation with their mental health conditions.

Brief description of the working environment: Great learning experience working in this company. Our mentor not only explained the work and technological needs of the company but also helped us gain some industry-related knowledge about start-ups. Helped us throughout our journey .

Academic courses relevant to the project: Computer Programming

Learning Outcome: Vocal acoustic features are a potential biomarker in the diagnosing depressing and possess a great potential for further research

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PS-I station: Manodayam Pvt. Ltd. , Lucknow

Student

Name: PARTH GUPTA .(2021AAPS2053H)

Student Write-up:

PS-I Project Title: ASSESSMENT OF MENTAL HEALTH AND DEPRESSION SYMPTOMS USING MACHINE LEARNING (ML) AND ARTIFICIAL INTELLIGENCE (AI) BASED VOICE MODEL

Short Summary of work done: Under the PS-1 project, I had to work on preparing a detailed work report on the foundation of a voice based biomarker which can be used to assess and diagnose depression in individuals using the vocal acoustic features.

Objectives of the project: Design of a Voice based biomarker that can be used to diagnose depression using the vocal acoustic features of a person's voice. Implemented using various Machine learning algorithms, Artificial Intelligence concepts and python programming language.

Tool used: Google colab , Librosa

Details of Papers/patents:-

Brief description of the working environment: The duration of the whole Practice school went in online mode. The online meets were actually very interesting and a lot of new advancements related to ML and AI were discussed in the company.

Academic courses relevant to the project: Machine Learning
Artificial intelligence

Learning Outcome: Machine Learning algorithms
Artificial Intelligence concepts.
Python programming language
Basics of corporate finance

PS-I station: Manodayam Pvt. Ltd. , Lucknow

Student

Name: UJJWAL PANDEY .(2021B1A12744P)

Student Write-up:

PS-I Project Title: MENTAL HEALTH ASSESSMENT USING ML AND AI ENGINE VIA A DATASET OF VOICE SAMPLES

Short Summary of work done: Voice samples are taken from the individual seeking help. Using the software Librosa, we derive the 13 features from the MFCC and using a database of hundreds of samples from Kaggle, we analyze the voice samples and determine the person's mental state. This method is very cost and time effective and eliminates the stigma against seeking help for mental problems as the automated system reachable to everyone is very quick and accurate to assess the state of people.

Objectives of the project: The project aimed to develop a way to assess the mental health using AI and ML engine via voice samples

Tool used: Majorly python was used and we did needed our laptops throughout the project.

Details of Papers/patents: None

Brief description of the working environment: The PS -1 Station members were all very supportive and were present to assist us at every step of our project, I was pleased when the project progressed in the expected manner and we were able to learn quite a lot. Also the things I have learnt were really interesting and kept me engaged throughout the project, it was an experience that I will never forget.

Academic courses relevant to the project: CP which I did in my first year, though not directly but was helpful in cultivating and developing the basic idea of programming.

Learning Outcome: I learnt about Artificial Intelligence and using it to develope new ML models, in addition to this I was able to discover new and interesting python libraries like librosa, panda and numpy, besides the technical knowledge that I gained, sanjay sir, who guided us throughout the project, provided us with a brief introduction to entrepreneur and the world of business.

PS-I station: Manodayam Pvt. Ltd. , Lucknow

Student

Name: AYUSH SRIVASTAVA .(2021B5AA0772P)

Student Write-up:

PS-I Project Title: Voice Biomarkers using Ai

Short Summary of work done: We would take voice sample and phq-9 test score and then put them in our pandas data frame and then using some deep learning algorithm to classify them into depressed and not depressed classes

Objectives of the project: Detect depression in person using his voice sample

Tool used: S/w- python, different python libraries, google colab

Details of Papers/patents: Zhao Q, Fan H-Z, Li Y-L, Liu L,
Wu Y-X, Zhao Y-L, Tian Z-X,
Wang Z-R, Tan Y-L and Tan S-P
(2022) Vocal Acoustic Features as
Potential Biomarkers
for Identifying/Diagnosing Depression:
A Cross-Sectional Study.
Front. Psychiatry 13:815678.
doi: 10.3389/fpsy

Brief description of the working environment: My work was online so usually we would have Google meet twice a week and in that Google meet he would give work and on the next meeting he would take the updates on the amount of work done. Overall working environment was good and very helpful

Academic courses relevant to the project: Machine Learning and Deep learning

Learning Outcome: ML algorithms, data processing, pandas data frame

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PS-I station: Mormugao Port Trust , Goa

Student

Name: Y E MAYURESH .(2021A2PS3055H)

Student Write-up:

PS-I Project Title: CONSTRUCTION OF CRUISE TERMINAL & ROAD NETWORK NH 17B

Short Summary of work done: This was a wonderful experience.

Objectives of the project: STUDY REPORT

Tool used: ANSYS, StaadPro

Details of Papers/patents: Report on Construction Project

Brief description of the working environment: It was wonderful experience

Academic courses relevant to the project: Construction Planning and technology, Mechanics of solids, Analysis of structures, Highway Engineering

Learning Outcome: MANAGEMENT AND CONSTRUCTION TECHNOLOGY

PS-I station: Mormugao Port Trust , Goa

Student

Name: TANMAY AGRAWAL .(2021A4PS2409P)

Student Write-up:

PS-I Project Title: Dust mitigation

Short Summary of work done: It was a project based on dust suppression at port. Since transport of materials generate large amount of dust particles which is harmful for the health of the workers as well as for the people living nearby. We found some useful and

feasible measures to be incorporated in MPA which includes basic measures as well as some technologies.

Objectives of the project: Improvising the existing dust mitigation technologies at MPA as well as suggest new measures

Tool used: Power point

Details of Papers/patents:N/A

Brief description of the working environment: Working environment was friendly and all the senior employees as well as our mentor were cooperative and guided us nicely.

Academic courses relevant to the project: EVS

Learning Outcome: Learnt about HMC, port functioning and environmental control.

PS-I station: Mormugao Port Trust , Goa

Student

Name: KARTIK SHARMA .(2021A4PS3085H)

Student Write-up:

PS-I Project Title: Improvements on Dust Mitigation System

Short Summary of work done: Understanding and analysing the already existing dust suppression system at the port and finding any places where improvement can be done and suggesting new technologies that can be used to further improve the existing dust suppression system, also considering the costs involved and budget that the organization have.

Objectives of the project: To improve the working condition at port and working under the govt. regulation by dust suppression

Tool used: Software - Internet, research papers, journals and conference papers; hardware - all the dust suppresion systems

Details of Papers/patents: Report submitted to PSD

Brief description of the working environment: It is like an average govt. Office, but the port area had very high security and a very dusty environment. Officials are excited to help, but they said "Actually we don't have many things to teach as we've leased most of the port to private corporations". as of now they only have a HMC crane, dust mitigation system and a fire fighting system.

Academic courses relevant to the project: nil

Learning Outcome:
Research and Analytics, Understanding dust suppression.

PS-I station: National Aluminium Company -A central PSU , Angul

Student

Name: SATWIK RATH .(2021A4PS2668P)

Student Write-up:

PS-I Project Title: Analysis and Improving of SMCD in furnace

Short Summary of work done: We were introduced to the basic aspects of the production on aluminium and the commercial aspect of it. Worked on the modification of the furnace to be more efficient and reduce carbon footprint. Also worked on organising the storage of the cast house using excel.

Objectives of the project: Improving efficiency and reducing carbon footprint

Tool used: NA

Details of Papers/patents:NA

Brief description of the working environment: Gave us a taste of how a PSU works and the various levels of organisation within the company. Understood about how various different departments need to be involved for production.

Academic courses relevant to the project: Thermodynamics, Heat Transfer, Manufacturing Processes

Learning Outcome: Learnt about the commercial large scale production of aluiminium

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PS-I station: National Aluminium Company -A central PSU , Angul

Student

Name: AUM JENA .(2021A4PS2759H)

Student Write-up:

PS-I Project Title: Analysis and Improving of SMCD in furnace

Short Summary of work done: We were introduced to the basic aspects of the production on aluminium and the commercial aspect of it. Worked on the modification of the furnace to be more efficient and reduce carbon footprint. Also worked on organising the storage of the cast house using excel.

Objectives of the project: Improving efficiency and reducing carbon footprint

Tool used: N/A

Details of Papers/patents:N/A

Brief description of the working environment: Gave us a taste of how a PSU works and the various levels of organisation within the company. Understood about how various different departments need to be involved for production.

Academic courses relevant to the project: Manufacturing Process

Learning Outcome: Learnt about the commercial large scale production of aluiminium

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PS-I station: National Aluminium Company -A central PSU , Angul

Student

Name: SASWAT SUMAN DWIBEDI .(2021B1A42293P)

Student Write-up:

PS-I Project Title: Analysis and Improving of SMCD in furnace

Short Summary of work done: We were introduced to the basic aspects of the production on aluminium and the commercial aspect of it. Worked on the modification of the furnace to be more efficient and reduce carbon footprint. Also worked on organising the storage of the cast house using excel.

Objectives of the project: Improving efficiency and reducing carbon footprint

Tool used: N/A

Details of Papers/patents:N/A

Brief description of the working environment: Gave us a taste of how a PSU works and the various levels of organisation within the company. Understood about how various different departments need to be involved for production.

Academic courses relevant to the project: Manufacturing Process

Learning Outcome: Learnt about the commercial large scale production of aluminum

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PS-I station: National Metallurgical Laboratory Madras Centre , Chennai

Student

Name: SHRINANDHAN B .(2021A1PS1680P)

Student Write-up:

PS-I Project Title: Coal flotation using biodiesel as a collector

Short Summary of work done: The project allotted to me was on 'Coal flotation using biodiesel as a collector'. Bio-diesel is a biodegradable material, produced typically from vegetable oils / animal fats. Our main aim was to produce biodiesel, the principle behind the process being 'Trans-esterification', and to find its use as a potential collector in flotation experiments. After producing biodiesel, we then have to make sure that the produced biodiesel is then suitable for use as a collector in coal flotation. This is done by characterisation tests such as FFA. FFA (Free Fatty Acid) test is a test where we measure the amount of FFA content in a given oil. If it lies within the prescribed range, then it is suitable as a collector. After ensuring the above, we then proceed for the actual flotation process, where we use reagents like collectors and frothers to help remove the coal sample from its impurities. We then proceed for proximate analysis of coal, to measure its volatile matter, ash, fixed carbon and moisture contents. For this particular experiment, coking coal was used. The coal sample and the biodiesel produced earlier were both analyzed using FTIR spectroscopy.

Objectives of the project: To produce biodiesel from animal fats, and to check its efficiency as a collector in coal flotation.

Tool used: Mendeley Reference Manager, FTIR Spectrophotometer

Details of Papers/patents:-

Brief description of the working environment: There specifically were no pre-requisites mandated by the company. The staffs and scientists-in charge were very helpful, who guided me in each and every step. I was able to improve and learn on a lot of aspects, including but not limited to things such as the work environment inside a research facility, improved social and presentation skills, etc. With respect to the project, we were introduced to the various instruments used to analyze a material's composition, such as XRF, FTIR and UV spectrophotometers. The working of a Particle Size Analyzer was also shown. I was introduced to flotation cells, which are used in froth flotation processes. The project revolved around an economic and environment-friendly method of purifying minerals/materials.

Academic courses relevant to the project: -

Learning Outcome: 1. Use of bio-diesel as a potential collector for flotation experiments.
2. Working of a mechanical flotation cell, and spectrophotometers.
3. procedure of coal flotation, and production of bio-diesel from wastes.

PS-I station: National Metallurgical Laboratory Madras Centre , Chennai

Student

Name: ADITI CHANDRAMOULEE .(2021A1PS2584P)

Student Write-up:

PS-I Project Title: ESTIMATION OF IRON ORE GRADE AND RECOVERY BY FLOTATION PROCESS, FTIR SPECTROSCOPY AND ANALYTICAL INSTRUMENTATION TECHNIQUES

Short Summary of work done: Flotation: Experimental Design: Researchers design flotation experiments to evaluate the effectiveness of different flotation parameters, such as reagent types, dosage, pH, and pulp density. They vary these parameters systematically to optimize the flotation process for specific iron ore samples. Reagent Selection: Different types of collectors, frothers, modifiers, and activators are tested to determine their performance in selectively separating iron ore minerals from gangue minerals. Researchers assess the efficiency and selectivity of each reagent and their interactions to identify the most suitable reagent combinations. Kinetic Studies: Researchers perform kinetic studies to understand the flotation kinetics of iron ore minerals. They investigate the rate at which the minerals attach to the air bubbles and the factors affecting the flotation rate. This information helps optimize the flotation process and develop mathematical models for process simulation. FTIR Spectroscopy: Mineral Identification: FTIR spectroscopy is used to identify the mineral composition of iron ore samples. By analyzing the infrared absorption spectrum of the minerals, researchers can identify specific mineral phases present in the ore. This helps in understanding the mineralogical characteristics and their influence on the flotation behavior. Surface Analysis: FTIR spectroscopy is employed to study the surface properties of iron ore minerals. By analyzing the functional groups and surface functionalization, researchers can gain insights into the hydrophobic or hydrophilic nature of the minerals. This information aids in understanding the mineral-water interactions during flotation. Chemical Analysis: Elemental Composition: Chemical analysis techniques, such as X-ray fluorescence (XRF) or inductively coupled plasma (ICP), are used to determine the elemental composition of iron ore samples. This analysis provides information about the major and trace elements present in the ore, which can influence its behavior during flotation. Impurity Assessment: Chemical analysis helps identify impurities in iron ore, such as silica, alumina, phosphorous, sulfur, and trace elements. These impurities can affect the quality of the iron ore concentrate produced. By quantifying the impurities,

researchers can optimize the flotation process to reduce their detrimental effects. Quality Control: Chemical analysis is crucial for quality control purposes during the mining and processing of iron ore. It ensures that the ore meets the required specifications and allows for adjustments in the beneficiation process to maintain product quality and consistency. Overall, the work conducted in flotation, FTIR spectroscopy, and chemical analysis of iron ore aims to understand the flotation behavior, mineralogical characteristics, and chemical composition of the ore. This knowledge helps in developing effective flotation strategies, optimizing process parameters, and ensuring the production of high-quality iron ore concentrates.

Objectives of the project: To minimize silica, alumina and clay content in low-quality BHQ Iron Ore by beneficiation through Froth Flotation

Tool used: Mendeley Reference Manager Software, Elsevier research tool, Minitab, ANOVA, FTIR Spectroscopy

Details of Papers/patents:-

Brief description of the working environment: Team Collaboration: Interns usually work in teams alongside experienced professionals. We collaborate on projects, contribute to discussions, and receive guidance from their mentors. The working environment promotes teamwork, communication, and knowledge sharing.
Learning Opportunities: Companies provide interns with opportunities to learn and develop their skills. They may offer training programs, workshops, and access to industry-specific tools and technologies. Interns are encouraged to actively participate in learning activities and take on challenging tasks to expand their knowledge.
Professionalism: The working environment expects interns to conduct themselves professionally. This includes being punctual, meeting deadlines, respecting confidentiality, and adhering to the company's policies and procedures. Interns are expected to demonstrate a strong work ethic, initiative, and a willingness to learn.
Networking: Internships provide an opportunity to build professional networks. Interns can interact with colleagues, supervisors, and professionals from various departments. Networking enables interns to gain insights into different roles, establish connections, and potentially open doors for future career opportunities.
Expectations from the company:
Active Participation: Expect interns to actively engage in the assigned tasks and contribute to the projects they are working on. Interns are encouraged to share their ideas, ask questions, and seek clarification when needed.
Learning and Growth: Expect interns to have a thirst for learning and a desire to grow professionally. Interns should be open to feedback, take initiative in acquiring new skills, and demonstrate a willingness to step out of their comfort zones.
Professional Conduct: Expect interns to maintain professionalism throughout their internship. This includes being respectful to colleagues, adhering to company policies, maintaining confidentiality, and acting with integrity.

Dedication and Accountability: Expect interns to be dedicated to their work and take ownership of their assigned tasks. Interns should demonstrate a sense of responsibility, meet deadlines, and communicate any challenges or issues proactively.

Academic courses relevant to the project: -

Learning Outcome: Understanding the principles of froth flotation: Froth flotation is a widely used method for the concentration of minerals. By studying the preparation of reagents for iron ore froth flotation, I could gain a deeper understanding of the underlying principles and mechanisms of this separation technique.
Knowledge of reagent selection: I learnt about the different types of reagents used in iron ore froth flotation and their roles in the process. This includes collectors, frothers, modifiers, and activators. Understanding the selection criteria for these reagents helped make informed decisions when designing and optimizing flotation processes.
Understanding ore characteristics: To effectively prepare reagents for iron ore froth flotation, you need to consider the characteristics of the ore, such as mineral composition, particle size distribution, and surface properties. Studying the process helped me learn how to assess ore characteristics and adapt the reagent formulation accordingly.
Practical laboratory skills: Preparation of reagents involves practical laboratory skills, such as accurate measuring and mixing of reagents, dissolution processes, and pH adjustment. By practicing these skills, I enhanced my laboratory techniques and developed proficiency in executing chemical procedures.
Process optimization: Learning about the preparation of reagents in iron ore froth flotation enabled me to understand the importance of process optimization. I explored various reagent combinations, concentrations, and pH levels to improve flotation efficiency, selectivity, and overall performance.
Environmental considerations: Froth flotation processes require the use of reagents, and understanding their preparation involves considering environmental factors. I could learn about the potential environmental impacts associated with the use of certain reagents and explore sustainable alternatives or mitigation measures.
Analytical techniques: The study of reagent preparation for froth flotation may involve the use of analytical techniques for characterizing ore samples, assessing reagent performance, and monitoring process parameters. This allows to develop proficiency in analytical techniques relevant to mineral processing and flotation.
Overall, studying the preparation of reagents for iron ore froth flotation provides a comprehensive understanding of the fundamental principles, practical aspects, and optimization strategies involved in this mineral processing technique. It equips one with knowledge and skills applicable to various stages of mineral processing and enables you to contribute to the efficient and sustainable extraction of valuable minerals from ore.

PS-I station: National Metallurgical Laboratory Madras Centre , Chennai

Student

Name: ABEL OOMMEN LUKE(2021A1PS2963G)

Student Write-up:

PS-I Project Title: Iron Ore Flotataion

Short Summary of work done: Iron flotation is a mineral processing method where iron ore particles are separated from impurities. The ore is crushed, mixed with water and a flotation reagent, and then air bubbles are introduced. The hydrophobic iron-bearing minerals attach to the bubbles, forming a froth that is collected as concentrate.

Objectives of the project: Estimation of iron content in the ore and minimisation of silica content

Tool used: Flotation device

Details of Papers/patents: Araujo, A. C., P. R. M. Viana and A. E. C. Peres (2005). "Reagents in iron ores
flotation." Minerals Engineering 18(2 SPEC. ISS.): 219-224.
Bhavan, I. (2023). Indian Minerals Yearbook 2021 60 th Edition IRON, STEEL & SCRAP
AND SLAG (ADVANCE

Brief description of the working environment: Clean Technology Development: NML focuses on the creation and use of energy-saving,
environmentally friendly technology in metallurgical processes. In order to cut greenhouse gas emissions and lessen environmental effects, this involves optimising current operations, looking into alternative raw materials, and using renewable energy sources.
Expectations: Increasing the minimization of silica content in the provided iron ore and to get the most possible iron content in the ore.
Learnt how to use a flotation device and the analysis of iron content in the ore

Academic courses relevant to the project: Metallurgy

Learning Outcome: flotation of iron ore and analysis of the iron content in the ore

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PS-I station: NCC(Nagarjuna Construction Company) , Hyderabad

Student

Name: MUDUNURI SUBRAMANYA VARAHA ABHIRAM VARMA(2021A2PS2361P)

Student Write-up:

PS-I Project Title: Insights and Innovations in Highway engineering (Roads and Bridges)

Short Summary of work done: The highway engineering sector has made significant advancements in pavement design methodologies, including traditional approaches like IRC 37 and IRC 58. These methodologies have played a crucial role in shaping the country's infrastructure landscape. However, alongside these traditional methods, the industry has also embraced cutting-edge technologies and innovative practices like 3D asphalt printers, Composite bridges - FRP, Reclaimed Asphalt Pavement (RAP), and Geographic Information System (GIS) technology. By incorporating Reclaimed Asphalt Pavement (RAP), the industry has demonstrated its commitment to sustainability, reducing waste and environmental impact while providing economic benefits for future road projects. The integration of GIS technology has transformed highway planning and management, empowering informed decision-making and optimizing routes for more efficient and well-connected road networks. Together with the advancements in bridge engineering and the introduction of Composite bridges - Fiber-Reinforced Polymers (FRP), Ultra-High-Performance Concrete (UHPC), and Ultra-High-Performance Fiber-Reinforced Concrete (UHPFRC), these innovations have paved the way for a safer, more connected, and prosperous future for India.

Objectives of the project: The main objective of this project is to understand the logic, Guidelines, and efforts that go into the design of Pavements, and Bridges and how new methodologies would give a better payoff by making the construction of these structures more sustainable,

Tool used: Software tools like IITPAVE, IITRIGID, and necessary IRC codes.

Details of Papers/patents:IRC 6 2017 & IRC 5 2015- STANDARD SPECIFICATIONS AND CODE OF PRACTICE
FOR ROAD BRIDGES, IRC 37 2018 - Design of Flexible Pavements, IRC 58 2015 - Design of Rigid Pavements.

Brief description of the working environment: I've been given a desk to work, and a bunch of experienced people to guide me on the advancements and insights in Highway engineering and I got connected to the head engineer for more insights. I expected to get a good exposure to know how employees are different from students to know how the company works. I've learned the behaviour, Etiquette, and the knowledge on advancements of Bridge and Road engineering which helped me see the constructions around me in a Civil engineer's perspective.

Academic courses relevant to the project: Yes, highway engineering, Mechanics of Solids and soil, Surveying, Civil engineering materials, and Analysis of Structures are the courses enhanced my understanding of my project.

Learning Outcome: Advancements in highway engineering have transformed India's infrastructure landscape, from traditional methods like IRC 37 and IRC 58 to innovative technologies such as 3D asphalt printers and Composite bridges - FRPs. Innovations in bridge engineering, including IRC 5 and IRC 6 guidelines, ensure robust and safe designs. Composite bridges with FRP, UHPC, and UHPFRC offer lightweight and durable structures for aggressive environments. Additionally, the Reclaimed Asphalt Pavement (RAP) and Geographic Information System (GIS) technology can be great resources and cost savers.

PS-I station: NCC(Nagarjuna Construction Company) , Hyderabad

Student

Name: NARKULA SRIKAR REDDY .(2021A2PS2624P)

Student Write-up:

PS-I Project Title: Water treatment plant

Short Summary of work done: We need drinking and fresh water in our daily to daily life but our world is consist of fresh water less than 3 percent but available for usuable is only 0.5% so how we can use the most of it without any headloss and how many steps are involved while we treating the raw water like aeration, dosing,rapid mixing, sedimentation, filtration, storage , filtration and distribution so these are the overview of the process in WTP

Objectives of the project: Study on WTP process

Tool used: Nothing

Details of Papers/patents: Nothing

Brief description of the working environment: Overall company is good but mentor is not friendly with me it's my personal feeling and overall about the work and learning insight were good .

Academic courses relevant to the project: Hydraulics

Learning Outcome: How drinking water is extracted from raw water

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PS-I station: NHPC Ltd , Faridabad

Student

Name: SHRUTI CHATURVEDI .(2021A2PS2637P)

Student Write-up:

PS-I Project Title: Preliminary Static Stability Analysis of Concrete Gravity Dam and Structural Analysis of Control Block

Short Summary of work done: This project conducted a comprehensive analysis of a gravity dam and its accompanying control block. The stability of the dam was assessed manually using Excel and AutoCAD, while the control block was analyzed using STAAD Pro software. The results indicated satisfactory resistance against overturning and sliding, with induced stresses within acceptable limits. The study's outcomes contribute valuable insights to the design and construction of dams and control blocks, improving engineering practices in this field.

Objectives of the project: Preliminary static stability analysis of Dam(NOF) block Preliminary structural Analysis of Dam Control Block

Tool used: STAAD.Pro,Excel,AutoCAD

Details of Papers/patents:no

Brief description of the working environment:
The office environment for this project was characterized by a pleasant and welcoming atmosphere. The workspace was thoughtfully designed, with comfortable seating arrangements, ample natural light, and a well-organized layout. The office was equipped with modern infrastructure and facilities, including advanced computer systems and software tools, fostering a technologically enabled work environment. Additionally, the office had a collaborative and supportive culture, with colleagues and mentors readily available for guidance and assistance. The overall ambiance was conducive to concentration, creativity, and effective teamwork, creating an ideal setting for successful project execution.

Academic courses relevant to the project:
The academic relevance of this project lies in its contribution to the field of civil engineering. By conducting a detailed static stability analysis of a gravity dam and structural analysis of its control block, the project enhances understanding of imp

Learning Outcome: -Proficiency in conducting static stability analysis and structural analysis of gravity dams and control blocks.
-Familiarity with using software tools like Excel, AutoCAD, and STAAD Pro for modeling and analysis.
-Understanding established methodologies and standards for stability assessment and design of dams and control blocks.
-Enhanced problem-solving and critical thinking skills in the field of civil engineering, particularly in the context of dam and control block design and construction.

PS-I station: NHPC Ltd , Faridabad

Student

Name: ANUSHA SINGH(2021A4PS2312G)

Student Write-up:

PS-I Project Title: Hydropower generation and power potential study

Short Summary of work done: Learnt details about hydropower generation. Visited labs and did testing relevant to these studies. Also learnt application of this study in real time dam projects.

Objectives of the project: To learn about hydropower generation and power potential study of a particular dam

Tool used: Data from excel, machinery from labs for testing

Details of Papers/patents: No details

Brief description of the working environment: Very cooperative work environment. Helped us kearn a lot. Always gave us new perspective to think about concepts and their applications as well.

Academic courses relevant to the project: Fluid mechanics

Learning Outcome: Can calculate and find the total capacity and various other factors about setting up of dam from the data given about the area

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PS-I station: NHPC Ltd , Faridabad

Student

Name: KESHAV PRANSHU(2021A4PS2410G)

Student Write-up:

PS-I Project Title: Basics and Fundamentals of Hydroelectric Power Generation from Mechanical Engineering Perspective

Short Summary of work done: We learnt about the work ethic of an office. Further, we learnt about the mechanical aspects of a hydro power plant while scratching the surface of electrical and electronic aspects of the hydro power generation. We explored the components of power plants, various factors accounted for before deciding the specifics of the equipment and their installation. Finally, the various challenges faced during hydro power generation.

Objectives of the project: Understanding about the basics of hydroelectric power generation: the idea, the working principle, the components, the selection of components, their implementation and the challenges faced.

Tool used: NA

Details of Papers/patents:NA

Brief description of the working environment: The working environment was subtle and mellow. We were checked in on continually and any doubts that we asked were addressed. Our general manager was very understanding and curious about the BITS culture. He guided us well with our work and understanding.

Academic courses relevant to the project: Fluid Mechanics

Learning Outcome: 1. Working principle for electricity generation
2. Components used in a hydro power plant.
3. Power Potential study
4. Classification of turbines
5. Various fluid phenomena and their significance in a power plant.

PS-I station: NHPC Ltd , Faridabad

Student

Name: ADITYA SRIVASTAVA .(2021A4PS2436H)

Student Write-up:

PS-I Project Title: Dhauliganga Power Plant

Short Summary of work done: Firstly, we were taught how vast each component of a hydropower station is and how it is used depending on different situations. We were taught about generators, turbines, runners, how to operate it and call for maintenance of a power station. We were shown how the OnM department handles, operates and supervises all power stations from the head office.

Objectives of the project: To know about the specific power station, its operation and components used in its making

Tool used: None

Details of Papers/patents:NA

Brief description of the working environment: The PS station I was allotted was a corporate office. Though I expected a field trip to an actual power station from the company so that we learn better, the station helped us in every possible way to teach us from the theoretical side. Working environment and people in the company were mostly friendly and helping.

Academic courses relevant to the project: Thermodynamics, Mechanisms and Machines

Learning Outcome: Working of hydro power plant in generating electricity

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PS-I station: NHPC Ltd , Faridabad

Student

Name: ADITYA BANDYOPADHYAY(2021A4PS2708G)

Student Write-up:

PS-I Project Title: Proposal for Renovation & Maintenance of the Loktak Hydro-Power Station

Short Summary of work done: During the first four weeks of my internship, I immersed myself in the operation & maintenance department, gaining proficient knowledge of various subjects such as Dams, Turbines, budgeting & tariffs, and Power scheduling with GRID India. In the subsequent four weeks, I initiated a dedicated project to create a comprehensive report on the Life extension proposal for the Loktak Station. The project was unique as it involved non-conventional tasks not previously established by the department. Collaborating closely with experienced personnel, I delved into the intricacies of the proposal and studied the science behind a Hydro-Power Plant's operations, from the reservoir to Power Distribution. Extensive discussions and interviews with experts provided valuable insights, while collecting confidential data and conducting thorough internet research deepened my understanding. Applying my academic knowledge, I analyzed challenges and devised effective solutions. The collaborative nature of the project enhanced my teamwork and communication skills, enabling me to work harmoniously with diverse professionals. The experience not only solidified my technical expertise but also honed my problem-solving abilities. In conclusion, my internship provided a strong foundation in the operation & maintenance department and offered a rewarding experience through the dedicated project. This hands-on approach

empowered me to apply theoretical knowledge to real-world scenarios, preparing me for future success in related industries.

Objectives of the project: My Objective for the first half of the PS was to gain substantial knowledge on *Study on the Operation&Maintainace (O&M) Division; For the second half of the training i extensivly studied on making a proposal for "Renovation & Modernisation of the Loktak

Tool used: I gained soft skills like - Communication with experienced people, Teamwork, Problem-Solving Leadership, Time Management & improved my Presentation Skills; HARD SKILLS & softwares: O&M doesnt use any software (unlike Design) i did gain some skills like K

Details of Papers/patents: As I previously mentioned, all the references are either by employees while meeting them via word of mouth, and I noted the details down. Still, the majority of the portion came from a DPR (Detailed Project Report), which NHPC is working on right now, whi

Brief description of the working environment: Employees are friendly yet professional, and they are helpful, especially my mentor Mr Sachin Aggarwal & also the Head of HRD Mr Anirudh Singh; Office Environment was overwhelming at the start and we may feel that we are overlooked but this what its like Working in a typical government MNC, which has its own caveats. My experience though, has been very good, to say the least.

Academic courses relevant to the project: A mechanical Course, POWERPLANT ENGINEERING; THERMODYNAMICS; TRW (technical Report writing); Fluid Mechanics & Mechanics of Solids are a few which I can think of the top of my head.

Learning Outcome: NHPC has contributed immensely to my academic and professional life these last eight weeks. I hope I was able to provide my level best to the company. I learnt a lot about the company and how the department functions, and more importantly i also learnt a lot of soft skills as well.

PS-I station: NHPC Ltd , Faridabad

Student

Name: AMAN ALI .(2021B1A22735P)

Student Write-up:

PS-I Project Title: Stability anaylsis of NOF block of Concrete Gravity Dam

Short Summary of work done: During my PS 1, I focused on the stability analysis of a gravity dam under dynamic conditions. I conducted extensive research and analysis to understand the behavior of the dam structure when subjected to dynamic loads such as seismic forces and water pressure fluctuations. I utilized specialized software and mathematical models to simulate and evaluate the dam's response to these dynamic forces. This involved considering factors such as soil-structure interaction, reservoir hydrodynamics, and material properties. Through my work, I identified critical areas that required reinforcement to ensure the dam's stability during dynamic events. I also studied and analyzed case studies of similar dam structures to gain further insights and validation for my findings. The work done during PS 1 provided valuable knowledge and experience in the field of dam stability analysis in dynamic conditions, equipping me with practical skills and a deeper understanding of the challenges and considerations involved in ensuring the safety and stability of gravity dams.

Objectives of the project: The project aims to analyze dam stability, evaluating failure modes, structural behavior, weaknesses, and safety factors. Recommendations for design enhancements will be provided, while considering existing standards and regulations. Strategies for long-t

Tool used: MS - Excel, Auto Cad, CADAM 3D

Details of Papers/patents:IS Code 6512 - 1984, IS Code1893 -1984

Brief description of the working environment: During my internship, I worked in a professional and collaborative environment, focusing on stability analysis of gravity dams and spillways. I visited geological labs to gain insights into geotechnical aspects and site-specific considerations for dam construction. Understanding the national power grid of India provided me with knowledge of power generation, transmission, and distribution. The company emphasized continuous learning and expected a strong work ethic, proactive problem-solving, and effective communication. I engaged in team discussions and decision-making processes, seeking guidance from experienced professionals. Overall, the internship provided a professional and collaborative work environment, allowing me to contribute to meaningful projects. I developed a strong work ethic, problem-solving abilities, and effective communication skills. The internship enhanced my skills in civil engineering principles, project management, teamwork, and adaptability.

Academic courses relevant to the project: Soil mechanics, Hydrology

Learning Outcome: The project will foster an in-depth understanding of dam stability, applying analysis techniques to identify weaknesses and propose design enhancements. Compliance with standards and regulations will be emphasized, alongside the

development of long-term stability strategies and informed decision-making skills in risk assessment.

PS-I station: NHPC Ltd , Faridabad

Student

Name: AMRIT KUMAR JHA .(2021B4A41696P)

Student Write-up:

PS-I Project Title: Dhauliganga Hydroelectric Power Plant

Short Summary of work done: Got to learn about the working of a hydro power plant

Objectives of the project: To learn about functioning of a Hydroelectric Power Plant

Tool used: NA

Details of Papers/patents: None

Brief description of the working environment: Very supportive friendly people

Academic courses relevant to the project: Na

Learning Outcome: Knowledge of turbines generators etc

PS-I station: Nirmaan Organization , Madhapur

Student

Name: MANTRI SRI KRUTHI .(2021A7PS2006P)

Student Write-up:

PS-I Project Title: Designing blood bank website

Short Summary of work done: I have been allotted to the blood bank project where the admin and donor can access the website admin will be redirected to one page and donor will be redirected to another page , i have created and designed both the pages and also created a page called register for donor.

Objectives of the project: We had to create an interface for the website blood bank and also create and design the pages of the website.

Tool used: Html,Css,JavaScript,ReactJs and VSCode

Details of Papers/patents:-

Brief description of the working environment: The environment was very good to work in we had our mentor's support all the time and also the tech team members always helped us if we had any doubts. they were very patient with us and helped in learning new things.

Academic courses relevant to the project: Nothing

Learning Outcome: Learned how to create a website, pages and learnt how to design a website.

PS-I station: Pilani AtmaNirbhar Resource Center (PARC) , Pilani

Student

Name: OM MISHRA .(2021A8PS3123H)

Student Write-up:

PS-I Project Title: village survey and rural development

Short Summary of work done: The visits to three different villages in Rajasthan, namely Jherli, Bajawa, and Mohansingh, was a good experience with lots of interactions within these communities, it was not what we expected - we expected a very backward class village like under constructed roads to our surprise . In Jherli, the village appeared more developed with technologies like cars and mobiles, but there were still issues with power outages and signal connectivity. The healthcare services were limited, and major emergencies required traveling to hospitals in nearby towns. Cancer cases in the village were linked to heavy tobacco consumption. Bajawa showcased a well-equipped government school and a PARC tinkering lab, indicating efforts towards education and technology adoption. Healthcare services were limited to a local clinic, and major emergencies required traveling to hospitals in Mandela, Jaipur, or Bikaner. The village had a relatively low rate of cancer cases. Mohansingh, a relatively isolated village, lacked technology, and electricity was scarce during the visit. The villagers' primary occupation was farming, and their lifestyle was relatively simple. Healthcare services were limited, and the village relied on the support of a teacher for certain activities. Overall, the experiences in these villages highlighted the challenges and strengths of rural communities in Rajasthan, offering insights into their lives and healthcare access which we found during our visit. How biases were involved in our visit: The first visit to the village was involved with a hesitant attitude from our side as we were nervous to approach them but as talked to more people, but as we further talked to more people and had more visits our person and urban bias got shed. Initially we didn't take up the option of visiting during June as this is the hottest month, but we gave up our seasonal bias and went our visit

Objectives of the project: to survey the villages , gather date and help other organizations to proceed accordingly

Tool used: microsoft excel and word

Details of Papers/patents:none

Brief description of the working environment: work environment was good, the institute was very great

Academic courses relevant to the project: none

Learning Outcome: increased communication skills along with insights of rural lifesytle.

PS-I station: Preto Tooling Systems , Hyderabad

Student

Name: LAKSHMI NARAYANA KOCHERLAKOTA .(2021A2PS2150P)

Student Write-up:

PS-I Project Title: Analysis of material selection in the sheet metal forming industry and the design development of press tools and dies.

Short Summary of work done: We were divided into three groups and worked on three projects in total and. We first had to understand the manufacturing industry and the mass manufacturing processes. Our main topic of interest was the press tool and die. I first looked into the press tool and its's uses and then started learning about the various components or parts of a press tool. Using a variety of resources such as research papers and engineering archives, I started looking for the design principles of a press tool. The area of research was very interesting and gave me an insight to the challenging decisions that had to be taken by an engineer while designing and allocating materials for a project. As a part of our project, we visited the workshop of the company which consisted of machines like CNC's, grinding, press tools, etc. We also worked on analyzing the materials used to manufacture these tools and what properties to look for while making these choices.

Objectives of the project: Learn about the various strategies of material selection for the sheet metal forming industry such as the ASHBY chart method and advantages and disadvantages of multi criteria decision making methods such as the weighted decision matrix. The study on desi

Tool used: -

Details of Papers/patents:-

Brief description of the working environment: The working environment we were offered was not the best. The conditions we were working in were the bare minimum requirements like a single table which was originally for the company's staff to have lunch. The faculty we got to interact with were helpful and supported us in our projects

Academic courses relevant to the project: Manufacturing processes

Learning Outcome: Working out the best decisions based on client needs and costs incurred.

PS-I station: Preto Tooling Systems , Hyderabad

Student

Name: VISHNU AKUMALLA .(2021A4PS1894H)

Student Write-up:

PS-I Project Title: 2) Die design and load calculations for sheet metal forming processe

Short Summary of work done: We had to research on our project topic and collect information regarding the same, compare how the process we selected is done in the industry in general with how it's done in our ps station, and look at the various areas where the company is either lagging or going ahead of their competition, and suggest changes if any.

Objectives of the project: • Choose the appropriate process for die design • Thorough literature survey of various aspects of die design for chosen process. • Understand the materials used for dies and possible alternative materials. • See the available die in the industries and

Tool used: Softwares included CAD

Details of Papers/patents: None

Brief description of the working environment: We got a separate room to sit in and brainstorm ideas with our group. Most days were spent on the Internet researching on our projects. The company doesn't put in effort in clearing your doubts unless you are proactive and approach them first with your questions, which is expected. They are understanding and explain without any inhibitions. The faculty are experienced and don't hesitate in explaining anything you have doubts in, and overall it was an excellent learning experience.

Academic courses relevant to the project: Manufacturing processes

Learning Outcome: To understand the load calculations that take place during sheet metal forming processes and understand the various heat treatment processes that go along with it

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PS-I station: Preto Tooling Systems , Hyderabad

Student

Name: BURUGU VENKATA SAKETHA REDDY .(2021A4PS1986H)

Student Write-up:

PS-I Project Title: Study on strategy for selection of materials for sheet metal forming industries, Die design and load calculations for sheet metal forming processes, Analysis of various defects in sheet metal forming processes and suggest possible troubleshooting.

Short Summary of work done: We basically had to prepare three reports based on the above project titles mentioned.

Objectives of the project: Thorough literature survey on various strategies adopted for material selection, Thorough literature survey of various aspects of die design for chosen process, Review the available literature related to the chosen process and defects involved during the

Tool used: -

Details of Papers/patents:NA

Brief description of the working environment: The working environment is amicable and encouraging. Just to let you know, staff are accommodating regarding any project doubts or while explaining the work. The workspace is very chill. We can expect to learn about working in an industry and root-level manufacturing. The way we deal with our clients and their expectations.

Academic courses relevant to the project: Manufacturing processes

Learning Outcome: I learnt about the selection of materials for sheet metal forming industries, Die design and load calculations for sheet metal forming processes and also the various defects in sheet metal forming processes and suggest possible troubleshooting.

PS-I station: Preto Tooling Systems , Hyderabad

Student

Name: ANKIT SHARMA .(2021A4PS2343H)

Student Write-up:

PS-I Project Title: Study on strategy for selection of materials for sheet metal forming industries ,Die design and load calculations for sheet metal forming processes and Analysis of various defects in sheet metal forming processes and suggest possible troubleshooting.

Short Summary of work done: We learnt about the defects that occur during the metal forming processes. The shearing and bending operations were looked into as they represented a wide variety working and process and help us understand the defects and how they occur. Various factors such as the material properties, tooling properties and cutting forces, clearances decide the finishing and the surface finishing of the metal sheet. Recognized the characteristic properties of each type of defects such as curls, burrs, etc.

Objectives of the project: Thorough literature survey on various strategies adopted for material selection. Understanding of basics of design for material selection using the ASHBY CHART method. Collection of available of Bill of materials of various automotive components in

Tool used: CNC machine

Details of Papers/patents:none

Brief description of the working environment: The company was very welcoming and we got a nice room where we could do all our work. The employees there were very helpful and explained a lot about how the company functions and how sheet metal forming industry functions.

Academic courses relevant to the project: Material science and manufacturing , Mechanics of Solids.

Learning Outcome: Learnt about how the sheet metal forming industry works.

PS-I station: Preto Tooling Systems , Hyderabad

Student

Name: PAVAN VISWANADHA .(2021A4PS2774H)

Student Write-up:

PS-I Project Title: Study on strategy for selection of materials for sheet metal forming industries, Die design and load calculations for sheet metal forming processes, Analysis of various defects in sheet metal forming processes and suggest possible troubleshooting.

Short Summary of work done: We basically had to prepare three reports based on the above project titles mentioned.

Objectives of the project: Thorough literature survey on various strategies adopted for material selection, Thorough literature survey of various aspects of die design for chosen process, Review the available literature related to the chosen process and defects involved during the

Tool used: Google docs , powerpoint, Solidworks , excel

Details of Papers/patents:none

Brief description of the working environment: The working environment is amicable and encouraging. Staff are accommodating regarding any project doubts or while explaining the work. The workspace is very chill. We can expect to learn about working in an industry and root-level manufacturing. The way we deal with our clients and their expectations.

Academic courses relevant to the project: Manufacturing processes

Learning Outcome: I learnt about the selection of materials for sheet metal forming industries, Die design and load calculations for sheet metal forming processes and also the various defects in sheet metal forming processes and suggest possible troubleshooting.

PS-I station: Preto Tooling Systems , Hyderabad

Student

Name: MARRI THANVITA REDDY .(2021B2A43134H)

Student Write-up:

PS-I Project Title: Study on strategy for selection of materials for sheet metal forming industries, Die design and load calculations for sheet metal forming processes, Analysis of various defects in sheet metal forming processes and suggest possible troubleshooting.

Short Summary of work done: We basically had to prepare three reports based on the above project titles mentioned.

Objectives of the project: Thorough literature survey on various strategies adopted for material selection, Thorough literature survey of various aspects of die design for chosen process, Review the available literature related to the chosen process and defects involved during the

Tool used: We didn't exactly use any tools. We just observed the company and the machines used their to produce press tools especially for the automobile industry.

Details of Papers/patents:NA

Brief description of the working environment: The working environment is amicable and encouraging. Just to let you know, staff are accommodating regarding any project doubts or while explaining the work. The workspace is very chill. We can expect to learn about working in an industry and root-level manufacturing. The way we deal with our clients and their expectations.

Academic courses relevant to the project: Manufacturing Process

Learning Outcome: I learnt about the selection of materials for sheet metal forming industries, Die design and load calculations for sheet metal forming processes and also the various defects in sheet metal forming processes and suggest possible troubleshooting.

PS-I station: Public Work Department , Lucknow

Student

Name: SHRIYANSH AGARWAL .(2021A2PS2381P)

Student Write-up:

PS-I Project Title: Development and advancement of road area in lucknow

Short Summary of work done: The project focused on the development and advancement of roads in and around the Lucknow area. Extensive work was undertaken to improve the road infrastructure and enhance connectivity within the region. The project began with a comprehensive survey and analysis of the existing road network to identify areas of improvement and formulate an effective plan. This involved assessing traffic patterns, identifying bottlenecks, and considering future growth projections. Based on the findings, several key initiatives were implemented. First, the project involved widening and strengthening existing roads to accommodate the increasing volume of vehicles and improve traffic flow. This included adding additional lanes, constructing new bypasses, and implementing modern traffic management systems. Furthermore, the project focused on integrating sustainable and environmentally friendly practices into road development. Efforts were made to incorporate green spaces, pedestrian walkways, and dedicated cycling lanes to promote alternative modes of transportation and reduce carbon emissions. In addition to physical improvements, the project emphasized the use of advanced technology and intelligent transportation systems. Smart traffic lights, surveillance cameras, and real-time information systems were installed to monitor traffic, improve safety, and optimize road efficiency.The project also addressed the issue of road safety. Measures such as road signage, speed calming devices, and designated pedestrian crossings were implemented to enhance safety for all road users. Public participation and stakeholder engagement played a vital role throughout the project. The opinions and concerns of local communities and businesses were taken into account to ensure the project's success and meet the needs of the area's residents. Overall, the development and advancement of roads in and around the Lucknow area aimed to provide a modern, efficient, and sustainable road network that supports economic growth, enhances connectivity, and improves the overall quality of life for the people of Lucknow.

Objectives of the project: To teach us about designing of road

Tool used: None

Details of Papers/patents: None

Brief description of the working environment: It was a fun working and learning environment with the PWD officials being supportive and understanding in nature for most of the time. The station for sure went beyond our expectations in terms of learning and experience as not only did we get to see the practical applications of what was taught to us in college but also got exposed to how the life of a working professional is.

Academic courses relevant to the project: Highway engineering

Learning Outcome: We got to learn alot about roads its design development and advancement where we went to sites, plants and laboratories.

PS-I station: Public Work Department , Lucknow

Student

Name: YASHWARDHAN RAI .(2021A2PS2517P)

Student Write-up:

PS-I Project Title: Development and advancement of roads in and around Lucknow

Short Summary of work done: The project focused on the development and advancement of roads in and around the Lucknow area. Extensive work was undertaken to improve the road infrastructure and enhance connectivity within the region. The project began with a comprehensive survey and analysis of the existing road network to identify areas of improvement and formulate an effective plan. This involved assessing traffic patterns, identifying bottlenecks, and considering future growth projections. Based on the findings, several key initiatives were implemented. First, the project involved widening and strengthening existing roads to accommodate the increasing volume of vehicles and improve traffic flow. This included adding additional lanes, constructing new bypasses, and implementing modern traffic management systems. Furthermore, the project focused on integrating sustainable and environmentally friendly practices into road development. Efforts were made to incorporate green spaces, pedestrian walkways, and dedicated cycling lanes to promote alternative modes of transportation and reduce carbon emissions. In addition to physical improvements, the project emphasized the use of advanced technology and intelligent transportation systems. Smart traffic lights, surveillance cameras, and real-time information systems were installed to monitor traffic, improve safety, and optimize road efficiency.The project also addressed the issue of road safety. Measures such as road signage, speed calming devices, and designated

pedestrian crossings were implemented to enhance safety for all road users. Public participation and stakeholder engagement played a vital role throughout the project. The opinions and concerns of local communities and businesses were taken into account to ensure the project's success and meet the needs of the area's residents. Overall, the development and advancement of roads in and around the Lucknow area aimed to provide a modern, efficient, and sustainable road network that supports economic growth, enhances connectivity, and improves the overall quality of life for the people of Lucknow.

Objectives of the project: To understand how PWD PRANTIYA KHAND develops roads

Tool used: Canva, MS Power Point, G-Docs, MS Word

Details of Papers/patents: None

Brief description of the working environment: It was a fun working and learning environment with the PWD officials being supportive and understanding in nature for most of the time. The station for sure went beyond our expectations in terms of learning and experience as not only did we get to see the practical applications of what was taught to us in college but also got exposed to how the life of a working professional is.

Academic courses relevant to the project: Highway Engineering

Learning Outcome: Practical understanding of Highway and Transport Engineering

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PS-I station: Public Work Department , Lucknow

Student

Name: SHIVENDRA PAL SINGH .(2021A2PS2618P)

Student Write-up:

PS-I Project Title: Development and Advancement of Road in and around Lucknow Area

Short Summary of work done: The project focused on the development and advancement of roads in and around the Lucknow area. Extensive work was undertaken to improve the road infrastructure and enhance connectivity within the region. The project began with a comprehensive survey and analysis of the existing road network to identify

areas of improvement and formulate an effective plan. This involved assessing traffic patterns, identifying bottlenecks, and considering future growth projections. Based on the findings, several key initiatives were implemented. First, the project involved widening and strengthening existing roads to accommodate the increasing volume of vehicles and improve traffic flow. This included adding additional lanes, constructing new bypasses, and implementing modern traffic management systems. Furthermore, the project focused on integrating sustainable and environmentally friendly practices into road development. Efforts were made to incorporate green spaces, pedestrian walkways, and dedicated cycling lanes to promote alternative modes of transportation and reduce carbon emissions. In addition to physical improvements, the project emphasized the use of advanced technology and intelligent transportation systems. Smart traffic lights, surveillance cameras, and real-time information systems were installed to monitor traffic, improve safety, and optimize road efficiency.The project also addressed the issue of road safety. Measures such as road signage, speed calming devices, and designated pedestrian crossings were implemented to enhance safety for all road users. Public participation and stakeholder engagement played a vital role throughout the project. The opinions and concerns of local communities and businesses were taken into account to ensure the project's success and meet the needs of the area's residents. Overall, the development and advancement of roads in and around the Lucknow area aimed to provide a modern, efficient, and sustainable road network that supports economic growth, enhances connectivity, and improves the overall quality of life for the people of Lucknow.

Objectives of the project: To analyse the ongoing projects in Lucknow which is meant to improve road conditions.

Tool used: Pages, MS Office, G-Suite

Details of Papers/patents: No patents.

Brief description of the working environment: Work Environment was rather fun and the learning environment was very fruitful.During the PS-I (Practice School - I) on the project related to the development and advancement of roads in and around the Lucknow area, the working environment was dynamic and collaborative. It involved working closely with a team of engineers, urban planners, and stakeholders from government agencies and private organizations. The work environment emphasized professionalism, teamwork, and effective communication to ensure the successful implementation of the project.

Expectations from the company were centered around delivering high-quality work, adhering to project timelines, and meeting the set objectives. The company emphasized the importance of attention to detail, accuracy in data analysis, and providing practical solutions to address the road infrastructure challenges in the Lucknow area. Adaptability and the ability to handle multiple tasks were also valued traits, as the project involved various aspects of road development, including planning, design, construction, and implementation.

During the PS-I, there were numerous opportunities for learning and growth. One of the key learning experiences was gaining practical knowledge about road design principles,

including traffic engineering, geometric design, and pavement design. This involved applying theoretical concepts to real-world scenarios and understanding the factors influencing road development decisions. Additionally, working with various stakeholders provided insights into the challenges and complexities of infrastructure projects and the importance of effective coordination and collaboration.Furthermore, the PS-I offered exposure to advanced technologies and tools used in road development projects. Learning to utilize software for traffic simulation, design optimization, and project management enhanced technical skills and familiarity with industry-standard software.
Overall, the PS-I experience provided a valuable learning platform to understand the intricacies of road development projects, develop practical skills, and gain a deeper understanding of the challenges and opportunities involved in the development and advancement of roads in the Lucknow area.

Academic courses relevant to the project: Highway Engineering

Learning Outcome: Got to learn about on-site implications of our learning from the course of Highway Engineering which was offered in 2-2 Sem.

PS-I station: Public Work Department , Lucknow

Student

Name: VAIBHAV KRISHNA .(2021A2PS3214H)

Student Write-up:

PS-I Project Title: Development and advancement of Road construction

Short Summary of work done: The project focused on the development and advancement of roads in and around the Lucknow area. Extensive work was undertaken to improve the road infrastructure and enhance connectivity within the region. The project began with a comprehensive survey and analysis of the existing road network to identify areas of improvement and formulate an effective plan. This involved assessing traffic patterns, identifying bottlenecks, and considering future growth projections. Based on the findings, several key initiatives were implemented. First, the project involved widening and strengthening existing roads to accommodate the increasing volume of vehicles and improve traffic flow. This included adding additional lanes, constructing new bypasses, and implementing modern traffic management systems. Furthermore, the project focused on integrating sustainable and environmentally friendly practices into road development.

Efforts were made to incorporate green spaces, pedestrian walkways, and dedicated cycling lanes to promote alternative modes of transportation and reduce carbon emissions. In addition to physical improvements, the project emphasized the use of advanced technology and intelligent transportation systems. Smart traffic lights, surveillance cameras, and real-time information systems were installed to monitor traffic, improve safety, and optimize road efficiency.The project also addressed the issue of road safety. Measures such as road signage, speed calming devices, and designated pedestrian crossings were implemented to enhance safety for all road users. Public participation and stakeholder engagement played a vital role throughout the project. The opinions and concerns of local communities and businesses were taken into account to ensure the project's success and meet the needs of the area's residents. Overall, the development and advancement of roads in and around the Lucknow area aimed to provide a modern, efficient, and sustainable road network that supports economic growth, enhances connectivity, and improves the overall quality of life for the people of Lucknow.

Objectives of the project: Learnt about practical application of knowledge gained about higways and new technologies used in the constitution

Tool used: PowerPoint presentation, laboratory tools

Details of Papers/patents: Nil

Brief description of the working environment: Work Environment was rather fun and the learning environment was very fruitful.It was a fun working and learning environment with the PWD officials being supportive and understanding in nature for most of the time. The station for sure went beyond our expectations in terms of learning and experience as not only did we get to see the practical applications of what was taught to us in college but also got exposed to how the life of a working professional is.

Academic courses relevant to the project: Higway engineering

Learning Outcome: New learning techniques used ,black spot and construction of new method of slowing down vehicle

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PS-I station: Putzmeister , Goa

Student

Name: KASAT HORAJI SUSHIL .(2021ABPS2560P)

Student Write-up:

PS-I Project Title: Sales forecast model using ML

Short Summary of work done: Learned basic working and about different departments of the company , got practical knowledge and got to know assumptions cannot be used in real life problems.

Objectives of the project: To create sales forecasting model for company using machine learning

Tool used: Machine learning

Details of Papers/patents: No

Brief description of the working environment: Working environment was really good.

Academic courses relevant to the project: Operation management

Learning Outcome: Overview of how a industry works , basics of machine learning and operation management

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PS-I station: Rajasthan Housing Board , Jaipur

Student

Name: LOHIT GAGGAR .(2021A2PS2617P)

Student Write-up:

PS-I Project Title: Reporting on running sites

Short Summary of work done: We had to go to the site, try to explore it, draw out major takeaways from it and make a report on the same.

Objectives of the project: To understand the site work and procedures

Tool used: None

Details of Papers/patents:n/a

Brief description of the working environment: Working environment was non-existent. Constant stalling of simplest of work because of government officials, no guidance whatsoever from anyone. No learning either obviously.

Academic courses relevant to the project: CEM, Soil Mechanics, HE

Learning Outcome: None

PS-I station: Rajasthan Housing Board , Jaipur

Student

Name: NAMAN SHARMA .(2021A2PS2632P)

Student Write-up:

PS-I Project Title: VISITING AND SUPERVISING VARIOUS RHB SITES

Short Summary of work done: During the internship at Rajasthan Housing Board in the domain of civil engineering, a wide range of practical experiences and knowledge was acquired, complementing the theoretical concepts learned during lectures. The internship provided valuable insights into various aspects of civil engineering field applications. One of the significant areas of learning during the internship was the tendering process. Interns gained exposure to the preparation, evaluation, and execution of tenders, which play a crucial role in the construction industry. Understanding this process helped in comprehending the intricacies of project management and contract administration. Foundational engineering was another key aspect explored during the internship. We learned about the importance of solid foundations for any structure and the methods involved in their design and implementation. This knowledge proved vital in grasping the fundamentals of construction practices and ensuring the safety and stability of buildings. We were also introduced to grid diagrams and columns. They learned about their design, analysis, and implementation, which are vital components in creating robust and efficient structures. Understanding these elements allowed the interns to appreciate the various structural considerations involved in real-world projects. Furthermore, the interns closely observed the roles and responsibilities of resident engineers and site engineers. we witnessed how these professionals oversee and manage construction activities, providing

instructions to the workforce on site. This hands-on experience shed light on the coordination and communication required to execute construction projects successfully. The internship also involved practical exposure to curing techniques, a critical process to ensure the strength and durability of concrete. By witnessing curing practices, we grasped the significance of proper construction practices in enhancing the longevity of structures. Moreover, we were introduced to various tests performed on soil and aggregates, such as compression tests and sieve analysis. These tests helped in understanding the properties of construction materials before their utilization in projects. Having knowledge of material testing is crucial in ensuring the quality and performance of the final product. Overall, the internship at Rajasthan Housing Board provided invaluable experience in applying theoretical knowledge to real-world scenarios in civil engineering. It equipped us with practical skills, insights into project management, and an understanding of essential construction processes, making them better-prepared professionals in the field of civil engineering.

Objectives of the project: Supervising and visiting different sites of the rajasthan housing board in jaipur

Tool used: NONE

Details of Papers/patents: None

Brief description of the working environment: The internship at Rajasthan Housing Board offered a practical and site-oriented working environment for the civil engineering interns. Over a span of 50 days, the interns were assigned to visit five different sites, allowing them to gain valuable exposure to various real-world construction scenarios.
We had a unique opportunity to apply the theoretical concepts they had learned in their previous semesters during academic studies to practical situations in the field.
no office work is there at all.
all 50 days we were required to visit to the sites everyday for 50 days except weekends.
One of the challenges faced during the internship was the difficulty in reaching the resident engineer at the sites. In such cases, the interns were directed to communicate with the site engineer instead. However, the site engineers were often preoccupied with their own responsibilities, making it challenging for them to dedicate time to guide the interns comprehensively.
As a consequence, much of the learning experience was driven by the interns' own efforts. They had to take the initiative to seek out opportunities to learn new concepts and techniques and thoroughly analyze the sites. This self-directed learning approach required determination and a proactive attitude from the interns to maximize the benefits of the internship.
Despite the challenges, some of the engineers on-site proved to be excellent mentors. They willingly imparted their knowledge, teaching the interns about various concepts and practical applications utilized in the execution of large-scale projects undertaken by Rajasthan Housing Board.
In conclusion, the internship at Rajasthan Housing Board provided a valuable hands-on experience in the civil engineering domain. Although the interns encountered challenges

with limited access to resident engineers and a substantial reliance on self-driven learning, they were able to learn how to analyze construction sites effectively and apply their academic knowledge in real-world scenarios. The mentorship from certain engineers further enriched their understanding of the practical aspects of civil engineering on significant projects.

Academic courses relevant to the project: Foundation engineering, soil mechanics, construction planning and technology, civil engineering materials

Learning Outcome: WORKINGS OF GOVERNMENT ORGANISATIONS. HOW THE THEORETICAL CONCEPTS ARE APPLIED IN THE REAL WORLD. SAW THE LARGE SCALE PROJECTS AND HOW THEIR EXECUTION TAKES PLACE.

PS-I station: Rajasthan Housing Board , Jaipur

Student

Name: ANIRUDH JAIMINI .(2021B3A21192P)

Student Write-up:

PS-I Project Title: Analysis of Multi Storey apartments

Short Summary of work done: We visited 5 construction sites, each for 10 days, tried to understand the processes undergoing and we learnt from site engineers.

Objectives of the project: To understand the process and engineering behind building large multi storey apartments

Tool used: None

Details of Papers/patents: None

Brief description of the working environment: The PS site being a government site had huge delays, because the officials were never on time also we could rarely meet the assigned engineers.

Academic courses relevant to the project: RCC, Soil Mechanics, other Civil courses

Learning Outcome: Practical experience of how government builds large multi-storey residential flats, what are the different processes of building flats.

PS-I station: Rajasthan State Industrial Development and Investment Corporation (RIICO) , Jaipur

Student

Name: ISHA SINGHAL .(2021A1PS2379P)

Student Write-up:

PS-I Project Title: Role of RIICO in the Socio-Economic Development of Rajasthan from a Civil Engineering Perspective

Short Summary of work done: We have studied the significant role played by RIICO in fostering the economic growth of Rajasthan. Additionally, we have explored the organizational structure and operational approach employed by RIICO to achieve its objectives. Lastly, our focus was on project management methodologies utilized by Civil Engineers to support and streamline RIICO's endeavors

Objectives of the project: 1) To shed light on some of the key aspects pertaining to RIICO's initiatives. 2) To foreground the role Civil engineers play in attaining RIICO's goals.

Tool used: Most of our work was pen-and-paper-based. However, we did gain insights on online tendering procedures.

Details of Papers/patents: We made a report.
Report title: Role of RIICO in the Socio-Economic Development of Rajasthan
from a Civil Engineering Perspective
Report description: The report sheds light on some of the key aspects pertaining to RIICO's initiatives. Additionally, it f

Brief description of the working environment: During my internship at RIICO's Jaipur (Rural) Office, I had the opportunity to work in different departments, including the legal cell and accounts department. The overall working environment exceeded my

expectations, and I was able to gather invaluable insights into the seamless coordination among these departments to support the organization's objectives.
Moreover, my experience provided me with in-depth knowledge about the diverse project management techniques utilised by Civil Engineers within RIICO to ensure the successful execution of various projects

Academic courses relevant to the project: CE F242

Learning Outcome: 1) The socio-economic impacts of industrialisation and their manifestations in several forms, like societal integration, changes in production and consumption patterns, etc.
2) Project Management techniques employed by Civil Engineers in order to efficiently and timely achieve our goals while using the least possible amount of resources
3) How teamwork and coordination among various departments are ensured so as to achieve organisational goals

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PS-I station: Rajasthan State Industrial Development and Investment Corporation (RIICO) , Jaipur

Student

Name: DIVYE SRIVASTAVA .(2021A2PS2135P)

Student Write-up:

PS-I Project Title: Role of RIICO in the Socio-Economic Development of Rajasthan from a Civil Engineering Perspective

Short Summary of work done: We learnt about RIICO's contribution to the economic development of Rajasthan. We also learnt about how RIICO is structured and what its modus operandi is. Lastly, we learnt about some project management techniques used by Civil Engineers in furtherance of facilitating RIICO's activities.

Objectives of the project: 1) To shed light on some of the key aspects pertaining to RIICO’s initiatives. 2) To foreground the role Civil engineers play in attaining RIICO’s goals.

Tool used: Most of our work was pen-and-paper based. However, we did gain insights on online tendering procedures.

Details of Papers/patents: We made a report.
Report title: Role of RIICO in the Socio-Economic Development of Rajasthan from a Civil Engineering Perspective
Report description: The report sheds light on some of the key aspects pertaining to RIICO's initiatives. Additionally, it f

Brief description of the working environment: I worked as an intern at the Jaipur (Rural) Office of RIICO, which comprises various departments such as the legal cell, the accounts department, etc.
The working environment met my expectations.
I gained several valuable insights on how various departments coordinate in furtherance of facilitating the organisation's activities, thereby attaining the organisational goals. I also learnt in detail about various project management techniques employed by Civil Engineers.

Academic courses relevant to the project: CE F242

Learning Outcome: 1) The socio-economic impacts of industrialisation and their manifestations in several forms, like societal integration, changes in production and consumption patterns, etc.
2) Project Management techniques employed by Civil Engineers in order to efficiently and timely achieve the goals while using the least possible amount of resources
3) How teamwork and coordination among various departments are ensured so as to achieve organisational goals

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PS-I station: Rajasthan State Industrial Development and Investment Corporation (RIICO) , Jaipur

Student

Name: PARTH RANA .(2021B1A12473P)

Student Write-up:

PS-I Project Title: Construction of Plug & Play Factory Complex

Short Summary of work done: We oversaw and understood the management of logistics, reading design plans made by the architects and understood in general what RIICO does. This is different in the sense that this complex is a rental space. We learnt how design plans are made, and applied.

Objectives of the project: The objective of the project was to verify and ensure the construction of this complex and oversee the logistics and work being done on the site

Tool used: CTM, Sieve Set, CTM mortar, Rod Bending Machine

Details of Papers/patents:-

Brief description of the working environment: The working environment was very good and interactive. They were well educated and knew what we wanted to ask and gave answers accordingly. They provided us with the perfect work environment by sending us to the Construction site of Plug & Play Factory Complex and helping us learn the different types of cement used, the management and arrangement of logistics, face-off against harsh weather to construct the complex and trying the very best to meet deadlines despite the monsoon season.

Academic courses relevant to the project: -

Learning Outcome: The major learning outcomes involved learning the different types of cement used

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PS-I station: Rajasthan State Industrial Development and Investment Corporation (RIICO) , Jaipur

Student

Name: TANISHQ RULANIA .(2021B4A21566P)

Student Write-up:

PS-I Project Title: Plug and play complex construction

Short Summary of work done: oversaw the construction, learned how to read and plan a design map, various construction techniques, how the industry works, Strength Testing etc.

Objectives of the project: Overseeing the work

Tool used: None

Details of Papers/patents:none

Brief description of the working environment: Good

Academic courses relevant to the project: Concrete technology book (course in 3-1 for me), building construction (3-1 course, i think)

Learning Outcome: A lot of civil engineering concepts and how the industry works.

PS-I station: Realty Advance Structure Enterprises Pvt Ltd , Patna

Student

Name: HARSH SINHA .(2021A2PS2609P)

Student Write-up:

PS-I Project Title: Road construction

Short Summary of work done: Got to know how roads are built

Objectives of the project: Road construction

Tool used: No

Details of Papers/patents: No

Brief description of the working environment: Good working environment,learn different material that are getting used in construction of road

Academic courses relevant to the project: Highway engineering

Learning Outcome: Got to know how to do management things

PS-I station: Realty Advance Structure Enterprises Pvt Ltd , Patna

Student

Name: ISHA APURVA .(2021A2PS2635P)

Student Write-up:

PS-I Project Title: Analysis of Model-Bidding document and Procedure for road construction.

Short Summary of work done: In this project, we initially began by familiarizing ourselves with the tender process in the construction field. As we progressed, we delved deeper into the documentation requirements and eligibility criteria necessary to submit a tender. Our focus then shifted to understanding rural road projects, encompassing the unique aspects and specific requirements of constructing roads in rural areas, which differ from other types of roads. We gained valuable insights into the significance of Detailed Project Reports (DPRs) and their role in ensuring successful road construction projects. Additionally, we enhanced our knowledge through site visits to various construction sites, providing practical exposure to real-world scenarios in the construction industry.

Objectives of the project: The objective of this project is to conduct a comprehensive analysis of the model-bidding document and the construction procedures for road projects.

Tool used: Since the project primarily revolved around on-site work, we didn't rely on any specialized tools. Instead, we utilized basic software applications like Excel and Word for documentation purposes. To grasp a comprehensive understanding of the construction

Details of Papers/patents:N/A

Brief description of the working environment: The working environment during was dynamic and collaborative, with a strong emphasis on hands-on learning. The company encouraged us to actively participate in projects, demonstrate eagerness to learn, and take on responsibilities with dedication. Clear communication and teamwork were encouraged, enabling us to contribute effectively to the team's objectives.

Academic courses relevant to the project: Construction Management,Transportation Engineering

Learning Outcome: The major learning outcome of the project includes a comprehensive understanding of construction project bidding processes, road construction procedures, and the critical role of documentation in ensuring transparency, efficiency, and adherence to regulatory requirements.

PS-I station: Roads and Buildings Department - Govt of Telangana , Hyderabad

Student

Name: R NISHANT .(2021A2PS2557H)

Student Write-up:

PS-I Project Title: Cable Bridge

Short Summary of work done: .

Objectives of the project: A study on the construction of the bridge

Tool used: Nil

Details of Papers/patents:.

Brief description of the working environment: .

Academic courses relevant to the project: Bridge engineering

Learning Outcome: Tender processing, and major project assessment

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PS-I station: Rourkela Steel Plant , Rourkela

Student

Name: VIDIT AGARWAL .(2021A4PS2408P)

Student Write-up:

PS-I Project Title: Tool Development of horizontal Boring and milling machine

Short Summary of work done: focused on designing and optimizing cutting tools and attachments used in the machining process. aim was to improve the machine's efficiency, accuracy, and versatility to meet various machining requirements. This involves creating and testing new tool designs. The goal is to enhance the machine's performance, reduce production time, and ensure precise machining of complex workpieces in industries like automotive, aerospace, and heavy machinery.

Objectives of the project: Revise the tooling system of milling machine

Tool used: Hardware: ISO 50,ISO60, SKoDAMilling machines, Cutters, spindles, tool holders

Details of Papers/patents:DANDRE'A tool heads manual and catalogue,Horizontal Boring and milling machine manual

Brief description of the working environment: The working environment is dynamic and challenging, with a focus on safety, teamwork, and continuous improvement. The company expects employees to be dedicated, responsible. They encourage learning and development opportunities, providing employees with a chance to enhance their skills and knowledge in various aspects of the steel industry. This fosters personal growth and helps individuals contribute effectively to the plant's success.

Academic courses relevant to the project: Workshop practice, Manufacturing processes

Learning Outcome: Industrial tooling system at steel manufacturing plant

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PS-I station: Rourkela Steel Plant , Rourkela

Student

Name: ARYAN THAKUR .(2021B2A22164P)

Student Write-up:

PS-I Project Title: Cement strengthening and decay prevention

Short Summary of work done: We went there had to wait till 14th June for PS to start. We went on the site for 2 days and then radio silence. I met the guide on the last day when he answered my doubts.

Objectives of the project: Prevention of acid decay or strengthening of cement

Tool used: 10 pages of report given to us

Details of Papers/patents: Not any

Brief description of the working environment: We went there had to wait till 14th June for PS to start. We went on the site for 2 days and then radio silence. I met the guide on the last day when he answered my doubts.

Academic courses relevant to the project: Cement, rubber and epoxy knowledge

Learning Outcome: Not much but use of sika products

PS-I station: SARDAR PATEL RENEWABLE ENERGY RESEARCH INSTITUTE (SPRERI) , Anand

Student

Name: PILLAI AYYAPPAN RAJESH(2021A1PS2370P)

Student Write-up:

PS-I Project Title: Bio-oil production through Banana tree waste Pyrolysis and Product characterisation

Short Summary of work done: Aim - Pyrolysis of Banana tree waste for Bio-oil production and Product characterisation. Multiple readings of the Pyrolysis experiment were taken using a 1 Kg Batch reactor. Proximate Analysis - Ash content, Volatile matter content, Moisture content and Fixed Carbon composition (in wt%) were calculated. As a part of the characterisation, Gravity separation was done along with Bull density analysis.

Objectives of the project: 1) Bio-oil production through Banana waste Pyrolysis 2) Product characterisation

Tool used: All mechanics related tools such as - Spanner, Hammer, Tongs, etc. Also since I had to do the servicing of the machine, I had to use many other small tools.

Details of Papers/patents: No papers published. Only report was submitted.

Brief description of the working environment: The working environment at my PS station - Sardar Patel Renewable Energy Research Institute was very positive. The research assistants and department heads at the institute are extremely encouraging and knowledgeable. The students are given hands-on experience of using the reactors and performing 100% of the experiments. The faculties are extremely responsible and properly correct the mistakes of the students, if any.

Academic courses relevant to the project: Heat Transfer

Learning Outcome: 1) Learnt about the functioning of Batch and Continuous reactors in industry
2) Also got to service the machines used by us, which helped me get to know the function of each of the individual parts of the machine.
3) Learnt to perform the literature review of multiple research papers and draw meaningful conclusions.

PS-I station: SARDAR PATEL RENEWABLE ENERGY RESEARCH INSTITUTE (SPRERI) , Anand

Student

Name: ARYAN DILLIWAR .(2021A4PS1580P)

Student Write-up:

PS-I Project Title: Testing and calculation of different solar devices

Short Summary of work done: We were given the task to find mathematically, using the empirical relations and conditions from literatures. Then we perform experiment for the particular solar device and check whether the results align with our calculations.

Objectives of the project: Testing of different solar devices

Tool used: CAD, presentation,etc

Details of Papers/patents:NA

Brief description of the working environment: We were given one solar device by our mentor and asked to perform calculations and test the solar device.

Academic courses relevant to the project: Applied thermodynamics, heat transfer

Learning Outcome: application of the learned theory

PS-I station: Starflex Sealing India Pvt. Ltd. , Goa

Student

Name: ADDYA KASHYAP .(2021A4PS2805H)

Student Write-up:

PS-I Project Title: Inventory Management

Short Summary of work done: The project involved designing a user-friendly spreadsheet with various sheets to record materials issued, stock balances, supplier information, and order tracking. We utilized formulas and functions to sort fast and slow moving items and forecast demands. The project was linked with the schedule planning project for more accurate inventory management.

Objectives of the project: Sort fast and slow moving items, demand forcasting, link with schedule planning

Tool used: Excel

Details of Papers/patents: No

Brief description of the working environment: Relaxed and understanding working environment, simple and easy project, company expects you to just complete the project and rarely communicates. The mentors are friendly and approachable but one has to initiate communication with the company even for basic and important information required for the project. The project gave us practical experience in implementing an Excel-based inventory management system, and we learnt inventory management principles, demand forecasting, supplier management, data analysis, problem-solving, and project management.

Academic courses relevant to the project: Manufacturing Processes, Principles of Economics, Probability and Statistics, Workshop Practice, Technical Report Writing

Learning Outcome:
Practical Application of Excel, Inventory Management Principles, Problem-Solving and Adaptability, Collaboration and Communication, Project Management Skills

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PS-I station: Starflex Sealing India Pvt. Ltd. , Goa

Student

Name: CHAITANYA SANJAY NAYAK .(2021A4PS3075H)

Student Write-up:

PS-I Project Title: Arduino Data Collector (ADC)

Short Summary of work done: My work was to retrieve data like machine run time, ideal time temperature fluctuations and other variables through respective sensors, then processing it through Arduino codes and finally relaying data through ESP8266 module and making it available through mail for the company officials.

Objectives of the project: Using Arduino, output of each machine, winding material used and machine runtime and ideal time to be recorded and shared using Wi-Fi to mobile / mail everyday

Tool used: Arduino IDE, Tinker CAD

Details of Papers/patents: None

Brief description of the working environment: My Practice School was an online remote basis. We used to communicate to the company and PS faculty regularly through google meets. They used to guide us regularly

Academic courses relevant to the project: Data Mining in Mechanical Sciences, Signals &Systems

Learning Outcome: Learning Arduino IDE in depth and Tinker CAD simulation for automation of Manufacturing unit of the company

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PS-I station: Starflex Sealing India Pvt. Ltd. , Goa

Student

Name: SIDDHANTH IYENGAR .(2021AAPS1181P)

Student Write-up:

PS-I Project Title: Arduino data collector

Short Summary of work done: Arduino data collector (ADC) Using Arduino, output of each machine, winding material used and machine run time and ideal time to be recorded and shared using Wifi to mobile / mail everyday

Objectives of the project: Use arduino microcontroller and sensors to help starflex track amount of material used and machining time for their assembly line

Tool used: TinkerCAD

Details of Papers/patents:-

Brief description of the working environment: Due to its online nature we used tinkercad for prototyping and the station provided videos of the working of the machine for us to understand how to rig the sensors

Academic courses relevant to the project: Robotics

Learning Outcome: Learnt how to use arduino and tinkercad for prototyping

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PS-I station: Starflex Sealing India Pvt. Ltd. , Goa

Student

Name: SIDDHARTH GUPTA .(2021ABPS1677P)

Student Write-up:

PS-I Project Title: Schedule Planning and Optimization

Short Summary of work done: During my PS-I, I had the privilege of working in an online environment with a supportive team at StarFlex. The company tasked me with optimizing gasket production scheduling and planning. Despite the remote nature of the internship, the team provided valuable guidance, feedback, and insights, creating a conducive learning environment. To achieve the project's goals, I applied my theoretical knowledge to real-world scenarios, utilizing Gantt charts and data analysis techniques. The focus was on improving production processes based on order priorities and machine utilization. This experience honed my project management, data analysis, and problem-solving skills. Throughout the internship, I learned how to effectively visualize project schedules with Gantt charts and make data-driven decisions. Additionally, I developed strong time management skills to meet project deadlines successfully. The internship offered valuable insights into industrial practices, allowing me to apply theoretical concepts to practical challenges. The collaborative work environment enhanced my communication and collaboration skills, contributing to my personal and professional growth. Overall, the PS-I internship at StarFlex was a transformative experience, enriching my technical expertise and providing a strong foundation for my future endeavors in industrial engineering and project management.

Objectives of the project: System optimization and planning with different sizes and material as input based on the priority of orders, based on each machine. Need to calculate * Max. Machine utilisation * Max output * Dispatch on time

Tool used: Microsoft Excel

Details of Papers/patents: None

Brief description of the working environment: During my PS-I , I had the opportunity to work in an online environment, collaborating with a team of helpful and supportive employees. Despite the remote nature of the internship, they went above and beyond to assist me at every step of the project, providing guidance, feedback, and valuable insights, creating a conducive learning environment.
Throughout the internship, I worked on an exciting project related to optimizing gasket production scheduling and planning at StarFlex. The company's expectations were to apply my theoretical knowledge to real-world scenarios and come up with practical solutions to enhance their production processes. I utilized Gantt charts and data analysis techniques to optimize the scheduling of gasket production based on order priorities and machine utilization.
The experience provided me with essential skills in project management, data analysis, and problem-solving. Working on the project taught me how to effectively use Gantt charts for visualizing project schedules and making data-driven decisions. Additionally, I honed my time management skills to meet project deadlines and ensure smooth progress.
Overall, the working environment was collaborative and conducive to learning. The internship offered valuable insights into the industry's practices and allowed me to apply theoretical concepts to real-world challenges. I am grateful for the guidance and support I received during my PS-I, as it significantly contributed to my personal and professional growth.

Academic courses relevant to the project: MF F219 - Operations Management

Learning Outcome: During my PS-I, I gained several major learning outcomes that have significantly contributed to my personal and professional growth:
Practical Application of Theoretical Knowledge: The internship provided me with an opportunity to apply the theoretical concepts I learned in my academic studies to real-world scenarios. Working on the project at StarFlex allowed me to see the direct impact of my knowledge on optimizing gasket production scheduling and planning.
Project Management Skills: Throughout the internship, I developed strong project management skills. I learned how to create and manage project schedules using Gantt charts, ensuring that tasks were completed efficiently and within deadlines. This experience helped me understand the importance of proper planning and organization in successful project execution.
Data Analysis Techniques: Analyzing data to optimize gasket production scheduling was a crucial aspect of my internship. I gained hands-on experience in data analysis techniques, which included identifying patterns, drawing insights, and making data-driven decisions to improve production processes.
Collaboration and Communication: The remote nature of the internship required effective collaboration and communication with the team members. Engaging with supportive

employees taught me the value of clear and concise communication, fostering a collaborative and productive work environment.
Problem-Solving and Critical Thinking: Tackling the challenges of optimizing production schedules demanded critical thinking and problem-solving skills. I learned to approach complex issues systematically, break them down into manageable parts, and devise practical solutions.
Time Management: Meeting project deadlines was crucial to the success of the internship. I developed strong time management skills to ensure smooth progress and timely completion of tasks, even in a remote setting.
Industry Insights: Working at StarFlex provided me with valuable insights into industrial practices and processes. Understanding the challenges and requirements of a real manufacturing environment has given me a deeper appreciation for the complexities involved in production management.
Conducive Learning Environment: The supportive and helpful team at StarFlex created a conducive learning environment. This positive experience boosted my confidence and motivation, encouraging me to explore new ideas and approaches during the internship.
Overall, my PS-I internship was a transformative experience that not only enriched my technical knowledge but also equipped me with essential soft skills. These major learning outcomes will undoubtedly serve as a strong foundation for my future endeavors in the field of industrial engineering and project management.

PS-I station: State Council of Education Research and Training , Thiruvananthapuram

Student

Name: RITHU PRAKASH .(2021ABPS1663P)

Student Write-up:

PS-I Project Title: Design and development of Educational Feedback System

Short Summary of work done: My partner and I created flexible and user friendly templates according to the suggestions of the mentor, after doing a week of research on the working of the institution. Then we made the features functioning by converting them into code and making necessary changes. The fuctioning live website was presented and the source code and necessary documents were shared with our mentor, so that they can merge our rudimentary website with CDIT when they develop the feedback system portal.

Objectives of the project: To create a feedback system where the entire process of project assignment, finances allocated and projects feedback are tracked and supervised.

Tool used: React, Next.js, HTML, CSS, figma

Details of Papers/patents: None

Brief description of the working environment: The working environment was similar to any government office in Kerala. It was relaxing and the people were very friendly and supportive. Our mentor used to check up on us every now and then to see how much the work is done. But technical guidance was a bit lacking, but we were given ample time to study the necessary skills ourselves.

Academic courses relevant to the project: None that I have taken as part of my coursework

Learning Outcome: Since I was not really knowledgeable in web development, I could learn HTML deeply and basics of CSS as well. Platforms like figma and plugins like anima and locofy were also tools that I hadn't explored much. JavaScript libraries like react and frameworks like next.js also came in handy. Although I was learning TensorFlow parallelly, the component that we had to implement it was scraped off due to lack of time, but in total my self exploration in tech was really eye opening.

PS-I station: State Council of Education Research and Training , Thiruvananthapuram

Student

Name: ATHREYAKRISHNA J N(2021B5A42705G)

Student Write-up:

PS-I Project Title: Development of Educational Feedback System

Short Summary of work done: The goal was to develop a comprehensive platform to efficiently manage and track educational projects. The project involved web development skills, including front-end and back-end technologies, to create an intuitive user interface

and a robust database system. Attention was given to user experience design, ensuring usability and accessibility. Regular client interaction and effective communication ensured that the website met SCERT's specific requirements. The project provided valuable learning experiences in web development, project management, collaboration, and client engagement.

Objectives of the project: To create a platform to track and manage all projects taking place in the organisation

Tool used: React, Chakra, Next.Js, Sql

Details of Papers/patents:-

Brief description of the working environment: The working environment at SCERT was a dynamic and collaborative one. There was consistent feedback from the mentor and other officials involved with the project. Regular meetings were conducted to discuss progress, address challenges, and gather feedback.
The organisation expected a quality deliverable in the form of the website and timely delivery on or before 21st July. There was significant emphasis on meeting their specific requirements with regards to the specifications of the website. The org also expected clear and precise communications from us with regards to any issues faced with the development of the website and we were expected to document and submit a report of the entire process towards the end.

Academic courses relevant to the project: -

Learning Outcome: As this was a full-stack project, I was able to improve my web development skills and familiarise myself with react and next.js - both front-end and back-end applications. It also helped me understand in detail on how to build an appealing UX design and manage the time and skills needed to develop an end to end web platform.

PS-I station: STEM4ALL , Usa

Student

Name: ANSHUL SINGH CHAUHAN .(2021A3PS2639P)

Student Write-up:

PS-I Project Title: Python

Short Summary of work done: My role was teaching python course to a student and prepare him for USACO olympiad.

Objectives of the project: Teaching Python to students.

Tool used: NA

Details of Papers/patents:NA

Brief description of the working environment: There was only a single student enrolled, I expected more students in python. As we were told to teach we learnt nothing new.

Academic courses relevant to the project: NA

Learning Outcome: Problem Solving.

PS-I station: STEM4ALL , Usa

Student

Name: AMAAN SAJID NALAKATH(2021A7PS0517G)

Student Write-up:

PS-I Project Title: AP Physics 1

Short Summary of work done: Gave the students a deep understanding of kinematics and dynamics and enriched and tested their analytical skills to their limit

Objectives of the project: Preparing Aspirants for the F = ma Olympiad

Tool used: A drawing tablet and Microsoft PowerPoint

Details of Papers/patents: Na

Brief description of the working environment: Online. Work from home. 9:30 pm to 11:30 pm

Academic courses relevant to the project: None

Learning Outcome: Well prepared aspirants for the Olympiad

PS-I station: STEM4ALL , Usa

Student

Name: SAUBHAGYA ANAND .(2021AAPS0539H)

Student Write-up:

PS-I Project Title: AI Chatbot - Quanta

Short Summary of work done: We were asked to build some innovative, hands-on projects on AI/ML using Python, to teach some High-school students of our organization, Stem4All. We were given various articles and resources to first study on our own, then use them to implement on projects and assignments. The projects which we chose were AI chatbot (Quanta), Image-Recognition model out of which we created a report and presentation on the chatbot model using NLP, NLTK and openAI. It was an enlightening experience for our group despite the fact that our classes were not conducted, but still our mentor made us to implement something useful for the organization.

Objectives of the project: Designing a desktop chatbot (Quanta) using AI/ML in Python.

Tool used: PyCharm, TensorFlow, Jupyter notebook, Python packages like NumPy, Pandas, etc.

Details of Papers/patents: No patents

Brief description of the working environment: The working mode was online, but still our mentor and us students made the most of it, holding regular meets day and night discussing about the non-profit, social-upliftment ideals of the organization, innovative new ideas emerging in the field of AI/ML, teaching us to use tools like PyCharm, Jupyter and Google Tensorflow, etc. Overall, the experience was pretty enlightening, where I got

to learn various new algorithms and built various new projects on AI/ML on a frequent basis, both on a small as well as a large scale.

Academic courses relevant to the project: 1. Microprocessor Programming & Interfacing
2. Computer Programming in C
3. Digital Design

Learning Outcome: AI/ML algorithms, Neural Networks, NLP, etc.

PS-I station: STEM4ALL , Usa

Student

Name: RAJDEEP SINGH(2021B4A72945G)

Student Write-up:

PS-I Project Title: Live Classes for Algebra I

Short Summary of work done: Taught US based K-12 students basics of Algebra, how these basic concepts leads to various core subjects and theories and journals in their future studies.

Objectives of the project: Learn and teach Algebra I

Tool used: NA

Details of Papers/patents: Reference Material : Big Ideas Algebra I

Brief description of the working environment: The work environment was good. The Stem4All mentor was supportive and helped if we faced any problems. I learned basic mentoring skills, improved my communication skills and revised Algebra I concepts.

Academic courses relevant to the project: Algebra I

Learning Outcome: Improved Communication skills, crowd handling, and basic mentoring skills

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PS-I station: STEM4ALL , Usa

Student

Name: SAYANTAN BERA(2021B4AA1828G)

Student Write-up:

PS-I Project Title: Algebra 2 Classes

Short Summary of work done: I taught Algebra 2 to one student throughout my PS1. There were 4 classes in a week from Tuesday to Friday starting from 5:30 AM and ending at 7:30 AM. I had only one student at my class. I covered 8 chapters of the Algebra 2 textbook which the organization provided. The experience has been immensely rewarding, and I am now equipped with valuable teaching experiences and insights.

Objectives of the project: Facilitate students' mastery of Algebra 2 topics, preparing them for future academic pursuits.

Tool used: Teach using Google Jamboard

Details of Papers/patents: No

Brief description of the working environment: The organization provided access to resources and technology to be implemented in the classroom. The organization expected my commitment to be regular in teaching. Besides, the organization valued open communication and encouraged sharing best practices with other educators to enhance the overall teaching experience. Throughout the project, I experienced significant personal and professional growth. I learned to engage student more effectively and foster a positive learning environment.

Academic courses relevant to the project: High School Algebra

Learning Outcome: I developed my communication and instructional skills as an educator.

PS-I station: STEM4ALL , Usa

Student

Name: PRADYUMAN SINGH .(2021B5A71204P)

Student Write-up:

PS-I Project Title: AP Physics-1

Short Summary of work done: We were required to teach AP Physics-1 to the interested students who were part of the STEM4ALL summer camp. We took online classes of a student as a group for three weeks.

Objectives of the project: To teach AP Physics-1 to the interested students

Tool used: none

Details of Papers/patents:none

Brief description of the working environment: The station was online and we worked from our homes. The working hours were told to us before the work of the regular classes started. We expected to learn some programming through many projects that we would do.

Academic courses relevant to the project: Physics

Learning Outcome: Communication and interpersonal skills

PS-I station: STEM4ALL , Usa

Student

Name: MANAV JAMUAR(2021B5A72638G)

Student Write-up:

PS-I Project Title: Ap physics -1

Short Summary of work done: I taught a kid from USA , two chapters from the syllabus of AP physics-1. Our objective was to teach him for F=ma olympiad. I taught him linear momentum and rotational motion

Objectives of the project: To teach student for F=ma olympiad

Tool used: nothing

Details of Papers/patents:.

Brief description of the working environment: Working environment was stress free and flexible. I expected company to give me some IT roles but rather i had to teach Physics which will be of no use in my career as such . This will be one disappointment from the company.

Academic courses relevant to the project: Physics

Learning Outcome: communication skill + presentation skill

PS-I station: Sud-Chemie India Pvt. Ltd - Engineering , Vadodara

Student

Name: ADITYA CHAKRANI .(2021A1PS1344P)

Student Write-up:

PS-I Project Title: Flue gas waste heat recovery

Short Summary of work done: I made the theoretical simulations for the processes that are happening in the industry and using the simulations, analyzed the quantity and quality of heat energy that is wasted through hot exhaust gasses. After calculating the amount of recoverable energy, appropriate heat recovery apparatus was chosen and the recovered heat was used appropriately.

Objectives of the project: To recover and reuse the heat energy wasted through the hot exhaust gasses released in the atmosphere

Tool used: Microsoft Excel

Details of Papers/patents:NA

Brief description of the working environment: All the staff members were very supportive and the company took care of any need that might arise in the duration of PS-I

Academic courses relevant to the project: Chemical Process Calculations
Heat Transfer
Chemical Engineering Thermodynamics
Numerical Methods for Chemical Engineers
Engineering Chemistry

Learning Outcome: Application of theoretical concepts like mass and energy balance to real-world applications

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PS-I station: Sud-Chemie India Pvt. Ltd - Engineering , Vadodara

Student

Name: SHIVANG RAI .(2021B2A42292P)

Student Write-up:

PS-I Project Title: Conversion of CO2 to value added chemicals

Short Summary of work done: Did research using google scholar, went through multiple reports related to project by which i could find the necessary information required. Worked on multiple machines used for characterizations of catalysts.

Objectives of the project: Finding catalysts for Conversion of CO2 to value added chemicals

Tool used: XRD, Google Scholar, Sci Hub, TPD, GC, TPR, Autoclave

Details of Papers/patents:-

Brief description of the working environment: The company provided apt opportunities and working environment required for the completion of my project. They were very helpful and made me aware about the industry. I got everything that was expected from the company.

Academic courses relevant to the project: Organic Chemistry 2, Nano chemistry, Chemical Experiments 1

Learning Outcome: Learnt how to do Literature Reviews, Operated multiple Industry Level Equipments.

PS-I station: Survey of India , Dehradun

Student

Name: SNEHA VIVEK .(2021A2PS2627P)

Student Write-up:

PS-I Project Title: Mapping made easy : Creating powerful mapping toolset

Short Summary of work done: Automating data entry in the excel with VLOOKUP and macro, fixing overexposed 360-degree images , building tools such as Jump to next geotagged image in ArcGIS software using python , studying a tool build in python to detect the missing pixels in the raster images.

Objectives of the project: To increasing the working efficiency of the NMCG project with the help of different python tools.

Tool used: ArcGIS Map, Python, Excel ,Adobe lightroom, Pyscripter , ADD-IN Wizard

Details of Papers/patents: No

Brief description of the working environment: The work environment was really good ,the mentors were really helpful and guided us really well. Co-workers were very interactive and helping us in every way possible.

Academic courses relevant to the project: Computer programming really helped as it made it easy to learn how to code and develop

Learning Outcome: Learned programming language like python

PS-I station: Survey of India , Dehradun

Student

Name: SHREYA NAG .(2021A2PS2636P)

Student Write-up:

PS-I Project Title: Identification of springs in Himalayan Region using ORI, DEM and LiDAR Point Cloud Data for NMCG

Short Summary of work done: My work was to create a Python code using Machine learning algorithms to predict the location of springs using LiDAR data. For completing my project I started with learning python and then read a lot of research papers and found about the Random Forest Regression, a machine learning algorithm that can predict the location of springs if trained using the already available dataset. And then started working on the code using various resources available to me.

Objectives of the project: The objective of the project is to identify the location of springs using the Lidar data that was in LAS. I worked on a python code that uses Machine learning Algorithm that can predict spring locations.

Tool used: VS code, ArcGIS pro, replit

Details of Papers/patents:NA

Brief description of the working environment: Working at Survey of India was a great experience. Everyone was very supportive and helped us in case of any doubts. Got to learn a lot from my mentor and peers.

Academic courses relevant to the project: Computer Programming

Learning Outcome: I learned Python coding language and the use of machine learning algorithms for training the dataset and extracting valuable information.

PS-I station: Survey of India , Dehradun

Student

Name: NAKUL SINGH ADHIKARI(2021A8PS2988G)

Student Write-up:

PS-I Project Title:

Short Summary of work done:

Objectives of the project:

Tool used:

Details of Papers/patents:

Brief description of the working environment:

Academic courses relevant to the project:

Learning Outcome:

PS-I station: Survey of India , Dehradun

Student

Name: PALAASH PATEL(2021B1A82938G)

Student Write-up:

PS-I Project Title: Upgradation of Programs for Monthly and Annual Mean Sea Level Calculation Using Tidal Data with Python and Java

Short Summary of work done: In the month of June, I was done with my PSI project, the one mentioned above. Then in July I was assigned the second part of this project, 'Prediction of tidal waves for the future'. Unfortunately due to time limitations I was unable to complete the second project but I am glad that the first project is highly helpful to the organization and they are running my programs for calculation.

Objectives of the project: To create a program in java and python that can extract data from the raw data file and calculate the mean sea level yearly and monthly

Tool used: IntelliJ IDEA, VS code

Details of Papers/patents:none

Brief description of the working environment: The work environment was very positive and welcoming. The seniors were very helpful and did constant checkup on us and made sure to help us out in every problem that we faced.

Academic courses relevant to the project: Computer Programming

Learning Outcome: How to make your programs in a way that they can run on any device, with or without python/java installed

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PS-I station: Survey of India , Dehradun

Student

Name: HARSHITH REDDY MAMIDI(2021B2A73017G)

Student Write-up:

PS-I Project Title: Error correction in gravity surveying

Short Summary of work done: First I brushed up my concepts in python and started going through the code of gs adjust developed by united states to understand the logic of how random error is minimised and devised a method and presented It to my mentor

Objectives of the project: To minimise random error in gravity surveying

Tool used: Gs adjust, visual studio code

Details of Papers/patents: No

Brief description of the working environment: Working environment was friendly, supportive and most of all they understood our needs.

Academic courses relevant to the project: Cp

Learning Outcome: Learned python, learned in depth about gravity surveying and how it is useful for other fields of science

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PS-I station: Suzlon Foundation , Pune

Student

Name: CHAITANYA JAIN(2021A3PS1155G)

Student Write-up:

PS-I Project Title: N/A

Short Summary of work done: Earlier part of internship included small small tasks that were pretty boring like cross verification of ppt, formatting, etc. Later on we were given tasks like making a short film of a clinic they had and making ppts/videos on sustainability and humans right that were not that bad. We also went on a field visit and the place was quite good. Also we automated a process which was assigned to us to do manually where we had to fill 17,000 rows in xcel sheet

Objectives of the project: Assisting the CSR team in their operations

Tool used: Ms office tools and video editing tools like DaVinci Resolve

Details of Papers/patents:N/A

Brief description of the working environment: Great campus, good corporate culture. The Suzlon Foundation handles CSR for Suzlon and you need to keep your expectations accordingly.

Academic courses relevant to the project: N/A

Learning Outcome: Majorly soft skills, basic video editing

PS-I station: Suzlon Foundation , Pune

Student

Name: DESHMUKH SANMESH MANISH .(2021B5A12576P)

Student Write-up:

PS-I Project Title: N.A.

Short Summary of work done: Involved calling managers, collecting and compiling info on excel, making a short film on their community clinic- filming and editing, conducting manure distribution and health awareness campaigns for employees

Objectives of the project: Making a video on their community clinic, compilation tasks, conducting drives

Tool used: MS. Excel, DaVinci Resolve

Details of Papers/patents:N.A.

Brief description of the working environment: The supervisors and co-workers are very nice and helpful, although, the scope of the work is very narrow and not technically difficult. The most technically difficult task was a new one on making a short film. Mostly, data filling and excel work is done.

Academic courses relevant to the project: TRW, Business Communication

Learning Outcome: People skills, professionalism, exercising creativity for the client

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PS-I station: Tamil Nadu eGovernance Agency , Chennai

Student

Name: YASH AGRAWAL .(2021A1PS2119P)

Student Write-up:

PS-I Project Title: Web Development

Short Summary of work done: Worked on chatbots, NLP and learned web development. By employing tools like HTML, CSS and Java script made a website supporting the requirements of a team in TNEGA

Objectives of the project: To create a webpage design which can be used by the Grains Team for their website.

Tool used: HTML, CSS, Javascript and Figma

Details of Papers/patents: None

Brief description of the working environment: A very healthy working environment. People quite helpful all around. Get exposed to working on a hands- on project was my expectation from PS-1. Learned new technologies, and life skills from PS-1

Academic courses relevant to the project: Courses in HTML, CSS and JavaScript

Learning Outcome: Learned team work, good communication skills, punctuality, discipline, new things like HTML, CSS and JavaScript. Got the industry experience and it helped transform my daily behaviour.

PS-I station: Tamil Nadu eGovernance Agency , Chennai

Student

Name: HIREMATH NIRANJAN VEERANNA .(2021A1PS2583P)

Student Write-up:

PS-I Project Title: Web Development

Short Summary of work done: We were a group of five members in the web development team who worked under Mr. Vivek Hariharan. We were assigned to develop a Tamil Nadu Animal Welfare Board website for the state government which took us about 3-4 weeks to complete.

Objectives of the project: To design and develop a government website

Tool used: HTML, CSS, JavaScript, Figma

Details of Papers/patents:none

Brief description of the working environment: Overall the work environment was very good. All the peers and the office employees were very friendly and helpful

Academic courses relevant to the project: None

Learning Outcome: Learned a lot of technical and professional skills, how to communicate and behave in professional life, team work and much more.

PS-I station: Tamil Nadu eGovernance Agency , Chennai

Student

Name: SWAHA PATI .(2021A3PS0027P)

Student Write-up:

PS-I Project Title: Developing the website for Tamil Nadu Animal Welfare Board

Short Summary of work done: The web development project for the Tamil Nadu Animal Welfare Board (TNAWB) aimed to create an online platform to support and promote animal welfare initiatives of the Board. The project's primary objective was to enhance awareness, facilitate donations, and streamline the process for Registration and recognition of pet shops, animal breeders, etc. The website featured a user-friendly interface, designed with a strong emphasis on user experience. It provided detailed information about the TNAWB's mission, activities, available services as well as details of acts and rules passed by the government. Throughout the project, the development team collaborated closely with the TNAWB officials, seeking their input and feedback to create a tailored solution that met their requirements and objectives. Regular testing and quality checks were performed to ensure a seamless user experience.

Objectives of the project: To Design and develop the website for Tamil Nadu Animal Welfare Board

Tool used: HTML, CSS, Javascript, Bootstrap,

Details of Papers/patents: None

Brief description of the working environment: The work environment in TNEGA, Chennai was dynamic and collaborative. The Team leads were constantly assessing our commitment and willingness to learn. They encouraged us to take ownership of our tasks, collaborate effectively, and deliver high-quality results. Professionalism and punctuality were emphasised, and we were expected to adhere to project timelines and complete assigned tasks.

Academic courses relevant to the project: None

Learning Outcome: Building a website from scratch helped me enhance my technical skills in programming languages, frameworks and other web development tools. I also learned how to effectively communicate and manage clients' demands

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PS-I station: Tamil Nadu eGovernance Agency , Chennai

Student

Name: REHAN NATH(2021A3PS0324P)

Student Write-up:

PS-I Project Title: Data Analytics and API

Short Summary of work done: OBJECTIVE: This project aimed to streamline data acquisition, gain insights through analysis, and create reliable prediction models to inform decision-making processes. METHODOLOGY: The project used multiple different algorithms and python libraries to implement data analysis procedures. Each procedure was rigorously tested with the help of expert data analysts and iterated upon until a satisfactory efficacy achieved while not sacrificing minimum automation. OUTCOMES: This report presents the successful execution of a comprehensive data analysis project that encompassed various domains and techniques. The project involved the development and implementation of advanced models, including a multivariate linear regression model for fertilizer sales prediction and a K Means Clustering algorithm for city performance assessment. Leveraging pre-trained weights, a text recognition system was also implemented. Furthermore, the report highlights the team's proficiency in data cleaning, organization, and exploratory data analysis specifically for the MSME and Agriculture departments. Overall, the project showcases the team's expertise in data analysis and visualization, which contributes to informed decision-making and generates impactful. CONCLUSIONS: Implemented a completely automated data analysis process, with web scraping, EDA, Data Visualization, MySQL and Data Prediction.

Objectives of the project: To make a fully automated data analysis system

Tool used: Pyhton, PowerBI and MySQL

Details of Papers/patents:NA

Brief description of the working environment: The Tamil Nadu CM Dashboard plays a crucial role as an internal governance tool within the state of Tamil Nadu. Its primary objective is to streamline the collection and analysis of data from multiple government departments, facilitating the creation of Key Performance Indicators (KPIs) that help evaluate the state's progress. These KPIs are regularly presented to the Chief Minister, providing a comprehensive overview of the state's performance and enabling informed decision-making.

The significance of this project cannot be understated, as it directly influences the overall performance and development of Tamil Nadu. By effectively gathering data from various department websites, such as the Integrated Fertilizer Management System (iFMS), the Ministry of Micro, Small, and Medium Enterprises (MSMEs), and the Health Management website, the project aims to analyze and extract valuable insights.

To achieve its goals, the project employs a range of methodologies and tools. Data scraping is utilized to extract relevant information from the government department websites, ensuring a comprehensive and up-to-date data collection process. Exploratory data analysis (EDA) techniques are then applied to gain initial insights and identify patterns or trends within the data.

To enhance data quality and accuracy, data preprocessing techniques are employed, which involve cleaning, transforming, and organizing the collected data. This ensures that the subsequent analysis is conducted on reliable and standardized data.
Data visualization techniques are utilized to present the analyzed data in a visually appealing and easily understandable format. By visualizing the data, key trends, correlations, and outliers can be effectively communicated, aiding in decision-making processes.
Additionally, the project incorporates machine learning algorithms to uncover hidden patterns or predictions within the data. These algorithms can identify significant factors, make accurate predictions, and provide valuable recommendations to improve governance and decision-making.
In summary, the Tamil Nadu CM Dashboard project leverages a diverse range of methodologies and tools, including data scraping, exploratory data analysis, data preprocessing, data visualization, machine learning, and optical character recognition. By utilizing these techniques, the project aims to provide comprehensive insights and support evidence-based decision-making processes for the benefit of Tamil Nadu's overall progress and development.

Academic courses relevant to the project: NA

Learning Outcome: Web Scraping, Basic Machine learning, EDA, MySQL and Data Modelling

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PS-I station: Tamil Nadu eGovernance Agency , Chennai

Student

Name: AAKASH GUDURU .(2021A4PS0797H)

Student Write-up:

PS-I Project Title: Web development frontend

Short Summary of work done: Initially we analysed all Govt official websites, we read all GIGW 3.0 norms, we designed 3 animal welfare websites and finally we made the website.

Objectives of the project: To make a official website for Tamil Nadu Animal Welfare board

Tool used: Figma, Bootstrap, HTML,CSS, JavaScript

Details of Papers/patents: No

Brief description of the working environment: The environment is very work friendly. All the teammates and our instructor are very kind. We distributed work among us and completed on time. We learnt all the languages and skills required for making a website.

Academic courses relevant to the project: No

Learning Outcome: We learnt HTM,L CSS, JavaScript, Figma, Bootstrap

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PS-I station: Tamil Nadu eGovernance Agency , Chennai

Student

Name: SANGARAJU VENKATA JAYESH RAJU(2021A7PS0533G)

Student Write-up:

PS-I Project Title: ONLINE TENDERING PORTAL DEVELOPMENT AND MANAGEMENT SYSTEM

Short Summary of work done: Create an efficient online portal for clients to submit company details while applying for tenders. The portal should have a user-friendly interface, secure data storage, and allow easy submission of required information. Ensuring compliance with tender requirements and providing prompt support for a smooth application process is crucial.

Objectives of the project: Our primary objective was to develop a secure online portal that allows clients to log in and generate tender documents. In parallel, we implemented an administrative user system to facilitate user management, including the addition of new users and the a

Tool used: JAVA, JAVA SCRIPT, MYSQL, JSON, SERVLET, JSP

Details of Papers/patents:NO

Brief description of the working environment: Tamil Nadu e-Governance Agency is a government office. It was nice experience working at such a prestigious location. The working environment is very good and and the employees are helpful.
The company expects to serve the people of Tamil Nadu through easing out the processes through online platforms like e-Sevai and GIS.
Since I was a member of Web-Dev(Full Stack) team I learnt about JavaScript, Java Servlets, HTML and CSS. I also learnt about Team Work and Co-ordination.

Academic courses relevant to the project: CS F212, CS F213

Learning Outcome: The implemented algorithm and approach successfully facilitated the user's input
through a form, capturing the required details efficiently. It then leveraged the provided entries to dynamically generate a Request for Quotation (RFQ) type document. The generated RFQ document seamlessly placed the user-entered information at the appropriate sections, ensuring accuracy and completeness. This streamlined process eliminated the need for manual document creation, saving time and minimizing errors. The user-friendly experience and automated RFQ generation significantly enhanced the overall efficiency and effectiveness of the tender application process.

PS-I station: Tamil Nadu eGovernance Agency , Chennai

Student

Name: RR ROSHAN KRISHNA .(2021AAPS2656H)

Student Write-up:

PS-I Project Title: CM Dashboard

Short Summary of work done: The Tamil Nadu CM Dashboard plays a crucial role as an internal governance tool within the state of Tamil Nadu. Its primary objective is to streamline the collection and analysis of data from multiple government departments, facilitating the creation of Key Performance Indicators (KPIs) that help evaluate the state's progress. These KPIs are regularly presented to the Chief Minister, providing a comprehensive overview of the state's performance and enabling informed decision-

making.To achieve its goals, the project employs a range of methodologies and tools. Data scraping is utilized to extract relevant information from the government department websites, ensuring a comprehensive and up-to-date data collection process. Exploratory data analysis (EDA) techniques are then applied to gain initial insights and identify patterns or trends within the data. To enhance data quality and accuracy, data preprocessing techniques are employed, which involve cleaning, transforming, and organizing the collected data. This ensures that the subsequent analysis is conducted on reliable and standardized data. Data visualization techniques are utilized to present the analyzed data in a visually appealing and easily understandable format. By visualizing the data, key trends, correlations, and outliers can be effectively communicated, aiding in decision-making processes. Additionally, the project incorporates machine learning algorithms to uncover hidden patterns or predictions within the data. These algorithms can identify significant factors, make accurate predictions, and provide valuable recommendations to improve governance and decision-making.

Objectives of the project: STREAMLINING THE PROCESS OF DATA ACQUISITION, FORMATTING, VISUALIZATION AND PREDICTION FOR CM DASHBOARD

Tool used: Power BI, Python, Mysql

Details of Papers/patents:nil

Brief description of the working environment: The Tamil Nadu CM Dashboard plays a crucial role as an internal governance tool within the state of Tamil Nadu. Its primary objective is to streamline the collection and analysis of data from multiple government departments, facilitating the creation of Key Performance Indicators (KPIs) that help evaluate the state's progress. These KPIs are regularly presented to the Chief Minister, providing a comprehensive overview of the state's performance and enabling informed decision-making.
The significance of this project cannot be understated, as it directly influences the overall performance and development of Tamil Nadu.

Academic courses relevant to the project: .

Learning Outcome: Data Analytics

PS-I station: Tamil Nadu eGovernance Agency , Chennai

Student

Name: RISHITA KHARE .(2021B2A12754P)

Student Write-up:

PS-I Project Title: Facial recognition system

Short Summary of work done: Built Android signed application for attendance using FRAS and did bug testing on it. Also analyzed the previous version that used raspberry Pi on retina face architecture instead of the new one that works in arcface.

Objectives of the project: To build and deploy facial recognition attendance system for Tamil nade state departments

Tool used: Python, tensorflow, pytorch, Android studio, raspberry Pi

Details of Papers/patents: Just the report

Brief description of the working environment: Working environment was conducive, we were allotted teams based on some skills and interest. The mentor have me daily tasks and regularly checked up on me. I was working on a live project so it was very enriching to do for the first time.

Academic courses relevant to the project: Artificial intelligence

Learning Outcome: Experience in working with various architecture and python libraries using AI/ML

PS-I station: Tamil Nadu eGovernance Agency , Chennai

Student

Name: AYUSH SHARMA .(2021B2A20842P)

Student Write-up:

PS-I Project Title: Animal Welfare Board Website

Short Summary of work done: I along with my team of 4 other students designed and developed a website for the Animal Welfare Board of Tamil Nadu. The website would be used to implement various schemes relating to animal Welfare in the state of Tamil Nadu.

Objectives of the project: To design and develop a website for the animal Welfare board of Tamil Nadu

Tool used: Figma, HTML, CSS, Javascript, Bootstrap

Details of Papers/patents: None

Brief description of the working environment: The working environment at the Tamil Nadu egovernance Agency was great. The employees and the team lead encouraged and helped us throughout the journey. Working environment was peaceful and I observed a increase in my productivity in the office.

Academic courses relevant to the project: Computer Programming

Learning Outcome: I learned how to design and prototype using Figma, I also learned how to code in HTML, CSS and JavaScript. I became more disciplined in the last 2 months and I learned how to work as a team.

PS-I station: Tamil Nadu eGovernance Agency , Chennai

Student

Name: SAI ANKIT SAHOO .(2021B3A82571P)

Student Write-up:

PS-I Project Title: Analyzing Coal Consumption and Stock Trends: Insights from Central Electricity Authority Data

Short Summary of work done: To gain an understanding of the coal consumption dataset, we employed popular libraries including NumPy, Pandas, Seaborn, and Matplotlib. These powerful tools enabled us to preprocess the data, perform exploratory analysis, and visualize the information. Through EDA, we aimed to reveal underlying patterns, relationships, and potential factors influencing coal consumption. To delve deeper into the dataset and make predictions, we turned to Scikit-learn (sklearn), a

widely-used machine learning library. Employing univariate and multivariate linear regression, and forecast outcomes related to coal consumption. By leveraging these machine learning techniques, we sought to identify significant factors and patterns contributing to coal consumption in the Southern region.

Objectives of the project: To collect, clean and visualize data.To learn about different types of Regression analysis and how to analyse a given data set by the use of exploratory data analysis and predictive data analysis.

Tool used: numpy, matplotlib, scikit library, pandas, seasonal exponential smoothing algorithm, powerBI

Details of Papers/patents:NA

Brief description of the working environment: the working environment was very friendly, everyone at the company was supporting and helped us in every step. we expected to work on data and therefore were disheartened to know that they can not share actual government data to work on as this might risk state security. we learnt a lot from ps, both how to work in a team and how to make deadlines meet. also in the process we were exposed to tools that helped us making our mere thoughts into reality. we learnt different python libraries like pandas, numpy, scikit lib etc and different regression techniques like linear, logistic, clustering, seasonal exponential smoothing etc to make our project which was very fun.

Academic courses relevant to the project: NA

Learning Outcome: 1. Proficiency in Diverse Data Visualization Techniques: We acquired a wide range of skills in data visualization techniques, enabling us to effectively present and communicate complex information.
2. Recognition of Domain Knowledge Significance: We acknowledged the importance of domain knowledge as a prerequisite for effective data handling. This understanding enhanced our ability to analyze and interpret data accurately within specific industry contexts.
3. Expertise in Graph Selection for Visualization: We developed expertise in identifying appropriate graph types for specific visualization scenarios. This allowed us to choose the most suitable graphs that accurately represented the data and delivered impactful insights.
4. Exploration of Python Libraries: Through extensive exploration, we discovered the streamlined functionalities offered by Python libraries. Leveraging their built-in functions, we simplified complex data operations, enabling efficient analysis and manipulation.
5. Expansion of Toolkit with Python Libraries: We expanded our toolkit by exploring various Python libraries, equipping ourselves with additional resources to tackle specific tasks more efficiently in future projects.

PS-I station: Tamil Nadu eGovernance Agency , Chennai

Student

Name: SARVAKSH SHUKLA .(2021B5A31378P)

Student Write-up:

PS-I Project Title: Online Tendering Portal Development and Management System

Short Summary of work done: I along with my team designed a tendering portal for generating the RFQ and RFP documents. We designed a login page

Objectives of the project: Our primary objective was to develop a secure online portal that allows clients to log in and generate tender documents. In parallel, we implemented an administrative user system to facilitate user management, including the addition of new users and the a

Tool used: We Used Java, HTML, CSS and Eclipse as IDE

Details of Papers/patents:NO PAPERS

Brief description of the working environment: Tamil Nadu e-Governance Agency is a government office. It was nice experience working at such a prestigious location. The working environment is very good and and the employees are helpful.
The company expects to serve the people of Tamil Nadu through easing out the processes through online platforms like e-Sevai and GIS.
Since I was a member of Web-Dev(Full Stack) team I learnt about JavaScript, Java Servlets, HTML, MySQL and CSS. I also learnt about Team Work and Co-ordination.

Academic courses relevant to the project: Computer Programming, DBMS and OOPS

Learning Outcome: Learned about servlet,JSP,JSON, and the working principle of MVC architecture.
Got a deep learning of how a portal works and how the data of a user is used for various purposes.
Got to learn about the functioning and linking of database using MySQL.
Learned to work in a team and to coordinate with other team members.

PS-I station: Tamil Nadu eGovernance Agency , Chennai

Student

Name: SUMIT KUMAR(2021D2TS2090P)

Student Write-up:

PS-I Project Title: CM Dashboard - ANALYZING COAL CONSUMPTION AND STOCK TRENDS: INSIGHTS FROM CENTRAL ELECTRICITY AUTHORITY DATA

Short Summary of work done: Predictive data analysis has been extensively employed to study coal consumption, offering valuable insights into future trends and impacts. Researchers have utilized historical consumption data and advanced analytical methods to develop predictive models, allowing for informed decision-making and policy formulation. Through time series analysis, researchers have identified patterns and seasonal variations in coal consumption data. These findings have enabled the creation of robust predictive models that forecast future coal demand in various regions and sectors. Such projections help policymakers, industries, and stakeholders anticipate energy needs and plan for sustainable energy transitions. Environmental impact assessments have also been integrated into predictive data analysis. By combining consumption forecasts with emission models, researchers can estimate the potential greenhouse gas emissions associated with coal usage. This aids in understanding the environmental consequences and designing measures to mitigate the negative effects on climate and air quality. Additionally, predictive data analysis has shed light on the economic implications of coal consumption. Researchers have examined how factors like energy prices, economic growth, and technological advancements influence coal demand. These insights assist in anticipating changes in energy markets and crafting strategies for a more resilient and balanced energy mix. Moreover, policy evaluation using predictive models has been instrumental in designing effective interventions. Researchers simulate different policy scenarios, such as carbon pricing or renewable energy incentives, to assess their potential impact on coal consumption. This allows policymakers to make evidence-based decisions to achieve energy and environmental objectives.

Objectives of the project: To collect, clean and visualize data.To learn about different types of Regression analysis and how to analyse a given data set by the use of exploratory data analysis and predictive data analysis.

Tool used: Python: Python is a widely used programming language for data analysis and machine learning. It offers numerous libraries, such as Pandas, NumPy, and Scikit-learn, which provide essential functionalities for data manipulation, statistical analysis, and pr

Details of Papers/patents:nil

Brief description of the working environment: the working environment is very good all the employees are very supportive and helpful. I learned how to work in a team, how to manage work etc.

Academic courses relevant to the project: cs(computer science)

Learning Outcome: 1. Proficiency in Diverse Data Visualization Techniques: We acquired a wide range of skills in data visualization techniques, enabling us to effectively present and communicate complex information.
2. Recognition of Domain Knowledge Significance: We acknowledged the importance of domain knowledge as a prerequisite for effective data handling. This understanding enhanced our ability to analyze and interpret data accurately within specific industry contexts.
3. Expertise in Graph Selection for Visualization: We developed expertise in identifying appropriate graph types for specific visualization scenarios. This allowed us to choose the most suitable graphs that accurately represented the data and delivered impactful insights.
4. Exploration of Python Libraries: Through extensive exploration, we discovered the streamlined functionalities offered by Python libraries. Leveraging their built-in functions, we simplified complex data operations, enabling efficient analysis and manipulation.
5. Expansion of Toolkit with Python Libraries: We expanded our toolkit by exploring various Python libraries, equipping ourselves with additional resources to tackle specific tasks more efficiently in future projects

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PS-I station: Tata Motors Ltd. , Pune

Student

Name: ARYAN RAVEENDRAN MENON .(2021A4PS2771H)

Student Write-up:

PS-I Project Title: INTEGRATED APPROACH FOR LAYOUT OPTIMIZATION, TECHNICAL TROUBLESHOOTING AND DEFECT RESOLUTION

Short Summary of work done: the work highlights the Validation and Modification of New Engine and Chassis Parts, a critical undertaking in maintaining the brand's

reputation for delivering high-performance luxury vehicles. Rigorous testing, collaboration with various teams, and strict quality assurance measures ensure that the new engine and chassis parts meet the highest standards of quality and compatibility.

Objectives of the project: To optimise the layout, to troubleshoot the current problems and over the technical side, we had to work on defects and problems faced by the industry.

Tool used: catia cad

Details of Papers/patents:project report showcases the department's commitment to excellence and continuous improvement in every aspect of luxury vehicle assembly. The projects undertaken reflect a collective pursuit of innovation and a dedication to delivering unparalleled drivin

Brief description of the working environment: the report highlights the Validation and Modification of New Engine and Chassis Parts, a critical undertaking in maintaining the brand's reputation for delivering high-performance luxury vehicles. Rigorous testing, collaboration with various teams, and strict quality assurance measures ensure that the new engine and chassis parts meet the highest standards of quality and compatibility.

Academic courses relevant to the project: amos, mnm, thermodynamics

Learning Outcome: industry experience

PS-I station: Tata Motors Ltd. , Pune

Student

Name: Gaurav Anand Godse(2021A4PS2977G)

Student Write-up:

PS-I Project Title: Industrial Research

Short Summary of work done: I worked on Perceived quality analysis of various Tata Products and also did a lot of research on the foreign car market

Objectives of the project: To aid in the product line planning division of Tata motors for the introduction of new features in Tata Motors' product lineup

Tool used: Powerpoint, Word

Details of Papers/patents:-

Brief description of the working environment: The working environment at the plant is very professional. Everyone speaks in English and everyone is soft spoken and punctual. HR department is very considerate towards the interns and respond promptly.

Academic courses relevant to the project: -

Learning Outcome: How a product planning division in an automotive company works

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PS-I station: Tata Motors Ltd. , Pune

Student

Name: AYUSH MAHAJAN .(2021A8PS1411P)

Student Write-up:

PS-I Project Title: Sigma Project,Launch Quality

Short Summary of work done: The aim of my project was to minimize the issues that arise in production of the newly launched electric vehicle.A detailed analysis of the issues raised was done and a structured method to classify them and therefore solve them step by step was given. In this project we get the idea of how the pre mass production stage of vehicle works and how the numerous issues that arise, are handled systematically. Moreover, a model related to the Advanced Driver Assistance Systems which can be implemented in near future is also worked upon. The idea of the model is that speed breakers should be detected by vehicles to avoid damages caused to the vehicle if the driver fails to slow down. My project also deals with an approach to reduce research and development cost of the new model by carry forwarding all possible parts of the old one.

Objectives of the project: Spotting and Analysing issues raised during production of the new model and finally suggesting solutions for them

Tool used: Autocad, Jupyter, Xcode, Microsoft Powerpoint, Microsoft excel, Microsoft Word.

Details of Papers/patents:N.A.

Brief description of the working environment: The PVBU has a very soothing work culture though the production demands are tending them to adopt some of the efficient and effective managerial concepts from the West and Japan, and a lot of emphasis is laid on the well-proven Japanese techniques. People respect each other a lot and the traditional Indian values of harmony and brotherhood in the plant are respected a lot. The people here are adjusting to the needs of the other people. They work for their co-workers in case of their absence too. The engineers and the supervisors have been seen working too on the shop floor when the production demands are high and the lunch times are also used for the production. In general, the mood on the shop floor and the plant as a whole is that of harmony. The managers are briefed in the start of every shift of the work done the previous day; the managers then direct the engineers and their sub- ordinates about the work to be done for the day. One of the greatest achievements is its totally indigenous technology. Emphasis has always been laid to orient the progress of the company through indigenously developed technology, keeping Indian traditions and values in mind on one hand and incorporating excellent foreign managerial concepts on other. Many of these values and concepts act as a morale booster not only for the company, but also for everybody attached to the company, either directly or indirectly.

Academic courses relevant to the project: Computer Programming, Principles of Economics, microelectronic circuits,Engineering Graphics, control systems, Microprocessing and interfacing, signals and systems.

Learning Outcome: Industry Analysis, On the spot Problem solving, team building and Communication Skills.

PS-I station: Tata Motors Ltd. , Pune

Student

Name: NAVIN VENKAT .(2021B3AA0983P)

Student Write-up:

PS-I Project Title: Warehouse Management

Short Summary of work done: In warehouse management, I optimized operations through a Just-in-Time (JIT) inventory system, reduced excess stock, and cut costs. Utilizing ABC analysis, I prioritized high-value items, improving efficiency. Visual cues enabled seamless communication between teams, and layout optimization enhanced productivity. Monitoring KPIs ensured continuous improvement and a well-organized, cost-effective warehouse.

Objectives of the project: Understand the workflow of the new project warehouse. Propose implementable changes to increase efficiency if possible.

Tool used: Figma, Excel

Details of Papers/patents:I made a report on the work I have done and helped in optimizing the inventory audits.

Brief description of the working environment: I had to work in the factory initially, but later I was shifted to an office. From then on, it was a smooth learning experience. My learning was not limited to just observation at the plant, but also included reading research papers on SCM.

Academic courses relevant to the project: Fundamentals of Finance and Accounting (Section on Inventory Management)

Learning Outcome: Learnt more about the manufacturing sector and supply chain management

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PS-I station: Tata Motors Ltd. , Pune

Student

Name: SHIVANGI PACHAURI .(2021B4A31708P)

Student Write-up:

PS-I Project Title: Optimising the efficiency of kuka krc2 robot controller

Short Summary of work done: I worked on various projects. Understanding the robot controller of KUKA KRC2 Robot Controller and then minimising the defects occurring in stud welding. So basically all the welding is done with the help of robotic arms and for different types of weldings different tools are required. Each tool and robot have it's separate controller and I was asked to study the same.

Objectives of the project: To decrease the errors in stud welding

Tool used: .

Details of Papers/patents:NA

Brief description of the working environment: The working environment was good and everyone was really helping and ready to answer the questions whenever asked.

Academic courses relevant to the project: Electric Machines

Learning Outcome: Understanding the robot controller

PS-I station: Tata Motors Ltd. , Pune

Student

Name: UMANG AGARWAL .(2021B5A40848P)

Student Write-up:

PS-I Project Title: Integrated Approach for Layout Optimization, Technical Troubleshooting, and Defect Resolution

Short Summary of work done: I had initiatives undertaken in the AME & CME Department at Tata Motors - Jaguar Land Rover Pune Plant to enhance efficiency, effectiveness, and safety in luxury vehicle assembly. Projects such as plant layout optimization, pan-roof alignment fixes, tracking parts using GSPAS, gauging air suspension filling equipment, and validation of new engine and chassis parts were executed. By fostering a culture of continuous improvement and implementing advanced manufacturing practices, the department achieved streamlined operations, reduced errors, improved productivity, and enhanced quality assurance. These projects

demonstrate the successful integration of 5S methodologies, DC tooling, and GSPAS, laying a strong foundation for further advancements in luxury vehicle production.

Objectives of the project: Efficiency Enhancement, Launch-related Activities, Equipment and Tooling Enhancement, Performance Testing and Validation

Tool used: AUTOCAD, CATIA

Details of Papers/patents:NONE

Brief description of the working environment: TML PVBU embraces a harmonious work culture while also adopting efficient managerial concepts from the West and Japan to meet production demands. Traditional Indian values of respect, harmony, and brotherhood are deeply ingrained, leading to mutual support and cooperation among team members. Engineers and supervisors willingly work on the shop floor, even during lunch breaks, when production demands are high. The managers receive briefings at each shift, guiding the team for the day. The company's remarkable achievement lies in its indigenous technology, blending Indian traditions and foreign managerial concepts. These values and concepts foster a strong foundation, boosting morale and progress for everyone involved with the company.

Academic courses relevant to the project: NA

Learning Outcome: Efficiency, Lean Manufacturing, Continuous Improvement, Production Optimization, Quality Assurance

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PS-I station: TerraServe Engineers and Contractors Pvt Ltd. , Bengaluru

Student

Name: ROHAN GUPTA(2021A4PS2661G)

Student Write-up:

PS-I Project Title: VACUUM HEAT TREATMENT, PRODUCTIVITY TECHNIQUES AND HAZARD REGULATIONS.

Short Summary of work done: In my final project, I explored the vital role of material properties and their modification techniques in the manufacturing industry. Understanding

how different materials behave and can be optimized enabled me to make informed decisions for better product outcomes. Lean manufacturing techniques were another focus, with an emphasis on waste elimination and process optimization. Implementing lean principles led to improved efficiency, cost savings, and reduced environmental impact. I also delved into the application of Kaizen ideologies, promoting continuous improvement in both professional and personal life. Creating a positive workplace environment fostered enhanced productivity and employee satisfaction. Observing pioneering work on vacuum heat treatment systems unveiled cutting-edge technology with vast potential for the industry. This experience sparked innovation possibilities for future projects. Moreover, I learned about managing hazardous substances responsibly, ensuring a safe and secure working environment. Overall, my final project enriched my understanding of material properties, lean manufacturing, Kaizen ideologies, and safety measures, empowering me to contribute to a more productive and sustainable manufacturing industry.

Objectives of the project: Enhancing Manufacturing Industry Knowledge: Material Properties, Lean Techniques, Kaizen Ideologies, and Hazard Management

Tool used: Vacuum Furnaces, Temperature Controllers, Cooling Systems, PLC (Programmable Logic Controller), HMI (Human-Machine Interface),

Details of Papers/patents: None

Brief description of the working environment: Throughout my final project, I embarked on a transformative journey in the manufacturing industry, seeking to understand the critical role of knowledge in material properties and their modification techniques. This project aimed to shed light on the impact of this knowledge on manufacturing processes and how it could be leveraged to enhance productivity and workplace environments.

One of the major learning outcomes was delving into the intricacies of material properties and their modification. Through rigorous research and expert interactions, I gained valuable insights into how different materials behave under varying conditions and how their properties can be tailored to meet specific manufacturing requirements. Armed with this knowledge, I now possess the expertise to make informed decisions in material selection and optimization, leading to more efficient and high-quality products.

Another key area of focus was lean manufacturing techniques and their importance in the industry. By understanding the principles of lean manufacturing, such as waste elimination and process optimization, I became adept at identifying inefficiencies in production processes. Implementing lean principles not only improved productivity but also contributed to cost savings and reduced environmental impact.

An eye-opening revelation came through the application of Kaizen ideologies. I learned how continuous improvement could be applied not only to manufacturing processes but also to personal life. The principles of Kaizen fostered a positive work environment, promoting employee satisfaction, and consequently, boosting overall productivity and creativity.

Witnessing pioneering work on vacuum heat treatment systems was a remarkable experience during the project. I explored cutting-edge technology that holds immense

potential for the manufacturing industry. Understanding its applications and benefits opened doors to innovation and possibilities for future projects.

Academic courses relevant to the project: BITS F221/BITS C221/BITS C231/BITS C241

Learning Outcome: Major Learning Outcomes:
Material Properties & Modification
Lean Manufacturing & Efficiency
Kaizen Ideologies for Productivity
Positive Workplace Environment
Pioneering Vacuum Heat Treatment
Hazardous Substance Awareness & Management.

PS-I station: TerraServe Engineers and Contractors Pvt Ltd. , Bengaluru

Student

Name: AMITESH SAHA(2021AAPS2340G)

Student Write-up:

PS-I Project Title: Developing a website-based system for displaying Gold, Silver and Repo rates

Short Summary of work done: We had the project of developing a website to display gold silver and repo rates and also display the rate graphs of the same for different times in a period. We used HTML, CSS and JAVASCRIPT to build the entire frontend part of the website(visual and basic functionality) we used chart.js to implement the charts we were asked for. We took the values to be displayed and plotted on the chart form an API; the API is however subscription based but we got a few free calls. We then hosted the website and presented it to the company officials. Aside from this, we also did a SWOT and Competitor Analysis for the company we were working at.

Objectives of the project: To replace existing segmented led display boards used in shops and banks with a more informative and robust system. To provide a single information bank to be displayed in shops and banks and also by general public.

Tool used: S/W: HTML, CSS, Javascript, Chart.JS

Details of Papers/patents:NA

Brief description of the working environment: Since this was an online station, most of the interaction with company officials happened over call. The people were very helpful despite the fact that the company is relatively new in this domain and the people are generally very busy. Our mentor was also extremely helpful, so much that he would ask us to directly call him in case of any doubts rather than texting him. The only expectation that wasn't met was acces to resources like AWS which could have been valuable training for us as well. During the PS term, i learned to interact effectively with industrialists, make proper reports and presentation for product and other software TechStack like HTML, CSS and JAVASCRIPT.

Academic courses relevant to the project: Technical report writing

Learning Outcome: Tech stack learnt : HTML, CSS, Javascript

PS-I station: TerraServe Engineers and Contractors Pvt Ltd. , Bengaluru

Student

Name: SHUBHRANSH SHARMA(2021B2A33016G)

Student Write-up:

PS-I Project Title: WEBSITE DEVELOPMENT FOR DISPLAYING REAL TIME GOLD AND SILVER RATES

Short Summary of work done: We were assigned to make a website whcih can display real time gold and silver rates. Apart from this as evaluatives we were also made to do SWOT and Competitor Analysis

Objectives of the project: Making a user-friendly website written using basic HTML, CSS, and JavaScript for the frontend, NodeJS is used as a development framework.

Tool used: S/w NodeJS,JavaScript,HTML,CSS,ChartJS

Details of Papers/patents: None

Brief description of the working environment: The company provided enough time and freedom to learn and adapt to new technologies and also supervised our work.

Academic courses relevant to the project: CS F111
BITS F112

Learning Outcome: Got to learn about latest website devlopment tools available and their use.

PS-I station: TerraServe Engineers and Contractors Pvt Ltd. , Bengaluru

Student

Name: NAMAN GOYAL(2021B2A33136H)

Student Write-up:

PS-I Project Title: Website Development for Displaying Real-Time Gold and Silver Rates

Short Summary of work done: I created a website to display real-time gold and silver rates. Also implemented a graph which showed the past five days' rate. The front end was made using HTML and CSS, and the site was made responsive using JavaScript to change elements on-screen and to fetch data from MetalsAPI.

Objectives of the project: The project's objective was to create a platform that provides up-to-date information on the prices of precious metals like Gold and Silver, catering to the needs of traders, investors, and individuals interested in monitoring market trends.

Tool used: HTML, CSS, JavaScript, Chartjs , MetalsAPI

Details of Papers/patents: None

Brief description of the working environment: The PS was online, and we had frequent meetings with the mentor and PS faculty to discuss the project and set goals. The company gave us enough time to learn the required technologies and did not place any deadlines regarding the completion of the project. I learned technologies like HTML, CSS, JavaScript and API through my PS project and soft skills like communication and teamwork.

Academic courses relevant to the project: CS F111

Learning Outcome: My major learning was working with APIs and Chartjs. I also used HTML, CSS and JavaScript to create the website.

PS-I station: TerraServe Engineers and Contractors Pvt Ltd. , Bengaluru

Student

Name: PREYAANSH BARARIA .(2021B2A43215H)

Student Write-up:

PS-I Project Title: Study on Various terminologies used in manufacturing sector

Short Summary of work done: We had learnt about the various terminologies like Kaizen, lean manufacturing and various processes of heat treatment in manufacturing sector and understood about the overall working of the organisation by the methods they use in manufacturing.

Objectives of the project: To learn about the various manufacturing terminologies and its application

Tool used: Microsoft Office

Details of Papers/patents: None

Brief description of the working environment: The mentors were helpful and were willing to help at all times.

Academic courses relevant to the project: Not sure as I am yet to start my mechanical degree.

Learning Outcome: Learnt about how manufacturing industries operate.

PS-I station: TerraServe Engineers and Contractors Pvt Ltd. , Bengaluru

Student

Name: VISHWAM BAGRODIA .(2021B3A70940P)

Student Write-up:

PS-I Project Title: Interest Rate Display System

Short Summary of work done: Created a web application for Interest Rate Display System using Python, FLask, HTML, CSS

Objectives of the project: To make a web application for Interest Rate Display System

Tool used: Python, FLask, HTML, CSS

Details of Papers/patents:NA

Brief description of the working environment: Nice and supportive work environment. Mentor is always ready tp help

Academic courses relevant to the project: OOP, CP and other cs courses

Learning Outcome: Python, FLask, HTML, CSS

PS-I station: TerraServe Engineers and Contractors Pvt Ltd. , Bengaluru

Student

Name: MALAVIKA VINAYACHANDRAN NAIR .(2021B4A42920H)

Student Write-up:

PS-I Project Title: Vacuum Heat Treatment, Lean Manufacturing and the Industrial Importance of Kaizen

Short Summary of work done: The mentor at our PS station made it clear that the main aim of our project was to absorb and learn, rather than to carry out a task for the company. So the work we did was learning new concepts and how to apply them in our careers. These concepts included the process of heat treatment, the role of vacuum in heat treatment, lean manufacturing and the 5S concept, the Kaizen concept and the RoHS directive. We also summarised our learnings within two reports and researched the previously mentioned concepts to know how to apply them in real life. We gained this knowledge through regular calls held by our mentor in Lakshmi Vacuum Technologies Pvt Ltd., where we had different discussions and took down notes related to what was presented to us.

Objectives of the project: The objectives of the project were mainly to learn more about the industry we will be moving into, and to experience what a work environment can be like. We were meant to gain more knowledge of lean manufacturing, the concept of Kaizen, vacuum heat treatm

Tool used: No such tools were used for our project.

Details of Papers/patents: No papers/patents of this kind were worked on during our project.

Brief description of the working environment: The working environment of TerraServe Engineers and Contractors was welcoming and open. Dr. Anil Sabaji, the CEO of TerraServe had us work in various companies affiliated w TerraServe based on our engineering branches, and this made the division of projects more efficient. Students in the mechanical branch worked in Lakshmi Vacuum Technologies, whose environment cultivated curiosity and interest. All the companies involved expected us to be eager to learn new things and co-operate with our mentors to get work done. We learnt how to manage our time and resources, and learnt more about the jobs and fields we're exploring in the future.

Academic courses relevant to the project: The relevant academic courses are: Thermodynamics (BITS F111), Technical Report Writing (BITS F112) and Materials Science and Engineering (ME F216).

Learning Outcome: Some major learning outcomes of the project have been that we are now able to understand more about heat treatment, the importance of vacuum in the same, and various industrial concepts. We are now also able to apply this knowledge in other spheres of mechanical engineering. We have learnt how to efficiently search for information and summarise our experiences in a readable report.

PS-I station: TEVA - Digitalization & Automation , Mumbai

Student

Name: SALIL SAMEER GODBOLE .(2021A7PS2004P)

Student Write-up:

PS-I Project Title: Digitalization and Automation

Short Summary of work done: Worked on two projects: 1. Creating an app for automating KPI reporting. Achieved this by using the framework Streamlit in python, and created a KPI reporting app to display the KPIs in an interactive and visually appealing form. 2.Value Stream Mapping. Created a blank value stream map (vsm) for a product, created a data collection template to be sent to the contract manufacturing organization (cmo), followed up with the cmo, and compiled the received data. But, due to a lack of the company's need for a future state vsm for the product, I was told by my mentor to discontinue working on this project.

Objectives of the project: Creating an app for automating KPI Reporting.

Tool used: Streamlit, pandas, plotly

Details of Papers/patents: None

Brief description of the working environment: The working environment was really good, everyone that I interacted with in the company was friendly and welcoming. My mentors supported me throughout the internship. I learned about OpEx, KPIs, flow of processes from manufacturing to dispatch of a drug, creating and deploying an app using Streamlit, value stream mapping.

Academic courses relevant to the project: DSA

Learning Outcome: Learned about OpEx, KPIs, flow of processes from manufacturing to dispatch of a drug, creating and deploying an app using Streamlit

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PS-I station: The Institute Of Minerals And Materials Technology , Bhubaneswar

Student

Name: MANAN TIWARI .(2020A1PS2293H)

Student Write-up:

PS-I Project Title: EFFECT OF TEMPERATURE AND CONCENTRATION OF ORGANIC ADDITIVES ON THE DESILICATION OF BAYER'S LIQUOR

Short Summary of work done: The established process in the aluminium production industry to obtain alumina from aluminium is the Bayer's process, which involves the digestion of bauxite using hot caustic soda solution. This solution, along with the dissolved bauxite, is called Bayer's Liquor, and is subjected to various unit operations to obtain alumina. However, sodium hydroxide reacts with the silica to form insoluble sodium aluminosilicates, called desilication products (DSP), which corrode the equipment used the unit operations the Bayer's liquor is subjected to. This project aims to prevent the formation and precipitation of DSP using organic additives, mainly trimethyl amine.

Objectives of the project: To prevent the formation and precipitation of desilication products in the Bayer's process

Tool used: MS Word, MS Excel

Details of Papers/patents:US Patent 5,314,626

Brief description of the working environment: Being a student from BITS, the expectations are gigantic. However, regularly talking to the mentor and expressing interest in the work allotted goes a long way. The facilities are state-of-the-art, and the staff is kind and helpful. The research that is done here is multi-disciplinary, and I also got to interact with students from other colleges and other domains, which lent me some perspective on approaching the problem statement in a few unorthodox ways.

Academic courses relevant to the project: Heat Transfer, Materials Science & Engineering

Learning Outcome: The project gave me a lot of details about the aluminium production industry, the various processes used throughout the world to refine bauxite and the challenges that the major players in the industry face in the refining processes.

PS-I station: The Institute Of Minerals And Materials Technology , Bhubaneswar

Student

Name: MRINAL RATHO .(2021A1PS2588P)

Student Write-up:

PS-I Project Title: Development of a carbonaceous material using Indian Blackberry seeds waste by Microwave Pyrolysis and KOH Chemical Activation for the adsorption of Carcinogenic Dyes.

Short Summary of work done: The project focuses upon the production of biochar by microwave pyrolysis of Indian blackberry seeds (Syzygium Cumini) at 500 C. Further the pore size was increased by chemical activation through KOH impregnation and two types of dye namely Methylene blue and Congo Red was used for batch adsorption experiments to understand activated carbon interaction with cationic and anionic dyes and study its surface properties.It was found that methylene blue adsorption rate was greater compared to congo red in untreated biochar however KOH activated biochar showed 20-25% improvements in Congo Red dye adsorption as compared to raw biochar sample.The various principles of microwave ,matter interactions was also highlighted and as possible chemical activation mechanism was concluded from literature survey. Further, the future applications were also studied which include carbon nanotubes and microbial fuel cell technologies.

Objectives of the project: Microwave pyrolysis of Jamun Seeds biomass at optimum temperature; Synthesising chemically activated biochar using KOH and washing with HCl followed by distilled water ; Batch Adsorption of Methylene blue dye and Congo Red dye; A comparative analysis betw

Tool used: -

Details of Papers/patents:-

Brief description of the working environment: Working environment was great, with experienced professionals for the necessary help required in any sophisticated equipment handling or procedures for machine operations. Expectations from the company was to get the necessary amount of guidance and support in order to push my own boundaries beyond the curriculum of textbooks and explore a new project by having hands on

experience and practice while working in a chemical laboratory on a project which would solve problems of the wider public through scientific innovations.
My learning during PS : To encourage, be enthusiastic myself and have enormous amount of patience to get the necessary guidance from senior mentors.Have full faith on yourself while being left alone to handle sophisticated equipments in my case the microwave furnace and UV Vis Spectrophotometer. To keep one's eyes and ears open to learning anything and everything going on in a research laboratory. Understanding the mechanisms through literature in order to proceed with meticulous experiments. Discovering something new and different from the current research literature and try to achieve and optimise the elements used for achieving the target.

Academic courses relevant to the project: Engineering Chemistry, General Chemistry, Heat Transfer, Technical Report Writing

Learning Outcome: Sophisticated equipment handling and protocols
Hands on experience in research laboratory and application of advanced reactions, Stoichiometry.
Literature review and report writing skill enhancement

PS-I station: The Institute Of Minerals And Materials Technology , Bhubaneswar

Student

Name: KUSHAGRA TIWARI .(2021A1PS2590P)

Student Write-up:

PS-I Project Title: Modelling of fluidized bed biomass gasification

Short Summary of work done: - Developed differential equations representing biomass behaviour in a bubbling fluidized bed gasifier. Developed various equation subroutines to solve.

Objectives of the project: Modelling of fluidized bed biomass gasification

Tool used: MATLAB pdepe

Details of Papers/patents:N/A

Brief description of the working environment: The working environment at CSIR-IMMT was very encouraging for someone like me inclined towards materials science.

Academic courses relevant to the project: Reaction engineering
Chemical Process Calculations

Learning Outcome: Learnt to develop partial differential equation based models using the pdepe solver on MATLAB.

PS-I station: The Institute Of Minerals And Materials Technology , Bhubaneswar

Student

Name: GARGI MISHRA(2021A1PS2804G)

Student Write-up:

PS-I Project Title: Wastewater Treatment Analysis

Short Summary of work done: I reviewed several research papers to understand the nature of shrimp shells as an adsorbent, the dependence of its activity on various factors and also how to make the most active derivative of shrimp shells based on different impurities. I then proposed a hypothesis that can improve the adsorption of chromium from water. To explore this hypothesis, I then conducted an experiment along with my research scholar. When the results of the experiment arrived, my hypothesis was proved valid.

Objectives of the project: To study the nature of shrimp shells as adsorbing material for purification of water and dependence of adsorption process on various factors.

Tool used: spectrophotometer, oven, pyrolysis reactor

Details of Papers/patents: According to the literature review, chitosan proved to be a better adsorbent than others. Therefore, the paper focuses on increasing chitosan's

efficiency as an adsorbent. First, chitosan is obtained from shrimp shells, and then based on the type of impur

Brief description of the working environment: The working environment was quite friendly. Both my mentor and research scholar helped me a lot in clarifying my doubts and also in performing my experiment. They also considered the ideas suggested by me to enhance the activity of shrimp shells as adsorbing material and even implemented my idea.

Academic courses relevant to the project: Material science, thermodynamics, chemical kinetics and environmental chemistry

Learning Outcome: Biowaste materials like shrimp shells can be used for adsorbing contaminants from wastewater. This is a sustainable way of utilising resources as this not only cleanses the environment but also purifies wastewater.

PS-I station: The Institute Of Minerals And Materials Technology , Bhubaneswar

Student

Name: RAHUL DAS(2021A8PS2060H)

Student Write-up:

PS-I Project Title: Time Series Forecasting using Machine Learning and Image Processing for Improved Size Analysis of Iron Ore Pellets in the Pelletization Industry

Short Summary of work done: The project consists of two separate parts, image processing and machine learning. In part 1, the project focuses on image processing and analysis. It utilizes the OpenCV library for image manipulation and the Tkinter library for creating a graphical user interface. The code begins by allowing the user to select an image file and specify a CSV file for output. The selected image is then loaded and processed. Several image processing techniques are applied, including rescaling, grayscale conversion, Gaussian blur, adaptive thresholding, and Canny edge detection. The resulting image is then subjected to circle detection using the Hough Circle Transform. The detected circles are categorized into small, medium, and large sizes based on their radii. The circles are visualized on the original image, and the number of

circles in each category is displayed. Additionally, the circle data is exported to the specified CSV file. Finally, a histogram is plotted to show the distribution of circle diameters. In part 2, the project shifts focus to data analysis and predictive modeling. It relies on various libraries, including pandas, numpy, matplotlib, scikit-learn, seaborn, statsmodels, and Prophet. The code loads multiple CSV files into pandas DataFrames and concatenates them into a single DataFrame. The data is filtered based on specific time points, and features (x, y, and time) and a target variable (diameter) are prepared. The data is split into training and test sets. Four different regression models (Linear Regression, Random Forest Regressor, Support Vector Regressor, and SARIMAX) are trained and used for prediction. Additionally, two different forecasting models (ARIMA and Prophet) are trained and used for prediction. The models are evaluated using mean squared error. The total number of pellets is calculated based on the predicted values for each model. The predicted values are visualized using histograms, with each model represented by a different color. Overall, the project combines image processing techniques with data analysis and predictive modeling. It aims to detect and categorize circles in images while also predicting the size distribution of pellets based on historical data. The project provides a comprehensive workflow, from image loading and processing to model training and evaluation, culminating in visualizations of the predicted size distributions.

Objectives of the project: objective of my project is to improvise the previously patented Eye-On-Pellet system on real-time video capturing followed by Image Processing and add a layer of Data Analysis on size of pellets through best fitting Machine Learning Algorithms

Tool used: Python, pandas, numpy, matplotlib, opencv, scikit-learn, seaborn, statsmodels, Prophet

Details of Papers/patents: The link to the ieee published research paper contains the previous project on which I am expanding in ML aspect of it. Currently in progress for fruitful publication.
https://ieeexplore.ieee.org/document/9487768

Brief description of the working environment: Worked closely with my guide and his project associates, PhD students and other interns on the project objectives. Got a glimpse of research and development atmosphere. Got to know how R&D labs in India function specially those which are under Ministry of Science and Technology, Govt. of India. Had an exposure on the dynamics of publishing of a research paper or how newly built technologies or industrial software programs are parsed and further patented and how the technological know how is transferred to industrial engineers and technicians/operators on field. Had a relatively free research experience rather than narrow deadlines.

Academic courses relevant to the project: Machine Learning, Digital Image Processing, Python for ML and computer vision

Learning Outcome: Got thorough with various Machine Learning models and large sets of data. Learned to implement libraries such as scikit learn for machine learning and opencv for image processing on python. Used other libraries of python such as pandas, matplotlib, numpy and tensorflow, pytorch to some bit

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PS-I station: The Institute Of Minerals And Materials Technology , Bhubaneswar

Student

Name: AYUSHMAN KAR .(2021AAPS1983H)

Student Write-up:

PS-I Project Title: Network Monitoring System

Short Summary of work done: Created a full stack responsive website

Objectives of the project: To create a full stack web application to be used by the campus for network monitoring and mangement

Tool used: C# and .NET FRAMEWORK

Details of Papers/patents: No patents

Brief description of the working environment: Working environment was giod , everyone was very helpful amd supportive

Academic courses relevant to the project: DBMS

Learning Outcome: Leatnt how to work with C# and .NET framework

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PS-I station: Tienda Advanced materials Pvt. Ltd. , Goa

Student

Name: DEV SANGHRAJKA .(2021B5A43162H)

Student Write-up:

PS-I Project Title: CNC Dust Collector

Short Summary of work done: We were assigned to build a dust collector mechanism for a factory which deals in the production of a material by the name Radome. The production process itself produces a lot of dust inside the factory premises which needs to be collected and disposed off somewhere. For that, we built a centralized vacuum system which consisted of a blower(to remove air and carry the remaining dust particles), a flexible pipe system(which can contract and expand during drilling but still stay in shape), a filter machine(to filter out bigger dust particles from smaller ones) and a bucket with water so that the dust particles settle down. We tried to make the system as cost effective as possible by building it in a centralized manner with only two pipe connections and a single blower.

Objectives of the project: To build a dust collector mechanism for a factory where a lot of dust is produced during the drilling process by the cnc machines which can penetrate the skin and cause diseases

Tool used: AutoCAD, Ansys, Solidworks, Latex

Details of Papers/patents: None

Brief description of the working environment: The working environment was pretty good, we were made to sit in an air-conditioned conference room with all the basic amenities such as a television for presentation purposes, office chairs and a large table to work on, a table fan and refilled water bottles. We were given lunch on a regular basis. The factory premises were well equipped with the latest machinery and state-of-the-art equipments to work with which exceeded my expectations. I learned to use simulation softwares which I'd be using in my 3rd year mechanical courses such as Ansys, Solidworks, AutoCAD etc. I also got hands-on experience with the factory equipment and learnt the production process of Radome in detail.

Academic courses relevant to the project: 3rd year Mechanical courses

Learning Outcome: Learnt about a few simulation softwares such as AutoCAD, Ansys, Solidworks/Fusion 360 and Latex for typesetting the report.

PS-I station: T-Works Foundation - Mechanical/Mechatronics , Hyderabad

Student

Name: SHUSWABHIT SHADANGI .(2021A4PS2405P)

Student Write-up:

PS-I Project Title: Learn the Manufacturing Machines and Develop a Auto Spool Rewinding Device for the 3D Printing Lab

Short Summary of work done: The project was divided into two parts. The first half (pre-midsem) was the learning phase. The post-midsem was planned as a project yet to be discussed, depending on our growth in the first 4 weeks. During the learning phase, I learnt about all the rapid prototyping services (laser-cutting and 3D Printing). I tried to learn a bit from metal shop in the form of plasma welding as well, but it was getting too hectic to learn all three at once. I continued with laser cutting and 3D printing, laser cutting being my main forte. I learned about the workings of the machine in 1.5 weeks and started helping out with orders from both internal team and external public. As a new routing machine (ZUND) was being setup, Kalmesh sir (my mentor, and laser cutting incharge) was busy with handling orders there. I took care of laser cutting orders on my own for quite a while. This continues post midsem as well, along with the project. For project, we discussed and chose a Automatic Filament Rewinding Tool, which will distribute the 3D Printing Filament into different spools depending on the weight. I designed a model, a measuring tool and created an algorithm in the first week after assignment. Then I started with the electronics and figuring out the code for a control panel to make the machine work. All components were made and while testing, some new challenges came which are being tackled (as of 15th July). I am confident it will get resolved and the machine will be a success, making the 3D printing lab more efficient.

Objectives of the project: Learn laser cutting machine; Learn 3D printer working; Produce a device for the 3D printing lab, enabling filament division from one spool to another.

Tool used: Fusion 360, Epilog Laser Cutter, SIL 80W Laser Cutter, Prusa Slicer, Prusa 3D Printer, Bambu Labs 3D Printer

Details of Papers/patents:-

Brief description of the working environment: T-Works as a company was great. I got to interact with people on various designations. Even the people in the topmost positions were very chill to talk with and seek advice from. All the members were extremely supportive.
There are many workshops/events for internal members which keep the team integrated. They partake in exhibitions and public interactions quite frequently. It was a great opportunity to learn not only about the technical stuff but also about the corporate world and managerial things.
The company also has a policy to allow the team members use the machines for their personal projects, which keep motivation and curiosity alive and pushes us to work harder.
I filled this station as my top preference with some expectations being learning more about manufacturing and prototyping while meeting with industry people and make connections. It has surely exceeded my expectations of not only learning about those things on a surface level, but into much more depth. And when it comes to connections, I got to hear so many fascinating stories that I am at awe with the progress I have made personally.

Academic courses relevant to the project: -

Learning Outcome: Learnt the backend of the manufacturing processes; What goes into developing a product; How Prototyping is a must in today's world; How we can rapidly prototype stuff;

PS-I station: T-Works Foundation - Mechanical/Mechatronics , Hyderabad

Student

Name: MANISHA JAKKULA .(2021A4PS3064H)

Student Write-up:

PS-I Project Title: Automatic Filament Rewinder By Weight

Short Summary of work done: Our project began with an initial report defining the purpose. Then Mechanical Systems were completed, including CAD designs and 3D printing. Electronic Assembly faced unexpected issues but was resolved by July 12th. Software preparation and testing followed, and design changes were made. The final version of parts was Rapidly Prototyped, and testing was conducted on July 18th, with some glitches but an overall successful operation. The project is progressing well despite challenges. Moreover, I have learned so much about 3d printing by printing and slicing prototyping orders .I also helped people with 3d printing. I did troubleshooting of 3d printers whenever print fails .

Objectives of the project: To make a rewinder device which speed up the printing process, calculate the amount of filament needed and distribute it among empty spools. • This helps by using multiple printers to work simultaneously. • Simply measure the filament, redistribute it

Tool used: Fusion 350 ,prusaslicer,ideamaker,formlabs tinkercad etc

Details of Papers/patents:NA

Brief description of the working environment: I have had an informative and enriching experience during my
internship at T-Works. As someone with a background in Mechanical Engineering, I
have gained valuable knowledge in various technical aspects.I had this chance to work with corporate project proposals at T-Works, which
provided a fantastic opportunity to enhance my skills. We successfully finished the project as we planned accordingly, and we adapted to any
challenges that came our way. We began by creating a clear project document to understand our goals. Then, we worked on the Mechanical Systems by making CAD designs,
printing parts, and testing everything to make sure it fits together correctly. Even though
we faced unexpected issues with the Electronic Assembly, we managed to solve them and
still completed the project on time. The Software part went smoothly after some improvements, and it worked well. We made design changes to incorporate electronic updates,
and our step-by-step approach helped us put everything together seamlessly. In the end, the machine operated just as we wanted it to during the final testing.

Academic courses relevant to the project: Useful for any SOP ,DOP and LOP .we had a real time experience of building a physical object .

Learning Outcome: Building a physical object using my theoretical knowledge has been an enjoyable experience. I had this chance to work with corporate project proposals at T-Works, which
provided a fantastic opportunity to enhance my skills. I have exposed to electronics arduino programming and all .The most challenging aspect was

troubleshooting during the project, which encouraged me to think critically and be adaptable to changes. Overall I have had an informative and exciting experience during my
internship at T-Works

PS-I station: UltraTech Cement Limited, Maihar Cement Works , Maihar

Student

Name: YASH HRISHIKESH PHADKE(2021A4PS1942G)

Student Write-up:

PS-I Project Title: Detailed study of pyro-processes and it's maintenance

Short Summary of work done: Site visits were done to further my understanding of pyro-processes. These were supplemented with meetings with senior engineers and making of floorplans for the company. There was also the learning responsibility of maintenance.

Objectives of the project: To understand the importance of pyro-processes in cement manufacturing and understand it's working principles and its maintenance

Tool used: AutoCAD

Details of Papers/patents: None

Brief description of the working environment: The learning was good and the working environment was very helpful in providing us with the necessary tools to complete the project. My expectations regarding the project were thoroughly exceeded and was very excited to work on my project.

Academic courses relevant to the project: Heat transfer

Learning Outcome: Deep understanding of machine working and it's maintenance and it's cost impact on the final product. Also a deeper understanding of application of heating systems and it's implementation on a large scale industrial setup.

PS-I station: UltraTech Cement Limited, Maihar Cement Works , Maihar

Student

Name: PURSHOTTAM SRIVASTAVA .(2021A4PS3076H)

Student Write-up:

PS-I Project Title: Raw Mill

Short Summary of work done: Understanding functioning of different departments in cement plant and understanding different processes in raw Mill and it's maintenance

Objectives of the project: Understanding grinding process in cement manufacturing

Tool used: None

Details of Papers/patents: Project Report On Raw Mill

Brief description of the working environment: Working environment was good.

Academic courses relevant to the project: Machenism and machine

Learning Outcome: I understood the mill circuit and feed circuit in mill and basic principles involved in cement grinding process

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PS-I station: UltraTech Cement Limited, Maihar Cement Works , Maihar

Student

Name: VISHAL ANAND .(2021B5A42890H)

Student Write-up:

PS-I Project Title: Thermal power plant.

Short Summary of work done: Did site visit, learned about types of boiler and its accessories and its operation .

Objectives of the project: Learn about operation of thermal power plant

Tool used: None

Details of Papers/patents: None

Brief description of the working environment: Rigid work environment, with helpful personals. The PS station can provide projects for mechanical, chemical and electrical.

Academic courses relevant to the project: Mechanical.

Learning Outcome: We're made familiar with operation involved and department, for manufacturing of cement.

PS-I station: Vikram Cement Works , Neemuch

Student

Name: VEDAANT RATH(2020A8PS1755G)

Student Write-up:

PS-I Project Title: Designing tool for conveying organizational effectiveness

Short Summary of work done: Made videos about processes machines involved in cement manufacturing. I also made plant visits.

Objectives of the project: To make videos regarding processes and machines involved in cement manufacturing

Tool used: Wondershare Filmora

Details of Papers/patents:NA

Brief description of the working environment: It is a pleasant working environment. Expected to work well. Our mentor was flexible.

Academic courses relevant to the project: NA

Learning Outcome: Learning more about cement manufacturing and video editing

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PS-I station: Vikram Cement Works , Neemuch

Student

Name: ABHAY GARG(2021A4PS3050G)

Student Write-up:

PS-I Project Title: Training Calendar

Short Summary of work done: I did a lot of small projects some were learned the power apps and made small applications, building the 2023-24 training calendar, learned sway and made a detailed orientation presentation on it, made staff fill their "my development programme" and made pictorial questionnaires for illiterate Labours, other than this I went for the industrial visits at mines, thermal power plant, many parts of the cement plant were i learned about different machines.

Objectives of the project: 1) Identify training needs 2) Align Training with Organizational Goals 3) optimize resource allocation 4) Ensure Consistency and continuity 5) Enhance Employee Engagement and retention 6) monitor and evaluate training effectiveness

Tool used: Microsoft Sway, Microsoft power apps, canva, Microsoft word, Microsoft excel, Purnaata, Vr device

Details of Papers/patents: No paper or patents

Brief description of the working environment:

Academic courses relevant to the project: No coarse

Learning Outcome: I learned Microsoft Sway and Microsoft Power Apps and i also learnd about the thermal power plant and how hr department works and its importance in the institution.

PS-I station: Vikram Cement Works , Neemuch

Student

Name: KISLAY TIWARI .(2021ABPS2566P)

Student Write-up:

PS-I Project Title: Oe newsletter

Short Summary of work done: It was nice overall, some of the work was really boring and monotonous but the organisation was good though.

Objectives of the project: Making a newsl

Tool used: Microsoft powerpoint

Details of Papers/patents: No

Brief description of the working environment: It was good. Infrastructure is nice but they really have to improve their hospitality

Academic courses relevant to the project: Operations management and some others management courses too.

Learning Outcome: Learning about working of an organisation and some skills on microsoft 365

PS-I station: Wadia Institute Of Himalayan Geology , Dehradun

Student

Name: MANASVITA PENUJURI .(2021A2PS1669H)

Student Write-up:

PS-I Project Title: Calculation of b-values in Uttarakhand

Short Summary of work done: I went through various research papers and watched tutorials in order to compute the b-value in various regions. I plotted the earthquake distribution map on ZMAP and used it to calculate various parameters. My conclusions about the seismic activity in the region were based on this.

Objectives of the project: Using different softwares to find the b-value distribution in Himachal-Haridwar. This can be used to find the seismic activity and earthquake distribution, which is useful information for disaster management and further research.

Tool used: QGIS, ZMAP (a MATLAB software)

Details of Papers/patents:-

Brief description of the working environment: Our internship was online, and we had daily meetings so our mentors could keep track of our projects and help us if we ran into any difficulties. Our projects were mostly self-guided (we got the opportunity to choose our own projects within the domain of the institute). We were expected to stay on track and be punctual, but we never felt stressed out or under pressure. This internship helped us develop our research and presentation skills.

Academic courses relevant to the project: CE F213 (surveying)

Learning Outcome: Learnt about the geology of the Himalayas through research. Learnt how to use various softwares to calculate parameters and plot maps. Learnt how to format a research paper properly.

PS-I station: Wadia Institute Of Himalayan Geology , Dehradun

Student

Name: HASIT GARG .(2021A2PS2382P)

Student Write-up:

PS-I Project Title: vulnerability of j&k region to earthquake damage

Short Summary of work done: i basically found the data of all the earthquakes that occured in the j&k region and then processed the data to find b value and then further sliced the data on various parameters to get insight on it variation with time depth and region

Objectives of the project: to find and analyse b value in the region

Tool used: zmap , matlab , qgis

Details of Papers/patents:na

Brief description of the working environment: it was online ps so we only interacted with Dr Rohella from the company he was very friendly and provided valuable insights to our work the only issue we faced was the regularity of the meets as meets often changed times and the meets often started late then scheduled. other then then that it was a great working experience and i learned a lot about how research happens at the govt. institutes in the country.

Academic courses relevant to the project: soil mechanics

Learning Outcome: i learned about the significance of b value of a earthquake sensitive zone

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PS-I station: Wadia Institute Of Himalayan Geology , Dehradun

Student

Name: BHUMAN PANDITA .(2021A2PS2605P)

Student Write-up:

PS-I Project Title: Flood Warning System using Machine Learning Techniques

Short Summary of work done: In this project, the focus was on flood prediction in the state of Kerala using machine learning algorithms. The objective was to develop accurate models that can forecast the occurrence of floods based on various features such as rainfall data. Six different algorithms, including Logistic Regression, k-Nearest Neighbors, Decision Trees, Random Forests, Naive Bayes, and Support Vector Machines, were implemented and evaluated. The project involved several key steps. Firstly, the dataset was gathered, consisting of relevant features related to flood occurrences. The dataset was then preprocessed, which included cleaning the data, transforming features, and performing feature selection. The data was also split into training and testing sets for model evaluation. Each machine learning algorithm was implemented using R programming language and relevant libraries. The algorithms were trained on the training dataset and their performance was assessed using various evaluation metrics, including accuracy, precision, recall, and F1-score. The results were compared to identify the most suitable algorithm for flood prediction in Kerala. Based on the analysis, it was found that models such as Decision Trees, Random Forests, and Naive Bayes showed promising results with higher accuracy and F1-score, indicating their effectiveness in flood prediction. On the other hand, Logistic Regression exhibited lower accuracy and F1-score, suggesting its limitations in this specific dataset. The findings of this project have significant implications for flood management and disaster preparedness in Kerala. By accurately predicting flood occurrences, stakeholders can implement timely preventive measures and allocate resources efficiently. The project contributes to the advancement of flood prediction techniques using machine learning algorithms. In conclusion, this project highlights the successful implementation and evaluation of various machine learning algorithms for flood prediction in Kerala. The findings provide valuable insights into the performance of different models and can guide future research and practical applications in flood forecasting and mitigation efforts.

Objectives of the project: To evaluate the performance of different machine learning algorithms for flood prediction in Kerala and compare each ML model's accuracy and effectiveness.

Tool used: RStudio, MS Excel, Tableau

Details of Papers/patents:NA

Brief description of the working environment: The working environment was casual, as it was an online internship. We had Google meetings with the coordinator and the FIC every day for 1 hour regarding the work progress, and the topics we would work on were chosen ourselves. My mentor helped me increase my models' accuracy by setting some particular criteria for the models.

Academic courses relevant to the project: Machine Learning

Learning Outcome: -> In the given dataset, Decision Trees, Random Forest, and Naive Bayes all outperformed Logistic Regression regarding accuracy and F1-score predicting flood forecasts.
->Based on numerous indicators, such as rainfall data, machine learning algorithms showed promise in correctly forecasting the occurrence of floods.
->The study of performance indicators provided insights into the usefulness of the models in accurately identifying floods and non-floods.
->Feature scaling and pre-processing approaches had a vital role in increasing the performance of the models.

PS-I station: Wadia Institute Of Himalayan Geology , Dehradun

Student

Name: ADITYA UPADHYAY .(2021A4PS2397H)

Student Write-up:

PS-I Project Title: Structure and Cost Analysis of Earthquake Resistant Materials

Short Summary of work done: We analysed 3 materials and compared it with the traditional material used for making houses,the analysis was in terms of structure integrity and cost.

Objectives of the project: To get better idea of cost vs performance of various materials that can be used for building earthquake resistant houses.

Tool used: STAAD.Pro, Canva,Sheets,Slides,Word, Various construction calculators

Details of Papers/patents: None

Brief description of the working environment: The working environment was good, the mentor was very supportive and helped us in all small problems.

Academic courses relevant to the project: Civil, Designing

Learning Outcome: The key factor is Awareness, the materials that we analysed are well known and the cost also was in budget but the only thing that was missing is Awareness about their use and benefits.

PS-I station: Wadia Institute Of Himalayan Geology , Dehradun

Student

Name: SAMARTH NASNODKAR .(2021A7PS3204H)

Student Write-up:

PS-I Project Title: Dataset Cleaning using Machine Learning

Short Summary of work done: Went through multiple machine learning models and Neural networks to find the best ones to predict anomalies and the best ones to predict correct values for those anomalies.

Objectives of the project: Using advanced machine learning models and neural networks to filter out anomalies in a dataset and predict new values for them.

Tool used: Google Colab, Python, Tensorflow, NumPy, Pandas

Details of Papers/patents: None

Brief description of the working environment: Learning environment was quite good. We were given tasks and were regularly assessed. Help and guidance was available anytime, as we could contact the faculty through whatsapp or email. We were asked to choose topics of our preference and the faculties helped us with anything they could.

Academic courses relevant to the project: Machine Learning

Learning Outcome: Got better at using tools like Google Colab. Also saw how hardware affects the training of machine learning models and the importance of GPUs to train massive models.

PS-I station: Wadia Institute Of Himalayan Geology , Dehradun

Student

Name: ANUSHKA AGGARWAL .(2021B1A32269P)

Student Write-up:

PS-I Project Title: Seismic Phase Detection in Himalayan Region: A comparative study using PhaseNet

Short Summary of work done: Implemented .SAC files from over 800 seismogram readings onto various published ML models and analysed the difficulties and results posed by each implementation.

Objectives of the project: To detect seismic phases from data of Himalayan region in Uttarakhand and do a comparative analysis of past models

Tool used: Python 3.11 Software and multiple packages involved like obspy. Hardware- Lenovo IdeaPad i5 gen

Details of Papers/patents: Data was received by the help of Wadia Institute of Himalayan Geology. Various past published models GIT hub repository was accessed.

Brief description of the working environment: There used to be regular meeting of 1 hour every weekday. There was only one faculty member from the organisation who interacted with 13 of us so the there was a bit of communication gap. The work was interesting but completing a remote research in 45 days can be a little difficult.

Academic courses relevant to the project: Computer Programming. (Machine Learning)
Technical Report Writing

Learning Outcome: Using python language to build ML models.
The danger posed by time interval between P and S wave of seismic data

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PS-I station: Wadia Institute Of Himalayan Geology , Dehradun

Student

Name: PRIYANSHU SHEKHAR SINGH .(2021B5AB2797P)

Student Write-up:

PS-I Project Title: Landslide Susceptibility Mapping

Short Summary of work done: As a student, my experience at the Wadia Institute of Himalayan Geology internship was truly exceptional and unforgettable. From the moment I stepped into the institute, I could sense the atmosphere of scientific curiosity and enthusiasm that permeated every aspect of the institute's activities. The internship provided a well-structured and comprehensive program that allowed me to immerse myself in the fascinating world of Himalayan geology. Under the guidance of experienced researchers and mentors, I was exposed to cutting-edge research, advanced laboratory techniques, and insightful fieldwork. The practical knowledge and skills I gained during this period were invaluable and complemented my academic studies perfectly. One of the most rewarding aspects of the internship was the opportunity to collaborate with other like-minded students and professionals from diverse backgrounds. The exchange of ideas and experiences not only enriched my understanding of geology but also broadened my perspective on various geological phenomena. The institute's location in the picturesque Himalayan region further added to the allure of the experience. Field trips to various geological sites allowed me to witness the awe-inspiring landscapes firsthand and observe geological processes in action. The breathtaking vistas of snow-capped peaks, serene valleys, and ancient rock formations served as a constant reminder of the significance and relevance of our research. Moreover, the supportive and encouraging environment at the institute fostered a sense of belonging and camaraderie among all interns. Regular seminars, workshops, and interactive sessions enhanced our learning and encouraged us to think critically and independently. Apart from the academic and research aspects, the institute also emphasized the importance of maintaining a work-life balance. Social events, cultural gatherings, and adventure activities allowed us to unwind and forge lasting friendships with fellow interns. In conclusion, my experience at the Wadia Institute of Himalayan Geology internship was truly great. It not only deepened my passion for geology but also instilled in me a greater appreciation for the Himalayan region and its geological significance. The knowledge, skills, and memories I gained from this internship will undoubtedly shape my academic and professional journey, and I am immensely grateful for the opportunity to be a part of such an enriching and inspiring experience.

Objectives of the project: Landslide Susceptibility Mapping

Tool used: QGIS, Machine Learning

Details of Papers/patents:no

Brief description of the working environment: It was good

Academic courses relevant to the project: None

Learning Outcome: Landslide Susceptibility Mapping

PS-I station: Western Irrigaation System Pvt. Ltd. - Core , Rajkot

Student

Name: SANKHARVA SHYAM RAMESHBHAI(2021A1PS2563G)

Student Write-up:

PS-I Project Title: Research/Testing of Pipes

Short Summary of work done: I had researched about the company from their website, furthermore we were given tasks weekly basis, know about the types of pipes and irrigation systems used in India, make reports/ppts on all these research done and to be submitted by the each weekend. Further we were given videos to watch on testing techniques that are used to test the quality and various other parameters of these pipes and then manufacturing process of HDPE and PVC pipes. In the end we had to prepare and submit a full assignment given the the company poc.

Objectives of the project: To research on types of HDPE pipes, irrigation systems, various testing methods of these pipes and how such pipes are made.

Tool used: Canva for making reports and ppts.

Details of Papers/patents: None

Brief description of the working environment: Working environment was quite flexible and the faculty Coordinator was very supportive of us. Company also was regularly updated on giving tasks. Overall it was a pleasant experience working with PS1 and the company itself.

Academic courses relevant to the project: None that come to my mind

Learning Outcome: I learnt several types of HDPE pipes that the company produces, irrigation systems used in the Indian subcontinent , various testing methods of these pipes and how HDPE & PVC pipes are made.

PS-I station: Zuari Industries Ltd (Paradeep Phosphates Ltd) , Goa

Student

Name: MISHIKA NARULA .(2021A1PS1327P)

Student Write-up:

PS-I Project Title: Waste Heat recovery from stripper effluent

Short Summary of work done: My project aimed to address the challenge of reducing the load on the compressor in the ammonia plant by implementing a sequential approach of designing a Vapour Absorption Machine (VAM) and a heat exchanger. By utilizing waste heat from the urea plant's stripping column, the VAM will produce chilled water, which will then be used to cool the ammonia using a heat exchanger along with a compressor. This integrated solution will improve energy efficiency, optimize the cooling process, and ultimately relieve the workload on the compressor, enhancing the overall performance of the ammonia plant. For this project I designed(by performing calculations) a Vapour Absorption Machine and a Heat Exchanger for the plant

Objectives of the project: Design a Vapour Absorption Machine (VAM) ,Design a Heat Exchanger, Reduce Load on the Compressor, Enhance Energy Efficiency, Optimize Cooling Process

Tool used: excel

Details of Papers/patents:none

Brief description of the working environment:
The PS1 course, has provided me with invaluable takeaways that have enriched both my academic and professional journeys.

Having studied heat transfer, thermodynamics, mass transfer, and material science, I was able to apply these theoretical principles during my project. This hands-on experience broadened and solidified my understanding of these topics. I saw how these concepts are applied in real-world scenarios, which improved my problem-solving and critical-thinking skills.

Aside from the technical aspects, the PS1 course gave me insight into the inner workings of a company. I became acquainted with various departments and developed a better understanding of organisational structures. I also delved into finance and supply chain management with my peers, broadening my knowledge of how different aspects of a company function and intertwine.

Furthermore, the course assisted me in improving my communication skills. I learned to effectively communicate my ideas, actively listen, and collaborate with diverse individuals through interactions with colleagues, supervisors, and team members. Furthermore, I gained a better understanding of professional ethics and the significance of upholding ethical standards in the workplace.

Overall, the PS1 course was a transformative experience that bridged the theoretical knowledge and practical application gap. It has provided me with a diverse skill set that includes technical proficiency, a better understanding of organisational dynamics, improved communication abilities, and a solid ethical foundation. These takeaways will undoubtedly shape my future endeavours and contribute to my development as a chemical engineering professional.

Academic courses relevant to the project: All chemical CDCs

Learning Outcome: Information about VAM and designing of a heat exchanger

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PS-I station: Zuari Industries Ltd (Paradeep Phosphates Ltd) , Goa

Student

Name: SACHIKA LALA .(2021A1PS2375P)

Student Write-up:

PS-I Project Title: Optimization of Supply Chain Management in Paradeep Phoshphates Limited

Short Summary of work done: In our project, our primary responsibility was to develop the production plan for the next two quarters. Our mentor shared previous data with us, specifically the consumption ratios and production plans of various grades of NPK, which served as crucial input for our analysis. We visited the plant facilities, where we gained firsthand insights into the WAS System and the storage capacities of various raw materials. These plant visits provided us with a deeper understanding of the operational processes and allowed us to witness the intricate logistics involved in the supply chain management at Paradeep Phosphates Limited. We also created an estimation of the raw material demand.

Objectives of the project: To develop the production plan of the grades of NPK for the next two quarters and the estimation of the raw material demand

Tool used: MS Excel

Details of Papers/patents:-

Brief description of the working environment: I was very excited to start my Practice School program all the more because it was an offline station, especially after reading about my seniors' experiences.
I was very interested in understanding the company culture and what the work system is like. I also expected to work closely with the industry mentors and see what their work looked like on a daily basis. I looked at it as a great opportunity to network with professionals and learn their problem-solving skills and approaches. I was looking forward to gaining more exposure to the industry and real-world applications.
The Practice School - 1 program has been extremely beneficial to me in multiple ways. Firstly, it has helped me understand the ways and working of organizations and the kind of culture they wish to instill in their employees and their work ethic. It has been extremely beneficial to understand how to work and maintain yourself in this sort of
environment. Understanding the hierarchy of the organizational structure and knowing whom to approach for what exactly has been another key takeaway. Secondly, I've been able to apply my theoretical knowledge in the field of Chemical Engineering practically. Last, but not the least, I've been introduced to a new domain of Supply Chain Management altogether. Understanding the workflow of a plant that is this old while reading about methods followed in other industries and trying to apply them here have been the major takeaways with respect to the Supply Chain.
The program has also enhanced my myopic vision of the working environment and other problem-solving approaches. I am now able to better critically analyze the problems and find better solutions to them.

Academic courses relevant to the project: - Supply Chain Management
- Principles of Management
- Principles of Economics
- Chemical Process Calculations

Learning Outcome: - Learnt the already existing model of the supply chain system here and devised new strategies and calculations for the next quarter of the company.
- The knowledge and experience gained through this project have enhanced my understanding of supply chain management and its importance in the fertilizer manufacturing industry.
- The Lean Principles of SCM.
- All in all, the Practice School - I program has taught me how to approach the industry to experientially learn from it and structure my learning by documenting it.

PS-I station: Zuari Industries Ltd (Paradeep Phosphates Ltd) , Goa

Student

Name: Vaibhav Biresh Prasad(2021B5AB2271P)

Student Write-up:

PS-I Project Title: Optimisation of supply chain management

Short Summary of work done: I was assigned to Mr Pankaj V. Naik who was the DGM of Phosphates and off-site and under him we visited the storage plants of the company and then we had to make production plan for coming 6 months .After the production plan was made we were asked to make the consumption plans for the raw materials used for the coming 6 months according to our plan. After completion of the consumption plan we were asked to make plans for raw material orders accordingly so that the raw material are always available and keeping in mind to reduce storage cost and lagistics cost.

Objectives of the project: Make production plan, consumption of raw materials according to production plan and then place raw material orders according according to requirements in optimized form

Tool used: Ms excel

Details of Papers/patents: No patents

Brief description of the working environment: It was a healthy working environment and all the staff were collaborative and helpful and measures of safety to create a safe working environment.

Expected to learn about working of a huge scale industry and learn about supply chain management.
Expectations were met.

Academic courses relevant to the project: M-1, Thermodynamics, General Chemistry , Technical Report writing,Manufacturing processes

Learning Outcome: Learnt about applications of optimisation and got in depth knowledge and application of supply chain management

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